

# HSE Service API Reference Manual

For S32K3X2  
v0.2.55.0



Revision 801e201f4e  
Jan 2025

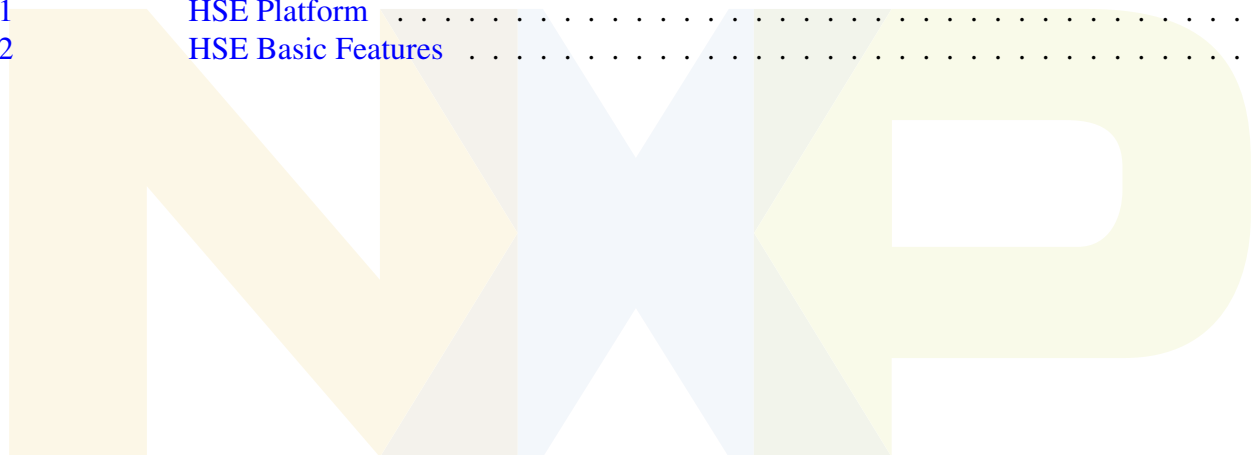
**NXP Semiconductors**



# Contents

|          |                                                        |            |
|----------|--------------------------------------------------------|------------|
| <b>1</b> | <b>Introduction</b>                                    | <b>1</b>   |
| 1.1      | HSE Messages Guidelines                                | 1          |
| <b>2</b> | <b>Host Interface</b>                                  | <b>2</b>   |
| 2.1      | About the Host Interface                               | 2          |
| 2.2      | HSE Service Descriptor                                 | 2          |
| 2.3      | HSE GPR Status                                         | 15         |
| 2.4      | HSE Service Responses                                  | 17         |
| 2.5      | HSE Errors                                             | 23         |
| 2.6      | Host Events To HSE                                     | 25         |
| 2.7      | HSE Status                                             | 25         |
| <b>3</b> | <b>Administration Services</b>                         | <b>30</b>  |
| 3.1      | HSE Utility Services                                   | 30         |
| 3.2      | HSE Set/Get Attribute Services                         | 33         |
| 3.3      | HSE System Authorization Services                      | 62         |
| 3.4      | HSE Boot Images Signature Generate/Verify              | 68         |
| 3.5      | HSE Firmware Update Service                            | 72         |
| 3.6      | Secure_BAF Firmware update service                     | 73         |
| <b>4</b> | <b>Cryptographic Services</b>                          | <b>75</b>  |
| 4.1      | HSE MAC Service                                        | 75         |
| 4.2      | HSE Symmetric Cipher Service                           | 78         |
| 4.3      | HSE HASH Service                                       | 80         |
| 4.4      | HSE AEAD Service                                       | 83         |
| 4.5      | HSE Digital Signature Service                          | 86         |
| 4.6      | HSE RSA Cipher Service                                 | 89         |
| <b>5</b> | <b>Key Management Services</b>                         | <b>91</b>  |
| 5.1      | HSE Key Management Common Types                        | 91         |
| 5.2      | HSE Key Management Utility Services                    | 118        |
| 5.3      | HSE MSC Key Management Services                        | 127        |
| 5.4      | HSE Key Import/Export Services                         | 131        |
| 5.5      | HSE Key Generate service                               | 144        |
| 5.6      | HSE Key Derivation Service                             | 149        |
| <b>6</b> | <b>Boot and Memory Verification Services</b>           | <b>173</b> |
| 6.1      | HSE Core Reset And Secure Memory Region (SMR) Services | 173        |

| Section number | Title                                                   | Page       |
|----------------|---------------------------------------------------------|------------|
| <b>7</b>       | <b>SHE Specification</b> . . . . .                      | <b>194</b> |
| 7.1            | HSE SHE Specification Services . . . . .                | 194        |
| <b>8</b>       | <b>Monotonic Counters Services</b> . . . . .            | <b>197</b> |
| 8.1            | HSE Monotonic Counters . . . . .                        | 197        |
| <b>9</b>       | <b>Random Number Generator Services</b> . . . . .       | <b>199</b> |
| 9.1            | HSE Random Number Generator services . . . . .          | 199        |
| <b>10</b>      | <b>Network Protocol Acceleration Services</b> . . . . . | <b>201</b> |
| <b>11</b>      | <b>Common Types and Definitions</b> . . . . .           | <b>201</b> |
| 11.1           | HSE Common Types . . . . .                              | 201        |
| 11.2           | HSE Defines . . . . .                                   | 226        |
| <b>12</b>      | <b>Features Implementation</b> . . . . .                | <b>237</b> |
| 12.1           | HSE Platform . . . . .                                  | 237        |
| 12.2           | HSE Basic Features . . . . .                            | 237        |



# 1 Introduction

This document describes the parameters of the NXP Native services and is an addendum to the HSE Firmware Reference Manual (available at NXP Docstore) which contains details on how to install, configure and use the HSE subsystem.

## 1.1 HSE Messages Guidelines

- The address parameters can be passed as 32 or 40 bit addresses, depending on HSE firmware support (if 64bit addressing is enabled and if the device supports 40 bit addressing mode)
- A service request can be accessible in one-pass or streaming (SUF) mode. In case of streaming mode "three" steps (calls) are needed: START, UPDATE, FINISH. Note that for each streaming step (START, UPDATE or FINISH), some of the parameters are mandatory or optional.
- The streaming mode operation begins with the START step using a specific MU Instance ID and stream ID. The UPDATES and FINISH steps shall be sent on the same MU Instance ID and stream ID as the START step; otherwise an error will be signaled.
- If a streaming operation produces an error, the stream is to be considered invalid; a stream can always be reset by a new start command.

## 2 Host Interface

### 2.1 About the Host Interface

This section contains information on the available services accepted by the firmware.

The firmware accepts commands in the form of service descriptors. Data types and values relevant for the services are also listed. One-time settings or information about the state of the system are accessible via attributes. The attributes are also listed below.

### 2.2 HSE Service Descriptor

#### Data Structures

- struct [hseSrvDescriptor\\_t](#)
- union [hseSrvDescriptor\\_t.hseSrv](#)

#### Macros

| Type: <a href="#">hseSrvId_t</a>                             |                               |              |
|--------------------------------------------------------------|-------------------------------|--------------|
| Name                                                         |                               | Value        |
| <a href="#">HSE_SRV_ID_SET_ATTR</a>                          | <a href="#">HSE_SRV_VER_0</a> | 0x00000001UL |
| <a href="#">HSE_SRV_ID_GET_ATTR</a>                          | <a href="#">HSE_SRV_VER_0</a> | 0x00A50002UL |
| <a href="#">HSE_SRV_ID_CANCEL</a>                            | <a href="#">HSE_SRV_VER_0</a> | 0x00A50004UL |
| <a href="#">HSE_SRV_ID_FIRMWARE_UPDATE</a>                   | <a href="#">HSE_SRV_VER_0</a> | 0x00000005UL |
| <a href="#">HSE_SRV_ID_SYS_AUTH_REQ</a>                      | <a href="#">HSE_SRV_VER_0</a> | 0x00000006UL |
| <a href="#">HSE_SRV_ID_SYS_AUTH_RESP</a>                     | <a href="#">HSE_SRV_VER_0</a> | 0x00000007UL |
| <a href="#">HSE_SRV_ID_BOOT_DATA_IMAGE_SIGN</a>              | <a href="#">HSE_SRV_VER_0</a> | 0x00000008UL |
| <a href="#">HSE_SRV_ID_BOOT_DATA_IMAGE_VERIFY</a>            | <a href="#">HSE_SRV_VER_0</a> | 0x00000009UL |
| <a href="#">HSE_SRV_ID_IMPORT_EXPORT_STREAM_CTX</a>          | <a href="#">HSE_SRV_VER_0</a> | 0x00A5000AUL |
| <a href="#">HSE_SRV_ID_ERASE_HSE_NVM_DATA</a>                | <a href="#">HSE_SRV_VER_0</a> | 0x00000050UL |
| <a href="#">HSE_SRV_ID_ERASE_FW</a>                          | <a href="#">HSE_SRV_VER_0</a> | 0x00000057UL |
| <a href="#">HSE_SRV_ID_ACTIVATE_PASSIVE_BLOCK</a>            | <a href="#">HSE_SRV_VER_0</a> | 0x00000051UL |
| <a href="#">HSE_SRV_ID_SBAF_UPDATE</a>                       | <a href="#">HSE_SRV_VER_0</a> | 0x00000053UL |
| <a href="#">HSE_SRV_ID_FW_INTEGRITY_CHECK</a>                | <a href="#">HSE_SRV_VER_0</a> | 0x00000054UL |
| <a href="#">HSE_SRV_ID_PUBLISH_NVM_KEYSTORE_RAM_TO_FLASH</a> | <a href="#">HSE_SRV_VER_0</a> | 0x00000055UL |
| <a href="#">HSE_SRV_ID_CONFIG_COUNTER</a>                    | <a href="#">HSE_SRV_VER_0</a> | 0x00000052UL |
| <a href="#">HSE_SRV_ID_LOAD_ECC_CURVE</a>                    | <a href="#">HSE_SRV_VER_0</a> | 0x00000100UL |
| <a href="#">HSE_SRV_ID_FORMAT_KEY_CATALOGS</a>               | <a href="#">HSE_SRV_VER_0</a> | 0x00000101UL |
| <a href="#">HSE_SRV_ID_ERASE_KEY</a>                         | <a href="#">HSE_SRV_VER_0</a> | 0x00000102UL |
| <a href="#">HSE_SRV_ID_GET_KEY_INFO</a>                      | <a href="#">HSE_SRV_VER_0</a> | 0x00A50103UL |
| <a href="#">HSE_SRV_ID_IMPORT_KEY</a>                        | <a href="#">HSE_SRV_VER_0</a> | 0x00000104UL |
| <a href="#">HSE_SRV_ID_EXPORT_KEY</a>                        | <a href="#">HSE_SRV_VER_0</a> | 0x00000105UL |

|                                     |               |              |
|-------------------------------------|---------------|--------------|
| HSE_SRV_ID_KEY_GENERATE             | HSE_SRV_VER_0 | 0x00000106UL |
| HSE_SRV_ID_DH_COMPUTE_SHARED_SECRET | HSE_SRV_VER_0 | 0x00000107UL |
| HSE_SRV_ID_KEY_DERIVE               | HSE_SRV_VER_0 | 0x00000108UL |
| HSE_SRV_ID_KEY_DERIVE_COPY          | HSE_SRV_VER_0 | 0x00000109UL |
| HSE_SRV_ID_KEY_VERIFY               | HSE_SRV_VER_0 | 0x0000010BUL |
| HSE_SRV_ID_SHE_LOAD_KEY             | HSE_SRV_VER_0 | 0x0000A101UL |
| HSE_SRV_ID_SHE_LOAD_PLAIN_KEY       | HSE_SRV_VER_0 | 0x0000A102UL |
| HSE_SRV_ID_SHE_EXPORT_RAM_KEY       | HSE_SRV_VER_0 | 0x0000A103UL |
| HSE_SRV_ID_SHE_GET_ID               | HSE_SRV_VER_0 | 0x0000A104UL |
| HSE_SRV_ID_SHE_BOOT_OK              | HSE_SRV_VER_0 | 0x0000A105UL |
| HSE_SRV_ID_SHE_BOOT_FAILURE         | HSE_SRV_VER_0 | 0x0000A106UL |
| HSE_SRV_ID_HASH                     | HSE_SRV_VER_0 | 0x00A50200UL |
| HSE_SRV_ID_MAC                      | HSE_SRV_VER_0 | 0x00A50201UL |
| HSE_SRV_ID_FAST_CMAC                | HSE_SRV_VER_0 | 0x00A50202UL |
| HSE_SRV_ID_SYM_CIPHER               | HSE_SRV_VER_0 | 0x00A50203UL |
| HSE_SRV_ID_AEAD                     | HSE_SRV_VER_0 | 0x00A50204UL |
| HSE_SRV_ID_SIGN                     | HSE_SRV_VER_0 | 0x00000206UL |
| HSE_SRV_ID_RSA_CIPHER               | HSE_SRV_VER_0 | 0x00000207UL |
| HSE_SRV_ID_GET_RANDOM_NUM           | HSE_SRV_VER_0 | 0x00000300UL |
| HSE_SRV_ID_INCREMENT_COUNTER        | HSE_SRV_VER_0 | 0x00A50400UL |
| HSE_SRV_ID_READ_COUNTER             | HSE_SRV_VER_0 | 0x00A50401UL |
| HSE_SRV_ID_SMR_ENTRY_INSTALL        | HSE_SRV_VER_0 | 0x00000501UL |
| HSE_SRV_ID_SMR_VERIFY               | HSE_SRV_VER_0 | 0x00000502UL |
| HSE_SRV_ID_CORE_RESET_ENTRY_INSTALL | HSE_SRV_VER_0 | 0x00000503UL |
| HSE_SRV_ID_ON_DEMAND_CORE_RESET     | HSE_SRV_VER_0 | 0x00000504UL |
| HSE_SRV_ID_SMR_ENTRY_ERASE          | HSE_SRV_VER_0 | 0x00000505UL |
| HSE_SRV_ID_CORE_RESET_ENTRY_ERASE   | HSE_SRV_VER_0 | 0x00000506UL |

## HSE service descriptor details

Each service is identified by a unique ID (called service ID). Each service ID identifies a service from the `hseSrvDescriptor_t::hseSrv` union. The service ID contains 4 bytes that specify the following:

- byte[0]: service index (0..255)
- byte[1]: service class index (0..255)(see more details below)
- byte[2]: 0x00 - service can be canceled; 0xA5 - service can not be canceled
- byte[3]: service version (0..255)

The following service classes are defined:

- Administrative services (e.g set/get an HSE attribute, self-test, cancel service etc.)
- Key management services (e.g key generation, Diffie-Hellman shared secret computation, import/export key etc.)
- Crypto services (e.g. HASH, MAC generate/verify, encryption/decryption, signature generate/verify)
- Random number

## Host Interface

- Monotonic counters
- Secure boot and memory checking services (Secure Memory Regions (SMR) and Core reset(CR) services)
- Network Crypto services (IPsec ).

### Note

- The services guarded by HSE\_SPT\_FLASHLESS\_DEV macro are available only for HSE\_H/M (flashless devices).
- The services guarded by HSE\_SPT\_INTERNAL\_FLASH\_DEV macro are available only for HSE\_B (devices with internal flash).

## Data Structure Documentation

### struct hseSrvDescriptor\_t

#### Data Fields

| Type                                            | Name        | Description                                                    |
|-------------------------------------------------|-------------|----------------------------------------------------------------|
| <a href="#">hseSrvId_t</a>                      | srvId       | The service ID of the HSE message.                             |
| <a href="#">hseSrvMetaData_t</a>                | srvMetaData | The service metadata (e.g. priority)                           |
| <a href="#">union hseSrvDescriptor_t.hseSrv</a> | hseSrv      | The service ID will identify a service in the following union. |

### union hseSrvDescriptor\_t.hseSrv

The service ID will identify a service in the following union.

#### Data Fields

| Type                                         | Name                 | Description                                                              |
|----------------------------------------------|----------------------|--------------------------------------------------------------------------|
| <a href="#">hseSetAttrSrv_t</a>              | setAttrReq           | Request to set a HSE attribute (note that some attributes are read only) |
| <a href="#">hseGetAttrSrv_t</a>              | getAttrReq           | Request to get a HSE attribute.                                          |
| <a href="#">hseCancelSrv_t</a>               | cancelSrvReq         | Request to cancel a one-pass or streaming service on a specific channel. |
| <a href="#">hseFirmwareUpdateSrv_t</a>       | firmwareUpdateReq    | Request to HSE firmware update.                                          |
| <a href="#">hseSysAuthorizationReqSrv_t</a>  | sysAuthorizationReq  | Perform an SYS Authorization Request.                                    |
| <a href="#">hseSysAuthorizationRespSrv_t</a> | sysAuthorizationResp | Send the SYS Authorization Response.                                     |



## Data Fields

| Type                                                 | Name                            | Description                                                                                                                                              |
|------------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseBootDataImageSignSrv_t</a>            | bootDataImageSignReq            | Request to generate the Signature for Boot Data images (e.g. for HSE_H/M, IVT/DCD/ST/LPDDR4(ZSE devices)/AppBSB image; for HSE_B, IVT/XRDC/AppBSB image) |
| <a href="#">hseBootDataImageVerifySrv_t</a>          | bootDataImageSigVerifyReq       | Request to verify the Signature for Boot Data images (e.g. for HSE_H/M, IVT/DCD/ST/LPDDR4(ZSE devices)/AppBSB image; for HSE_B, IVT/XRDC/AppBSB image)   |
| <a href="#">hseImportExportStreamCtxSrv_t</a>        | importExportStreamCtx           | Request to import/export a streaming context.                                                                                                            |
| <a href="#">hseEraseNvmDataSrv_t</a>                 | eraseNvmDataReq                 | Request to reset HSE data flash. Only allowed in CUST_DEL LC.                                                                                            |
| <a href="#">hsePublishNvmKeystoreRamToFlashSrv_t</a> | publishNvmKeystoreRamToFlashReq | Request to publish the NVM keystore in RAM to NVM SYS-IMAGE.                                                                                             |
| <a href="#">hseSbafUpdateSrv_t</a>                   | sbafUpdateReq                   | Request to SBAF firmware update.                                                                                                                         |
| <a href="#">hseEraseFwSrv_t</a>                      | eraseFwReq                      | Request to erase sys-img, backup, current firmware. Only allowed in CUST_DEL LC.                                                                         |
| <a href="#">hseLoadEccCurveSrv_t</a>                 | loadEccCurveReq                 | Request to load an ECC curve.                                                                                                                            |
| <a href="#">hseFormatKeyCatalogsSrv_t</a>            | formatKeyCatalogsReq            | Format the key catalogs.                                                                                                                                 |
| <a href="#">hseEraseKeySrv_t</a>                     | eraseKeyReq                     | Request to erase NVM/RAM key(s).                                                                                                                         |
| <a href="#">hseGetKeyInfoSrv_t</a>                   | getKeyInfoReq                   | Request to get key information (flags)                                                                                                                   |
| <a href="#">hseImportKeySrv_t</a>                    | importKeyReq                    | Request to import a key.                                                                                                                                 |
| <a href="#">hseExportKeySrv_t</a>                    | exportKeyReq                    | Request to export a key.                                                                                                                                 |
| <a href="#">hseKeyVerifySrv_t</a>                    | verifyKeyReq                    | Request to verify a key.                                                                                                                                 |
| <a href="#">hseKeyGenerateSrv_t</a>                  | keyGenReq                       | Request to generate a key (e.g. sym random key, rsa key pair etc.) .                                                                                     |
| <a href="#">hseDHComputeSharedSecretSrv_t</a>        | dhComputeSecretReq              | Request a ECC Diffie-Hellman Compute shared secret.                                                                                                      |
| <a href="#">hseKeyDeriveSrv_t</a>                    | keyDeriveReq                    | Request key derivation function.                                                                                                                         |
| <a href="#">hseKeyDeriveCopyKeySrv_t</a>             | keyDeriveCopyKeyReq             | Request to copy a key from the derived key material.                                                                                                     |

## Host Interface

### Data Fields

| Type                                        | Name               | Description                                                                       |
|---------------------------------------------|--------------------|-----------------------------------------------------------------------------------|
| <a href="#">hseSheLoadKeySrv_t</a>          | sheLoadKeyReq      | Request to load a SHE key using memory update protocol (as per SHE specification) |
| <a href="#">hseSheLoadPlainKeySrv_t</a>     | sheLoadPlainKeyReq | Request to load the SHE RAM key from plain text (as per SHE specification)        |
| <a href="#">hseSheExportRamKeySrv_t</a>     | sheExportRamKeyReq | Request to export the SHE RAM key (as per SHE specification)                      |
| <a href="#">hseSheGetIdSrv_t</a>            | sheGetIdReq        | Request to get UID (as per SHE specification)                                     |
| <a href="#">hseHashSrv_t</a>                | hashReq            | Request a HASH.                                                                   |
| <a href="#">hseMacSrv_t</a>                 | macReq             | Request to generate/verify a MAC.                                                 |
| <a href="#">hseFastCMACSrv_t</a>            | fastCmacReq        | Request to FAST generate/verify a CMAC.                                           |
| <a href="#">hseSymCipherSrv_t</a>           | symCipherReq       | Request a Symmetric Cipher operation.                                             |
| <a href="#">hseAeadSrv_t</a>                | aeadReq            | Request an AEAD operation.                                                        |
| <a href="#">hseSignSrv_t</a>                | signReq            | Request a Digital Signature Generation/Verification.                              |
| <a href="#">hseRsaCipherSrv_t</a>           | rsaCipherReq       | Request a RSA Cipher (Encryption/Decryption) operation.                           |
| <a href="#">hseGetRandomNumSrv_t</a>        | getRandomNumReq    | Request to random number generation.                                              |
| <a href="#">hseIncrementCounterSrv_t</a>    | incCounterReq      | Request to increment a monotonic counter.                                         |
| <a href="#">hseReadCounterSrv_t</a>         | readCounterReq     | Request to read a monotonic counter.                                              |
| <a href="#">hseConfigSecureCounterSrv_t</a> | configSecCounter   | Request to configure a secure counter.                                            |
| <a href="#">hseSmrEntryInstallSrv_t</a>     | smrEntryInstallReq | Request to install a Secure Memory Region (SMR) table entry.                      |
| <a href="#">hseSmrVerifySrv_t</a>           | smrVerifyReq       | Request to verify a Secure Memory Region (SMR) table entry.                       |
| <a href="#">hseSmrEntryEraseSrv_t</a>       | smrEntryEraseReq   | Request to erase a Secure Memory Region (SMR) table entry.                        |
| <a href="#">hseCrEntryInstallSrv_t</a>      | crEntryInstallReq  | Request to install a Core Reset (CR) table entry.                                 |
| <a href="#">hseCrOnDemandBootSrv_t</a>      | crOnDemandBootReq  | Request to release a Core Reset (CR) table entry.                                 |
| <a href="#">hseCrEntryEraseSrv_t</a>        | crEntryEraseReq    | Request to erase a Core Reset (CR) table entry.                                   |

## Macro Definition Documentation

## HSE\_SRV\_ID\_SET\_ATTR

```
#define HSE_SRV_ID_SET_ATTR ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000001UL))
```

Set HSE attribute. Data structure used: [hseSetAttrSrv\\_t](#).

## HSE\_SRV\_ID\_GET\_ATTR

```
#define HSE_SRV_ID_GET_ATTR ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00A50002UL))
```

Get HSE attribute. Data structure used: [hseGetAttrSrv\\_t](#).

## HSE\_SRV\_ID\_CANCEL

```
#define HSE_SRV_ID_CANCEL ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00A50004UL))
```

Cancel a one-pass or streaming service on a specific channel. Data structure used: [hseCancelSrv\\_t](#).

## HSE\_SRV\_ID\_FIRMWARE\_UPDATE

```
#define HSE_SRV_ID_FIRMWARE_UPDATE ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000005UL))
```

HSE firmware update. Data structure used: [hseFirmwareUpdateSrv\\_t](#).

## HSE\_SRV\_ID\_SYS\_AUTH\_REQ

```
#define HSE_SRV_ID_SYS_AUTH_REQ ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000006UL))
```

Perform a SYS Authorization request. Data structure used: [hseSysAuthorizationReqSrv\\_t](#).

## HSE\_SRV\_ID\_SYS\_AUTH\_RESP

```
#define HSE_SRV_ID_SYS_AUTH_RESP ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000007UL))
```

Send the SYS Authorization response. Data structure used: [hseSysAuthorizationRespSrv\\_t](#).

## Host Interface

### HSE\_SRV\_ID\_BOOT\_DATA\_IMAGE\_SIGN

```
#define HSE_SRV_ID_BOOT_DATA_IMAGE_SIGN ((hseSrvId_t) ( HSE_SRV_VER_0 |  
0x00000008UL))
```

Boot Data image sign (e.g. for HSE\_H/M, IVT/DCD/ST/LPDDR4(S32Z/E devices)/AppBSB image; for HSE\_B, IVT/AppBSB image). Data structure used: [hseBootDataImageSignSrv\\_t](#).

### HSE\_SRV\_ID\_BOOT\_DATA\_IMAGE\_VERIFY

```
#define HSE_SRV_ID_BOOT_DATA_IMAGE_VERIFY ((hseSrvId_t) ( HSE_SRV_VER_0 |  
0x00000009UL))
```

Boot Data images verify (e.g. for HSE\_H/M, IVT/DCD/ST/LPDDR4(S32Z/E devices)/AppBSB image; for HSE\_B, IVT/AppBSB image). Data structure used: [hseBootDataImageVerifySrv\\_t](#).

### HSE\_SRV\_ID\_IMPORT\_EXPORT\_STREAM\_CTX

```
#define HSE_SRV_ID_IMPORT_EXPORT_STREAM_CTX ((hseSrvId_t) ( HSE_SRV_VER_0 |  
0x00A5000AUL))
```

Import/Export Streaming Context. Data structure used: [hseImportExportStreamCtxSrv\\_t](#).

### HSE\_SRV\_ID\_ERASE\_HSE\_NVM\_DATA

```
#define HSE_SRV_ID_ERASE_HSE_NVM_DATA ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000050UL))
```

Erase HSE Data Flash (only for HSE\_B). This service is only allowed in CUST\_DEL LC. Data structure used: [hseEraseNvmDataSrv\\_t](#).

### HSE\_SRV\_ID\_ERASE\_FW

```
#define HSE_SRV_ID_ERASE_FW ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000057UL))
```

Erase Sys-Img, Backup FW as well Current running HSE FW from code flash. Data Structure used: [hseEraseFwSrv\\_t](#).

**HSE\_SRV\_ID\_ACTIVATE\_PASSIVE\_BLOCK**

```
#define HSE_SRV_ID_ACTIVATE_PASSIVE_BLOCK ((hseSrvId_t) ( HSE_SRV_VER_0 |  
0x00000051UL))
```

Application request to switch passive flash block area (only for HSE\_B). This service requires SuperUser rights. No data structure used.

**HSE\_SRV\_ID\_SBAF\_UPDATE**

```
#define HSE_SRV_ID_SBAF_UPDATE ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000053UL))
```

SBAF firmware update request. Data structure used: [hseSbafUpdateSrv\\_t](#).

**HSE\_SRV\_ID\_FW\_INTEGRITY\_CHECK**

```
#define HSE_SRV_ID_FW_INTEGRITY_CHECK ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000054UL))
```

Application request to checks the integrity of HSE FW and SYS-IMG inside HSE (only for HSE\_B). No data structure used.

**HSE\_SRV\_ID\_PUBLISH\_NVM\_KEYSTORE\_RAM\_TO\_FLASH**

```
#define HSE_SRV_ID_PUBLISH_NVM_KEYSTORE_RAM_TO_FLASH ((hseSrvId_t) ( HSE_SRV_VER_0 |  
0x00000055UL))
```

Application requests the firmware to write the NVM keys from RAM mirrored keystore into the data flash. Data structure used: [hsePublishNvmKeystoreRamToFlashSrv\\_t](#).

**HSE\_SRV\_ID\_CONFIG\_COUNTER**

```
#define HSE_SRV_ID_CONFIG_COUNTER ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000052UL))
```

Configure the secure counter (only for HSE\_B). This service requires SuperUser rights. Data structure used: [hseConfigSecCounterSrv\\_t](#).

## Host Interface

### HSE\_SRV\_ID\_LOAD\_ECC\_CURVE

```
#define HSE_SRV_ID_LOAD_ECC_CURVE ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000100UL))
```

Load the parameters for a Weierstrass ECC curve. Data structure used: [hseLoadEccCurveSrv\\_t](#).

### HSE\_SRV\_ID\_FORMAT\_KEY\_CATALOGS

```
#define HSE_SRV_ID_FORMAT_KEY_CATALOGS ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000101UL))
```

Format key catalogs (NVM or RAM). Data structure used: [hseFormatKeyCatalogsSrv\\_t](#).

### HSE\_SRV\_ID\_ERASE\_KEY

```
#define HSE_SRV_ID_ERASE_KEY ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000102UL))
```

Erase NVM/RAM key(s). Data structure used: [hseEraseKeySrv\\_t](#).

### HSE\_SRV\_ID\_GET\_KEY\_INFO

```
#define HSE_SRV_ID_GET_KEY_INFO ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00A50103UL))
```

Get key information header. Data structure used: [hseGetKeyInfoSrv\\_t](#).

### HSE\_SRV\_ID\_IMPORT\_KEY

```
#define HSE_SRV_ID_IMPORT_KEY ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000104UL))
```

Import a key. Data structure used: [hseImportKeySrv\\_t](#).

### HSE\_SRV\_ID\_EXPORT\_KEY

```
#define HSE_SRV_ID_EXPORT_KEY ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000105UL))
```

Export a key. Data structure used: [hseExportKeySrv\\_t](#).

**HSE\_SRV\_ID\_KEY\_GENERATE**

```
#define HSE_SRV_ID_KEY_GENERATE ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000106UL))
```

Key Generation (e.g. rsa key pair, ecc key pair etc.). Data structure used: [hseKeyGenerateSrv\\_t](#).

**HSE\_SRV\_ID\_DH\_COMPUTE\_SHARED\_SECRET**

```
#define HSE_SRV_ID_DH_COMPUTE_SHARED_SECRET ((hseSrvId_t) ( HSE_SRV_VER_0 |  
0x00000107UL))
```

ECC Diffie-Hellman Compute Key (shared secret). Data structure used: [hseDHComputeSharedSecretSrv\\_t](#).

**HSE\_SRV\_ID\_KEY\_DERIVE**

```
#define HSE_SRV_ID_KEY_DERIVE ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000108UL))
```

Perform a key derivation function. Data structure used: [hseKeyDeriveSrv\\_t](#).

**HSE\_SRV\_ID\_KEY\_DERIVE\_COPY**

```
#define HSE_SRV_ID_KEY_DERIVE_COPY ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000109UL))
```

Copy a key from the derived key material. Data structure used: [hseKeyDeriveCopyKeySrv\\_t](#).

**HSE\_SRV\_ID\_KEY\_VERIFY**

```
#define HSE_SRV_ID_KEY_VERIFY ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x0000010BUL))
```

Perform a verification for CMAC and SHA256/384/512. Data structure used: [hseKeyVerifySrv\\_t](#).

**HSE\_SRV\_ID\_SHE\_LOAD\_KEY**

```
#define HSE_SRV_ID_SHE_LOAD_KEY ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x0000A101UL))
```

Load a SHE key using the SHE memory update protocol. Data structure used: [hseSheLoadKeySrv\\_t](#).

## Host Interface

### HSE\_SRV\_ID\_SHE\_LOAD\_PLAIN\_KEY

```
#define HSE_SRV_ID_SHE_LOAD_PLAIN_KEY ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x0000A102UL))
```

Load the SHE RAM key as plain text. Data structure used: [hseSheLoadPlainKeySrv\\_t](#).

### HSE\_SRV\_ID\_SHE\_EXPORT\_RAM\_KEY

```
#define HSE_SRV_ID_SHE_EXPORT_RAM_KEY ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x0000A103UL))
```

Export the SHE RAM key. Data structure used: [hseSheExportRamKeySrv\\_t](#).

### HSE\_SRV\_ID\_SHE\_GET\_ID

```
#define HSE_SRV_ID_SHE_GET_ID ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x0000A104UL))
```

Get UID as per SHE specification. Data structure used: [hseSheGetIdSrv\\_t](#).

### HSE\_SRV\_ID\_SHE\_BOOT\_OK

```
#define HSE_SRV_ID_SHE_BOOT_OK ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x0000A105UL))
```

BOOT\_OK as per SHE specification. No data structure used.

### HSE\_SRV\_ID\_SHE\_BOOT\_FAILURE

```
#define HSE_SRV_ID_SHE_BOOT_FAILURE ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x0000A106UL))
```

BOOT\_FAILURE as per SHE specification. No data structure used.

### HSE\_SRV\_ID\_HASH

```
#define HSE_SRV_ID_HASH ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00A50200UL))
```

HASH service ID. Data structure used: [hseHashSrv\\_t](#).



## HSE\_SRV\_ID\_MAC

```
#define HSE_SRV_ID_MAC ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00A50201UL))
```

MAC generate/verify. Data structure used: [hseMacSrv\\_t](#).

## HSE\_SRV\_ID\_FAST\_CMACH

```
#define HSE_SRV_ID_FAST_CMACH ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00A50202UL))
```

CMAC fast generate/verify. Data structure used: [hseFastCMACHSrv\\_t](#).

## HSE\_SRV\_ID\_SYM\_CIPHER

```
#define HSE_SRV_ID_SYM_CIPHER ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00A50203UL))
```

Symmetric encryption/decryption. Data structure used: [hseSymCipherSrv\\_t](#).

## HSE\_SRV\_ID\_AEAD

```
#define HSE_SRV_ID_AEAD ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00A50204UL))
```

AEAD encryption/decryption. Data structure used: [hseAeadSrv\\_t](#).

## HSE\_SRV\_ID\_SIGN

```
#define HSE_SRV_ID_SIGN ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000206UL))
```

Digital Signature. Data structure used: [hseSignSrv\\_t](#).

## HSE\_SRV\_ID\_RSA\_CIPHER

```
#define HSE_SRV_ID_RSA_CIPHER ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000207UL))
```

RSA Cipher ID. Data structure used: [hseRsaCipherSrv\\_t](#).

## Host Interface

### HSE\_SRV\_ID\_GET\_RANDOM\_NUM

```
#define HSE_SRV_ID_GET_RANDOM_NUM ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000300UL))
```

Get random number. Data structure used: [hseGetRandomNumSrv\\_t](#).

### HSE\_SRV\_ID\_INCREMENT\_COUNTER

```
#define HSE_SRV_ID_INCREMENT_COUNTER ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00A50400UL))
```

Increment a monotonic counter. Data structure used: [hseIncrementCounterSrv\\_t](#).

### HSE\_SRV\_ID\_READ\_COUNTER

```
#define HSE_SRV_ID_READ_COUNTER ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00A50401UL))
```

Read a monotonic counter. Data structure used: [hseReadCounterSrv\\_t](#).

### HSE\_SRV\_ID\_SMR\_ENTRY\_INSTALL

```
#define HSE_SRV_ID_SMR_ENTRY_INSTALL ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000501UL))
```

Install a Secure memory region (SMR) table entry. Data structure used: [hseSmrEntryInstallSrv\\_t](#).

### HSE\_SRV\_ID\_SMR\_VERIFY

```
#define HSE_SRV_ID_SMR_VERIFY ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000502UL))
```

Verify a Secure memory region (SMR) table entry. Data structure used: [hseSmrVerifySrv\\_t](#).

### HSE\_SRV\_ID\_CORE\_RESET\_ENTRY\_INSTALL

```
#define HSE_SRV_ID_CORE_RESET_ENTRY_INSTALL ((hseSrvId_t) ( HSE_SRV_VER_0 |  
0x00000503UL))
```

Install a Core Reset (CR) table entry. Data structure used: [hseCrEntryInstallSrv\\_t](#).

**HSE\_SRV\_ID\_ON\_DEMAND\_CORE\_RESET**

```
#define HSE_SRV_ID_ON_DEMAND_CORE_RESET ((hseSrvId_t) ( HSE_SRV_VER_0 |  
0x00000504UL))
```

On demand release a core from reset after loading and verification. Data structure used: [hseCrOnDemandBootSrv\\_t](#).

**HSE\_SRV\_ID\_SMR\_ENTRY\_ERASE**

```
#define HSE_SRV_ID_SMR_ENTRY_ERASE ((hseSrvId_t) ( HSE_SRV_VER_0 | 0x00000505UL))
```

Erase a Secure memory region (SMR) table entry. Data structure used: [hseSmrEntryEraseSrv\\_t](#).

**HSE\_SRV\_ID\_CORE\_RESET\_ENTRY\_ERASE**

```
#define HSE_SRV_ID_CORE_RESET_ENTRY_ERASE ((hseSrvId_t) ( HSE_SRV_VER_0 |  
0x00000506UL))
```

Erase a Core Reset (CR) table entry. Data structure used: [hseCrEntryEraseSrv\\_t](#).

## 2.3 HSE GPR Status

### Macros

| Type: (implicit C type)                |              |
|----------------------------------------|--------------|
| Name                                   | Value        |
| <a href="#">HSE_GPR_STATUS_ADDRESS</a> | 0x4039C02CUL |

| Type: <a href="#">hseTamperConfigStatus_t</a>     |          |
|---------------------------------------------------|----------|
| Name                                              | Value    |
| <a href="#">HSE_CMU_TAMPER_CONFIG_STATUS</a>      | 1U << 0U |
| <a href="#">HSE_PHYSICAL_TAMPER_CONFIG_STATUS</a> | 1U << 1U |

| Type: <a href="#">hseEccErrorStatus_t</a>        |           |
|--------------------------------------------------|-----------|
| Name                                             | Value     |
| <a href="#">HSE_ECC_KEYSTORE_STATUS</a>          | 1U << 16U |
| <a href="#">HSE_ECC_CONFIG_DATA_STATUS</a>       | 1U << 17U |
| <a href="#">HSE_ECC_MONOTONIC_COUNTER_STATUS</a> | 1U << 18U |

## Host Interface

### Typedefs

- typedef uint32\_t [hseEccErrorStatus\\_t](#)
- typedef uint32\_t [hseTamperConfigStatus\\_t](#)

## Macro Definition Documentation

### HSE\_GPR\_STATUS\_ADDRESS

```
#define HSE_GPR_STATUS_ADDRESS (0x4039C02CUL)
```

CONFIG\_REG4 is in Configuration GPR Description (refer to [hseTamperConfigStatus\\_t](#) and [hseEccErrorStatus\\_t](#)). This register is to inform the application that there is a loss of data due to ECC error and user needs necessary action.

### HSE\_ECC\_KEYSTORE\_STATUS

```
#define HSE_ECC_KEYSTORE_STATUS ((hseEccErrorStatus_t)1U << 16U)
```

CONFIG\_REG4[16] - This bit is set when the key store area is erased.

### HSE\_ECC\_CONFIG\_DATA\_STATUS

```
#define HSE_ECC_CONFIG_DATA_STATUS ((hseEccErrorStatus_t)1U << 17U)
```

CONFIG\_REG4[17] - This bit is set when the config data area is erased.

### HSE\_ECC\_MONOTONIC\_COUNTER\_STATUS

```
#define HSE_ECC_MONOTONIC_COUNTER_STATUS ((hseEccErrorStatus_t)1U << 18U)
```

CONFIG\_REG4[18] - This bit is set when the monotonic counter area is erased.

### HSE\_CMU\_TAMPER\_CONFIG\_STATUS

```
#define HSE_CMU_TAMPER_CONFIG_STATUS ((hseTamperConfigStatus_t)1U << 0U)
```

HSE-GPR REG3[0]- this bit is set when the CMU tamper is configured:

- For HSE\_H, the clock must be configured in this range: 10Mhz < clock frequency < 420Mhz.
- For HSE\_B, the clock must be configured in this range: 3Mhz < clock frequency < 126Mhz.

- For HSE\_M, the clock must be configured in this range:
  - s32r41x: 45.6Mhz < clock frequency < 420Mhz.

## HSE\_PHYSICAL\_TAMPER\_CONFIG\_STATUS

```
#define HSE_PHYSICAL_TAMPER_CONFIG_STATUS ((hseTamperConfigStatus_t)1U << 1U)
```

HSE-GPR REG3[1]- this bit is set when the physical tamper is configured. Note that the application must configure SIUL2 Pads before enabling the tamper.

## Typedef Documentation

### hseEccErrorStatus\_t

```
typedef uint32_t hseEccErrorStatus_t
```

HSE SYS-IMG Dataset Erase Status bits (register address is [HSE\\_GPR\\_STATUS\\_ADDRESS](#))

This status is updated when a particular dataset has been erased by HSE Firmware because of corruption due to ECC error (this can happen if a reset occurred during a flash programming/erase operation), or because some invalid data was found.

### hseTamperConfigStatus\_t

```
typedef uint32_t hseTamperConfigStatus_t
```

HSE Tamper Config Status bits (register address is [HSE\\_GPR\\_STATUS\\_ADDRESS](#))

This status is updated when a tamper is configured by HSE during initialization or via attribute. The host can get the HSE Tamper Config Status reading the [HSE\\_GPR\\_STATUS\\_ADDRESS](#) register. In this way, the host to check what tampers are configured. The [HSE\\_GPR\\_STATUS\\_ADDRESS](#) register is read-only.

Note

- For HSE\_H/S32R41, HSE-GPR REG3 used.
- For HSE\_B, CONFIG\_REG4 used.

## 2.4 HSE Service Responses

### Macros

| Type: <a href="#">hseSrvResponse_t</a> |       |
|----------------------------------------|-------|
| Name                                   | Value |

## Host Interface

|                                     |              |
|-------------------------------------|--------------|
| HSE_SRV_RSP_OK                      | 0x55A5AA33UL |
| HSE_SRV_RSP_VERIFY_FAILED           | 0x55A5A164UL |
| HSE_SRV_RSP_INVALID_ADDR            | 0x55A5A26AUL |
| HSE_SRV_RSP_INVALID_PARAM           | 0x55A5A399UL |
| HSE_SRV_RSP_NOT_SUPPORTED           | 0xAA55A11EUL |
| HSE_SRV_RSP_NOT_ALLOWED             | 0xAA55A21CUL |
| HSE_SRV_RSP_NOT_ENOUGH_SPACE        | 0xAA55A371UL |
| HSE_SRV_RSP_READ_FAILURE            | 0xAA55A427UL |
| HSE_SRV_RSP_WRITE_FAILURE           | 0xAA55A517UL |
| HSE_SRV_RSP_STREAMING_MODE_FAILURE  | 0xAA55A6B1UL |
| HSE_SRV_RSP_KEY_NOT_AVAILABLE       | 0xA5AA51B2UL |
| HSE_SRV_RSP_KEY_INVALID             | 0xA5AA52B4UL |
| HSE_SRV_RSP_KEY_EMPTY               | 0xA5AA5317UL |
| HSE_SRV_RSP_KEY_WRITE_PROTECTED     | 0xA5AA5436UL |
| HSE_SRV_RSP_KEY_UPDATE_ERROR        | 0xA5AA5563UL |
| HSE_SRV_RSP_MEMORY_FAILURE          | 0x33D6D136UL |
| HSE_SRV_RSP_CANCEL_FAILURE          | 0x33D6D261UL |
| HSE_SRV_RSP_CANCELED                | 0x33D6D396UL |
| HSE_SRV_RSP_GENERAL_ERROR           | 0x33D6D4F1UL |
| HSE_SRV_RSP_COUNTER_OVERFLOW        | 0x33D6D533UL |
| HSE_SRV_RSP_SHE_NO_SECURE_BOOT      | 0x33D6D623UL |
| HSE_SRV_RSP_SHE_BOOT_SEQUENCE_ERROR | 0x33D7D83AUL |
| HSE_SRV_RSP_RNG_INIT_IN_PROGRESS    | 0x33D7D92AUL |
| HSE_SRV_RSP_FUSE_WRITE_FAILURE      | 0xBB4456E7UL |
| HSE_SRV_RSP_FUSE_VDD_GND            | 0xBB4457F3UL |
| HSE_SRV_RSP_SBAF_UPDATE_REQUIRED    | 0xCC66FEADUL |

## Typedefs

- typedef uint32\_t [hseSrvResponse\\_t](#)

## Macro Definition Documentation

### HSE\_SRV\_RSP\_OK

```
#define HSE_SRV_RSP_OK ((hseSrvResponse\_t)0x55A5AA33UL)
```

HSE service successfully executed with no error.

**HSE\_SRV\_RSP\_VERIFY\_FAILED**

```
#define HSE_SRV_RSP_VERIFY_FAILED ((hseSrvResponse_t) 0x55A5A164UL)
```

HSE signals that a verification request fails (e.g. MAC and Signature verification).

**HSE\_SRV\_RSP\_INVALID\_ADDR**

```
#define HSE_SRV_RSP_INVALID_ADDR ((hseSrvResponse_t) 0x55A5A26AUL)
```

The address parameters are invalid.

**HSE\_SRV\_RSP\_INVALID\_PARAM**

```
#define HSE_SRV_RSP_INVALID_PARAM ((hseSrvResponse_t) 0x55A5A399UL)
```

The HSE request parameters are invalid.

**HSE\_SRV\_RSP\_NOT\_SUPPORTED**

```
#define HSE_SRV_RSP_NOT_SUPPORTED ((hseSrvResponse_t) 0xAA55A11EUL)
```

The operation or feature not supported.

**HSE\_SRV\_RSP\_NOT\_ALLOWED**

```
#define HSE_SRV_RSP_NOT_ALLOWED ((hseSrvResponse_t) 0xAA55A21CUL)
```

The operation is not allowed because of some restrictions (in attributes, life-cycle dependent operations, key-management, etc.).

**HSE\_SRV\_RSP\_NOT\_ENOUGH\_SPACE**

```
#define HSE_SRV_RSP_NOT_ENOUGH_SPACE ((hseSrvResponse_t) 0xAA55A371UL)
```

There is no enough space to perform service (e.g. format key store)

## Host Interface

### HSE\_SRV\_RSP\_READ\_FAILURE

```
#define HSE_SRV_RSP_READ_FAILURE ((hseSrvResponse_t) 0xAA55A427UL)
```

The service request failed because read access was denied. For HSE\_B, it can be returned if Host Flash Programming/Erase operation was in progress at the time of giving the command.

### HSE\_SRV\_RSP\_WRITE\_FAILURE

```
#define HSE_SRV_RSP_WRITE_FAILURE ((hseSrvResponse_t) 0xAA55A517UL)
```

The service request failed because write access was denied.

### HSE\_SRV\_RSP\_STREAMING\_MODE\_FAILURE

```
#define HSE_SRV_RSP_STREAMING_MODE_FAILURE ((hseSrvResponse_t) 0xAA55A6B1UL)
```

The service request that uses streaming mode failed (e.g. UPDATES and FINISH steps do not use the same HSE interface ID and channel ID as START step).

### HSE\_SRV\_RSP\_KEY\_NOT\_AVAILABLE

```
#define HSE_SRV_RSP_KEY_NOT_AVAILABLE ((hseSrvResponse_t) 0xA5AA51B2UL)
```

This error code is returned if a key is locked due to failed boot measurement or an active debugger.

### HSE\_SRV\_RSP\_KEY\_INVALID

```
#define HSE_SRV_RSP_KEY_INVALID ((hseSrvResponse_t) 0xA5AA52B4UL)
```

The key usage flags (provided using the key handle) don't allow to perform the requested crypto operation (the key flags don't match the crypto operation; e.g. the key is configured to be used for decryption, and the host requested an encryption). In SHE, the key ID provided is either invalid or non-usable due to some flag restrictions.

### HSE\_SRV\_RSP\_KEY\_EMPTY

```
#define HSE_SRV_RSP_KEY_EMPTY ((hseSrvResponse_t) 0xA5AA5317UL)
```



Specified key slot is empty.

### **HSE\_SRV\_RSP\_KEY\_WRITE\_PROTECTED**

```
#define HSE_SRV_RSP_KEY_WRITE_PROTECTED ((hseSrvResponse_t) 0xA5AA5436UL)
```

Key slot to be loaded is protected with WRITE PROTECTION restriction flag.

### **HSE\_SRV\_RSP\_KEY\_UPDATE\_ERROR**

```
#define HSE_SRV_RSP_KEY_UPDATE_ERROR ((hseSrvResponse_t) 0xA5AA5563UL)
```

Used only in the context of SHE specification: specified key slot cannot be updated due to errors in verification of the parameters.

### **HSE\_SRV\_RSP\_MEMORY\_FAILURE**

```
#define HSE_SRV_RSP_MEMORY_FAILURE ((hseSrvResponse_t) 0x33D6D136UL)
```

Detect physical errors, flipped bits etc., during memory read or write operations.

### **HSE\_SRV\_RSP\_CANCEL\_FAILURE**

```
#define HSE_SRV_RSP_CANCEL_FAILURE ((hseSrvResponse_t) 0x33D6D261UL)
```

The service can not be canceled.

### **HSE\_SRV\_RSP\_CANCELED**

```
#define HSE_SRV_RSP_CANCELED ((hseSrvResponse_t) 0x33D6D396UL)
```

The service has been canceled.

### **HSE\_SRV\_RSP\_GENERAL\_ERROR**

```
#define HSE_SRV_RSP_GENERAL_ERROR ((hseSrvResponse_t) 0x33D6D4F1UL)
```

## Host Interface

This error code is returned if an error not covered by the error codes above is detected inside HSE.

### HSE\_SRV\_RSP\_COUNTER\_OVERFLOW

```
#define HSE_SRV_RSP_COUNTER_OVERFLOW ((hseSrvResponse_t) 0x33D6D533UL)
```

The monotonic counter overflows.

### HSE\_SRV\_RSP\_SHE\_NO\_SECURE\_BOOT

```
#define HSE_SRV_RSP_SHE_NO_SECURE_BOOT ((hseSrvResponse_t) 0x33D6D623UL)
```

HSE did not perform SHE based secure Boot.

### HSE\_SRV\_RSP\_SHE\_BOOT\_SEQUENCE\_ERROR

```
#define HSE_SRV_RSP_SHE_BOOT_SEQUENCE_ERROR ((hseSrvResponse_t) 0x33D7D83AUL)
```

Received SHE\_BOOT\_OK or SHE\_BOOT\_FAILURE more than one time.

### HSE\_SRV\_RSP\_RNG\_INIT\_IN\_PROGRESS

```
#define HSE_SRV_RSP_RNG_INIT_IN_PROGRESS ((hseSrvResponse_t) 0x33D7D92AUL)
```

This error code is returned if RNG Initialization is in Progress.

### HSE\_SRV\_RSP\_FUSE\_WRITE\_FAILURE

```
#define HSE_SRV_RSP_FUSE_WRITE_FAILURE ((hseSrvResponse_t) 0xBB4456E7UL)
```

This error code is returned, if fuse write operation fail.

### HSE\_SRV\_RSP\_FUSE\_VDD\_GND

```
#define HSE_SRV_RSP_FUSE_VDD_GND ((hseSrvResponse_t) 0xBB4457F3UL)
```

This error code is returned, if EFUSE\_VDD connected to ground during fuse write operation.

**HSE\_SRV\_RSP\_SBAF\_UPDATE\_REQUIRED**

```
#define HSE_SRV_RSP_SBAF_UPDATE_REQUIRED ((hseSrvResponse_t) 0xCC66FEADUL)
```

This error code is returned, if operation is dependent on Secure BAF version, which on the device happens to be old.

**Typedef Documentation****hseSrvResponse\_t**

```
typedef uint32_t hseSrvResponse_t
```

HSE Service response.

The Service response is provided by MUB\_RRx register after the service execution.

**2.5 HSE Errors****Macros**

| Type: <a href="#">hseError_t</a>                 |            |
|--------------------------------------------------|------------|
| Name                                             | Value      |
| <a href="#">HSE_ERR_GENERAL</a>                  | 1UL << 0U  |
| <a href="#">HSE_WA_SMR_PERIODIC_CHECK_FAILED</a> | 1UL << 8U  |
| <a href="#">HSE_WA_DATA_FLASH_INTEGRITY_FAIL</a> | 1UL << 9U  |
| <a href="#">HSE_WA_RNG_NOT_INIT</a>              | 1UL << 10U |

**Typedefs**

- typedef uint32\_t [hseError\\_t](#)

**HSE Errors Details**

These error events are reported when some kind of intrusion/violation is detected in the system. The most significant 16 bits are reserved for NXP internal errors and less significant 16 bits indicate the source of violation as defined below.

Note

- If the MU General Purpose Interrupt is enabled on the host-side, any bit set to "1" (on MUB\_GSR register) triggers an interrupt.

## Host Interface

- The host must read the MUB\_GSR register and write back the register value to clear the bits (W1C - write one to clear).
- The bits[0..7] (listed below) are fatal errors that trigger an HSE shutdown (HSE enters in the secure failure state, all MU are disabled).
- The bits[8..15] (listed below) are warning events (something failed, but it is not fatal).

## Macro Definition Documentation

### HSE\_ERR\_GENERAL

```
#define HSE_ERR_GENERAL ((hseError_t)1UL << 0U)
```

Internal fatal error detected by HSE. The HSE system shutdowns.

### HSE\_WA\_SMR\_PERIODIC\_CHECK\_FAILED

```
#define HSE_WA_SMR_PERIODIC_CHECK_FAILED ((hseError_t)1UL << 8U)
```

The verification of periodic check SMR ([hseSmrEntry\\_t::checkPeriod](#) !=0) failed. The application can read [HSE\\_SMR\\_CORE\\_BOOT\\_STATUS\\_ATTR\\_ID](#) attribute to see what SMR failed.

### HSE\_WA\_DATA\_FLASH\_INTEGRITY\_FAIL

```
#define HSE_WA_DATA_FLASH_INTEGRITY_FAIL ((hseError_t)1UL << 9U)
```

HSE Data flash memory integrity check failed.

### HSE\_WA\_RNG\_NOT\_INIT

```
#define HSE_WA_RNG_NOT_INIT ((hseError_t)1UL << 10U)
```

RNG is not initialized. Services depending on the RNG may be delayed as HSE attempts RNG re-initialization.

## Typedef Documentation

### hseError\_t

```
typedef uint32_t hseError_t
```

## 2.6 Host Events To HSE

## 2.7 HSE Status

### Macros

| Type: <a href="#">hseStatus_t</a>                            |           |
|--------------------------------------------------------------|-----------|
| Name                                                         | Value     |
| <a href="#">HSE_SHE_STATUS_SECURE_BOOT</a>                   | 1U << 1U  |
| <a href="#">HSE_SHE_STATUS_SECURE_BOOT_INIT</a>              | 1U << 2U  |
| <a href="#">HSE_SHE_STATUS_SECURE_BOOT_FINISHED</a>          | 1U << 3U  |
| <a href="#">HSE_SHE_STATUS_SECURE_BOOT_OK</a>                | 1U << 4U  |
| <a href="#">HSE_STATUS_RNG_INIT_OK</a>                       | 1U << 5U  |
| <a href="#">HSE_STATUS_HOST_DEBUGGER_ACTIVE</a>              | 1U << 6U  |
| <a href="#">HSE_STATUS_HSE_DEBUGGER_ACTIVE</a>               | 1U << 7U  |
| <a href="#">HSE_STATUS_INIT_OK</a>                           | 1U << 8U  |
| <a href="#">HSE_STATUS_INSTALL_OK</a>                        | 1U << 9U  |
| <a href="#">HSE_STATUS_BOOT_OK</a>                           | 1U << 10U |
| <a href="#">HSE_STATUS_CUST_SUPER_USER</a>                   | 1U << 11U |
| <a href="#">HSE_STATUS_OEM_SUPER_USER</a>                    | 1U << 12U |
| <a href="#">HSE_STATUS_FW_UPDATE_IN_PROGRESS</a>             | 1U << 13U |
| <a href="#">HSE_STATUS_PUBLISH_NVM_KEYSTORE_RAM_TO_FLASH</a> | 1U << 14U |

### Typedefs

- typedef uint16\_t [hseStatus\\_t](#)

### HSE Status Details

HSE status can be read by the HOST and represents the most significant 16 bits in MUB.FSR register. The least significant 16 bits in MUB.FSR register identifies the status of each channel:

- 0b - channel idle and it can accept service requests
- 1b - channel busy

### Macro Definition Documentation

#### HSE\_SHE\_STATUS\_SECURE\_BOOT

```
#define HSE_SHE_STATUS_SECURE_BOOT ((hseStatus_t)1U << 1U)
```

This bit is set when the SHE based secure boot process has been started by HSE firmware. This bit is only set when SMR0 entry has been installed by the user and its authentication key is set as SHE based BOOT\_MAC\_KEY

## Host Interface

### HSE\_SHE\_STATUS\_SECURE\_BOOT\_INIT

```
#define HSE_SHE_STATUS_SECURE_BOOT_INIT ((hseStatus_t)1U << 2U)
```

This bit is set when BOOT\_MAC personalization has been completed by HSE firmware. It means that the BOOT\_MAC slot was empty and SHE-based secure boot is performed the first time. In that case, if BOOT\_MAC\_KEY is present, then HSE firmware calculates the BOOT\_MAC of the SMR image present in the SMR0 (using the BOOT\_MAC\_KEY) and store it as part of sys image.

### HSE\_SHE\_STATUS\_SECURE\_BOOT\_FINISHED

```
#define HSE_SHE_STATUS_SECURE_BOOT_FINISHED ((hseStatus_t)1U << 3U)
```

This bit is set when the HSE firmware has completed the secure boot process with a failure status. (the image verification failed).

### HSE\_SHE\_STATUS\_SECURE\_BOOT\_OK

```
#define HSE_SHE_STATUS_SECURE_BOOT_OK ((hseStatus_t)1U << 4U)
```

This bit is set when the HSE firmware has successfully completed the secure boot process (the image verification was successful).

### HSE\_STATUS\_RNG\_INIT\_OK

```
#define HSE_STATUS_RNG_INIT_OK ((hseStatus_t)1U << 5U)
```

This bit is set when HSE FW has successfully initialized the RNG.

### HSE\_STATUS\_HOST\_DEBUGGER\_ACTIVE

```
#define HSE_STATUS_HOST_DEBUGGER_ACTIVE ((hseStatus_t)1U << 6U)
```

This bit is set when debugger on HOST side is active as well as enabled.

### HSE\_STATUS\_HSE\_DEBUGGER\_ACTIVE

```
#define HSE_STATUS_HSE_DEBUGGER_ACTIVE ((hseStatus_t)1U << 7U)
```

This bit is set when debugger on HSE side is active as well as enabled.

**HSE\_STATUS\_INIT\_OK**

```
#define HSE_STATUS_INIT_OK ((hseStatus_t)1U << 8U)
```

This bit is set when the HSE initialization has been successfully completed (HSE service requests can be sent over MUs). If this bit is cleared, the host can NOT perform any service request (MUs are disabled).

**HSE\_STATUS\_INSTALL\_OK**

```
#define HSE_STATUS_INSTALL_OK ((hseStatus_t)1U << 9U)
```

This flag signals the application that needs to format the key catalogs (NVM and RAM).

- When it is clear, the application shall format the key catalogs;
- When it is set, the HSE installation phase has been successfully completed. (e.g HSE is in normal state and the application can install the NVM key, configure the SMR entries etc).

Note

This step is MANDATORY.

**HSE\_STATUS\_BOOT\_OK**

```
#define HSE_STATUS_BOOT_OK ((hseStatus_t)1U << 10U)
```

This bit is set when the HSE booting phase has been successfully completed. This bit is cleared if the HSE booting phase is still in execution or failed.

Note

- HSE set this bit only when the secure boot is configured (BOOT\_SEQ = 1).
- This bit represents the status of booting phase which includes the PRE\_BOOT SMR verification (without POST\_BOOT SMRs) and cores un-gating.
- The HSE FW signals the end of the POST\_BOOT phase along with additional peripherals initialization via [HSE\\_STATUS\\_INIT\\_OK](#) flag.

**HSE\_STATUS\_CUST\_SUPER\_USER**

```
#define HSE_STATUS_CUST_SUPER_USER ((hseStatus_t)1U << 11U)
```

After reset, if the Life Cycle = CUST\_DEL, this bit is set (SuperUser rights are granted).

During run-time:

- it is set if the authorization request for CUST SuperUser rights are granted using an CUST authorization key.
- it is cleared for USER rights.

## Host Interface

### Note

If CUST\_START\_AS\_USER policy attribute is set (TRUE), the device will always start having User rights.

## HSE\_STATUS\_OEM\_SUPER\_USER

```
#define HSE_STATUS_OEM_SUPER_USER ((hseStatus_t)1U << 12U)
```

After reset: if the Life Cycle = OEM\_PROD, this bit is set (SuperUser rights are granted).

During run-time:

- it is set if the authorization request for OEM SuperUser rights are granted using an OEM authorization key.
- it is cleared for USER rights.

### Note

If OEM\_START\_AS\_USER policy attribute is set (TRUE), the device will always start having User rights.

## HSE\_STATUS\_FW\_UPDATE\_IN\_PROGRESS

```
#define HSE_STATUS_FW_UPDATE_IN_PROGRESS ((hseStatus_t)1U << 13U)
```

This bit is set when the HSE FW update is in progress. This bit is cleared after HSE FW update completion.

## HSE\_STATUS\_PUBLISH\_NVM\_KEYSTORE\_RAM\_TO\_FLASH

```
#define HSE_STATUS_PUBLISH_NVM_KEYSTORE_RAM_TO_FLASH ((hseStatus_t)1U << 14U)
```

This flag signals the application to publish the NVM KEYSTORE to Secure flash Region.

- This feature can be enabled via [HSE\\_ENABLE\\_PUBLISH\\_KEY\\_STORE\\_RAM\\_TO\\_FLASH\\_ATTR\\_ID](#) attribute.
- When this flag is set, the host must trigger a PUBLISH\_KEYSTORE request via [HSE\\_SRV\\_ID\\_PUBLISH\\_NVM\\_KEYSTORE\\_RAM\\_TO\\_FLASH](#).

### Note

This flag is set whenever the HSE NVM KEYSTORE has been updated in the HSE internal RAM indicating that it is not safe to reset the device.

- Once NVM KEYSTORE via [HSE\\_SRV\\_ID\\_PUBLISH\\_NVM\\_KEYSTORE\\_RAM\\_TO\\_FLASH](#), it is written on secure region in data flash and this bit is cleared.
- If this bit is set, the application must call the [HSE\\_SRV\\_ID\\_PUBLISH\\_NVM\\_KEYSTORE\\_RAM\\_TO\\_FLASH](#) service before issuing the Firmware Update. Otherwise, the HSE\_SRV\_RSP\_NOT\_ALLOWED response status will be returned.



## Typedef Documentation

### **hseStatus\_t**

```
typedef uint16_t hseStatus_t
```



## 3 Administration Services

### 3.1 HSE Utility Services

#### Data Structures

- struct [hseCancelSrv\\_t](#)
- struct [hseImportExportStreamCtxSrv\\_t](#)
- struct [hseEraseNvmDataSrv\\_t](#)
- struct [hsePublishNvmKeystoreRamToFlashSrv\\_t](#)
- struct [hseEraseFwSrv\\_t](#)

#### Macros

| Type: (implicit C type)                    |       |
|--------------------------------------------|-------|
| Name                                       | Value |
| <a href="#">MAX_STREAMING_CONTEXT_SIZE</a> | 372UL |

| Type: <a href="#">hseStreamContextOp_t</a>   |       |
|----------------------------------------------|-------|
| Name                                         | Value |
| <a href="#">HSE_IMPORT_STREAMING_CONTEXT</a> | 1U    |
| <a href="#">HSE_EXPORT_STREAMING_CONTEXT</a> | 2U    |

#### Typedefs

- typedef uint8\_t [hseStreamContextOp\\_t](#)

#### Data Structure Documentation

##### struct [hseCancelSrv\\_t](#)

HSE Cancel service.

This service cancels a HSE one-pass and streaming service that was sent on a specific channel.

Note

- The requests with the service ID that starts with 0x00A5XXXX can not be canceled.
- Cancel requests cannot be canceled (by a subsequent request);

## Data Fields

| Type    | Name         | Description                                                                                                                                                                                                                                                                           |
|---------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint8_t | muChannelIdx | INPUT: The channel Index of MU interface [0.. <a href="#">HSE_NUM_OF_CHANNELS_PER_MU</a> ).<br>The muChannelIdx and the MU channel on which the service is sent, must belong to the same MU Interface. Otherwise an <a href="#">HSE_SRV_RSP_INVALID_PARAM</a> error will be reported. |
| uint8_t | reserved[3]  |                                                                                                                                                                                                                                                                                       |

**struct hseImportExportStreamCtxSrv\_t**

HSE Import/Export Streaming Context service.

This service allows import/export of a streaming context used in an on-going streaming operation (e.g. Hash, MAC, Cipher, AEAD, etc).

The streaming context will be imported/exported as a blob (encrypted with a device specific key).

## Data Fields

| Type                                 | Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseStreamContextOp_t</a> | operation      | INPUT: Specifies the operation to be performed with the streaming context: Import/Export.                                                                                                                                                                                                                                                                                                                                                                 |
| <a href="#">hseStreamId_t</a>        | streamId       | INPUT: Specifies the stream to be exported or overwritten if imported. Note that each interface supports up to <a href="#">HSE_STREAM_COUNT</a> streams per interface.                                                                                                                                                                                                                                                                                    |
| uint8_t                              | reserved[2]    |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| uint32_t                             | pStreamContext | OUTPUT/INPUT: The output buffer where the streaming context will be copied (export) or the input buffer from which HSE will copy the streaming context (import). Length of the buffer should be at least <a href="#">MAX_STREAMING_CONTEXT_SIZE</a> bytes. A streaming context can be imported or exported on the same MU instance on which the streaming START step was called (e.g. the steaming context is allocated when the START step is called).". |

**struct hseEraseNvmDataSrv\_t**

Prepare the security subsystem (BootROM + HSE) for Stand-By.

This service is used for updating the internal state of HSE before system goes in Stand-By mode. Applicable only for flashless devices (HSE\_H/M variants). This service can be called only once per running state, otherwise HSE will return [HSE\\_SRV\\_RSP\\_NOT\\_ALLOWED](#).

Erase SYS-IMG in Data Flash.

## Administration Services

This service is used for erasing SYS-IMG in HSE Data Flash. The service is available for flash based devices only (HSE\_B variant). Can be performed only in CUST\_DEL life cycle, otherwise [HSE\\_SRV\\_RSP\\_NOT\\_ALLOWED](#) error will be reported.

### Data Fields

| Type    | Name        | Description |
|---------|-------------|-------------|
| uint8_t | reserved[4] |             |

## struct hsePublishNvmKeystoreRamToFlashSrv\_t

Publish NVM Keystore from RAM to Flash.

This service is used to push all the keys of NVM keystore stored temporarily in RAM to secure flash. The service is available for flash based devices only (HSE\_B variant). Can be performed only in CUST\_DEL or OEM\_PROD life cycles, otherwise [HSE\\_SRV\\_RSP\\_NOT\\_ALLOWED](#) error will be reported.

### Note

If the status [HSE\\_STATUS\\_PUBLISH\\_NVM\\_KEYSTORE\\_RAM\\_TO\\_FLASH](#) is not set, [HSE\\_SRV\\_RSP\\_NOT\\_ALLOWED](#) error will be reported.

### Data Fields

| Type    | Name        | Description |
|---------|-------------|-------------|
| uint8_t | reserved[4] |             |

## struct hseEraseFwSrv\_t

Erase HSE Firmware from the device.

This service is used for erasing the HSE Firmware. This service also erases the SYS-IMG and backup (if present) in the secure flash from the device. The service is available for flash based devices only (HSE\_B variant). Can be performed only in CUST\_DEL life cycle, otherwise [HSE\\_SRV\\_RSP\\_NOT\\_ALLOWED](#) error will be reported.

### Data Fields

| Type    | Name        | Description |
|---------|-------------|-------------|
| uint8_t | reserved[4] |             |

## Macro Definition Documentation

**MAX\_STREAMING\_CONTEXT\_SIZE**

```
#define MAX_STREAMING_CONTEXT_SIZE (372UL)
```

The maximum size of the streaming context for any operation.

**HSE\_IMPORT\_STREAMING\_CONTEXT**

```
#define HSE_IMPORT_STREAMING_CONTEXT ((hseStreamContextOp_t)1U)
```

Import streaming context.

**HSE\_EXPORT\_STREAMING\_CONTEXT**

```
#define HSE_EXPORT_STREAMING_CONTEXT ((hseStreamContextOp_t)2U)
```

Export streaming context.

**Typedef Documentation****hseStreamContextOp\_t**

```
typedef uint8_t hseStreamContextOp_t
```

Streaming Context Operation: Import/Export.

**3.2 HSE Set/Get Attribute Services****Data Structures**

- struct [hseSetAttrSrv\\_t](#)
- struct [hseGetAttrSrv\\_t](#)
- struct [hseAttrFwVersion\\_t](#)
- struct [hseFwBuildInfo\\_t](#)
- struct [hseAttrSmrCoreStatus\\_t](#)
- struct [hseAttrMUInstanceConfig\\_t](#)
- struct [hseAttrMUConfig\\_t](#)
- struct [hseAttrMemRegion\\_t](#)
- struct [hseAttrMuMemRegions\\_t](#)
- struct [hseAttrAllMuMemRegions\\_t](#)
- struct [hseAttrExtendCustSecurityPolicy\\_t](#)
- struct [hseAttrExtendOemSecurityPolicy\\_t](#)
- struct [hseAttrPhysicalTamper\\_t](#)
- struct [hseAttrPhysicalTamperConfig\\_t](#)

## Macros

| Type: (implicit C type)                    |                    |
|--------------------------------------------|--------------------|
| Name                                       | Value              |
| <a href="#">HSE_ALGO_CAP_MASK</a> (capIdx) | (1ULL << (capIdx)) |
| <a href="#">HSE_FILTER_DURATION_MAX</a>    | ((uint32_t)128U)   |

| Type: <a href="#">hseMemRegAccess_t</a>       |                                                                                          |
|-----------------------------------------------|------------------------------------------------------------------------------------------|
| Name                                          | Value                                                                                    |
| <a href="#">HSE_MEM_REG_ACCESS_MASK_IN</a>    | 0x00003C96UL                                                                             |
| <a href="#">HSE_MEM_REG_ACCESS_MASK_OUT</a>   | 0x5A690000UL                                                                             |
| <a href="#">HSE_MEM_REG_ACCESS_MASK_INOUT</a> | <a href="#">HSE_MEM_REG_ACCESS_MASK_IN</a>   <a href="#">HSE_MEM_REG_ACCESS_MASK_OUT</a> |

| Type: <a href="#">hseMUConfig_t</a> |       |
|-------------------------------------|-------|
| Name                                | Value |
| <a href="#">HSE_MU_ACTIVATED</a>    | 0xA5U |
| <a href="#">HSE_MU_DEACTIVATED</a>  | 0x5AU |

| Type: <a href="#">hseAttrRamPubKeyImportPolicy_t</a>   |              |
|--------------------------------------------------------|--------------|
| Name                                                   | Value        |
| <a href="#">HSE_KM_POLICY_DEFAULT</a>                  | 0x4E8BD124UL |
| <a href="#">HSE_KM_POLICY_ALLOW_RAM_PUB_KEY_IMPORT</a> | 0xB1742EDBUL |

| Type: <a href="#">hseAttrCfg_t</a> |              |
|------------------------------------|--------------|
| Name                               | Value        |
| <a href="#">HSE_CFG_NO</a>         | 0x0UL        |
| <a href="#">HSE_CFG_YES</a>        | 0xB7A5C365UL |

| Type: <a href="#">hseTamperOutputClock_t</a> |       |
|----------------------------------------------|-------|
| Name                                         | Value |
| <a href="#">HSE_TAMPER_ACTIVE_CLOCK_16HZ</a> | 0U    |
| <a href="#">HSE_TAMPER_ACTIVE_CLOCK_8HZ</a>  | 1U    |
| <a href="#">HSE_TAMPER_ACTIVE_CLOCK_4HZ</a>  | 2U    |
| <a href="#">HSE_TAMPER_ACTIVE_CLOCK_2HZ</a>  | 3U    |

| Type: <a href="#">hseAttrCoreResetRelease_t</a> |              |
|-------------------------------------------------|--------------|
| Name                                            | Value        |
| <a href="#">HSE_CR_RELEASE_ALL_AT_ONCE</a>      | 0xA5556933UL |
| <a href="#">HSE_CR_RELEASE_ONE_BY_ONE</a>       | 0xA5557555UL |

| Type: <a href="#">hseOutputPinConfig_t</a> |       |
|--------------------------------------------|-------|
| Name                                       | Value |
| <a href="#">HSE_TAMPER_PASSIVE</a>         | 0U    |
| <a href="#">HSE_TAMPER_ACTIVE_ONE</a>      | 1U    |
| <a href="#">HSE_TAMPER_ACTIVE_TWO</a>      | 2U    |

| Type: <a href="#">hseFircDivConfig_t</a>  |       |
|-------------------------------------------|-------|
| Name                                      | Value |
| <a href="#">HSE_FIRC_NO_CONFIG</a>        | 0U    |
| <a href="#">HSE_FIRC_DIV_BY_1_CONFIG</a>  | 1U    |
| <a href="#">HSE_FIRC_DIV_BY_2_CONFIG</a>  | 2U    |
| <a href="#">HSE_FIRC_DIV_BY_16_CONFIG</a> | 16U   |

| Type: <a href="#">hseAttrConfigBootAuth_t</a> |       |
|-----------------------------------------------|-------|
| Name                                          | Value |
| <a href="#">HSE_IVT_NO_AUTH</a>               | 0x0U  |
| <a href="#">HSE_IVT_AUTH</a>                  | 0x1U  |

| Type: <a href="#">hseAttrDebugAuthMode_t</a> |       |
|----------------------------------------------|-------|
| Name                                         | Value |
| <a href="#">HSE_DEBUG_AUTH_MODE_PW</a>       | 0x0U  |
| <a href="#">HSE_DEBUG_AUTH_MODE_CR</a>       | 0x1U  |

| Type: <a href="#">hseAttrId_t</a>                       |       |
|---------------------------------------------------------|-------|
| Name                                                    | Value |
| <a href="#">HSE_NONE_ATTR_ID</a>                        | 0U    |
| <a href="#">HSE_FW_VERSION_ATTR_ID</a>                  | 1U    |
| <a href="#">HSE_CAPABILITIES_ATTR_ID</a>                | 2U    |
| <a href="#">HSE_SMR_CORE_BOOT_STATUS_ATTR_ID</a>        | 3U    |
| <a href="#">HSE_FW_BUILD_INFO_ATTR_ID</a>               | 4U    |
| <a href="#">HSE_DEBUG_AUTH_MODE_ATTR_ID</a>             | 10U   |
| <a href="#">HSE_APP_DEBUG_KEY_ATTR_ID</a>               | 11U   |
| <a href="#">HSE_SECURE_LIFECYCLE_ATTR_ID</a>            | 12U   |
| <a href="#">HSE_ENABLE_BOOT_AUTH_ATTR_ID</a>            | 13U   |
| <a href="#">HSE_EXTEND_CUST_SECURITY_POLICY_ATTR_ID</a> | 14U   |
| <a href="#">HSE_MU_CONFIG_ATTR_ID</a>                   | 20U   |
| <a href="#">HSE_EXTEND_OEM_SECURITY_POLICY_ATTR_ID</a>  | 21U   |

## Administration Services

|                                                                   |      |
|-------------------------------------------------------------------|------|
| <a href="#">HSE_FAST_CMACE_MIN_TAG_BIT_LEN_ATTR_ID</a>            | 22U  |
| <a href="#">HSE_CORE_RESET_RELEASE_ATTR_ID</a>                    | 23U  |
| <a href="#">HSE_RAM_PUB_KEY_IMPORT_POLICY_ATTR_ID</a>             | 24U  |
| <a href="#">HSE_PHYSICAL_TAMPER_ATTR_ID</a>                       | 30U  |
| <a href="#">HSE_MEM_REGIONS_PROTECT_ATTR_ID</a>                   | 31U  |
| <a href="#">HSE_DISABLE_PAIRWISE_CONSISTENCY_TEST_ATTR_ID</a>     | 32U  |
| <a href="#">HSE_FIRC_DIVIDER_CONFIG_ATTR_ID</a>                   | 600U |
| <a href="#">HSE_SECURE_RECOVERY_CONFIG_ATTR_ID</a>                | 601U |
| <a href="#">HSE_ENABLE_PUBLISH_KEY_STORE_RAM_TO_FLASH_ATTR_ID</a> | 602U |

| Type: <a href="#">hseTamperConfig_t</a>      |       |
|----------------------------------------------|-------|
| Name                                         | Value |
| <a href="#">HSE_TAMPER_CONFIG_DEACTIVATE</a> | 0U    |
| <a href="#">HSE_TAMPER_CONFIG_ACTIVATE</a>   | 1U    |

| Type: <a href="#">hseAttrSecureLifecycle_t</a> |       |
|------------------------------------------------|-------|
| Name                                           | Value |
| <a href="#">HSE_LC_CUST_DEL</a>                | 0x4U  |
| <a href="#">HSE_LC_OEM_PROD</a>                | 0x8U  |
| <a href="#">HSE_LC_IN_FIELD</a>                | 0x10U |
| <a href="#">HSE_LC_PRE_FA</a>                  | 0x14U |
| <a href="#">HSE_LC_SIMULATED_OEM_PROD</a>      | 0xA6U |
| <a href="#">HSE_LC_SIMULATED_IN_FIELD</a>      | 0xA7U |

| Type: <a href="#">hseAttrConfigSecureRecovery_t</a> |       |
|-----------------------------------------------------|-------|
| Name                                                | Value |
| <a href="#">HSE_SECURE_RECOVERY_DISABLE</a>         | 0x0U  |
| <a href="#">HSE_SECURE_RECOVERY_ENABLE</a>          | 0x1U  |

| Type: <a href="#">hseTamperPolarity_t</a>  |       |
|--------------------------------------------|-------|
| Name                                       | Value |
| <a href="#">HSE_TAMPER_POL_ACTIVE_LOW</a>  | 0U    |
| <a href="#">HSE_TAMPER_POL_ACTIVE_HIGH</a> | 1U    |

## Typedefs

- typedef uint32\_t [hseAttrCfg\\_t](#)
- typedef uint16\_t [hseAttrId\\_t](#)



- typedef uint64\_t hseAttrCapabilities\_t
- typedef uint32\_t hseAttrCoreResetRelease\_t
- typedef uint8\_t hseAttrDebugAuthMode\_t
- typedef uint8\_t hseAttrApplDebugKey\_t[16]
- typedef hseKeyHandle\_t hseAttrSecureApplDebugKey\_t
- typedef uint8\_t hseAttrSecureLifecycle\_t
- typedef uint8\_t hseAttrConfigBootAuth\_t
- typedef uint8\_t hseMUConfig\_t
- typedef uint32\_t hseMemRegAccess\_t
- typedef uint32\_t hseAttrRamPubKeyImportPolicy\_t
- typedef uint8\_t hseAttrFastCmacMinTagBitLen\_t
- typedef uint8\_t hseTamperConfig\_t
- typedef uint8\_t hseTamperPolarity\_t
- typedef uint8\_t hseOutputPinConfig\_t
- typedef uint8\_t hseTamperOutputClock\_t
- typedef uint8\_t hseFircDivConfig\_t
- typedef uint8\_t hseAttrConfigSecureRecovery\_t
- typedef hseAttrCfg\_t hsePublishNvmKeystoreRamtToFlash\_t
- typedef hseAttrCfg\_t hseDisablePairWiseConsistencyTest\_t

## Data Structure Documentation

### struct hseSetAttrSrv\_t

Set HSE attribute service.

Note

SuperUser rights (for NVM Configuration) are needed to perform this service.

Data Fields

| Type                        | Name        | Description                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseAttrId_t</a> | attrId      | INPUT: Specifies the HSE attribute ID.                                                                                                                                                                                                                                                                                                                                                                    |
| uint8_t                     | reserved[2] |                                                                                                                                                                                                                                                                                                                                                                                                           |
| uint32_t                    | attrLen     | INPUT: Specifies the attribute length (in bytes). The size of the memory location must be equal to the length of attribute structure.                                                                                                                                                                                                                                                                     |
| uint32_t                    | pAttr       | INPUT: The address of the attribute. The attribute must have the format of the corresponding attributes structure (see attributes definition)<br><br>Note<br><br>The comment for each attribute ID provides the name for the attribute data structure. E.g:The <a href="#">HSE_MU_CONFIG_ATTR_ID</a> definition includes the following comment: "NVM-RW-ATTR; MU configuration (see #hseAttrMUConfig_t)". |

## Administration Services

### struct hseGetAttrSrv\_t

Get HSE attribute service.

Data Fields

| Type                        | Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseAttrId_t</a> | attrId      | INPUT: Specifies the HSE attribute ID.                                                                                                                                                                                                                                                                                                                                                                                         |
| uint8_t                     | reserved[2] |                                                                                                                                                                                                                                                                                                                                                                                                                                |
| uint32_t                    | attrLen     | INPUT: Specifies the attribute length (in bytes).The size of the memory location must be bigger than or equal to the length of attribute structure.                                                                                                                                                                                                                                                                            |
| uint32_t                    | pAttr       | OUTPUT: The address where the attribute will be stored. The attribute must be stored in the format of the corresponding attribute Id (see the attributes definition).<br><br>Note<br><br>The comment for each attribute ID provides the name for the attribute data structure. E.g: The <a href="#">HSE_FW_VERSION_ATTR_ID</a> definition includes the following comment: "RO-ATTR; HSE FW version (see #hseAttrFwVersion_t)". |

### struct hseAttrFwVersion\_t

HSE FW version attribute (RO-ATTR attribute; refer to [hseAttrId\\_t](#)).

Data Fields

| Type     | Name         | Description                                                                                                                                                                                                                           |
|----------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint8_t  | reserved     | For HSE_B, it is used for OTA Config: 0 = Full Mem Config; 1 = AB Swap Config. For other SOC type: Reserved, expected to be 0.                                                                                                        |
| uint8_t  | socTypeId    | Identifies the SoC Type ID; same as HSE_PLATFORM from hse_target.h.                                                                                                                                                                   |
| uint16_t | fwTypeId     | Identifies the FW type: <ul style="list-style-type: none"><li>– 0 - Standard FW targeting all customers</li><li>– 1 - Premium FW targeting all customers</li><li>– 2-7 - Reserved</li><li>– 8 &gt;= Custom1, Custom2... etc</li></ul> |
| uint8_t  | majorVersion | Major revision. <ul style="list-style-type: none"><li>– 0 - Pre-stabilization releases</li><li>– 1 - at first stable interface release, and increased later if breaking changes were introduced</li></ul>                             |
| uint8_t  | minorVersion | Minor revision, bumped on new compatible changes added; reset to 0 on majorVersion bump, if majorVersion > 0.                                                                                                                         |

## Data Fields

| Type     | Name         | Description                                                                                                                   |
|----------|--------------|-------------------------------------------------------------------------------------------------------------------------------|
| uint16_t | patchVersion | Hotfix release (patch version, bug fix releases).<br>After majorVersion > 0, reset to 0 on majorVersion or minorVersion bump. |

**struct hseFwBuildInfo\_t**

HSE FW build information attribute (RO-ATTR attribute; refer to [hseAttrId\\_t](#)).

## Data Fields

| Type     | Name                  | Description                                            |
|----------|-----------------------|--------------------------------------------------------|
| uint64_t | buildUniqueIdentifier | Build unique identifier (8 bytes of SHA1 of the HEAD ) |
| uint32_t | buildDate             | Build date in hexadecimal (hex(YYYYMMDD))              |
| uint32_t | buildTime             | Build time in hexadecimal (hex(HHMMSS))                |

**struct hseAttrSmrCoreStatus\_t**

The SMR and Core Boot status (RO-ATTR attribute; refer to [hseAttrId\\_t](#)).

Provides the following information:

- SMR entry installation status corresponding to the entries present in SMR table (refer to [smrEntryInstallStatus](#))
- SMR verification status corresponding to the entries present in SMR table (refer to [smrStatus\[\]](#))
- Provides Core Boot status (refer to [coreBootStatus\[\]](#))
- In case Basic Secure Boot (BSB) is performed, it provides the Core Boot status and the location of loaded application (primary/backup, refer to [coreBootStatus\[\]](#))

## Data Fields

| Type     | Name          | Description                                                                                                                                                                                                                                                                                                                                       |
|----------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | smrStatus[2U] | 0-31 bit will represent 32 SMR table entries (applicable when SMR is present/enabled). <ul style="list-style-type: none"> <li>– smrStatus[0].bit : 0 - SMR Not verified</li> <li>– smrStatus[0].bit : 1 - SMR verified</li> <li>– smrStatus[1].bit : 0 - SMR verification fail</li> <li>– smrStatus[1].bit : 1 - SMR verification pass</li> </ul> |

## Administration Services

### Data Fields

| Type     | Name                  | Description                                                                                                                                                                                                                                                                                                                                                     |
|----------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | coreBootStatus[2U]    | 0-31 bit will represent CORE-ID (0-31): <ul style="list-style-type: none"><li>– coreBootStatus[0].bit : 1 - Core booted</li><li>– coreBootStatus[0].bit : 0 - Core Not booted</li><li>– coreBootStatus[1].bit : 1 - Core booted with pass/primary reset address</li><li>– coreBootStatus[1].bit : 0 - Core booted with alternate/backup reset address</li></ul> |
| uint32_t | smrEntryInstallStatus | 0-31 bit will represent 32 SMR table entries (applicable when SMR is present/enabled). <ul style="list-style-type: none"><li>– bit : 0 - SMR entry not installed</li><li>– bit : 1 - SMR entry installed</li></ul>                                                                                                                                              |

### struct hseAttrMUInstanceConfig\_t

MU Configuration and XRDC configuration definition for a MU interface.

Configures a MU interface and XRDC configuration for the HOST Interface Memory.

#### Note

If the device does have (or use) any Host Interface memory, the `xrdcDomainId` and `sharedMemChunkSize` can be set zero.

### Data Fields

| Type                          | Name               | Description                                                                                                                                                                                                                                                                                                                      |
|-------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseMUConfig_t</a> | muConfig           | This value specifies MU interface state. <ul style="list-style-type: none"><li>– <a href="#">HSE_MU_ACTIVATED</a> : MU interface activated</li><li>– <a href="#">HSE_MU_DEACTIVATED</a> : MU interface deactivated</li></ul> <p>Note</p> <p>It is not allowed to deactivate the MU0 interface</p>                                |
| uint8_t                       | xrdcDomainId       | Domain Id to access the Host Interface memory chunk reserved for the MU interface. Must have a value between interval [0, 7]. The <code>xrdcDomainId</code> field is not taken into account when the <a href="#">sharedMemChunkSize</a> field is equal to 0.                                                                     |
| uint16_t                      | sharedMemChunkSize | Specifies what chunk of host interface memory to reserve for the specific MU interface. For a value of 0 there is no memory reserved for the MU interface. If the <a href="#">sharedMemChunkSize</a> field is equal to 0 for all MU interfaces, the XRDC is disabled and there are no restrictions on the host interface memory. |
| uint8_t                       | reserved[60]       |                                                                                                                                                                                                                                                                                                                                  |

**struct hseAttrMUConfig\_t**

MU Configurations and XRDC configuration definition (NVM-RW-ATTR attribute; refer to [hseAttrId\\_t](#)).  
Configures the MU interfaces and XRDC configurations for the HOST Interface Memory.

@note:  
- SU rights are needed

## Data Fields

| Type                                      | Name              | Description                                        |
|-------------------------------------------|-------------------|----------------------------------------------------|
| <a href="#">hseAttrMUInstanceConfig_t</a> | muInstances[(2U)] | Contains the configurations for all MU interfaces. |

**struct hseAttrMemRegion\_t**

HSE Memory region.

Defines base address and length of a region

## Data Fields

| Type                              | Name       | Description                                     |
|-----------------------------------|------------|-------------------------------------------------|
| <a href="#">hseMemRegAccess_t</a> | accessType | INPUT: Access type on which the region applies. |
| uint32_t                          | length     | INPUT: Length of memory region.                 |
| uint32_t                          | pBaseAddr  | INPUT: Start address of memory region.          |

**struct hseAttrMuMemRegions\_t**

HSE Memory region attribute for a single MU.

Defines the number of regions and their start address and sizes for a single MU

## Data Fields

| Type                               | Name                | Description                                                                                    |
|------------------------------------|---------------------|------------------------------------------------------------------------------------------------|
| uint8_t                            | numofMemRegions     | INPUT: Specify the number of memory regions for one MU.<br><br>Note<br>Set to zero if not used |
| uint8_t                            | reserved[3]         |                                                                                                |
| <a href="#">hseAttrMemRegion_t</a> | memRegionList[(6U)] | INPUT: Specifies the memory regions for one MU.                                                |

### struct hseAttrAllMuMemRegions\_t

HSE Memory regions protection attribute for all HSE MUs (SET-ONLY-ONCE-ATTR attribute; refer to [hseAttrId\\_t](#)).

HSE Memory regions protection is a service used to prevent memory accesses between disallowed bus masters through HSE MUs. HSE uses these regions to validate the input/output parameters for each service received on the corresponding MU.

#### Note

- The attribute is not persistent and can only be set once. A reset is necessary for this configuration to be settable again.
- Input and output data linked via pointers in the service descriptor (these are typically pointers to SRAM or DRAM) can be isolated between hosts using the HSE\_MEM\_REGIONS\_PROTECT\_ATTR\_ID attribute. The host can communicate to HSE the memory ranges that are associated with each MU instance. If provided, the HSE dismisses the data that falls outside the ranges for a particular MU instance.

#### Data Fields

| Type                                  | Name               | Description                                   |
|---------------------------------------|--------------------|-----------------------------------------------|
| <a href="#">hseAttrMuMemRegions_t</a> | muMemRegions[(2U)] | INPUT: Array with memory regions for all MUs. |

### struct hseAttrExtendCustSecurityPolicy\_t

HSE extend CUST security policies attribute definition (OTP-ATTR & NVM-RW-ATTR attribute; refer to [hseAttrId\\_t](#)).

Determines whether certain security policies are extended in HSE Firmware or not; applies only for CUST\_DEL LC.

- Read: Tells which extended security policies are set or not.
- Write:
  - If a given policy is not set to be TRUE, there is no change on security policy extension.
  - If a given policy is set to be TRUE, security policy is extended on successful operation.
  - Write operation is allowed only for users with CUST SU rights in CUST\_DEL LC.

## Data Fields

| Type    | Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bool_t  | enableADKm  | <p>Application Debug Key/Password (attribute) diversified with UID before being written in fuse. The supplied 128-bit value for ADK/P attribute will be interpreted as ADKPm (customer's master key/ password). If needed, this policy must be set before setting ADK/P attribute.</p> <p>If set, the following logic must be used at customer's end for debug-authorization:</p> <ul style="list-style-type: none"> <li>– hUID = SHA2_256(UID)</li> <li>– hADKPm = SHA2_256(ADKPm)</li> <li>– ADKP {for debugger} = AES256-ECB(hUID(16 bytes..0 to 15)), key = hADKPm; {ADKPm = customer's master key/ password}. The hash of ADKPm (set using ADKP attribute) will be used as the key in the derivation of the application password. An error will be returned if the value of this attribute is given as 0 from host interface.</li> </ul> <p>Note</p> <p>For HSE_H/M, the UID is read by HSE from system fuses. The application must provide read access (xRDC restriction) to HSE in order to be able to read the UID.</p> |
| bool_t  | startAsUser | <p>Host starts with User rights in LC = CUST_DEL.</p> <p>Note</p> <p>Setting this attribute will take effect only after publishing the SYS Image and issuing a reset.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| uint8_t | reserved[2] | HSE reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**struct hseAttrExtendOemSecurityPolicy\_t**

HSE extend OEM security policies attribute definition (NVM-RW-ATTR attribute; refer to [hseAttrId\\_t](#)).

Determines whether certain security policies are extended in HSE Firmware or not in OEM\_PROD LC.

- Read: Tells which extended security policies are set or not.
- Write:
  - If a given policy is not set to be TRUE, there is no change on security policy extension.
  - If a given policy is set to be TRUE, security policy is extended on successful operation.
  - Write operation is allowed only for users with OEM SU rights in OEM\_PROD LC.

## Administration Services

### Data Fields

| Type    | Name        | Description                                                                                                                                                        |
|---------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bool_t  | startAsUser | Host starts with User rights in LC = OEM_PROD.<br><br>Note<br><br>Setting this attribute will take effect only after publishing the SYS Image and issuing a reset. |
| uint8_t | reserved[3] | HSE reserved.                                                                                                                                                      |

### struct hseAttrPhysicalTamper\_t

Enables the tamper violation in HSE subsystem for all physical tampers supported by the SOC.

This service only enables the tamper violation in HSE subsystem for all physical tampers supported by the SOC. Once violation is active it cannot be disabled until next reset.

Physical tamper feature can be configured in following two ways:

1. Active Tamper Configuration
2. Passive tamper configuration

Note

User must configure the GPIO pins for tamper functionality before calling this service; otherwise, a false violation can be triggered by HSE. User is also recommended to protect the tamper GPIO configuration using register protection, virtual wrapper and XRDC configuration against further modification by any application running on host side.

### Data Fields

| Type                                 | Name               | Description                                                                                                                                                                                                                                                                                        |
|--------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseTamperConfig_t</a>    | tamperConfig       | This field indicates the tamper configuration to be enable or not.                                                                                                                                                                                                                                 |
| <a href="#">hseOutputPinConfig_t</a> | tamperOutputConfig | This parameter tells which type (Active or Passive) of input is connected to external tamper input. If it is an active input, up to 2 tamper options can be selected as input source for external tamper input. Based on the value of this parameter, the clock will be driven on this pad by HSE. |



## Data Fields

| Type                                   | Name              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint8_t                                | filterDuration    | Configures the length of the digital glitch filter for the external tamper pin between 128 and 32640 SIRC clock cycles. Any assertion on external tamper that is equal to or less than the value of the digital glitch filter is ignored. The length of the glitches filtered out is:<br>– $128 + ((\text{FilterDuration} - 1) \times 256)$ , where $\text{FilterDuration} = 1, \dots, 128$ .<br>If the FilterDuration value is 0, then the glitch filter will not be enabled. Filter Duration is a must requirement for Active Tamper and optional for Passive Tamper. |
| <a href="#">hseTamperPolarity_t</a>    | tamperPolarity    | This field indicates the polarity of the tamper to be configured. It can be "Active LOW" or "Active HIGH". This parameter is considered only when the tamper source in tamperOutputConfig is selected as passive.                                                                                                                                                                                                                                                                                                                                                       |
| <a href="#">hseTamperOutputClock_t</a> | tamperActiveClock | Determines the clock to be driven on the output pad of the tamper. This parameter is considered only when the tamper source in tamperOutputConfig is selected as active.                                                                                                                                                                                                                                                                                                                                                                                                |
| uint8_t                                | reserved[3]       | HSE reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**struct hseAttrPhysicalTamperConfig\_t**

Physical Tamper Configurations (SET-ONLY-ONCE-ATTR attribute; refer to [hseAttrId\\_t](#)).

Configures all available physical tamper instances.

## Data Fields

| Type                                    | Name                  | Description                                                        |
|-----------------------------------------|-----------------------|--------------------------------------------------------------------|
| <a href="#">hseAttrPhysicalTamper_t</a> | tamperInstances[(1U)] | Contains the configuration for all the physical temper interfaces. |

**Macro Definition Documentation****HSE\_CFG\_NO**

```
#define HSE_CFG_NO ((hseAttrCfg_t) (0x0UL))
```

NO, deactivate the configuration.

## Administration Services

### HSE\_CFG\_YES

```
#define HSE_CFG_YES ((hseAttrCfg_t) (0xB7A5C365UL))
```

YES, activate the configuration

### HSE\_NONE\_ATTR\_ID

```
#define HSE_NONE_ATTR_ID ((hseAttrId_t) 0U)
```

### HSE\_FW\_VERSION\_ATTR\_ID

```
#define HSE_FW_VERSION_ATTR_ID ((hseAttrId_t) 1U)
```

RO-ATTR; HSE FW version (see [hseAttrFwVersion\\_t](#))

### HSE\_CAPABILITIES\_ATTR\_ID

```
#define HSE_CAPABILITIES_ATTR_ID ((hseAttrId_t) 2U)
```

RO-ATTR; HSE capabilities (see [hseAttrCapabilities\\_t](#))

### HSE\_SMR\_CORE\_BOOT\_STATUS\_ATTR\_ID

```
#define HSE_SMR_CORE_BOOT_STATUS_ATTR_ID ((hseAttrId_t) 3U)
```

RO-ATTR; SMR verification & Core-boot status (see [hseAttrSmrCoreStatus\\_t](#))

### HSE\_FW\_BUILD\_INFO\_ATTR\_ID

```
#define HSE_FW_BUILD_INFO_ATTR_ID ((hseAttrId_t) 4U)
```

RO-ATTR; HSE FW Build information (see [hseFwBuildInfo\\_t](#))

**HSE\_DEBUG\_AUTH\_MODE\_ATTR\_ID**

```
#define HSE_DEBUG_AUTH_MODE_ATTR_ID ((hseAttrId_t)10U)
```

OTP-ATTR; Debug Authorization mode (see [hseAttrDebugAuthMode\\_t](#))

**HSE\_APP\_DEBUG\_KEY\_ATTR\_ID**

```
#define HSE_APP_DEBUG_KEY_ATTR_ID ((hseAttrId_t)11U)
```

OTP-ATTR; Application Debug Key / Password (see [hseAttrApplDebugKey\\_t](#) and [hseAttrSecureApplDebugKey\\_t](#))

**HSE\_SECURE\_LIFECYCLE\_ATTR\_ID**

```
#define HSE_SECURE_LIFECYCLE_ATTR_ID ((hseAttrId_t)12U)
```

OTP-ADVANCE-ATTR; Secure Life-cycle (see [hseAttrSecureLifecycle\\_t](#))

**HSE\_ENABLE\_BOOT\_AUTH\_ATTR\_ID**

```
#define HSE_ENABLE_BOOT_AUTH_ATTR_ID ((hseAttrId_t)13U)
```

OTP-ATTR; If this attribute bit is set, all the artifacts handled by BootROM(e.g IVT, DCD etc) must be authenticated. (see [hseAttrConfigBootAuth\\_t](#))

**HSE\_EXTEND\_CUST\_SECURITY\_POLICY\_ATTR\_ID**

```
#define HSE_EXTEND_CUST_SECURITY_POLICY_ATTR_ID ((hseAttrId_t)14U)
```

OTP-ATTR & NVM-RW-ATTR; HSE security policies extension in CUST\_DEL lifecycle for user with CUST SU rights (see [hseAttrExtendCustSecurityPolicy\\_t](#)). \ Note that this attribute also enables the ADKPM in OTP (ADKP diversified with UID), along with the START\_AS\_USER setting for CUST\_DEL lifecycle.

**HSE\_MU\_CONFIG\_ATTR\_ID**

```
#define HSE_MU_CONFIG_ATTR_ID ((hseAttrId_t)20U)
```

## Administration Services

NVM-RW-ATTR; MU configuration (see [hseAttrMUConfig\\_t](#))

### HSE\_EXTEND\_OEM\_SECURITY\_POLICY\_ATTR\_ID

```
#define HSE_EXTEND_OEM_SECURITY_POLICY_ATTR_ID ((hseAttrId_t)21U)
```

NVM-RW-ATTR; HSE security policies extension in OEM\_PROD lifecycle for user with OEM SU rights (see [hseAttrExtendOemSecurityPolicy\\_t](#))

### HSE\_FAST\_CMACE\_MIN\_TAG\_BIT\_LEN\_ATTR\_ID

```
#define HSE_FAST_CMACE_MIN_TAG_BIT_LEN_ATTR_ID ((hseAttrId_t)22U)
```

NVM-RW-ATTR; The minimum tag bit length that can be used for Fast CMACE verify/generate (see [hseAttrFastCmacMinTagBitLen\\_t](#))

### HSE\_CORE\_RESET\_RELEASE\_ATTR\_ID

```
#define HSE_CORE_RESET_RELEASE_ATTR_ID ((hseAttrId_t)23U)
```

NVM-RW-ATTR; Specifies Core Reset table parsing strategy (see [hseAttrCoreResetRelease\\_t](#))

### HSE\_RAM\_PUB\_KEY\_IMPORT\_POLICY\_ATTR\_ID

```
#define HSE_RAM_PUB_KEY_IMPORT_POLICY_ATTR_ID ((hseAttrId_t)24U)
```

NVM-RW-ATTR; Specifies RAM public keys import policy in advanced LCs (see [hseAttrRamPubKeyImportPolicy\\_t](#))

### HSE\_PHYSICAL\_TAMPER\_ATTR\_ID

```
#define HSE_PHYSICAL_TAMPER_ATTR_ID ((hseAttrId_t)30U)
```

SET-ONLY-ONCE-ATTR; Enables the physical tamper violation in HSE. Once the violation is enabled in HSE, it can not be cleared until next reset. There are two tamper related functions available on PADS: Input (TAMPER\_IN), Output (TAMPER\_OUT). To support protection against physical tampering, connect TAMPER\_OUT to TAMPER\_IN. Any physical tamper that breaks this connectivity sets off an alarm at HSE (if enabled using this attribute). User can optionally lock those pads configuration for

further modification using virtual wrapper (refer to [hseAttrPhysicalTamper\\_t](#)). The configuration status is provided by reading the HSE\_GPR\_STATUS\_ADDRESS register (refer to [hseTamperConfigStatus\\_t](#)).

### **HSE\_MEM\_REGIONS\_PROTECT\_ATTR\_ID**

```
#define HSE_MEM_REGIONS_PROTECT_ATTR_ID ((hseAttrId_t) 31U)
```

SET-ONLY-ONCE-ATTR; Configures memory regions accessible through each MU (refer to [hseAttrAllMuMemRegions\\_t](#))

### **HSE\_DISABLE\_PAIRWISE\_CONSISTENCY\_TEST\_ATTR\_ID**

```
#define HSE_DISABLE_PAIRWISE_CONSISTENCY_TEST_ATTR_ID ((hseAttrId_t) 32U)
```

SET-ONLY-ONCE-ATTR; Disable the pair wise consistency test when calling import RSA/ECC key pair (see [hseDisablePairWiseConsistencyTest\\_t](#))

### **HSE\_FIRC\_DIVIDER\_CONFIG\_ATTR\_ID**

```
#define HSE_FIRC_DIVIDER_CONFIG_ATTR_ID ((hseAttrId_t) 600U)
```

RAM-RW; FIRC Divider Configuration by HSE Firmware from HSE\_GPR (see [hseFircDivConfig\\_t](#))

### **HSE\_SECURE\_RECOVERY\_CONFIG\_ATTR\_ID**

```
#define HSE_SECURE_RECOVERY_CONFIG_ATTR_ID ((hseAttrId_t) 601U)
```

OTP-ATTR; Secure Recovery Configuration by HSE Firmware (see [hseAttrConfigSecureRecovery\\_t](#))

### **HSE\_ENABLE\_PUBLISH\_KEY\_STORE\_RAM\_TO\_FLASH\_ATTR\_ID**

```
#define HSE_ENABLE_PUBLISH_KEY_STORE_RAM_TO_FLASH_ATTR_ID ((hseAttrId_t) 602U)
```

RAM-RW; Allow to publish the NVM keystore from secure NVM keystore into the data flash (see [hsePublishNvmKeystoreRamToFlash\\_t](#))

## Administration Services

### HSE\_ALGO\_CAP\_MASK

```
#define HSE_ALGO_CAP_MASK( capIdx ) (1ULL << (capIdx))
```

Provided the bit (used in `hseAttrCapabilities_t`) based on the algorithm capability index (see [hseAlgoCapIdx\\_t](#))

### HSE\_CR\_RELEASE\_ALL\_AT\_ONCE

```
#define HSE_CR_RELEASE_ALL_AT_ONCE ((hseAttrCoreResetRelease_t)0xA5556933UL)
```

Cores are released all-at-once after the pre-boot verification phase is over.

### HSE\_CR\_RELEASE\_ONE\_BY\_ONE

```
#define HSE_CR_RELEASE_ONE_BY_ONE ((hseAttrCoreResetRelease_t)0xA5557555UL)
```

Cores are released from reset one-by-one after their respective pre-boot phase has finalized successfully \ (i.e. the SMR entries linked to the core via CR table have been loaded and verified).

\ The cores are released in ascending order of their indices in the Core Reset table.

\ Flashless devices (e.g. HSE\_H/M) limitations: .

- Only the first Core Reset entry can be booted from SD/MMC. \
- The system clocks and QSPI configurations shall not be changed by the core(s) booted until [HSE\\_STATUS\\_BOOT\\_OK](#) status is set.

### HSE\_DEBUG\_AUTH\_MODE\_PW

```
#define HSE_DEBUG_AUTH_MODE_PW ((hseAttrDebugAuthMode_t)0x0U)
```

Password based application debug authorization mode. .

- Read: Application debug authorization will be password based. \
- Write: Does not affect application debug authorization mode at all.

### HSE\_DEBUG\_AUTH\_MODE\_CR

```
#define HSE_DEBUG_AUTH_MODE_CR ((hseAttrDebugAuthMode_t)0x1U)
```

Challenge-Response based application debug authorization mode. .

- Read: Application debug authorization will be challenge-response based. \

- Write: Enables challenge-response application debug authorization mode. \ Once this mode is enabled, it cannot be disabled. \ Operation allowed in CUST\_DEL, OEM\_PROD and IN\_FIELD LCs only.

## HSE\_LC\_CUST\_DEL

```
#define HSE_LC_CUST_DEL ((hseAttrSecureLifecycle_t)0x4U)
```

Customer Delivery Lifecycle. .

- Read: The current LC is CUST\_DEL. \
- Write: Advancement to this LC is not allowed (through HSE Firmware).

## HSE\_LC\_OEM\_PROD

```
#define HSE_LC_OEM_PROD ((hseAttrSecureLifecycle_t)0x8U)
```

OEM Production Lifecycle. .

- Read: The current LC is OEM\_PROD. \
- Write: Advancement to this LC is allowed only once (from CUST\_DEL LC). \ The key catalogs MUST be configured before advancing to this lifecycle.

## HSE\_LC\_IN\_FIELD

```
#define HSE_LC_IN_FIELD ((hseAttrSecureLifecycle_t)0x10U)
```

In-Field Lifecycle. .

- Read: The current LC is IN\_FIELD. \
- Write: Advancement to this LC is allowed only once (from CUST\_DEL, OEM\_PROD LCs). \ The key catalogs MUST be configured before advancing to this lifecycle.

## HSE\_LC\_PRE\_FA

```
#define HSE_LC_PRE_FA ((hseAttrSecureLifecycle_t)0x14U)
```

Pre-Failure Analysis Lifecycle. .

- Read: The current LC is Pre-FA. \
- Write: Advancement from/to this LC is NOT allowed (through HSE Firmware).

### HSE\_LC\_SIMULATED\_OEM\_PROD

```
#define HSE_LC_SIMULATED_OEM_PROD ((hseAttrSecureLifecycle_t)0xA6U)
```

Simulated OEM\_PROD to avoid writing in FUSE/UTEST. A system reset will revert LC to FUSE/UTEST value. .

- Read: The current LC is OEM\_PROD. \
- Write: Advancement to this LC is allowed only once (from CUST\_DEL LC). \ The key catalogs MUST be configured before advancing to this lifecycle.

### HSE\_LC\_SIMULATED\_IN\_FIELD

```
#define HSE_LC_SIMULATED_IN_FIELD ((hseAttrSecureLifecycle_t)0xA7U)
```

Simulated IN\_FIELD to avoid writing in FUSE/UTEST. A system reset will revert LC to FUSE/UTEST value. .

- Read: The current LC is IN\_FIELD. \
- Write: Advancement to this LC is allowed only once (from CUST\_DEL, SIMULATED\_OEM\_PROD LCs). \ The key catalogs MUST be configured before advancing to this lifecycle.

### HSE\_IVT\_NO\_AUTH

```
#define HSE_IVT_NO_AUTH ((hseAttrConfigBootAuth_t)0x0U)
```

\ For HSE\_H/M, the IVT/DCD/ST is not authenticated by BootROM: \

- Read: IVT/DCD/ST is not authenticated by BootROM. \
- Write: Does not affect IVT/ DCD authentication value at all. \ \ For HSE\_B, the IVT configuration is not authenticated by Secure BAF: \
- Read: IVT is not authenticated by Secure BAF. \
- Write: Does not affect IVT configuration authentication value at all. \

### HSE\_IVT\_AUTH

```
#define HSE_IVT_AUTH ((hseAttrConfigBootAuth_t)0x1U)
```

\ For HSE\_H/M, the IVT/DCD/ST to be authenticated by BootROM: \

- Read: IVT/DCD/ST is authenticated by BootROM. \
- Write: Sets IVT/DCD/ST authentication value. \ Once this value is set, it cannot be cleared back. \ Operation allowed in CUST\_DEL, OEM\_PROD & IN\_FIELD LCs only. \ \ For HSE\_B, the IVT to be authenticated by Secure BAF: \
- Read: IVT will be authenticated by Secure BAF. \



- Write: Sets IVT authentication value. \ Once this value is set, it cannot be cleared back. \ Operation allowed in CUST\_DEL, OEM\_PROD & IN\_FIELD LCs only. \

**HSE\_MU\_ACTIVATED**

```
#define HSE_MU_ACTIVATED ((hseMUConfig_t) (0xA5U))
```

HSE enables the receive interrupt on the MU interface.

**HSE\_MU\_DEACTIVATED**

```
#define HSE_MU_DEACTIVATED ((hseMUConfig_t) (0x5AU))
```

HSE disables the receive interrupt on the MU interface.

**HSE\_MEM\_REG\_ACCESS\_MASK\_IN**

```
#define HSE_MEM_REG_ACCESS_MASK_IN ((hseMemRegAccess_t) (0x00003C96UL))
```

**HSE\_MEM\_REG\_ACCESS\_MASK\_OUT**

```
#define HSE_MEM_REG_ACCESS_MASK_OUT ((hseMemRegAccess_t) (0x5A690000UL))
```

**HSE\_MEM\_REG\_ACCESS\_MASK\_INOUT**

```
#define HSE_MEM_REG_ACCESS_MASK_INOUT ((hseMemRegAccess_t) ( HSE_MEM_REG_ACCESS_MASK_IN | HSE_MEM_REG_ACCESS_MASK_OUT ))
```

**HSE\_KM\_POLICY\_DEFAULT**

```
#define HSE_KM_POLICY_DEFAULT ((hseAttrRamPubKeyImportPolicy_t) (0x4E8BD124UL))
```

## Administration Services

### HSE\_KM\_POLICY\_ALLOW\_RAM\_PUB\_KEY\_IMPORT

```
#define  
HSE_KM_POLICY_ALLOW_RAM_PUB_KEY_IMPORT ((hseAttrRamPubKeyImportPolicy_t) (0xB1742EDBUL))
```

### HSE\_TAMPER\_CONFIG\_DEACTIVATE

```
#define HSE_TAMPER_CONFIG_DEACTIVATE ((hseTamperConfig_t) (0U))
```

HSE Tamper Deactivate.

### HSE\_TAMPER\_CONFIG\_ACTIVATE

```
#define HSE_TAMPER_CONFIG_ACTIVATE ((hseTamperConfig_t) (1U))
```

HSE Tamper Activate.

### HSE\_TAMPER\_POL\_ACTIVE\_LOW

```
#define HSE_TAMPER_POL_ACTIVE_LOW ((hseTamperPolarity_t) (0U))
```

HSE Tamper Active low polarity.

### HSE\_TAMPER\_POL\_ACTIVE\_HIGH

```
#define HSE_TAMPER_POL_ACTIVE_HIGH ((hseTamperPolarity_t) (1U))
```

HSE Tamper Active high polarity.

### HSE\_FILTER\_DURATION\_MAX

```
#define HSE_FILTER_DURATION_MAX ((uint32_t)128U)
```

Filter Duration.

This macro describes the maximum filter duration that is possible for the physical tamper. The clock frequency used in the glitch filter is 32 KHz.

**HSE\_TAMPER\_PASSIVE**

```
#define HSE_TAMPER_PASSIVE ((hseOutputPinConfig_t) (0U))
```

**HSE\_TAMPER\_ACTIVE\_ONE**

```
#define HSE_TAMPER_ACTIVE_ONE ((hseOutputPinConfig_t) (1U))
```

**HSE\_TAMPER\_ACTIVE\_TWO**

```
#define HSE_TAMPER_ACTIVE_TWO ((hseOutputPinConfig_t) (2U))
```

**HSE\_TAMPER\_ACTIVE\_CLOCK\_16HZ**

```
#define HSE_TAMPER_ACTIVE_CLOCK_16HZ ((hseTamperOutputClock_t) (0U))
```

**HSE\_TAMPER\_ACTIVE\_CLOCK\_8HZ**

```
#define HSE_TAMPER_ACTIVE_CLOCK_8HZ ((hseTamperOutputClock_t) (1U))
```

**HSE\_TAMPER\_ACTIVE\_CLOCK\_4HZ**

```
#define HSE_TAMPER_ACTIVE_CLOCK_4HZ ((hseTamperOutputClock_t) (2U))
```

**HSE\_TAMPER\_ACTIVE\_CLOCK\_2HZ**

```
#define HSE_TAMPER_ACTIVE_CLOCK_2HZ ((hseTamperOutputClock_t) (3U))
```

**HSE\_FIRC\_NO\_CONFIG**

```
#define HSE_FIRC_NO_CONFIG ((hseFircDivConfig_t) 0U)
```

## Administration Services

No Configuration.

### HSE\_FIRC\_DIV\_BY\_1\_CONFIG

```
#define HSE_FIRC_DIV_BY_1_CONFIG ((hseFircDivConfig_t)1U)
```

HSE enables the FIRC divider by 1.

### HSE\_FIRC\_DIV\_BY\_2\_CONFIG

```
#define HSE_FIRC_DIV_BY_2_CONFIG ((hseFircDivConfig_t)2U)
```

HSE enables the FIRC divider by 2.

### HSE\_FIRC\_DIV\_BY\_16\_CONFIG

```
#define HSE_FIRC_DIV_BY_16_CONFIG ((hseFircDivConfig_t)16U)
```

HSE enables the FIRC divider by 16.

### HSE\_SECURE\_RECOVERY\_DISABLE

```
#define HSE_SECURE_RECOVERY_DISABLE ((hseAttrConfigSecureRecovery_t)0x0U)
```

- \
- Secure Recovery is disabled by HSE Firmware. \
  - Write: It does not affect the value at all. \
  - Read: Secure Recovery OTP flag is not programmed. \

### HSE\_SECURE\_RECOVERY\_ENABLE

```
#define HSE_SECURE_RECOVERY_ENABLE ((hseAttrConfigSecureRecovery_t)0x1U)
```

- \
- Secure Recovery is enabled by HSE Firmware. \
  - Write: It enables the Secure Recovery mode. \
  - Read: Secure Recovery OTP flag is programmed. \

## Typedef Documentation

### **hseAttrCfg\_t**

```
typedef uint32_t hseAttrCfg_t
```

Activate or not a specific configuration.

Tells whether the HSE activate or not a specific configuration.

### **hseAttrId\_t**

```
typedef uint16_t hseAttrId_t
```

HSE attribute IDs.

The following attribute types are defined:

- RO-ATTR - Read-Only attribute
- OTP-ATTR - One Time Programmable; can be written only once (set FUSE/UTEST area)
- OTP-ADVANCE-ATTR - One Time Programmable attribute that can only be advanced (e.g. LifeCycle)
- NVM-RW-ATTR - System NVM attributes; can be read or written.
- SET-ONCE-ATTR- Once the attribute is set, it can not be changed until next reset (e.g. can be set once at initialization time)

#### Note

- To set/update the OTP-ATTR or NVM-RW-ATTR attributes (except SET-ONCE-ATTR and OTP-ADVANCE-ATTR), the host needs SuperUser rights.
- After setting the HSE\_SECURE\_LIFECYCLE\_ATTR\_ID attribute, a destructive reset is needed.
- For HSE\_H/M devices, if the NVM-RW-ATTR attributes were updated, the SYS-IMAGE must be published and stored in external flash.
- For HSE\_H/M devices, in order to program fuses (OTP-ATTR and OTP-ADVANCE-ATTR) during development and potentially in the field, the application must supply power to the VDD\_EFUSE pin of the device. This allows the programming operation to take place. If the VDD\_EFUSE is not powered and the application tries to write a fuse, the HSE returns an error (HSE\_SRV\_RSP\_FUSE\_VDD\_GND). The VDD\_EFUSE state is checked before the fuse write by reading the NCSPD\_STAT register of the on-chip PMC module. The application must provide read access (xRDC restriction) to HSE in order to be able to read the NCSPD\_STAT register. For more details, refer to HSE FW Reference Manual.

### **hseAttrCapabilities\_t**

```
typedef uint64_t hseAttrCapabilities_t
```

HSE capabilities bits definition (RO-ATTR attribute; refer to [hseAttrId\\_t](#)).

## Administration Services

Provides information about the capabilities of HSE security blocks (list of what algorithms are supported). Each bit specifies an supported algorithm. The index for each bit in the attribute is defined by [hseAlgoCapIdx\\_t](#).

### **hseAttrCoreResetRelease\_t**

```
typedef uint32_t hseAttrCoreResetRelease_t
```

The Core Reset release from reset method (NVM-RW-ATTR attribute; refer to [hseAttrId\\_t](#)).

Specifies the startup method for releasing the application core from reset.

### **hseAttrDebugAuthMode\_t**

```
typedef uint8_t hseAttrDebugAuthMode_t
```

Debug Authorization Mode bit (OTP-ATTR attribute; refer to [hseAttrId\\_t](#)).

Tells whether the Application debug authorization will be password based or challenge-response based.

### **hseAttrApplDebugKey\_t**

```
typedef uint8_t hseAttrApplDebugKey_t[16]
```

Application Debug Key/ Password definition (OTP-ATTR attribute; refer to [hseAttrId\\_t](#)).

It is an 128-bit Application Debug Key/ Password to be set by the host in CUST\_DEL LifeCycle.

- Read: Not allowed if ADKP has not been written yet. After it has been written, first 16 bytes of SHA2\_224(ADKP) can be requested via get ADKP attribute service.
- Write: ADKP can be updated only once. The operation allowed only in CUST\_DEL LifeCycle.

Note

Provided ADKP, if containing all 0x00 bytes or all 0xFF bytes will be rejected by HSE Firmware with the response [HSE\\_SRV\\_RSP\\_INVALID\\_PARAM](#).

### **hseAttrSecureApplDebugKey\_t**

```
typedef hseKeyHandle_t hseAttrSecureApplDebugKey_t
```

Secure Application Debug Key/ Password definition (OTP-ATTR attribute; refer to [hseAttrId\\_t](#)).

It is the key handle referencing a key already installed in HSE. It must be an AES 128-bits key from RAM or NVM key catalogs.

- Read: Allowed only as the hash over the ADKP (see Read from [hseAttrApplDebugKey\\_t](#)).
- Write:

- ADKP can be updated only once. The operation allowed only in CUST\_DEL LifeCycle.
- The key referenced must be installed in HSE a priori. After the key is written successfully in the fuse as ADK/P, it will be erased from the RAM/NVM key catalog.

### **hseAttrSecureLifecycle\_t**

```
typedef uint8_t hseAttrSecureLifecycle_t
```

HSE secure lifecycle definition (OTP-ADVANCE-ATTR attribute; refer to [hseAttrId\\_t](#)).

Represents HSE secure lifecycle. The lifecycle can be advanced only in forward direction. Warnings:

- The lifecycle is read/scanned by hardware during the reset phase. Hence, a reset is recommended after each LC write-advance operation.
- The lifecycle can be advanced to OEM\_PROD/IN\_FIELD only if the [HSE\\_APP\\_DEBUG\\_KEY\\_ATTR\\_ID](#) attribute was set before.

### **hseAttrConfigBootAuth\_t**

```
typedef uint8_t hseAttrConfigBootAuth_t
```

Boot Authentication bit (OTP-ATTR attribute; refer to [hseAttrId\\_t](#)).

Value used by Boot ROM to check whether the IVT data needs be authenticated.

### **hseMUConfig\_t**

```
typedef uint8_t hseMUConfig_t
```

MU configuration byte (HSE\_H/M/B attribute).

Tells whether the HSE enables the receive interrupt on the configured MU interface.

### **hseMemRegAccess\_t**

```
typedef uint32_t hseMemRegAccess_t
```

Access types for [HSE\\_SPT\\_MEM\\_REGION\\_PROTECT](#) service regions.

### **hseAttrRamPubKeyImportPolicy\_t**

```
typedef uint32_t hseAttrRamPubKeyImportPolicy_t
```

## Administration Services

HSE key management policy regarding RAM public keys import (NVM-RW-ATTR attribute; refer to [hseAttrId\\_t](#)).

Determines whether public keys can be imported without authentication in advanced LCs.

Default value is HSE\_KM\_POLICY\_DEFAULT, i.e. HSE does not allow public key import in RAM, when having User rights, if they are not an authenticated key container.

Otherwise, if set to HSE\_KM\_POLICY\_ALLOW\_RAM\_PUB\_KEY\_IMPORT, RAM public keys are allowed to be imported without authentication, regardless of the access rights.

SU access rights with configuration privileges are required to update this attribute value.

Note

- :
- SU rights are needed

### **hseAttrFastCmacMinTagBitLen\_t**

```
typedef uint8_t hseAttrFastCmacMinTagBitLen_t
```

Minimal tag bit length for Fast CMAC service (NVM-RW-ATTR attribute; refer to [hseAttrId\\_t](#)).

By default, the minimal tag bit length that can be used for the Fast CMAC service (see [hseFastCMACSvc\\_t](#)) is [HSE\\_DEFAULT\\_MIN\\_FAST\\_CMAC\\_TAG\\_BITLEN](#) bits. This attribute can be set to be able to use the Fast CMAC service with the tag bit length less than [HSE\\_DEFAULT\\_MIN\\_FAST\\_CMAC\\_TAG\\_BITLEN](#) bits. The value to be set must be provided in bits.

### **hseTamperConfig\_t**

```
typedef uint8_t hseTamperConfig_t
```

Activate or Deactivate a tamper.

Tells whether tamper needs to be activated or deactivated.

### **hseTamperPolarity\_t**

```
typedef uint8_t hseTamperPolarity_t
```

Tamper Polarity.

Specifies the polarity to activate the tamper. This configuration is applicable only for passive tamper configuration. User must set the default state of the tamper input pin accordingly on the board. For example: If the tamper polarity is set "ACTIVE\_HIGH" then the default state on the tamper input pin must be "ACTIVE LOW".



**hseOutputPinConfig\_t**

```
typedef uint8_t hseOutputPinConfig_t
```

Tamper routing configuration.

This configuration defines the type of tamper (i.e. active or passive).

- In case of active tamper, the clock is derived on GPIO pad which should be routed back to the input tamper pin on the ECU. User must configure the alternate functionality of GPIO pin to tamper output so that the clock can be routed on that pin.
- In case of passive tamper, HSE senses the change in polarity of the input pin. In this case, there is no need to configure the active tamper pin. Only external tamper pin should be configured.
- User is recommended to refer the SIUL chapter in SOC reference manual to configure the correct GPIO pin. For some SOC types, only one active tamper can be supported. Please refer to [HSE\\_NUM\\_OF\\_PHYSICAL\\_TAMPER\\_INSTANCES](#) to see how many active tamper are supported.

Note

[HSE\\_TAMPER\\_ACTIVE\\_TWO](#) is not supported.

**hseTamperOutputClock\_t**

```
typedef uint8_t hseTamperOutputClock_t
```

Tamper clock that needs to be driven on the tamper output pad.

Tamper clock that needs to be driven on the tamper output pad. Please note that the alternate functionality of GPIO pin must be configured (for the tamper functionality) so that below the mentioned clock can be driven on that pad. Not applicable for passive tamper configuration

**hseFircDivConfig\_t**

```
typedef uint8_t hseFircDivConfig_t
```

FIRC Divider Configuration by HSE Firmware from HSE GPR.

**hseAttrConfigSecureRecovery\_t**

```
typedef uint8_t hseAttrConfigSecureRecovery_t
```

Secure Recovery bit (OTP-ATTR attribute).

This setting is used by SecureBAF/HSE Firmware to check whether the firmware enters in the Secure Recovery state or not. Setting the attribute multiple times is allowed (HSE\_SRV\_RSP\_OK is returned). The UTEST programming is done only the first time and skipped for subsequent calls.

## Administration Services

### **hsePublishNvmKeystoreRamtToFlash\_t**

```
typedef hseAttrCfg_t hsePublishNvmKeystoreRamtToFlash_t
```

HSE Publish NVM Keystore RAM to Flash.

This service can be used to reduce the number of write operations in the data flash, and increase the performance when the key store is updated. At start-up, the HSE FW loads the NVM key from data flash into the secure RAM (NVM keys are mirrored in RAM). After loading, the NVM keys are used only from RAM memory. At key update/erase, both the mirrored RAM area and the data flash for the keys are updated.

- By default, the attribute is set to [HSE\\_CFG\\_NO](#); this means that during key import (or load key) service, HSE updates the NVM keys to both the mirrored RAM area and the data flash.
- By setting this attribute to [HSE\\_CFG\\_YES](#), the HSE FW will update the NVM keys only in the mirror RAM memory. To perform the flash write operation, the application must call the [HSE\\_SRV\\_ID\\_PUBLISH\\_NVM\\_KEYSTORE\\_RAM\\_TO\\_FLASH](#) service.

#### Note

This attribute is available in Cust-Del and Oem-Prod LC only.

### **hseDisablePairWiseConsistencyTest\_t**

```
typedef hseAttrCfg_t hseDisablePairWiseConsistencyTest_t
```

Disable the pair wise consistency test at import for RSA/ECC/DH key pairs (SET-ONLY-ONCE-ATTR attribute; refer to [hseAttrId\\_t](#)).

By default, when importing a RSA/ECC/DH key pair, HSE checks the pair wise consistency of the provided public and private keys. Since this operation is time consuming, the host can disable this check to speed up the key import operation. To disable the pair wise consistency test, the host must set this attribute to [HSE\\_CFG\\_YES](#). By default, this attribute is set to [HSE\\_CFG\\_NO](#) (pair wise consistency is checked).

#### Note

: This is a "SET-ONLY-ONCE-ATTR" attribute. The attribute can only be set from [HSE\\_CFG\\_NO](#) to [HSE\\_CFG\\_YES](#), using the set attribute service. Once the attribute is set to [HSE\\_CFG\\_YES](#), it cannot be set back to [HSE\\_CFG\\_NO](#) in the current power cycle.

## 3.3 HSE System Authorization Services

### Data Structures

- struct [hseSysAuthorizationReqSrv\\_t](#)
- struct [hseSysAuthorizationRespSrv\\_t](#)

## Macros

| Type: (implicit C type)                       |                                                                                         |
|-----------------------------------------------|-----------------------------------------------------------------------------------------|
| Name                                          | Value                                                                                   |
| <a href="#">HSE_SYS_AUTH_ALL</a>              | ( <a href="#">HSE_SYS_AUTH_KEY_MGMT</a> )   ( <a href="#">HSE_SYS_AUTH_NVM_CONFIG</a> ) |
| <a href="#">HSE_SYS_AUTH_CHALLENGE_LENGTH</a> | 32UL                                                                                    |

| Type: <a href="#">hseSysRights_t</a>  |       |
|---------------------------------------|-------|
| Name                                  | Value |
| <a href="#">HSE_RIGHTS_SUPER_USER</a> | 1U    |
| <a href="#">HSE_RIGHTS_USER</a>       | 2U    |

| Type: <a href="#">hseSysAuthOption_t</a> |          |
|------------------------------------------|----------|
| Name                                     | Value    |
| <a href="#">HSE_SYS_AUTH_KEY_MGMT</a>    | 1U << 0U |
| <a href="#">HSE_SYS_AUTH_NVM_CONFIG</a>  | 1U << 1U |

## Typedefs

- typedef uint8\_t [hseSysRights\\_t](#)
- typedef uint8\_t [hseSysAuthOption\\_t](#)

## Data Structure Documentation

### struct hseSysAuthorizationReqSrv\_t

HSE SYS Authorization Request service.

During run-time, the User rights can be temporarily elevated to SuperUser(CUST/OEM) using HSE Authorization Request/Response.

- CUST SuperUser rights are granted using an authorization key owned by CUST.
- OEM SuperUser rights are granted using an authorization key owned by OEM.
- The User rights (non privilege rights) can be requested without authorization. In this case, HSE\_SYS\_Authorization\_Resp shall not be used.

#### Note

- After reset, the default access rights are used (see [hseSysRights\\_t](#)).
- If no authorization key is installed during CUST\_DEL or OEM\_PROD life cycle, the keys can be updated only having USER rights.
- HSE FW can perform only one SYS Authorization Request at a time. A second request will overwrite the first request.
- An authorization key is a NVM key that can only be used for verify.

## Administration Services

- If authorization succeeds, it will be opened on the MU Interface on which the request was performed, and the services that needs authorization (e.g. key import/generate/derive/export) must be performed on the same MU Interface.
- The system authorization procedure can be used to emulate the SHE CMD\_DEBUG using the MASTER\_ECU\_KEY key (as per SHE specification). In this case, if SU access rights are requested for Key Management services (see [hseSysAuthOption\\_t](#)), the authorization using MASTER\_ECU\_KEY cannot be performed if any SHE key has the WRITE\_PROTECTED flag set.

Access rights requested only for NVM Configuration services (see [hseSysAuthOption\\_t](#)) are not bound to this condition. Note that SHE keys can be erased only if the authorization was performed with the MASTER\_ECU\_KEY (refer to [hseEraseKeySrv\\_t](#)).

### Data Fields

| Type                               | Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseSysAuthOption_t</a> | sysAuthOption  | INPUT: Authorization option: Key management/NVM configuration/Both.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <a href="#">hseSysRights_t</a>     | sysRights      | INPUT: Requested system rights: SuperUser (CUST/OEM) or User rights.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| uint8_t                            | reserved[2]    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">hseKeyHandle_t</a>     | ownerKeyHandle | INPUT: The owner key handle: <ul style="list-style-type: none"><li>– if sysRights = HSE_RIGHTS_SUPER_USER, it shall be a CUST or OEM key used for only for signature verification.</li><li>– if sysRights = HSE_RIGHTS_USER, the key handle is not used.</li></ul>                                                                                                                                                                                                                                                                                                                                                |
| <a href="#">hseAuthScheme_t</a>    | authScheme     | INPUT: Authentication scheme. ONLY RSA, ECDSA, EDDSA and CMAC schemes are supported.<br>If sysRights = HSE_RIGHTS_USER, authScheme is not used.<br><br>Note <ul style="list-style-type: none"><li>– EDDSA scheme with user provided context (eddsa.contextLength != 0) is NOT supported.</li></ul>                                                                                                                                                                                                                                                                                                                |
| uint32_t                           | pChallenge     | OUTPUT: The output challenge that needs to be signed by the HOST.<br>In case SHE MASTER_ECU_KEY is used, the returned challenge is HSE_SYS_AUTH_CHALLENGE_LENGTH - 1 byte long and is formed from 16 random bytes concatenated with SHE UID: (RANDOM(16 bytes)    SHE_UID(15 bytes)).<br>Otherwise, for any other key type, the challenge size is HSE_SYS_AUTH_CHALLENGE_LENGTH bytes. If sysRights = HSE_RIGHTS_USER, pChallenge is not used.<br><br>Note<br><br>For HSE_H/M, the UID is read by HSE from system fuses. The application must provide read-only access (xRDC restriction) to HSE to read the UID. |

**struct hseSysAuthorizationRespSrv\_t**

HSE SYS Authorization Response service.

Provides the signature for the requested challenge (using [hseSysAuthorizationReqSrv\\_t](#) service).

**Note**

- In case SHE MASTER\_ECU key is used, the HSE will return the HSE\_SRV\_RSP\_VERIFY\_FAILED status as the equivalent of ERC\_NO\_DEBUGGING status as specified by the SHE spec (returned when the tag over the challenge is not correct).

**Data Fields**

| Type     | Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint16_t | authLen[2] | INPUT: Byte length(s) of the authentication tag(s).<br><br>Note <ul style="list-style-type: none"> <li>– For RSA signature and CMAC only authLen[0] is used.</li> <li>– Both lengths are used for (R,S) (ECC).</li> <li>– The MAC tag size must be minimum 16 bytes.</li> <li>– RSA signature size must be <a href="#">HSE_BYTES_TO_BITS(keyBitLength)</a>;</li> <li>– R or S size for ECDSA/EDDSA signature must be <a href="#">HSE_BYTES_TO_BITS(keyBitLength)</a></li> </ul> |
| uint32_t | pAuth[2]   | INPUT: Address(es) to authentication tag.<br><br>Note <ul style="list-style-type: none"> <li>– For RSA signature and CMAC only pAuth[0] is used.</li> <li>– Both pointers are used for (R,S) (ECC).</li> <li>– If SHE MASTER_ECU_KEY is used, the CMAC must be computed over the challenge (31 bytes) using a derived key (as per SHE specification).</li> </ul>                                                                                                                |

**Macro Definition Documentation****HSE\_RIGHTS\_SUPER\_USER**

```
#define HSE_RIGHTS_SUPER_USER ((hseSysRights_t)1U)
```

SuperUser rights: less restrictions apply for key installation/updates or/and HSE configuration.

The SU rights can be CUST or OEM depending on the life cycle (on reset) or the owner of the authorization key (when SYS authorization is used) .

## Administration Services

### HSE\_RIGHTS\_USER

```
#define HSE_RIGHTS_USER ((hseSysRights_t)2U)
```

User rights: high restrictions apply for key installation/updates or HSE configuration.

User restrictions are specific to IN\_FILED life cycle or when the CUST/OEM START\_AS\_USER policy attribute is set (refer to [hseAttrExtendCustSecurityPolicy\\_t](#) and [hseAttrExtendOemSecurityPolicy\\_t](#)).

### HSE\_SYS\_AUTH\_KEY\_MGMT

```
#define HSE_SYS_AUTH_KEY_MGMT ((hseSysAuthOption_t)(1U << 0U))
```

Request SuperUser rights for Key Management services (e.g. import/export/erase/key generate/key derive).

If SuperUser rights are granted, Key Management services can be performed using less restrictions.

### HSE\_SYS\_AUTH\_NVM\_CONFIG

```
#define HSE_SYS_AUTH_NVM_CONFIG ((hseSysAuthOption_t)(1U << 1U))
```

Request SuperUser rights to update/install the HSE NVM tables/attributes which are stored in SYS-IMAGE(HSE\_H/M)/internal flash(HSE\_B) (e.g. SMR, CR, OTFAD, NVM attributes).

If SuperUser rights are granted, updates of NVM configuration is permitted.

### HSE\_SYS\_AUTH\_ALL

```
#define HSE_SYS_AUTH_ALL (( HSE_SYS_AUTH_KEY_MGMT ) | ( HSE_SYS_AUTH_NVM_CONFIG ))
```

Request SuperUser rights for both Key Management services and NVM configuration updates.

### HSE\_SYS\_AUTH\_CHALLENGE\_LENGTH

```
#define HSE_SYS_AUTH_CHALLENGE_LENGTH (32UL)
```

Challenge length: Length of the challenge (in bytes) returned by a successful authorization request.

## Typedef Documentation

**hseSysRights\_t**

```
typedef uint8_t hseSysRights_t
```

HSE System Access rights.

After reset (default access rights):

| Life Cycle | Exec rights |
|------------|-------------|
| CUST_DEL   | SU/U*       |
| OEM_PROD   | SU/U**      |
| IN_FIELD   | U           |

Note

- :
- U\* means that the CUST\_START\_AS\_USER policy attribute is set (refer to [hseAttrExtendCustSecurityPolicy\\_t.startAsUser](#)).
- U\*\* means that the OEM\_START\_AS\_USER policy attribute is set (refer to [hseAttrExtendOemSecurityPolicy\\_t.startAsUser](#)).
- After reset, the SU rights are granted for both key management and HSE configuration updates (see [HSE\\_SYS\\_AUTH\\_ALL](#)).

After reset, the SYS rights are synchronized with Life cycle (LC) and CUST/OEM START\_AS\_USER policy attributes (see CUST/OEM policy attributes).

- if LC = CUST\_DEL:
  - if CUST\_START\_AS\_USER policy = FALSE, CUST SuperUser rights are granted for both key management and HSE configuration updates (refer to [hseSysAuthOption\\_t](#))
  - otherwise User rights are granted (U\* in the above table)
- if LC = OEM\_PROD:
  - if OEM\_START\_AS\_USER policy = FALSE, OEM SuperUser rights are granted for both key management and HSE configuration updates (refer to [hseSysAuthOption\\_t](#))
  - otherwise User rights are granted (U\*\* in the above table)
- if LC = IN\_FIELD, User rights are granted.

**hseSysAuthOption\_t**

```
typedef uint8_t hseSysAuthOption_t
```

HSE System Authorization options.

Specifies the services for which the system authorization is performed.

## 3.4 HSE Boot Images Signature Generate/Verify

### Data Structures

- struct [hseAppHeader\\_t](#)
- struct [hseBootDataImageSignSrv\\_t](#)
- struct [hseBootDataImageVerifySrv\\_t](#)

### Data Structure Documentation

#### struct hseAppHeader\_t

The Application Image header that keeps information about the Basic Secure Booting (BSB) (e.g. header information, source and destination addresses, app code length, tag location).

#### Note

- If both SMR and BSB are configured, HSE executes the secure boot using SMR only. If the SYS-IMG is not loaded because it is corrupted (the SMRs are not present), HSE executes the secure boot using BSB. In this case, the App BSB can be seen as a recovery image (to recover the SYS-IMG). Note that the App image can be booted without loading the SYS-IMG.
- For SAF86XX devices, if the flashless boot mode is configured (for the BSB feature), the HSE firmware authenticates the image in place (without copying the image) when the AppBL address in the IVT is equal to the RAM start pointer in AppBL image header.

#### Data Fields

| Type                         | Name            | Description                                                                                                                     |
|------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------|
| uint8_t                      | hdrTag          | App header tag shall be 0xD5.                                                                                                   |
| uint8_t                      | reserved1[2]    | Reserved field has no impact. Set to all zeroes.                                                                                |
| uint8_t                      | hdrVersion      | App header version shall be 0x60.                                                                                               |
| uint32_t                     | pAppDestAddress | The destination address where the application is copied.<br><br>Note<br>For HSE_B, it is NULL (the code is executed from flash) |
| uint32_t                     | pAppStartEntry  | The address of the first instruction to be executed.                                                                            |
| uint32_t                     | codeLength      | Length of application image.                                                                                                    |
| <a href="#">hseAppCore_t</a> | coreId          | The application core ID that is un-gated.<br><br>Note<br>Valid for HSE_B devices only. For HSE_H/M core id defined in IVT       |
| uint8_t                      | reserved2[47]   | Reserved field has no impact. Set to all zeroes.                                                                                |



**struct hseBootDataImageSignSrv\_t**

HSE Boot Data Image GMAC generation.

This service is used to generate the GMAC tag along with the random IV (for new device revisions; see notes below) for different Boot Data images.

For HSE\_H/M, the following Boot Data Images can be signed:

- IVT, DCD, SELF-TEST and Application Image (also referred below as App BSB Image).
- LPDDR4 Flash image for S32Z/E (HSE\_H) devices.
- the Flashless Boot Image Header for SAF86XX (HSE\_M) devices.

For HSE\_B, the following Boot Data Images can be signed:

- IVT and Application Image (also referred below as App BSB Image). The computed random IV and GMAC tag must be placed/copied at the end of the image. The 12 bytes of random IV and 16 bytes of GMAC are generated by HSE Firmware. The random IV is also part of GMAC calculation (for images format, refer to HSE FW Reference Manual).

Note

SuperUser rights (for NVM Configuration) are needed to perform this service.

- For new device revisions (see table below), the service provides in the pOutTagAddr a random IV (12byte) followed by the GMAC tag (16 bytes). The IV and GMAC tag must be placed at the end of the image (with one exception for LPDDR4 FLash image; see below). Note that the GMAC tag generation is also done over the random IV. The GMAC tag and random IV offsets in the image are specified in the HSE Firmware Reference Manual.
- For older device revisions (the part revision is smaller than the revision specified in the table below), a static IV is used that is not placed in the image. For the static IV value, refer to HSE Firmware Reference Manual.
- The application can check the device revision information reading the MAJOR\_MASK and MINOR\_MASK fields of SIUL2\_4 for S32ZE or SIUL2\_0 for the others devices
- S32K3XX devices support only random IV

| Device  | New part revision |
|---------|-------------------|
| S32G2   | rev2.1 or higher  |
| S32G3   | rev1.1 or higher  |
| S32ZE   | rev1.1 or higher  |
| S32R45  | rev2.1 or higher  |
| S32R41  | rev1.1 or higher  |
| SAF85XX | rev2.0 or higher  |
| S32K3XX | new rev only      |

### Data Fields

| Type     | Name     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pInImage | <p>INPUT: The address of the Boot Data Image. The Boot Data Image can be:</p> <ul style="list-style-type: none"> <li>– For HSE_H/M, IVT or DCD or SELF-TEST or App BSB or LPDDR4(for S32Z/E devices) image; the address may be a QSPI-FLASH (external flash) or system RAM address.</li> <li>– For SAF86XX devices (HSE_M), the Flashless Boot Image Header address must be system RAM address.</li> <li>– For HSE_B, the IVT or App BSB image; the address can be a flash or system RAM address.</li> </ul> <p>The length of the pInImage is not provided. HSE uses the information from the provided pInImage to compute the image length.<br/>The length of each image is computed in the below manner:</p> <ol style="list-style-type: none"> <li>1. For HSE_H/M new device revisions: <ul style="list-style-type: none"> <li>– the IVT Image length must be 256 bytes (IVT Image header (4bytes) + IVT Image data (224 bytes) + Random IV (12bytes) + GMAC(16 bytes))</li> <li>– For S32Z/E devices (HSE_H), DCD/SELF-TEST Image length must be maximum 32768 bytes (DCD/ST Image header(4 bytes) + maximum DCD/ST Image data (32764 byte))</li> <li>– For other devices,DCD/SELF-TEST Image length must be maximum 8192 bytes (DCD/ST Image header(4 bytes) + maximum DCD/ST Image data (8188 byte))</li> <li>– For S32Z/E devices (HSE_H), the maximum length of the LPDDR4 Flash image must be smaller or equal to (7MB + 336bytes)(Image header(336 bytes) + code length(maximum 7MB))</li> <li>– For SAF86XX devices (HSE_M), the maximum length of the Flashless Boot Image Header must be smaller or equal to 1024 bytes</li> </ul> </li> <li>– pInImage can point to the App BSB Image that contains the App header and App code: <ul style="list-style-type: none"> <li>· App image header shall be specified as <a href="#">hseAppHeader_t</a>. It has a fixed size of 64 bytes.</li> <li>· App image code shall follow the App image header and has a variable length specified by "codelength" parameter.</li> <li>· The computed GMAC tag for App BSB Image includes both App header, App code and Random IV.</li> </ul> </li> </ol> <p>...</p> |

## Data Fields

| Type     | Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pInImage    | <p>For old device revision (static IV):</p> <ul style="list-style-type: none"> <li>– For IVT, the IV bytes are reserved (set to zero)</li> <li>– For DCD/SELF-TEST and APP BSB, the IV bytes are not included at all.</li> <li>– For S32Z/E devices, the image does not include any IV in the image header.</li> <li>– For S32Z/E devices (HSE_H), DCD/SELF-TEST Image length must be maximum 8192 bytes (DCD/ST Image header(4 bytes) + maximum DCD/ST Image data (8188 byte))</li> </ul> <p>2. For HSE_B:</p> <ul style="list-style-type: none"> <li>– The IVT image length must be 256 bytes (IVT Image header (4bytes) + IVT Image data (224 bytes) + IV (12 bytes) + GMAC(16 bytes)). The computed GMAC tag is over IVT Image header and data (228 bytes) and IV (12 bytes).</li> <li>– pInImage can point to the App BSB Image that contains the App header and App code: <ul style="list-style-type: none"> <li>· App image header shall be specified as <a href="#">hseAppHeader_t</a>. It has a fixed size of 64 bytes.</li> <li>· App image code shall follow the App image header and has a variable length specified by "codelength" parameter.</li> <li>· The computed GMAC tag for App BSB Image includes App header, App code and IV (12 bytes)</li> </ul> </li> </ul> |
| uint32_t | inTagLength | <p>INPUT: The length in bytes of the IV + GMAC tag. This length must be equal to or greater than.</p> <p>1. For HSE_H/M:</p> <ul style="list-style-type: none"> <li>– new device revisions (random IV): 28 bytes</li> <li>– old device revisions (static IV): 16 bytes</li> </ul> <p>2. For HSE_B, 28 bytes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## Administration Services

### Data Fields

| Type     | Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pOutTagAddr | <p>OUTPUT: 1. For HSE_H/M, the output address:</p> <ul style="list-style-type: none"><li>– new device revisions: random IV (12 bytes) followed by the GMAC tag (16 bytes)</li><li>– old device revisions: GMAC tag (16 bytes)</li></ul> <p>For HSE_B: The address where the random IV (12 bytes), followed by the GMAC tag (16 bytes) are generated. It must be a system RAM address.</p> <p>Note</p> <p>The computed output data shall be copied at the end of boot data image.</p> <p>Exception: For S32Z/E devices (HSE_H), computed random IV shall be copied to LPDDR4 image header (at 0x144 offset) and the computed GMAC tag shall be copied at the end of boot data image.</p> |

### struct hseBootDataImageVerifySrv\_t

HSE Boot Data Image GMAC verification.

This service can be used to verify the GMAC tag generated using the [hseBootDataImageSignSrv\\_t](#) service.

### Data Fields

| Type     | Name     | Description                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pInImage | <p>INPUT: The address of the HSE Boot Data Image (for more details about the HSE Boot Data Images refer to pInImage parameter from <a href="#">hseBootDataImageSignSrv_t</a> service).</p> <p>Note</p> <ul style="list-style-type: none"><li>– HSE uses the Boot Data Image information (provided by <a href="#">pInImage</a>) to compute the length of the image and to verify the authentication TAG.</li></ul> |

## 3.5 HSE Firmware Update Service

### Data Structures

- struct [hseFirmwareUpdateSrv\\_t](#)

## Data Structure Documentation

### struct hseFirmwareUpdateSrv\_t

HSE\_B Firmware Update Service.

This service is used to update the HSE firmware into the HSE internal flash memory.

The table below summarizes which fields are used by each access mode. Unused fields are ignored by the HSE.

| Field / Mode | One-pass | Start | Update | Finish |
|--------------|----------|-------|--------|--------|
| accessMode   | *        | *     | *      | *      |
| streamLength |          | *     | *      | *      |
| pInFwFile    | *        | *     | *      | *      |

#### Data Fields

| Type                            | Name         | Description                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseAccessMode_t</a> | accessMode   | INPUT: Specifies the access mode: ONE-PASS, START, UPDATE, FINISH.                                                                                                                                                                                                                                                                                                |
| uint8_t                         | reserved[3]  |                                                                                                                                                                                                                                                                                                                                                                   |
| uint32_t                        | streamLength | INPUT: The length in bytes of a chunk. It is used only for STREAMING mode. It must be at least 64 bytes or multiple of 64 bytes; otherwise, an HSE error is returned. <ul style="list-style-type: none"> <li>– START mode: must be multiple of 64bytes.</li> <li>– UPDATE mode: must be multiple of 64bytes.</li> <li>– FINISH mode: can be any value.</li> </ul> |
| uint32_t                        | pInFwFile    | INPUT: ONE-PASS USAGE: The address of new version of HSE Firmware file to be updated into the HSE internal flash memory.<br>STREAMING USAGE: The address of chunk to be updated into the HSE internal flash memory.                                                                                                                                               |

## 3.6 Secure\_BAF Firmware update service

### Data Structures

- struct [hseSbafUpdateSrv\\_t](#)

## Data Structure Documentation

## Administration Services

### struct hseSbafUpdateSrv\_t

SBAF Update Service.

This service is used to update the SBAF firmware into the HSE internal flash memory.\ Sbaf update supports both One-pass and streaming mode, We recommend to use One-pass\ mode for sbaf update.

The table below summarizes which fields are used by each access mode. Unused fields are ignored by the HSE.

| Field / Mode | One-pass | Start | Update | Finish |
|--------------|----------|-------|--------|--------|
| accessMode   | *        | *     | *      | *      |
| streamLength |          | *     | *      | *      |
| pInFwFile    | *        | *     | *      | *      |

#### Note

After successful SBAF update, application should issue a reset to the device.

#### Data Fields

| Type                            | Name         | Description                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseAccessMode_t</a> | accessMode   | INPUT: Specifies the access mode: ONE-PASS, START, UPDATE, FINISH.                                                                                                                                                                                                                                                                                         |
| uint8_t                         | reserved[3]  |                                                                                                                                                                                                                                                                                                                                                            |
| uint32_t                        | streamLength | INPUT: The length in bytes of a chunk. It is used only for STREAMING mode. It must be at least 64bytes or multiple of 64bytes, otherwise an HSE error is returned. <ul style="list-style-type: none"><li>– START mode: must be multiple of 64bytes.</li><li>– UPDATE mode: must be multiple of 64bytes.</li><li>– FINISH mode: can be any value.</li></ul> |
| uint32_t                        | pInFwFile    | INPUT: ONE-PASS USAGE:The address of new version of SBAF Firmware file to be updated into the HSE internal flash memory.<br>STREAMING USAGE: The address of chunk to be updated into the HSE internal flash memory.                                                                                                                                        |

## 4 Cryptographic Services

### 4.1 HSE MAC Service

#### Data Structures

- struct [hseMacSrv\\_t](#)
- struct [hseFastCMACSrv\\_t](#)

#### Data Structure Documentation

##### struct hseMacSrv\_t

MAC service.

MAC algorithms are symmetric key cryptographic techniques to provide message authentication codes (MACs), also known as tags. These can be used to verify both the integrity and authenticity of a message.

This service can be accessible in one-pass or streaming (SUF) mode. In case of streaming mode, three steps (calls) will be used: START, UPDATE, FINISH. START and FINISH are mandatory; UPDATE is optional. Not all fields are used by each access mode.

The table below summarizes which fields are used by each access mode. Unused fields are ignored by the HSE.

| Field \ Mode | One-pass | Start | Update | Finish |
|--------------|----------|-------|--------|--------|
| accessMode   | *        | *     | *      | *      |
| streamId     | *        | *     | *      | *      |
| authDir      | *        | *     |        |        |
| sgtOption    | *        | *     | *      | *      |
| macScheme    | *        | *     |        |        |
| keyHandle    | *        | *     |        |        |
| inputLength  | *        | *     | *      | *      |
| pInput       | *        | *     | *      | *      |
| pTagLength   | *        |       |        | *      |
| pTag         | *        |       |        | *      |

#### Data Fields

| Type                            | Name       | Description                                                                                               |
|---------------------------------|------------|-----------------------------------------------------------------------------------------------------------|
| <a href="#">hseAccessMode_t</a> | accessMode | INPUT: Specifies the access mode: ONE-PASS, START, UPDATE, FINISH.<br>STREAMING USAGE: Used in all steps. |

## Cryptographic Services

### Data Fields

| Type                           | Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseStreamId_t</a>  | streamId    | INPUT: Specifies the stream to use for START, UPDATE, FINISH access modes. Each interface supports a limited number of streams per interface, up to <a href="#">HSE_STREAM_COUNT</a> .<br>STREAMING USAGE: Used in all steps.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">hseAuthDir_t</a>   | authDir     | INPUT: Specifies the direction: generate/verify.<br>STREAMING USAGE: Used in START.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <a href="#">hseSGTOption_t</a> | sgtOption   | INPUT: Specify if pInput is provided as <a href="#">hseScatterList_t</a> list (the host address points to a <a href="#">hseScatterList_t</a> list). Ignored if SGT is not supported.<br><br>Note <ul style="list-style-type: none"> <li>– ONLY HSE_SGT_OPTION_INPUT can be used.</li> <li>– For HSE_B devices: <ul style="list-style-type: none"> <li>· The SGT for the HMAC scheme is not available for the following hash algorithms: <ul style="list-style-type: none"> <li>· SHA2_384/512 (not available in HW)</li> <li>· If the HMAC is requested with above hash algorithms, the <a href="#">HSE_SRV_RSP_NOT_SUPPORTED</a> error is returned.</li> </ul> </li> </ul> </li> <li>– If scatter option is selected (set), the length (e.g. <a href="#">inputLength</a>) shall specified the entire message length (sum of all <a href="#">hseScatterList_t</a> lengths).</li> <li>– The number for SGT entries shall be less then <a href="#">HSE_MAX_NUM_OF_SGT_ENTRIES</a>.</li> </ul> STREAMING USAGE: Used in all steps. |
| <a href="#">hseMacScheme_t</a> | macScheme   | INPUT: Specifies the MAC scheme.<br>STREAMING USAGE: Used in START.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <a href="#">hseKeyHandle_t</a> | keyHandle   | INPUT: The key to be used for the operation. STREAMING USAGE: Used in START.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| uint32_t                       | inputLength | INPUT: Length of the input message. Can be zero.<br>STREAMING USAGE: Used in all steps. <ul style="list-style-type: none"> <li>– START: Must be a multiple of block length (for HMAC-hash or AES), or zero. Cannot be zero for HMAC.</li> <li>– UPDATE: Must be a multiple of block length (for HMAC-hash or AES). Cannot be zero. Refrain from issuing the service request, instead of passing zero.</li> <li>– FINISH: Can be any value (For CMAC &amp; XCBC-MAC, zero length is invalid).</li> </ul> Algorithm block lengths (for STREAMING USAGE): <ul style="list-style-type: none"> <li>– CMAC, GMAC, XCBC-MAC: 16</li> <li>– HMAC, depends on underlying hash: <ul style="list-style-type: none"> <li>· SHA1, SHA2_224, SHA2_256: 64</li> <li>· SHA2_512_224, SHA2_512_256, SHA2_384, SHA2_512: 128</li> <li>· SHA3: not supported for HMAC</li> <li>· Miyaguchi-Preneel: not supported for HMAC</li> </ul> </li> </ul>                                                                                                  |



## Data Fields

| Type     | Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pInput     | <p>INPUT: The input message.</p> <p>Note</p> <p>The input message for GMAC is the AAD (as specified by AEAD-GCM).</p> <p>STREAMING USAGE: Used in all steps, but ignored when <a href="#">inputLength</a> is zero</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| uint32_t | pTagLength | <p>INPUT/OUTPUT: Holds the address to a memory location (an uint32_t variable) in which the tag length in bytes is stored.</p> <p>– GENERATE:</p> <ul style="list-style-type: none"> <li>· On calling service (input), this parameter shall contain the size of the buffer provided by <a href="#">pTag</a>.</li> <li>· For GMAC, valid tag lengths are 8, 12, 13, 14, 15 and 16. Tag-lengths greater than 16 will be truncated to 16.</li> <li>· For HMAC, valid tag lengths are [8, hash-length]. Tag-lengths greater than hash-length will be truncated to hash-length.</li> <li>· For CMAC &amp; XCBC-MAC, valid tag lengths are [8, cipher-block-length]. Tag-lengths greater than cipher-block-length will be truncated to cipher-block-length.</li> <li>· When the request has finished (output), the actual length of the returned value shall be stored.</li> </ul> <p>– VERIFY:</p> <ul style="list-style-type: none"> <li>· On calling service (input), this parameter shall contain the tag-length to be verified.</li> <li>· For GMAC, valid tag lengths are 8, 12, 13, 14, 15 and 16.</li> <li>· For HMAC, valid tag lengths are [8, hash-length].</li> <li>· For CMAC &amp; XCBC-MAC, valid tag lengths are [8, cipher block-length].</li> </ul> <p>STREAMING USAGE: Used in FINISH.</p> |
| uint32_t | pTag       | <p>OUTPUT/INPUT: The output tag for "generate"; the input tag for "verify".</p> <p>STREAMING USAGE: Used in FINISH.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**struct hseFastCMACSrv\_t**

Fast CMAC service.

CMAC algorithms are symmetric key cryptographic techniques to provide message authentication codes (MACs), also known as tags. These can be used to verify both the integrity and authenticity of a message.

This FAST CMAC version can provide improved performance for CAN frames and compared to the other MAC implementation is using bits representation for [pInput](#) and [pTag](#).

## Cryptographic Services

### Note

Bits are represented from left to right at byte level.

### Data Fields

| Type                           | Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a> | keyHandle      | INPUT: The key to be used for the operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| uint32_t                       | pInput         | INPUT: The input message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| uint32_t                       | inputBitLength | INPUT: Length of the input message.(in bits)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <a href="#">hseAuthDir_t</a>   | authDir        | INPUT: Specifies the direction: generate/verify.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| uint8_t                        | tagBitLength   | INPUT/OUTPUT: Holds tag length in bits. <ul style="list-style-type: none"><li>– GENERATE:<ul style="list-style-type: none"><li>· On calling service (input), this parameter shall contain the size of the buffer provided by <a href="#">pTag</a>.</li><li>· Recommended tag lengths are [32, 128]. Tag-lengths greater than 128 will be truncated to 128.</li></ul></li><li>– VERIFY:<ul style="list-style-type: none"><li>· On calling service (input), this parameter shall contain the tag-length to be verified.</li><li>· Recommended tag lengths are [32, 128].</li><li>· The <a href="#">HSE_FAST_CMAC_MIN_TAG_BIT_LEN_ATTR_ID</a> attribute can be used to overwrite the lower recommended tag bit length limit (minimum is 1).</li></ul></li></ul> |
| uint8_t                        | reserved[2]    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| uint32_t                       | pTag           | OUTPUT/INPUT: The output tag for "generate"; the input tag for "verify".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## 4.2 HSE Symmetric Cipher Service

### Data Structures

- struct [hseSymCipherSrv\\_t](#)

### Data Structure Documentation

#### struct hseSymCipherSrv\_t

Symmetric Cipher service.

To perform encryption/decryption with a block cipher in ECB or CBC mode, the length of the input must be an exact multiple of the block size. For all AES variants it is 16 bytes (128 bits). If the input plaintext

is not an exact multiple of block size, it must be padded by application (by adding a padding string). For other modes, such as counter mode (CTR) or OFB or CFB, padding is not required. In these cases, the ciphertext is always the same length as the plaintext. If the plaintext is always an exact multiple of the block length, padding can be avoided.

This service can be accessible in one-pass or streaming (SUF) mode. In case of streaming mode, three steps (calls) will be used: START, UPDATE, FINISH. START and FINISH are mandatory; UPDATE is optional. Not all fields are used by each access mode.

The table below summarizes which fields are used by each access mode. Unused fields are ignored by the HSE.

| Field \ Mode    | One-pass | Start | Update | Finish |
|-----------------|----------|-------|--------|--------|
| accessMode      | *        | *     | *      | *      |
| streamId        |          | *     | *      | *      |
| cipherAlgo      | *        | *     |        |        |
| cipherBlockMode | *        | *     |        |        |
| cipherDir       | *        | *     |        |        |
| sgtOption       | *        | *     | *      | *      |
| keyHandle       | *        | *     |        |        |
| pIV             | *        | *     |        |        |
| inputLength     | *        | *     | *      | *      |
| pInput          | *        | *     | *      | *      |
| pOutput         | *        | *     | *      | *      |

#### Data Fields

| Type                                 | Name            | Description                                                                                                                                                                                                                   |
|--------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseAccessMode_t</a>      | accessMode      | INPUT: Specifies the access mode: ONE-PASS, START, UPDATE, FINISH.<br>STREAMING USAGE: Used in all steps.                                                                                                                     |
| <a href="#">hseStreamId_t</a>        | streamId        | INPUT: Specifies the stream to use for START, UPDATE, FINISH access modes. Each interface supports a limited number of streams per interface, up to <a href="#">HSE_STREAM_COUNT</a> .<br>STREAMING USAGE: Used in all steps. |
| <a href="#">hseCipherAlgo_t</a>      | cipherAlgo      | INPUT: Specifies the cipher algorithm.<br>STREAMING USAGE: Used in START.                                                                                                                                                     |
| <a href="#">hseCipherBlockMode_t</a> | cipherBlockMode | INPUT: Specifies the cipher mode.<br>STREAMING USAGE: Used in START.                                                                                                                                                          |
| <a href="#">hseCipherDir_t</a>       | cipherDir       | INPUT: Specifies the cipher direction: encryption/decryption.<br>STREAMING USAGE: Used in START.                                                                                                                              |

## Cryptographic Services

### Data Fields

| Type                           | Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseSGTOption_t</a> | sgtOption   | INPUT: Specify if pInput/pOutput are provided as <a href="#">hseScatterList_t</a> list (the host address points to a <a href="#">hseScatterList_t</a> list). Ignored if SGT is not supported.<br><br>Note <ul style="list-style-type: none"><li>– If scatter option is selected (set), the length (e.g. <a href="#">inputLength</a>) shall specified the entire message length (sum of all <a href="#">hseScatterList_t</a> lengths).</li><li>– The number for SGT entries shall be less then <a href="#">HSE_MAX_NUM_OF_SGT_ENTRIES</a> .</li></ul> STREAMING USAGE: Used in all steps.                                                      |
| uint8_t                        | reserved[2] |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <a href="#">hseKeyHandle_t</a> | keyHandle   | INPUT: The key to be used for the operation.<br>STREAMING USAGE: Used in START step.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| uint32_t                       | pIV         | INPUT: Initialization Vector/Nonce. Ignored for NULL & ECB cipher block modes.<br>IV length is 16 bytes. (AES cipher block size).<br>STREAMING USAGE: Used in START.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| uint32_t                       | inputLength | INPUT: The plaintext and ciphertext length. For ECB, CBC & CFB cipher block modes, must be a multiple of block length. Cannot be zero.<br>STREAMING USAGE: MANDATORY for all steps. <ul style="list-style-type: none"><li>– START: Must be a multiple of block length. Can be zero.</li><li>– UPDATE: Must be a multiple of block length. Cannot be zero. Refrain from issuing the service request, instead of passing zero.</li><li>– FINISH: For ECB, CBC &amp; CFB cipher block modes, must be a multiple of block length. Cannot be zero. For remaining cipher block modes, can be any value except zero.</li></ul> AES block lengths: 16 |
| uint32_t                       | pInput      | INPUT: The plaintext for encryption or the ciphertext for decryption.<br>STREAMING USAGE: Used in START, UPDATE and FINISH.<br>Ignored in START if <a href="#">inputLength</a> is zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| uint32_t                       | pOutput     | OUTPUT: The plaintext for decryption or ciphertext for encryption.<br>STREAMING USAGE: Used in START, UPDATE and FINISH.<br>Ignored in START if <a href="#">inputLength</a> is zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## 4.3 HSE HASH Service

### Data Structures

- struct [hseHashSrv\\_t](#)

## Data Structure Documentation

### struct hseHashSrv\_t

HASH service.

The HASH service is used to map data of arbitrary size to data of fixed size. The values returned by a hash function are called hash values, hash codes, digests, or simply hashes.

The HASH service can be accessible in one-pass or streaming (SUF) mode. In case of streaming mode, three steps (calls) will be used: START, UPDATE, FINISH. START and FINISH are mandatory; UPDATE is optional. Not all fields are used by each access mode.

The table below summarizes which fields are used by each access mode. Unused fields are ignored by the HSE.

| Field \ Mode | One-pass | Start | Update | Finish |
|--------------|----------|-------|--------|--------|
| accessMode   | *        | *     | *      | *      |
| streamId     |          | *     | *      | *      |
| hashAlgo     | *        | *     |        |        |
| sgtOption    | *        | *     | *      | *      |
| inputLength  | *        | *     | *      | *      |
| pInput       | *        | *     | *      | *      |
| pHashLength  | *        |       |        | *      |
| pHash        | *        |       |        | *      |

#### Data Fields

| Type                            | Name       | Description                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseAccessMode_t</a> | accessMode | INPUT: Specifies the access mode: ONE-PASS, START, UPDATE, FINISH.<br>Note <ul style="list-style-type: none"> <li>– Miyaguchi-Preneel does not support streaming. For MP this parameter is ignored and considered default ONE-PASS.</li> </ul> STREAMING USAGE: Used in all steps.                                                                                     |
| <a href="#">hseStreamId_t</a>   | streamId   | INPUT: Specifies the stream to use for START, UPDATE, FINISH access modes. Each interface supports a limited number of streams per interface, up to <a href="#">HSE_STREAM_COUNT</a> .<br>Note <ul style="list-style-type: none"> <li>– Miyaguchi-Preneel does not support streaming. For MP this parameter is ignored. STREAMING USAGE: Used in all steps.</li> </ul> |

## Cryptographic Services

### Data Fields

| Type                           | Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseHashAlgo_t</a>  | hashAlgo    | INPUT: Specifies the hash algorithm.<br>STREAMING USAGE: Used in START.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <a href="#">hseSGTOption_t</a> | sgtOption   | <p>INPUT: Specify if pInput is provided as <a href="#">hseScatterList_t</a> list (the host address points to a <a href="#">hseScatterList_t</a> list). Ignored if SGT is not supported.</p> <p>Note</p> <ul style="list-style-type: none"> <li>– SGT is not available for the following hash algorithms and the parameter is ignored: <ul style="list-style-type: none"> <li>· Miyaguchi-Preneel</li> <li>· SHA3 (unless the targeted platform has #HSE_SPT_HW_SHA3 defined)</li> <li>· SHA2_384/512 for HSE_B devices (not available in hardware)</li> </ul> </li> <li>– ONLY HSE_SGT_OPTION_INPUT can be used. <a href="#">HSE_SGT_OPTION_OUTPUT</a> will be ignored if used, as output is always considered a buffer.</li> <li>– If scatter option is selected (set), the length (e.g. <a href="#">inputLength</a>) shall specified the entire message length (sum of all <a href="#">hseScatterList_t</a> lengths).</li> <li>– The number for SGT entries shall be less then <a href="#">HSE_MAX_NUM_OF_SGT_ENTRIES</a>.</li> </ul> <p>STREAMING USAGE: Used in all steps.</p> |
| <a href="#">uint32_t</a>       | inputLength | <p>INPUT: Length of the input message. Can be zero (except Miyaguchi-Preneel).<br/>For Miyaguchi-Preneel, inputLength must be multiple of 16 bytes and not equal to zero.</p> <p>STREAMING USAGE: Used in all steps.</p> <ul style="list-style-type: none"> <li>– START: Must be a multiple of block length, or zero.</li> <li>– UPDATE: Must be a multiple of block length. Cannot be zero. Refrain from issuing the service request, instead of passing zero.</li> <li>– FINISH: Can be any value.</li> </ul> <p>Algorithm block lengths:</p> <ul style="list-style-type: none"> <li>– Miyaguchi-Preneel: not supported in streaming mode</li> <li>– SHA1, SHA2_224, SHA2_256: 64</li> <li>– SHA2_384, SHA2_512, SHA2_512_224, SHA2_512_256: 128</li> <li>– SHA3-224: 144</li> <li>– SHA3-256: 136</li> <li>– SHA3-384: 104</li> <li>– SHA3-512: 72</li> <li>– SHA3: If the targeted platform does NOT have #HSE_SPT_HW_SHA3 defined, there is no limitation (input can be any size)</li> </ul>                                                                                  |

## Data Fields

| Type     | Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pInput      | <p>INPUT: Address of the input message.</p> <p>For Miyaguchi-Preneel, according to SHE specification, the input shall be (K   C   padding).</p> <p>Ignored if inputLength is zero.</p> <p>STREAMING USAGE: Used in all steps (except if inputLength is zero).</p> <p>Note</p> <p>If the HOST_ADDR is on 64 bits and the hash algorithm is not supported in HW (#HSE_SPT_HW_SHA3 macro is not defined), the address must fall within the 32-bit address range.</p>                                                                                                                                                                                                                                                                                                      |
| uint32_t | pHashLength | <p>INPUT/OUTPUT: Pointer to a uint32_t location in which the hash length in bytes is stored. On calling this service, this parameter shall contain the size of the buffer provided by host. When the request has finished, the actual length of the returned value shall be stored. If the buffer is smaller than the size of the hash, the hash will be truncated (not applicable for Miyaguchi Preneel).</p> <p>For Miyaguchi-Preneel, if the buffer is smaller than the size of the hash (16 bytes), parameter will be considered invalid. If the buffer is larger, pHashLength is adjusted to the size of the hash. A hash buffer length (i.e. a pHashLength) of zero makes no sense, and is considered invalid.</p> <p>STREAMING USAGE: MANDATORY for FINISH.</p> |
| uint32_t | pHash       | <p>OUTPUT: The address of the output buffer where the resulting hash will be stored.</p> <p>STREAMING USAGE: MANDATORY for FINISH.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## 4.4 HSE AEAD Service

### Data Structures

- struct [hseAeadSrv\\_t](#)

### Data Structure Documentation

#### struct hseAeadSrv\_t

AEAD service.

Authenticated Encryption with Associated Data (AEAD, also known as Authenticated Encryption) is a block cipher mode of operation which also allows integrity checks (e.g. AES-GCM). Additional

## Cryptographic Services

authenticated data (AAD) is optional additional input header which is authenticated, but not encrypted. Both confidentiality and message authentication is provided on the input plaintext.

This service can be accessible in one-pass or streaming (SUF) mode. In case of streaming mode, three steps (calls) will be used: START, UPDATE, FINISH. START and FINISH are mandatory; UPDATE is optional. Not all fields are used by each access mode.

### Note

1. Streaming mode is not supported for CCM.
2. The key usage flags used with AEAD operations:
  - [HSE\\_KF\\_USAGE\\_ENCRYPT](#) specifies that the key can be used for encryption and tag computation (note that the [HSE\\_KF\\_USAGE\\_SIGN](#) flag is not used).
  - [HSE\\_KF\\_USAGE\\_DECRYPT](#) specifies that the key can be used for decryption and tag verification (note that [HSE\\_KF\\_USAGE\\_VERIFY](#) flag is not used).

The table below summarizes which fields are used by each access mode. Unused fields are ignored by the HSE.

| Field \ Mode   | One-pass | Start | Update | Finish |
|----------------|----------|-------|--------|--------|
| accessMode     | *        | *     | *      | *      |
| streamId       |          | *     | *      | *      |
| authCipherMode | *        | *     |        |        |
| cipherDir      | *        | *     |        |        |
| keyHandle      | *        | *     |        |        |
| ivLength       | *        | *     |        |        |
| pIV            | *        | *     |        |        |
| aadLength      | *        | *     |        |        |
| pAAD           | *        | *     |        |        |
| sgtOption      | *        | *     | *      | *      |
| inputLength    | *        |       | *      | *      |
| pInput         | *        |       | *      | *      |
| tagLength      | *        |       |        | *      |
| pTag           | *        |       |        | *      |
| pOutput        | *        |       | *      | *      |

### Data Fields

| Type                            | Name       | Description                                                                                               |
|---------------------------------|------------|-----------------------------------------------------------------------------------------------------------|
| <a href="#">hseAccessMode_t</a> | accessMode | INPUT: Specifies the access mode: ONE-PASS, START, UPDATE, FINISH.<br>STREAMING USAGE: Used in all steps. |



## Data Fields

| Type                                | Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseStreamId_t</a>       | streamId       | INPUT: Specifies the stream to use for START, UPDATE, FINISH access modes. Each interface supports a limited number of streams per interface, up to <a href="#">HSE_STREAM_COUNT</a> .<br>STREAMING USAGE: Used in all steps.                                                                                                                                                                                                                                                                                                                                                 |
| <a href="#">hseAuthCipherMode_t</a> | authCipherMode | INPUT: Specifies the authenticated cipher mode.<br>STREAMING USAGE: Used in all steps.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <a href="#">hseCipherDir_t</a>      | cipherDir      | INPUT: Specifies the cipher direction: encryption/decryption.<br>STREAMING USAGE: Used in all steps.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <a href="#">hseKeyHandle_t</a>      | keyHandle      | INPUT: The key to be used for the operation.<br>STREAMING USAGE: Used in START step.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| uint32_t                            | ivLength       | INPUT: The length of the IV/Nonce (in bytes).<br>– CCM valid IV sizes 7, 8, 9, 10, 11, 12, 13 bytes<br>– GCM: $1 \leq \text{ivLength} \leq 2^{32}-1$ . Recommended 12 bytes or greater.<br>STREAMING USAGE: Used in START.                                                                                                                                                                                                                                                                                                                                                    |
| uint32_t                            | pIV            | INPUT: Initialization Vector/Nonce.<br>STREAMING USAGE: Used in START.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| uint32_t                            | aadLength      | INPUT: The length of AAD Header data (in bytes). Can be zero.<br>– CCM: Restricted to lengths less than or equal to $(2^{16} - 2^8)$ bytes.<br>STREAMING USAGE: Used in START. Any AAD is ignored in UPDATE or FINISH, and must be passed to the HSE in START.                                                                                                                                                                                                                                                                                                                |
| uint32_t                            | pAAD           | INPUT: The AAD Header data. Ignored if aadLength is zero.<br>STREAMING USAGE: Used in START. Any AAD is ignored in UPDATE or FINISH, and must be passed to the HSE in START.                                                                                                                                                                                                                                                                                                                                                                                                  |
| <a href="#">hseSGTOption_t</a>      | sgtOption      | INPUT: Specify if pInput/pOutput are provided as <a href="#">hseScatterList_t</a> list (the host address points to a <a href="#">hseScatterList_t</a> list). Ignored if SGT is not supported.<br><br>Note<br><ul style="list-style-type: none"> <li>– If scatter option is selected (set), the length (e.g. inputLength) shall specified the entire message length (sum of all <a href="#">hseScatterList_t</a> lengths).</li> <li>– The number for SGT entries shall be less then <a href="#">HSE_MAX_NUM_OF_SGT_ENTRIES</a>.</li> </ul> STREAMING USAGE: Used in all steps. |
| uint8_t                             | reserved[3]    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| uint32_t                            | inputLength    | INPUT: The length of the plaintext and ciphertext (in bytes). Can be zero (compute/verify the tag without input message).<br>STREAMING USAGE:<br><ul style="list-style-type: none"> <li>– START: The input length is ignored.</li> <li>– UPDATE: Must be a multiple of block length. Cannot be zero. Refrain from issuing the service request instead of passing zero.</li> <li>– FINISH: All lengths are allowed.</li> </ul>                                                                                                                                                 |

## Cryptographic Services

### Data Fields

| Type     | Name      | Description                                                                                                                                                                                                 |
|----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pInput    | INPUT: The plaintext for "authenticated encryption" or the ciphertext for "authenticated decryption".<br>STREAMING USAGE: Used in UPDATE and FINISH step. Ignored for START step or if inputLength is zero. |
| uint32_t | tagLength | INPUT: The length of tag (in bytes).<br>– CCM valid Tag sizes 4, 6, 8, 10, 12, 14, 16 bytes<br>– GCM valid Tag sizes 4, 8, 12, 13, 14, 15, 16 bytes<br>STREAMING USAGE: Used in FINISH step.                |
| uint32_t | pTag      | OUTPUT/INPUT: The output tag for "authenticated encryption" or the input tag for "authenticated decryption".<br>STREAMING USAGE: Used in FINISH step.                                                       |
| uint32_t | pOutput   | OUTPUT: The ciphertext for "authenticated encryption" or the plaintext for "authenticated decryption".<br>STREAMING USAGE: Used in UPDATE and FINISH step.                                                  |

## 4.5 HSE Digital Signature Service

### Data Structures

- struct [hseSignSrv\\_t](#)

### Data Structure Documentation

#### struct hseSignSrv\_t

Digital Signature service.

Uses the input parameters to perform the signature calculation and stores the signature in the memory location pointed by the output parameter.

This service can be accessible in one-pass or streaming (SUF) mode. In case of streaming mode, three steps (calls) will be used: START, UPDATE, FINISH. START and FINISH are mandatory; UPDATE is optional. Not all fields are used by each access mode.

The table below summarizes which fields are used by each access mode. Unused fields are ignored by the HSE.

| Field \ Mode | One-pass | Start | Update | Finish |
|--------------|----------|-------|--------|--------|
| accessMode   | *        | *     | *      | *      |
| streamId     |          | *     | *      | *      |
| signScheme   | *        | *     |        |        |

| Field \ Mode     | One-pass | Start | Update | Finish |
|------------------|----------|-------|--------|--------|
| authDir          | *        | *     |        |        |
| keyHandle        | *        | *     |        |        |
| sgtOption        | *        | *     | *      | *      |
| inputLength      | *        | *     | *      | *      |
| pInput           | *        | *     | *      | *      |
| pSignatureLength | *        |       |        | *      |
| pSignature       | *        |       |        | *      |

## Data Fields

| Type                            | Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseAccessMode_t</a> | accessMode     | INPUT: Specifies the access mode: ONE-PASS, START, UPDATE, FINISH.<br>STREAMING USAGE: Used in all steps.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <a href="#">hseStreamId_t</a>   | streamId       | INPUT: Specifies the stream to use for START, UPDATE, FINISH access modes. Each interface supports a limited number of streams per interface, up to <a href="#">HSE_STREAM_COUNT</a> .<br>STREAMING USAGE: Used in all steps.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <a href="#">hseAuthDir_t</a>    | authDir        | INPUT: Specifies the direction: generate/verify.<br>STREAMING USAGE: Used in START.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| bool_t                          | bInputIsHashed | INPUT: Specifies that the input is already hashed with the algorithm in specified in the sign scheme. Not valid for any signing scheme that does not perform prehashing (i.e. PureEDDSA)<br><br>Note<br><br>The hashing algorithm from signScheme and the input length must be:<br><ul style="list-style-type: none"> <li>– for RSA scheme: hashAlgo != HSE_HASH_ALGO_NULL, where the inputLength must be equal to the hash output length;</li> <li>– for ECDSA scheme: hashAlgo != HSE_HASH_ALGO_NULL, where the inputLength must be equal to the hash output length; hashAlgo == HSE_HASH_ALGO_NULL, where the inputLength must be within (0, 64];</li> <li>– for EDDSA scheme: hashAlgo is not used and the inputLength should be the output length of the hash associated with the curve; STREAMING USAGE: Not supported in streaming mode.</li> </ul> |
| <a href="#">hseSignScheme_t</a> | signScheme     | INPUT: Scheme for selected Signature algo.<br>STREAMING USAGE: Used in START.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <a href="#">hseKeyHandle_t</a>  | keyHandle      | INPUT: The key to be used for the operation.<br>STREAMING USAGE: Used in START.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## Cryptographic Services

### Data Fields

| Type                           | Name         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseSGTOption_t</a> | sgtOption    | <p>INPUT: Specify if pInput is provided as <a href="#">hseScatterList_t</a> list (the host address points to a <a href="#">hseScatterList_t</a> list). Ignored if SGT is not supported.</p> <p>Note</p> <ul style="list-style-type: none"> <li>– ONLY HSE_SGT_OPTION_INPUT can be used (the rest of the bits are ignored)</li> <li>– Scatter option is supported only for RSA and ECDSA. For EDDSA, it is not supported.</li> <li>– Scatter option is not supported for SHA3 hashes (unless the targeted platform has #HSE_SPT_HW_SHA3 defined).</li> <li>– Scatter option is not supported for pre-hashed inputs</li> <li>– If scatter option is selected (set), the length (e.g. <a href="#">inputLength</a>) shall specified the entire message length (sum of all <a href="#">hseScatterList_t</a> lengths).</li> <li>– The number for SGT entries shall be less then <a href="#">HSE_MAX_NUM_OF_SGT_ENTRIES</a>.</li> </ul> <p>STREAMING USAGE: Used in all steps.</p> |
| uint8_t                        | reserved[3U] |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| uint32_t                       | inputLength  | <p>INPUT: The length of the message. For RSA schemes, this must be the length of the original (not pre-hashed) input.</p> <p>STREAMING USAGE: Used in all steps.</p> <ul style="list-style-type: none"> <li>– START: Must be a multiple of block length of the hash, or zero.</li> <li>– UPDATE: Must be a multiple of block length of the hash. Cannot be zero. Refrain from issuing the service request, instead of passing zero.</li> <li>– FINISH: Can be any value.</li> </ul> <p>Algorithm block lengths:</p> <ul style="list-style-type: none"> <li>– SHA1, SHA2_224, SHA2_256: 64</li> <li>– SHA2_512_224, SHA2_512_256, SHA2_384, SHA2_512: 128</li> <li>– SHA3-224: 144</li> <li>– SHA3-256: 136</li> <li>– SHA3-384: 104</li> <li>– SHA3-512: 72</li> <li>– SHA3: If the targeted platform does NOT have #HSE_SPT_HW_SHA3 defined, there is no limitation (input can be any size)</li> </ul> <p>Note</p> <p>EDDSA does not support streaming</p>                 |

## Data Fields

| Type     | Name                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pInput              | <p>INPUT: The address of the message to be signed/verify. For RSA schemes, this is the actual (not pre-hashed) input.</p> <p>STREAMING USAGE: Used in all steps.</p> <p>Note</p> <p>If the HOST_ADDR is on 64 bits:</p> <ul style="list-style-type: none"> <li>– For PureEDDSA: the address must fall within the 32-bit address range.</li> <li>– If hash algorithm is not supported in HW (#HSE_SPT_HW_SHA3 macro is not defined) and input is not hashed, the address must fall within the 32-bit address range.</li> </ul>                                      |
| uint32_t | pSignatureLength[2] | <p>INPUT/OUTPUT: An array of two addresses of two uint32_t values containing signature lengths. It is input/output for "generate" and input for "verify". On calling "generate" service, these parameter shall contain the size of the signature buffers provided by the application. When the request has finished, the actual lengths of the signature components.</p> <p>STREAMING USAGE: Used in FINISH.</p>                                                                                                                                                   |
| uint32_t | pSignature[2]       | <p>OUTPUT: Where the signature components must be stored. It is output for "generate" and input for "verify".</p> <ul style="list-style-type: none"> <li>– RSA has a single signature component, at index 0, and the size of buffer must be at least the byteLength(public modulus n)</li> <li>– ECDSA and EDDSA signature format as (r,s), with r at index 0, and s at index 1. The buffer size for each component must be at least the length of the used curve in bytes (e.g. 32 bytes for a 256 bit curve).</li> </ul> <p>STREAMING USAGE: Used in FINISH.</p> |

## 4.6 HSE RSA Cipher Service

### Data Structures

- struct [hseRsaCipherSrv\\_t](#)

### Data Structure Documentation

## Cryptographic Services

### struct hseRsaCipherSrv\_t

RSA Cipher service.

Performs the RSA Cipher (Encryption/Decryption) (RSAEP) operation.

#### Data Fields

| Type                                 | Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseRsaCipherScheme_t</a> | rsaScheme     | INPUT: The RSA cipher scheme.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <a href="#">hseCipherDir_t</a>       | cipherDir     | INPUT: Specifies the cipher direction: encryption/decryption.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| uint8_t                              | reserved[3]   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <a href="#">hseKeyHandle_t</a>       | keyHandle     | INPUT: The key to be used for the operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| uint32_t                             | inputLength   | INPUT: The input length (plaintext or ciphertext): <ul style="list-style-type: none"><li>– The length of the ciphertext should be <a href="#">HSE_BITS_TO_BYTES(keyBitLen)</a> .</li><li>– The length of the plaintext (in bytes):<ul style="list-style-type: none"><li>· For RSAES NO PADDING, the Input Length must be less than or equal to <a href="#">HSE_BITS_TO_BYTES(keyBitLen)</a>, and <a href="#">pInput</a> is considered a big-endian integer.</li><li>· For RSAES-PKCS1-v1_5, the Input Length shall not be greater than <a href="#">HSE_BITS_TO_BYTES(keyBitLen)</a> -11 bytes.</li><li>· For RSAES-OAEP, Input Length shall not be greater than <a href="#">HSE_BITS_TO_BYTES(keyBitLen)</a> - 2 * hashLen - 2 bytes.</li></ul></li></ul> |
| uint32_t                             | pInput        | INPUT: The plaintext for encryption or the ciphertext for decryption.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| uint32_t                             | pOutputLength | INPUT/OUTPUT: Holds the address to a location (an uint32_t variable) in which the output length in bytes is stored.<br>On calling this service, this parameter shall contain the size of the buffer provided by the application. When the request has finished, the actual length of the returned value shall be stored.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| uint32_t                             | pOutput       | OUTPUT: The address of the Output. The plaintext for decryption or ciphertext for encryption. The size of output must be at least the <a href="#">HSE_BITS_TO_BYTES(keyBitLen)</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## 5 Key Management Services

### 5.1 HSE Key Management Common Types

#### Data Structures

- struct [hseKeyGroupCfgEntry\\_t](#)
- struct [hseKeyInfo\\_t](#)
- union [hseKeyInfo\\_t.specific](#)

#### Macros

| Type: (implicit C type)                                       |                                                                                                                                                       |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                                                          | Value                                                                                                                                                 |
| <a href="#">GET_KEY_HANDLE</a> (catalogId, groupIdx, slotIdx) | -                                                                                                                                                     |
| <a href="#">HSE_KF_USAGE_MASK</a>                             | -                                                                                                                                                     |
| <a href="#">HSE_KF_MSC_USAGE_MASK</a>                         | <a href="#">HSE_KF_USAGE_ENCRYPT</a>   <a href="#">HSE_KF_USAGE_DECRYPT</a>   <a href="#">HSE_KF_USAGE_SIGN</a>   <a href="#">HSE_KF_USAGE_VERIFY</a> |
| <a href="#">HSE_KF_ACCESS_MASK</a>                            | <a href="#">HSE_KF_ACCESS_WRITE_PROT</a>   <a href="#">HSE_KF_ACCESS_DEBUG_PROT</a>   <a href="#">HSE_KF_ACCESS_EXPORTABLE</a>                        |
| <a href="#">HSE_KF_MAX_KEY_COUNTER_VALUE</a>                  | ((uint32_t)0xFFFFFFFFUL - 1UL)                                                                                                                        |

| Type: <a href="#">hseKeyHandle_t</a>         |               |
|----------------------------------------------|---------------|
| Name                                         | Value         |
| <a href="#">HSE_INVALID_KEY_HANDLE</a>       | 0xFFFFFFFFFUL |
| <a href="#">HSE_ROM_KEY_AES256_KEY0</a>      | 0x00000000UL  |
| <a href="#">HSE_ROM_KEY_AES256_KEY1</a>      | 0x00000001UL  |
| <a href="#">HSE_ROM_KEY_RSA3072_PUB_KEY0</a> | 0x00000100UL  |
| <a href="#">HSE_ROM_KEY_ECC256_PUB_KEY0</a>  | 0x00000200UL  |

| Type: <a href="#">hseKeyGroupIdx_t</a>    |                             |
|-------------------------------------------|-----------------------------|
| Name                                      | Value                       |
| <a href="#">GET_GROUP_IDX</a> (keyHandle) | ((keyHandle) >> 8U) & 0xFFU |
| <a href="#">HSE_INVALID_GROUP_IDX</a>     | 0xFFU                       |

| Type: <a href="#">hseKeySlotIdx_t</a>    |                     |
|------------------------------------------|---------------------|
| Name                                     | Value               |
| <a href="#">GET_SLOT_IDX</a> (keyHandle) | (keyHandle) & 0xFFU |
| <a href="#">HSE_INVALID_SLOT_IDX</a>     | 0xFFU               |

|                                     |  |
|-------------------------------------|--|
| Type: <a href="#">hseSmrFlags_t</a> |  |
|-------------------------------------|--|

## Key Management Services

| Name                          | Value       |
|-------------------------------|-------------|
| <a href="#">HSE_KF_SMR_0</a>  | 1UL << 0UL  |
| <a href="#">HSE_KF_SMR_1</a>  | 1UL << 1UL  |
| <a href="#">HSE_KF_SMR_2</a>  | 1UL << 2UL  |
| <a href="#">HSE_KF_SMR_3</a>  | 1UL << 3UL  |
| <a href="#">HSE_KF_SMR_4</a>  | 1UL << 4UL  |
| <a href="#">HSE_KF_SMR_5</a>  | 1UL << 5UL  |
| <a href="#">HSE_KF_SMR_6</a>  | 1UL << 6UL  |
| <a href="#">HSE_KF_SMR_7</a>  | 1UL << 7UL  |
| <a href="#">HSE_KF_SMR_8</a>  | 1UL << 8UL  |
| <a href="#">HSE_KF_SMR_9</a>  | 1UL << 9UL  |
| <a href="#">HSE_KF_SMR_10</a> | 1UL << 10UL |
| <a href="#">HSE_KF_SMR_11</a> | 1UL << 11UL |
| <a href="#">HSE_KF_SMR_12</a> | 1UL << 12UL |
| <a href="#">HSE_KF_SMR_13</a> | 1UL << 13UL |
| <a href="#">HSE_KF_SMR_14</a> | 1UL << 14UL |
| <a href="#">HSE_KF_SMR_15</a> | 1UL << 15UL |
| <a href="#">HSE_KF_SMR_16</a> | 1UL << 16UL |
| <a href="#">HSE_KF_SMR_17</a> | 1UL << 17UL |
| <a href="#">HSE_KF_SMR_18</a> | 1UL << 18UL |
| <a href="#">HSE_KF_SMR_19</a> | 1UL << 19UL |
| <a href="#">HSE_KF_SMR_20</a> | 1UL << 20UL |
| <a href="#">HSE_KF_SMR_21</a> | 1UL << 21UL |
| <a href="#">HSE_KF_SMR_22</a> | 1UL << 22UL |
| <a href="#">HSE_KF_SMR_23</a> | 1UL << 23UL |
| <a href="#">HSE_KF_SMR_24</a> | 1UL << 24UL |
| <a href="#">HSE_KF_SMR_25</a> | 1UL << 25UL |
| <a href="#">HSE_KF_SMR_26</a> | 1UL << 26UL |
| <a href="#">HSE_KF_SMR_27</a> | 1UL << 27UL |
| <a href="#">HSE_KF_SMR_28</a> | 1UL << 28UL |
| <a href="#">HSE_KF_SMR_29</a> | 1UL << 29UL |
| <a href="#">HSE_KF_SMR_30</a> | 1UL << 30UL |
| <a href="#">HSE_KF_SMR_31</a> | 1UL << 31UL |

| Type: <a href="#">hseEccCurveId_t</a>            |       |
|--------------------------------------------------|-------|
| Name                                             | Value |
| <a href="#">HSE_EC_CURVE_NONE</a>                | 0U    |
| <a href="#">HSE_EC_SEC_SECP256R1</a>             | 1U    |
| <a href="#">HSE_EC_SEC_SECP384R1</a>             | 2U    |
| <a href="#">HSE_EC_SEC_SECP521R1</a>             | 3U    |
| <a href="#">HSE_EC_BRAINPOOL_BRAINPOOLP256R1</a> | 4U    |
| <a href="#">HSE_EC_BRAINPOOL_BRAINPOOLP320R1</a> | 5U    |
| <a href="#">HSE_EC_BRAINPOOL_BRAINPOOLP384R1</a> | 6U    |
| <a href="#">HSE_EC_BRAINPOOL_BRAINPOOLP512R1</a> | 7U    |



|                         |      |
|-------------------------|------|
| HSE_EC_25519_ED25519    | 9U   |
| HSE_EC_25519_CURVE25519 | 10U  |
| HSE_EC_448_ED448        | 11U  |
| HSE_EC_448_CURVE448     | 12U  |
| HSE_EC_USER_CURVE1      | 101U |
| HSE_EC_USER_CURVE2      | 102U |
| HSE_EC_USER_CURVE3      | 103U |

| Type: hseKeyBits_t   |         |
|----------------------|---------|
| Name                 | Value   |
| HSE_KEY_BITS_INVALID | 0xFFFFU |
| HSE_KEY_BITS_ZERO    | 0U      |
| HSE_KEY64_BITS       | 64U     |
| HSE_KEY128_BITS      | 128U    |
| HSE_KEY160_BITS      | 160U    |
| HSE_KEY192_BITS      | 192U    |
| HSE_KEY224_BITS      | 224U    |
| HSE_KEY240_BITS      | 240U    |
| HSE_KEY256_BITS      | 256U    |
| HSE_KEY320_BITS      | 320U    |
| HSE_KEY384_BITS      | 384U    |
| HSE_KEY512_BITS      | 512U    |
| HSE_KEY521_BITS      | 521U    |
| HSE_KEY638_BITS      | 638U    |
| HSE_KEY1024_BITS     | 1024U   |
| HSE_KEY2048_BITS     | 2048U   |
| HSE_KEY3072_BITS     | 3072U   |
| HSE_KEY4096_BITS     | 4096U   |

| Type: hseKeyType_t         |       |
|----------------------------|-------|
| Name                       | Value |
| HSE_KEY_TYPE_SHE           | 0x11U |
| HSE_KEY_TYPE_AES           | 0x12U |
| HSE_KEY_TYPE_HMAC          | 0x20U |
| HSE_KEY_TYPE_SHARED_SECRET | 0x30U |
| HSE_KEY_TYPE_SIPHASH       | 0x40U |
| HSE_KEY_TYPE_ECC_PAIR      | 0x87U |
| HSE_KEY_TYPE_ECC_PUB       | 0x88U |
| HSE_KEY_TYPE_ECC_PUB_EXT   | 0x89U |
| HSE_KEY_TYPE_RSA_PAIR      | 0x97U |
| HSE_KEY_TYPE_RSA_PUB       | 0x98U |
| HSE_KEY_TYPE_RSA_PUB_EXT   | 0x99U |
| HSE_KEY_TYPE_DH_PAIR       | 0xA7U |

## Key Management Services

|                     |       |
|---------------------|-------|
| HSE_KEY_TYPE_DH_PUB | 0xA8U |
|---------------------|-------|

| Type: <a href="#">hseKeyGroupOwner_t</a> |       |
|------------------------------------------|-------|
| Name                                     | Value |
| <a href="#">HSE_KEY_OWNER_ANY</a>        | 0U    |
| <a href="#">HSE_KEY_OWNER_CUST</a>       | 1U    |
| <a href="#">HSE_KEY_OWNER_OEM</a>        | 2U    |

| Type: <a href="#">hseKeyFlags_t</a>        |           |
|--------------------------------------------|-----------|
| Name                                       | Value     |
| <a href="#">HSE_KF_USAGE_ENCRYPT</a>       | 1U << 0U  |
| <a href="#">HSE_KF_USAGE_DECRYPT</a>       | 1U << 1U  |
| <a href="#">HSE_KF_USAGE_SIGN</a>          | 1U << 2U  |
| <a href="#">HSE_KF_USAGE_VERIFY</a>        | 1U << 3U  |
| <a href="#">HSE_KF_USAGE_EXCHANGE</a>      | 1U << 4U  |
| <a href="#">HSE_KF_USAGE_DERIVE</a>        | 1U << 5U  |
| <a href="#">HSE_KF_USAGE_KEY_PROVISION</a> | 1U << 6U  |
| <a href="#">HSE_KF_USAGE_AUTHORIZATION</a> | 1U << 7U  |
| <a href="#">HSE_KF_USAGE_SMR_DECRYPT</a>   | 1U << 8U  |
| <a href="#">HSE_KF_ACCESS_WRITE_PROT</a>   | 1U << 9U  |
| <a href="#">HSE_KF_ACCESS_DEBUG_PROT</a>   | 1U << 10U |
| <a href="#">HSE_KF_ACCESS_EXPORTABLE</a>   | 1U << 11U |
| <a href="#">HSE_KF_USAGE_XTS_TWEAK</a>     | 1U << 12U |
| <a href="#">HSE_KF_USAGE_OTFAD_DECRYPT</a> | 1U << 13U |

| Type: <a href="#">hseAesBlockModeMask_t</a> |                                                 |
|---------------------------------------------|-------------------------------------------------|
| Name                                        | Value                                           |
| <a href="#">HSE_KU_AES_BLOCK_MODE_ANY</a>   | 0U                                              |
| <a href="#">HSE_KU_AES_BLOCK_MODE_XTS</a>   | 1U << 0U                                        |
| <a href="#">HSE_KU_AES_BLOCK_MODE_CTR</a>   | 1U << <a href="#">HSE_CIPHER_BLOCK_MODE_CTR</a> |
| <a href="#">HSE_KU_AES_BLOCK_MODE_CBC</a>   | 1U << <a href="#">HSE_CIPHER_BLOCK_MODE_CBC</a> |
| <a href="#">HSE_KU_AES_BLOCK_MODE_ECB</a>   | 1U << <a href="#">HSE_CIPHER_BLOCK_MODE_ECB</a> |
| <a href="#">HSE_KU_AES_BLOCK_MODE_CFB</a>   | 1U << <a href="#">HSE_CIPHER_BLOCK_MODE_CFB</a> |
| <a href="#">HSE_KU_AES_BLOCK_MODE_OFB</a>   | 1U << <a href="#">HSE_CIPHER_BLOCK_MODE_OFB</a> |
| <a href="#">HSE_KU_AES_BLOCK_MODE_CCM</a>   | 1U << 6U                                        |
| <a href="#">HSE_KU_AES_BLOCK_MODE_GCM</a>   | 1U << 7U                                        |

| Type: <a href="#">hseKeyCatalogId_t</a> |       |
|-----------------------------------------|-------|
| Name                                    | Value |
| <a href="#">HSE_KEY_CATALOG_ID_ROM</a>  | 0U    |
| <a href="#">HSE_KEY_CATALOG_ID_NVM</a>  | 1U    |

|                                            |                              |
|--------------------------------------------|------------------------------|
| <a href="#">HSE_KEY_CATALOG_ID_RAM</a>     | 2U                           |
| <a href="#">GET_CATALOG_ID</a> (keyHandle) | ((keyHandle) >> 16U) & 0xFFU |

## Typedefs

- typedef uint8\_t [hseKeyCatalogId\\_t](#)
- typedef uint8\_t [hseKeyGroupOwner\\_t](#)
- typedef uint8\_t [hseKeyType\\_t](#)
- typedef uint16\_t [hseKeyFlags\\_t](#)
- typedef uint32\_t [hseSmrFlags\\_t](#)
- typedef uint8\_t [hseEccCurveId\\_t](#)
- typedef uint16\_t [hseKeyBits\\_t](#)
- typedef uint8\_t [hseAesBlockModeMask\\_t](#)

## Data Structure Documentation

### struct hseKeyGroupCfgEntry\_t

The entry of the Key Catalog Configuration.

The size of a key slot is computed internally based on keytype and maxKeyBitLen.

#### Note

A key group (catalog entry) contains keys that have the same key type and the keybitLen <= maxKeyBitLen.

#### Data Fields

| Type                               | Name           | Description                                                                                 |
|------------------------------------|----------------|---------------------------------------------------------------------------------------------|
| <a href="#">hseMuMask_t</a>        | muMask         | Specifies the MU Instance(s) for the key group. A key group can belong to one ore more MUs. |
| <a href="#">hseKeyGroupOwner_t</a> | groupOwner     | Specifies the key group owner.                                                              |
| <a href="#">hseKeyType_t</a>       | keyType        | The key type (see <a href="#">hseKeyType_t</a> ).                                           |
| uint8_t                            | numOfKeySlots  | The number of key slots.                                                                    |
| uint16_t                           | maxKeyBitLen   | The maximum length of the key (in bits). All stored keys have keyBitLen <= maxKeyBitLen.    |
| uint8_t                            | hseReserved[2] | HSE reserved.                                                                               |

### struct hseKeyInfo\_t

Key properties.

## Key Management Services

Each cryptographic key material will be based on key properties (info) and key data

### Data Fields

| Type                               | Name            | Description                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyFlags_t</a>      | keyFlags        | The key flags (see <a href="#">hseKeyFlags_t</a> )                                                                                                                                                                                                                                                                                                                                                   |
| uint16_t                           | keyBitLen       | The length of key in bits. <ul style="list-style-type: none"><li>– For RSA, bit length of modulus n</li><li>– For ECC, the bit length of the base point order.</li><li>– Any other key, the bit length of the key.</li></ul>                                                                                                                                                                         |
| uint32_t                           | keyCounter      | The key counter used to prevent the rollback attacks on the key. For NVM keys, the key counter must be between 0 and <a href="#">HSE_KF_MAX_KEY_COUNTER_VALUE</a> For RAM keys, the key counter is forced to 0xFFFFFFFF (not used).<br><br>Note<br><br>The key counter for SHE keys follows the SHE specification (e.g. key counter is 28bits; for SHE RAM keys, the key counter is forced to zero). |
| <a href="#">hseSmrFlags_t</a>      | smrFlags        | A set of flags that define which secure memory region (SMR), indexed from 0 to 31, should be verified before the key can be used. Set to zero means not used.<br>For RAM keys, the SMR flags are forced to zero (not used). Keys linked with SMR(s) that are not yet present in the system will be available until these SMR(s) are successfully installed.                                          |
| <a href="#">hseKeyType_t</a>       | keyType         | The key type (see <a href="#">hseKeyType_t</a> ).                                                                                                                                                                                                                                                                                                                                                    |
| union <a href="#">hseKeyInfo_t</a> | specific        |                                                                                                                                                                                                                                                                                                                                                                                                      |
| uint8_t                            | hseReserved[2U] |                                                                                                                                                                                                                                                                                                                                                                                                      |

### union hseKeyInfo\_t.specific

### Data Fields

| Type                                  | Name             | Description                                                                                                                        |
|---------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseEccCurveId_t</a>       | eccCurveId       | The ECC curve Id used with this key. This is used only for ECC key type.                                                           |
| uint8_t                               | pubExponentSize  | The size (in bytes) of the RSA public exponent (e); it should be less than 16 bytes.                                               |
| <a href="#">hseAesBlockModeMask_t</a> | aesBlockModeMask | The cipher mode usage for an AES key. This is used only for AES key type If aesBlockModeMask == 0, any AES block mode can be used. |

## Macro Definition Documentation

### HSE\_KEY\_CATALOG\_ID\_ROM

```
#define HSE_KEY_CATALOG_ID_ROM ((hseKeyCatalogId_t)0U)
```

ROM key catalog (NXP keys)

### HSE\_KEY\_CATALOG\_ID\_NVM

```
#define HSE_KEY_CATALOG_ID_NVM ((hseKeyCatalogId_t)1U)
```

NVM key catalog.

### HSE\_KEY\_CATALOG\_ID\_RAM

```
#define HSE_KEY_CATALOG_ID_RAM ((hseKeyCatalogId_t)2U)
```

RAM key catalog.

### GET\_KEY\_HANDLE

```
#define GET_KEY_HANDLE( catalogId, groupId, slotIdx )
```

**Value:**

```
((((hseKeyHandle_t)((hseKeyCatalogId_t)(catalogId))) « 16U) | \
 ((hseKeyHandle_t)((hseKeyGroupId_t)(groupId))) « 8U) | \
 ((hseKeyHandle_t)((hseKeySlotIdx_t)(slotIdx))))
```

All keys used in cryptographic operations are referenced by a unique key handle. The key handle is a 32-bit integer: the key catalog(byte2), group index in catalog (byte1) and key slot index (byte0). It can be retrieved based on the catalog ID, the group index and its slot index within the group. The group index is between 0 and (n-1), where n is the maximum number of groups defined in the catalog. The slot index is between 0 and (p-1), where p is the maximum number of keys defined in the group.

### GET\_CATALOG\_ID

```
#define GET_CATALOG_ID( keyHandle ) ((hseKeyCatalogId_t)((keyHandle) >> 16U) & 0xFFU)
```

Get key catalog Id.

### GET\_GROUP\_IDX

```
#define GET_GROUP_IDX( keyHandle ) ((hseKeyGroupIdx_t) (((keyHandle) >> 8U) & 0xFFU))
```

Get key group index.

### GET\_SLOT\_IDX

```
#define GET_SLOT_IDX( keyHandle ) ((hseKeySlotIdx_t) ((keyHandle) & 0xFFU))
```

Get key slot index.

### HSE\_INVALID\_KEY\_HANDLE

```
#define HSE_INVALID_KEY_HANDLE ((hseKeyHandle_t) 0xFFFFFFFFFUL)
```

HSE invalid key .

### HSE\_INVALID\_GROUP\_IDX

```
#define HSE_INVALID_GROUP_IDX ((hseKeyGroupIdx_t) 0xFFU)
```

HSE invalid key group index.

### HSE\_INVALID\_SLOT\_IDX

```
#define HSE_INVALID_SLOT_IDX ((hseKeySlotIdx_t) 0xFFU)
```

HSE invalid key slot index.

### HSE\_KEY\_OWNER\_ANY

```
#define HSE_KEY_OWNER_ANY ((hseKeyGroupOwner_t) 0U)
```

The key are owned by ANY owner. This applies only for RAM key groups. The RAM keys can be installed/updated by any owner (CUST or OEM) having SuperUser or User rights.

### HSE\_KEY\_OWNER\_CUST

```
#define HSE_KEY_OWNER_CUST ((hseKeyGroupOwner_t)1U)
```

The key are owned by OWNER\_CUST. This applies only for NVM key groups.

The CUST keys can be installed/updated as follow:

- using CUST SuperUser rights (if Life Cycle = CUST\_DEL or if the host was granted with CUST SuperUser rights).
- using User rights (Life Cycle = IN\_FIELD)

### HSE\_KEY\_OWNER\_OEM

```
#define HSE_KEY_OWNER_OEM ((hseKeyGroupOwner_t)2U)
```

The key groups owned by OWNER\_OEM. This applies only for NVM key groups.

The OEM keys can be installed/updated as follow:

- using OEM SuperUser rights (if Life Cycle = OEM\_PROD or if the host was granted with OEM SuperUser rights).
- using User rights (Life Cycle = IN\_FIELD)

### HSE\_KEY\_TYPE\_SHE

```
#define HSE_KEY_TYPE_SHE ((hseKeyType_t)0x11U)
```

Symmetric AES128 key used with SHE specification commands. It can be used with any AES block ciphering mode and AES MACs (same as any AES128 key).

### HSE\_KEY\_TYPE\_AES

```
#define HSE_KEY_TYPE_AES ((hseKeyType_t)0x12U)
```

Symmetric AES key or AES OTFAD key.

## Key Management Services

### HSE\_KEY\_TYPE\_HMAC

```
#define HSE_KEY_TYPE_HMAC ((hseKeyType_t) 0x20U)
```

Symmetric HMAC key.

### HSE\_KEY\_TYPE\_SHARED\_SECRET

```
#define HSE_KEY_TYPE_SHARED_SECRET ((hseKeyType_t) 0x30U)
```

Shared secret used by DH key exchange protocols.

### HSE\_KEY\_TYPE\_SIPHASH

```
#define HSE_KEY_TYPE_SIPHASH ((hseKeyType_t) 0x40U)
```

Symmetric SipHash key.

### HSE\_KEY\_TYPE\_ECC\_PAIR

```
#define HSE_KEY_TYPE_ECC_PAIR ((hseKeyType_t) 0x87U)
```

ECC key pair (private and public)

### HSE\_KEY\_TYPE\_ECC\_PUB

```
#define HSE_KEY_TYPE_ECC_PUB ((hseKeyType_t) 0x88U)
```

ECC Public key.

### HSE\_KEY\_TYPE\_ECC\_PUB\_EXT

```
#define HSE_KEY_TYPE_ECC_PUB_EXT ((hseKeyType_t) 0x89U)
```

ECC public keys, where the key value is stored in the application area (e.g. certificate)



**HSE\_KEY\_TYPE\_RSA\_PAIR**

```
#define HSE_KEY_TYPE_RSA_PAIR ((hseKeyType_t)0x97U)
```

RSA key pair (private and public key)

**HSE\_KEY\_TYPE\_RSA\_PUB**

```
#define HSE_KEY_TYPE_RSA_PUB ((hseKeyType_t)0x98U)
```

RSA Public key.

**HSE\_KEY\_TYPE\_RSA\_PUB\_EXT**

```
#define HSE_KEY_TYPE_RSA_PUB_EXT ((hseKeyType_t)0x99U)
```

RSA public keys, where the key value is stored in the application area (e.g. certificate)

**HSE\_KEY\_TYPE\_DH\_PAIR**

```
#define HSE_KEY_TYPE_DH_PAIR ((hseKeyType_t)0xA7U)
```

DH key pair.

**HSE\_KEY\_TYPE\_DH\_PUB**

```
#define HSE_KEY_TYPE_DH_PUB ((hseKeyType_t)0xA8U)
```

DH public key.

**HSE\_KF\_USAGE\_ENCRYPT**

```
#define HSE_KF_USAGE_ENCRYPT ((hseKeyFlags_t)1U << 0U)
```

Key is used to encrypt data. If the HSE\_KF\_USAGE\_KEY\_PROVISION flag is set, the key can not be used for data encryption (only for key encryption).

## Key Management Services

### HSE\_KF\_USAGE\_DECRYPT

```
#define HSE_KF_USAGE_DECRYPT ((hseKeyFlags_t)1U << 1U)
```

Key is used to decrypt data. If the HSE\_KF\_USAGE\_KEY\_PROVISION flag is set, the key can not be used for data decryption (only for key decryption).

### HSE\_KF\_USAGE\_SIGN

```
#define HSE_KF_USAGE_SIGN ((hseKeyFlags_t)1U << 2U)
```

Key is used to generate digital signatures or MACs of any data (including keys if HSE\_KF\_USAGE\_KEY\_PROVISION is set).

### HSE\_KF\_USAGE\_VERIFY

```
#define HSE_KF_USAGE_VERIFY ((hseKeyFlags_t)1U << 3U)
```

Key is used to verify digital signatures or MACs of any data (including keys if HSE\_KF\_USAGE\_KEY\_PROVISION is set).

### HSE\_KF\_USAGE\_EXCHANGE

```
#define HSE_KF_USAGE_EXCHANGE ((hseKeyFlags_t)1U << 4U)
```

Key is used for key exchange protocol (e.g. DH).

### HSE\_KF\_USAGE\_DERIVE

```
#define HSE_KF_USAGE_DERIVE ((hseKeyFlags_t)1U << 5U)
```

Key may be use as a base key for deriving other keys.

### HSE\_KF\_USAGE\_KEY\_PROVISION

```
#define HSE_KF_USAGE_KEY_PROVISION ((hseKeyFlags_t)1U << 6U)
```

Key used for key provisioning operation. The provision keys can only be NVM keys. This bit (if it is set) along with the encrypt/decrypt/sign/verify flags specifies which operations can be performed on a key using this key (provisioning key).

### **HSE\_KF\_USAGE\_AUTHORIZATION**

```
#define HSE_KF_USAGE_AUTHORIZATION ((hseKeyFlags_t)1U << 7U)
```

Key can be used for system authorization. Can be set only for NVM keys. This key should have the verify flag set, but the sign flag NOT set.

### **HSE\_KF\_USAGE\_SMR\_DECRYPT**

```
#define HSE_KF_USAGE_SMR_DECRYPT ((hseKeyFlags_t)1U << 8U)
```

The key is used for SMR decryption. If this bit is set during key installation, the HSE will set the HSE\_KF\_USAGE\_DECRYPT flag to zero.

### **HSE\_KF\_ACCESS\_WRITE\_PROT**

```
#define HSE_KF_ACCESS_WRITE_PROT ((hseKeyFlags_t)1U << 9U)
```

The key is write protected and cannot change anymore. For RAM keys, this flag is forced to zero.

### **HSE\_KF\_ACCESS\_DEBUG\_PROT**

```
#define HSE_KF_ACCESS_DEBUG_PROT ((hseKeyFlags_t)1U << 10U)
```

The key is disabled when a debugger is attached. For RAM keys, this flag is forced to zero.

### **HSE\_KF\_ACCESS\_EXPORTABLE**

```
#define HSE_KF_ACCESS_EXPORTABLE ((hseKeyFlags_t)1U << 11U)
```

The key can be exported or not in any format. Ignored when used in combination with HSE\_KF\_USAGE\_KEY\_PROVISION or HSE\_KF\_USAGE\_AUTHORIZATION (provision/authorization keys are NOT exportable).

## Key Management Services

### HSE\_KF\_USAGE\_XTS\_TWEAK

```
#define HSE_KF_USAGE_XTS_TWEAK ((hseKeyFlags_t)1U << 12U)
```

This is used as a tweak key in xts aes encryption; no other flag shall be set.

### HSE\_KF\_USAGE\_OTFAD\_DECRYPT

```
#define HSE_KF_USAGE_OTFAD_DECRYPT ((hseKeyFlags_t)1U << 13U)
```

The key is used just in OTFAD decryption; no other flag shall be set.

### HSE\_KF\_USAGE\_MASK

```
#define HSE_KF_USAGE_MASK
```

#### Value:

```
( HSE_KF_USAGE_ENCRYPT | HSE_KF_USAGE_DECRYPT | HSE_KF_USAGE_SIGN | HSE_KF_USAGE_VERIFY | HSE_KF_USAGE_EXCHANGE | \
  HSE_KF_USAGE_DERIVE | HSE_KF_USAGE_KEY_PROVISION | HSE_KF_USAGE_AUTHORIZATION | HSE_KF_USAGE_SMR_DECRYPT | \
  HSE_KF_USAGE_XTS_TWEAK | HSE_KF_USAGE_OTFAD_DECRYPT )
```

The Key Usage flags mask.

### HSE\_KF\_MSC\_USAGE\_MASK

```
#define HSE_KF_MSC_USAGE_MASK ( HSE_KF_USAGE_ENCRYPT | HSE_KF_USAGE_DECRYPT | HSE_KF_USAGE_SIGN | HSE_KF_USAGE_VERIFY )
```

The Key Usage flags mask for Managed Security Component (MSC) targeted keys. The accepted key flags for keys that are configured in Key Handle Translation Table (KHTT) for MSC keystore. These key flags can be configured only with HSE\_KF\_USAGE\_ENCRYPT, HSE\_KF\_USAGE\_DECRYPT, HSE\_KF\_USAGE\_SIGN or HSE\_KF\_USAGE\_VERIFY.

### HSE\_KF\_ACCESS\_MASK

```
#define HSE_KF_ACCESS_MASK ( HSE_KF_ACCESS_WRITE_PROT | HSE_KF_ACCESS_DEBUG_PROT | HSE_KF_ACCESS_EXPORTABLE )
```

The Key Access flags mask.

**HSE\_KF\_MAX\_KEY\_COUNTER\_VALUE**

```
#define HSE_KF_MAX_KEY_COUNTER_VALUE ((uint32_t)0xFFFFFFFFUL - 1UL)
```

The maximum value of key counter. Note that 0xFFFFFFFF is reserved for RAM keys.

**HSE\_ROM\_KEY\_AES256\_KEY0**

```
#define HSE_ROM_KEY_AES256_KEY0 ((hseKeyHandle_t)0x00000000UL)
```

This key can be used for data encryption/decryption, having the following usage restrictions:

HSE ROM key handles. The ROM key catalog references keys that are provisioned by NXP and can be used by the host.

## Note

- The ROM keys have the following access restriction flags set:  
(# HSE\_KF\_ACCESS\_WRITE\_PROT | # HSE\_KF\_ACCESS\_DEBUG\_PROT)
- This key is a device-specific secret
- This key can be used to encrypt/decrypt application data with a device-specific key  
(# HSE\_KF\_USAGE\_ENCRYPT | # HSE\_KF\_USAGE\_DECRYPT)

**HSE\_ROM\_KEY\_AES256\_KEY1**

```
#define HSE_ROM_KEY_AES256_KEY1 ((hseKeyHandle_t)0x00000001UL)
```

This key can be used for key derivation and key provisioning, having the following usage restrictions:

## Note

- This key is a shared secret owned by NXP
- It can be used during key provision to import an application key encrypted with an NXP secret
- This NXP key can be used to encrypt a customer key using an email service provided by NXP. In this way, the customer key can be injected in HSE sub-system in a secure manner. Contact NXP support team for more details.
- The service is used in pair with another RSA key. The email service provides a signature which is verified using the RSA key.  
(# HSE\_KF\_USAGE\_DERIVE | # HSE\_KF\_USAGE\_VERIFY | # HSE\_KF\_USAGE\_DECRYPT | # HSE\_KF\_USAGE\_KEY\_PROVISION)

## Key Management Services

### HSE\_ROM\_KEY\_RSA3072\_PUB\_KEY0

```
#define HSE_ROM_KEY_RSA3072_PUB_KEY0 ((hseKeyHandle_t)0x00000100UL)
```

This key can be used for RSA decryption and signature verify, having the following usage restrictions:

#### Note

- This key is a public RSA key owned by NXP; the corresponding private key is owned by NXP.
- It can be used during key provision to import an application key signed.
- This NXP key can be used to verify a signature on a customer key which is signed using an email service provided by NXP. In this way, the customer key can be injected in HSE sub-system in a secure manner. Contact NXP support team for more details.
- The service is used in pair with another ROM key i.e HSE\_ROM\_KEY\_AES256\_KEY1.  
(# HSE\_ KF\_ USAGE\_ DECRYPT | # HSE\_ KF\_ USAGE\_ VERIFY | # HSE\_ KF\_ USAGE\_ KEY\_ PROVISION)

### HSE\_ROM\_KEY\_ECC256\_PUB\_KEY0

```
#define HSE_ROM_KEY_ECC256_PUB_KEY0 ((hseKeyHandle_t)0x00000200UL)
```

This key can be used for key provisioning having the following usage restrictions:

#### Note

- This key is a public ECC key owned by NXP; the corresponding private key owned by NXP.
- It can be used during key provision to import an application key signed using an NXP ECC public key.
- This NXP key can be used to sign a customer key using an email service provided by NXP. In this way, the customer key can be injected in HSE sub-system in a secure manner. Contact NXP for more details.  
(# HSE\_ KF\_ USAGE\_ VERIFY | # HSE\_ KF\_ USAGE\_ KEY\_ PROVISION)

### HSE\_KF\_SMR\_0

```
#define HSE_KF_SMR_0 ((hseSmrFlags_t)1UL << 0UL)
```

### HSE\_KF\_SMR\_1

```
#define HSE_KF_SMR_1 ((hseSmrFlags_t)1UL << 1UL)
```

**HSE\_KF\_SMR\_2**

```
#define HSE_KF_SMR_2 ((hseSmrFlags_t)1UL << 2UL)
```

**HSE\_KF\_SMR\_3**

```
#define HSE_KF_SMR_3 ((hseSmrFlags_t)1UL << 3UL)
```

**HSE\_KF\_SMR\_4**

```
#define HSE_KF_SMR_4 ((hseSmrFlags_t)1UL << 4UL)
```

**HSE\_KF\_SMR\_5**

```
#define HSE_KF_SMR_5 ((hseSmrFlags_t)1UL << 5UL)
```

**HSE\_KF\_SMR\_6**

```
#define HSE_KF_SMR_6 ((hseSmrFlags_t)1UL << 6UL)
```

**HSE\_KF\_SMR\_7**

```
#define HSE_KF_SMR_7 ((hseSmrFlags_t)1UL << 7UL)
```

**HSE\_KF\_SMR\_8**

```
#define HSE_KF_SMR_8 ((hseSmrFlags_t)1UL << 8UL)
```

**HSE\_KF\_SMR\_9**

```
#define HSE_KF_SMR_9 ((hseSmrFlags_t)1UL << 9UL)
```

### HSE\_KF\_SMR\_10

```
#define HSE_KF_SMR_10 ((hseSmrFlags_t)1UL << 10UL)
```

### HSE\_KF\_SMR\_11

```
#define HSE_KF_SMR_11 ((hseSmrFlags_t)1UL << 11UL)
```

### HSE\_KF\_SMR\_12

```
#define HSE_KF_SMR_12 ((hseSmrFlags_t)1UL << 12UL)
```

### HSE\_KF\_SMR\_13

```
#define HSE_KF_SMR_13 ((hseSmrFlags_t)1UL << 13UL)
```

### HSE\_KF\_SMR\_14

```
#define HSE_KF_SMR_14 ((hseSmrFlags_t)1UL << 14UL)
```

### HSE\_KF\_SMR\_15

```
#define HSE_KF_SMR_15 ((hseSmrFlags_t)1UL << 15UL)
```

### HSE\_KF\_SMR\_16

```
#define HSE_KF_SMR_16 ((hseSmrFlags_t)1UL << 16UL)
```



**HSE\_KF\_SMR\_17**

```
#define HSE_KF_SMR_17 ((hseSmrFlags_t)1UL << 17UL)
```

**HSE\_KF\_SMR\_18**

```
#define HSE_KF_SMR_18 ((hseSmrFlags_t)1UL << 18UL)
```

**HSE\_KF\_SMR\_19**

```
#define HSE_KF_SMR_19 ((hseSmrFlags_t)1UL << 19UL)
```

**HSE\_KF\_SMR\_20**

```
#define HSE_KF_SMR_20 ((hseSmrFlags_t)1UL << 20UL)
```

**HSE\_KF\_SMR\_21**

```
#define HSE_KF_SMR_21 ((hseSmrFlags_t)1UL << 21UL)
```

**HSE\_KF\_SMR\_22**

```
#define HSE_KF_SMR_22 ((hseSmrFlags_t)1UL << 22UL)
```

**HSE\_KF\_SMR\_23**

```
#define HSE_KF_SMR_23 ((hseSmrFlags_t)1UL << 23UL)
```

**HSE\_KF\_SMR\_24**

```
#define HSE_KF_SMR_24 ((hseSmrFlags_t)1UL << 24UL)
```

### HSE\_KF\_SMR\_25

```
#define HSE_KF_SMR_25 ((hseSmrFlags_t)1UL << 25UL)
```

### HSE\_KF\_SMR\_26

```
#define HSE_KF_SMR_26 ((hseSmrFlags_t)1UL << 26UL)
```

### HSE\_KF\_SMR\_27

```
#define HSE_KF_SMR_27 ((hseSmrFlags_t)1UL << 27UL)
```

### HSE\_KF\_SMR\_28

```
#define HSE_KF_SMR_28 ((hseSmrFlags_t)1UL << 28UL)
```

### HSE\_KF\_SMR\_29

```
#define HSE_KF_SMR_29 ((hseSmrFlags_t)1UL << 29UL)
```

### HSE\_KF\_SMR\_30

```
#define HSE_KF_SMR_30 ((hseSmrFlags_t)1UL << 30UL)
```

### HSE\_KF\_SMR\_31

```
#define HSE_KF_SMR_31 ((hseSmrFlags_t)1UL << 31UL)
```

**HSE\_EC\_CURVE\_NONE**

```
#define HSE_EC_CURVE_NONE ((hseEccCurveId_t) 0U)
```

**HSE\_EC\_SEC\_SECP256R1**

```
#define HSE_EC_SEC_SECP256R1 ((hseEccCurveId_t) 1U)
```

**HSE\_EC\_SEC\_SECP384R1**

```
#define HSE_EC_SEC_SECP384R1 ((hseEccCurveId_t) 2U)
```

**HSE\_EC\_SEC\_SECP521R1**

```
#define HSE_EC_SEC_SECP521R1 ((hseEccCurveId_t) 3U)
```

**HSE\_EC\_BRAINPOOL\_BRAINPOOLP256R1**

```
#define HSE_EC_BRAINPOOL_BRAINPOOLP256R1 ((hseEccCurveId_t) 4U)
```

**HSE\_EC\_BRAINPOOL\_BRAINPOOLP320R1**

```
#define HSE_EC_BRAINPOOL_BRAINPOOLP320R1 ((hseEccCurveId_t) 5U)
```

**HSE\_EC\_BRAINPOOL\_BRAINPOOLP384R1**

```
#define HSE_EC_BRAINPOOL_BRAINPOOLP384R1 ((hseEccCurveId_t) 6U)
```

**HSE\_EC\_BRAINPOOL\_BRAINPOOLP512R1**

```
#define HSE_EC_BRAINPOOL_BRAINPOOLP512R1 ((hseEccCurveId_t) 7U)
```

### **HSE\_EC\_25519\_ED25519**

```
#define HSE_EC_25519_ED25519 ((hseEccCurveId_t) 9U)
```

### **HSE\_EC\_25519\_CURVE25519**

```
#define HSE_EC_25519_CURVE25519 ((hseEccCurveId_t) 10U)
```

### **HSE\_EC\_448\_ED448**

```
#define HSE_EC_448_ED448 ((hseEccCurveId_t) 11U)
```

### **HSE\_EC\_448\_CURVE448**

```
#define HSE_EC_448_CURVE448 ((hseEccCurveId_t) 12U)
```

### **HSE\_EC\_USER\_CURVE1**

```
#define HSE_EC_USER_CURVE1 ((hseEccCurveId_t) 101U)
```

### **HSE\_EC\_USER\_CURVE2**

```
#define HSE_EC_USER_CURVE2 ((hseEccCurveId_t) 102U)
```

### **HSE\_EC\_USER\_CURVE3**

```
#define HSE_EC_USER_CURVE3 ((hseEccCurveId_t) 103U)
```

**HSE\_KEY\_BITS\_INVALID**

```
#define HSE_KEY_BITS_INVALID ((hseKeyBits_t) 0xFFFFU)
```

**HSE\_KEY\_BITS\_ZERO**

```
#define HSE_KEY_BITS_ZERO ((hseKeyBits_t) 0U)
```

**HSE\_KEY64\_BITS**

```
#define HSE_KEY64_BITS ((hseKeyBits_t) 64U)
```

**HSE\_KEY128\_BITS**

```
#define HSE_KEY128_BITS ((hseKeyBits_t) 128U)
```

**HSE\_KEY160\_BITS**

```
#define HSE_KEY160_BITS ((hseKeyBits_t) 160U)
```

**HSE\_KEY192\_BITS**

```
#define HSE_KEY192_BITS ((hseKeyBits_t) 192U)
```

**HSE\_KEY224\_BITS**

```
#define HSE_KEY224_BITS ((hseKeyBits_t) 224U)
```

**HSE\_KEY240\_BITS**

```
#define HSE_KEY240_BITS ((hseKeyBits_t) 240U)
```

### HSE\_KEY256\_BITS

```
#define HSE_KEY256_BITS ((hseKeyBits_t)256U)
```

### HSE\_KEY320\_BITS

```
#define HSE_KEY320_BITS ((hseKeyBits_t)320U)
```

### HSE\_KEY384\_BITS

```
#define HSE_KEY384_BITS ((hseKeyBits_t)384U)
```

### HSE\_KEY512\_BITS

```
#define HSE_KEY512_BITS ((hseKeyBits_t)512U)
```

### HSE\_KEY521\_BITS

```
#define HSE_KEY521_BITS ((hseKeyBits_t)521U)
```

### HSE\_KEY638\_BITS

```
#define HSE_KEY638_BITS ((hseKeyBits_t)638U)
```

### HSE\_KEY1024\_BITS

```
#define HSE_KEY1024_BITS ((hseKeyBits_t)1024U)
```

**HSE\_KEY2048\_BITS**

```
#define HSE_KEY2048_BITS ((hseKeyBits_t)2048U)
```

**HSE\_KEY3072\_BITS**

```
#define HSE_KEY3072_BITS ((hseKeyBits_t)3072U)
```

**HSE\_KEY4096\_BITS**

```
#define HSE_KEY4096_BITS ((hseKeyBits_t)4096U)
```

**HSE\_KU\_AES\_BLOCK\_MODE\_ANY**

```
#define HSE_KU_AES_BLOCK_MODE_ANY ((hseAesBlockModeMask_t)0U)
```

Any block mode below.

**HSE\_KU\_AES\_BLOCK\_MODE\_XTS**

```
#define HSE_KU_AES_BLOCK_MODE_XTS ((hseAesBlockModeMask_t)(1U << 0U))
```

XTS mode (AES)

**HSE\_KU\_AES\_BLOCK\_MODE\_CTR**

```
#define HSE_KU_AES_BLOCK_MODE_CTR ((hseAesBlockModeMask_t)(1U << HSE_CIPHER_BLOCK_MODE_CTR))
```

CTR mode (AES)

**HSE\_KU\_AES\_BLOCK\_MODE\_CBC**

```
#define HSE_KU_AES_BLOCK_MODE_CBC ((hseAesBlockModeMask_t)(1U << HSE_CIPHER_BLOCK_MODE_CBC))
```

## Key Management Services

CBC mode (AES)

### HSE\_KU\_AES\_BLOCK\_MODE\_ECB

```
#define HSE_KU_AES_BLOCK_MODE_ECB ((hseAesBlockModeMask_t) (1U << HSE_CIPHER_BLOCK_MODE_ECB))
```

ECB mode (AES)

### HSE\_KU\_AES\_BLOCK\_MODE\_CFB

```
#define HSE_KU_AES_BLOCK_MODE_CFB ((hseAesBlockModeMask_t) (1U << HSE_CIPHER_BLOCK_MODE_CFB))
```

CFB mode (AES)

### HSE\_KU\_AES\_BLOCK\_MODE\_OFB

```
#define HSE_KU_AES_BLOCK_MODE_OFB ((hseAesBlockModeMask_t) (1U << HSE_CIPHER_BLOCK_MODE_OFB))
```

OFB mode (AES)

### HSE\_KU\_AES\_BLOCK\_MODE\_CCM

```
#define HSE_KU_AES_BLOCK_MODE_CCM ((hseAesBlockModeMask_t) (1U << 6U))
```

CCM mode (AES)

### HSE\_KU\_AES\_BLOCK\_MODE\_GCM

```
#define HSE_KU_AES_BLOCK_MODE_GCM ((hseAesBlockModeMask_t) (1U << 7U))
```

GCM mode (AES)

## Typedef Documentation



**hseKeyCatalogId\_t**

```
typedef uint8_t hseKeyCatalogId_t
```

HSE key catalog type.

A key catalog is a memory container that holds groups of keys. The catalog defines the type of storage (volatile / non-volatile) and the visibility to the application (host)

**hseKeyGroupOwner\_t**

```
typedef uint8_t hseKeyGroupOwner_t
```

HSE Key Group owner.

**hseKeyType\_t**

```
typedef uint8_t hseKeyType_t
```

HSE Key type. Specifies the Key type. It provides information about the interpretation of key data.

**hseKeyFlags\_t**

```
typedef uint16_t hseKeyFlags_t
```

The key flags specifies the operations or restrictions that can be apply to a key.

**hseSmrFlags\_t**

```
typedef uint32_t hseSmrFlags_t
```

The SMR flags.

A set of flags that define which secure memory region (SMR), shall be verified before the key can be used. For RAM keys, the SMR flags are forced to zero (not used).

**hseEccCurveId\_t**

```
typedef uint8_t hseEccCurveId_t
```

The ECC curve IDs.

## Key Management Services

### hseKeyBits\_t

```
typedef uint16_t hseKeyBits_t
```

Some default key bits values.

The below values are only a few possible values. Note that HSE supports key bit length different than those defined below (eg. TU Darmstadt curves 1 to 38).

### hseAesBlockModeMask\_t

```
typedef uint8_t hseAesBlockModeMask_t
```

Cipher modes flags for AES keys.

The values below are representing the cipher mode flags that an AES key can take.

## 5.2 HSE Key Management Utility Services

### Data Structures

- struct [hseLoadEccCurveSrv\\_t](#)
- struct [hseFormatKeyCatalogsSrv\\_t](#)
- struct [hseEraseKeySrv\\_t](#)
- struct [hseGetKeyInfoSrv\\_t](#)
- struct [hseKeyVerifySrv\\_t](#)

### Macros

| Type: (implicit C type)                              |       |
|------------------------------------------------------|-------|
| Name                                                 | Value |
| <a href="#">HSE_ERASE_NOT_USED</a>                   | 0U    |
| <a href="#">HSE_ERASE_ALL_RAM_KEYS_ON_MU_IF</a>      | 1U    |
| <a href="#">HSE_ERASE_ALL_NVM_SYM_KEYS_ON_MU_IF</a>  | 2U    |
| <a href="#">HSE_ERASE_ALL_NVM_ASYM_KEYS_ON_MU_IF</a> | 3U    |
| <a href="#">HSE_ERASE_ALL_NVM_KEYS_ON_MU_IF</a>      | 4U    |
| <a href="#">HSE_ERASE_KEYGROUP_ON_MU_IF</a>          | 5U    |

| Type: <a href="#">hseKeyVerAlgo_t</a> |                                        |
|---------------------------------------|----------------------------------------|
| Name                                  | Value                                  |
| <a href="#">HSE_KEY_VER_SHA256</a>    | <a href="#">HSE_HASH_ALGO_SHA2_256</a> |
| <a href="#">HSE_KEY_VER_SHA384</a>    | <a href="#">HSE_HASH_ALGO_SHA2_384</a> |
| <a href="#">HSE_KEY_VER_SHA512</a>    | <a href="#">HSE_HASH_ALGO_SHA2_512</a> |
| <a href="#">HSE_KEY_VER_CMAC</a>      | <a href="#">HSE_MAC_ALGO_CMAC</a>      |

|                          |       |
|--------------------------|-------|
| HSE_KEY_VER_PUB_EXT_HASH | 0xE5U |
|--------------------------|-------|

## Typedefs

- typedef uint8\_t [hseEraseKeyOptions\\_t](#)
- typedef uint8\_t [hseKeyVerAlgo\\_t](#)

## Data Structure Documentation

### struct hseLoadEccCurveSrv\_t

HSE Load ECC curve.

This service can be used to set the domain parameters for a Weierstrass ECC curve that is not supported by default. Twisted Edwards or Montgomery curve parameters cannot be loaded by this service.

Note

1. Loading a curve into the HSE modifies the SYS-IMAGE, making it necessary to publish it and store it in external flash on HSE\_H/M devices.
2. The host needs super-user rights to update the NVM configuration, in order to use this service.

Data Fields

| Type                            | Name        | Description                                                                                                                                                                                                                                                      |
|---------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseEccCurveId_t</a> | eccCurveId  | INPUT: The ECC curve ID. Must be a user allocated curve ID (i.e. HSE_ECC_CURVE <sub>x</sub> ).                                                                                                                                                                   |
| uint8_t                         | reserved[3] |                                                                                                                                                                                                                                                                  |
| <a href="#">hseKeyBits_t</a>    | pBitLen     | INPUT: The bit length of the prime p.                                                                                                                                                                                                                            |
| <a href="#">hseKeyBits_t</a>    | nBitLen     | INPUT: The bit length of the order n.                                                                                                                                                                                                                            |
| uint32_t                        | pA          | INPUT: Elliptic curve parameter a. Must be represented as a big endian number, in the form of a byte array of length <a href="#">HSE_BITS_TO_BYTES</a> ( <a href="#">pBitLen</a> ), e.g. 256 bit curves need 32 byte arrays, 521 bit curves need 66 byte arrays. |
| uint32_t                        | pB          | INPUT: Elliptic curve parameter b. Must be represented as a big endian number, in the form of a byte array of length <a href="#">HSE_BITS_TO_BYTES</a> ( <a href="#">pBitLen</a> ), e.g. 256 bit curves need 32 byte arrays, 521 bit curves need 66 byte arrays. |
| uint32_t                        | pP          | INPUT: Elliptic curve prime p. Must be represented as a big endian number, in the form of a byte array of length <a href="#">HSE_BITS_TO_BYTES</a> ( <a href="#">pBitLen</a> ), e.g. 256 bit curves need 32 byte arrays, 521 bit curves need 66 byte arrays.     |
| uint32_t                        | pN          | INPUT: Elliptic curve order n. Must be represented as a big endian number, in the form of a byte array of length <a href="#">HSE_BITS_TO_BYTES</a> ( <a href="#">nBitLen</a> ), e.g. 256 bit curves need 32 byte arrays, 521 bit curves need 66 byte arrays.     |

## Key Management Services

### Data Fields

| Type     | Name | Description                                                                                                                                                                                                                                                                                                        |
|----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pG   | INPUT: Elliptic curve generator point. The x and y coordinates of the generator, represented as big endian numbers, each in the form of a byte array of length <a href="#">HSE_BITS_TO_BYTES (pBitLen)</a> , then concatenated. The HSE expects an array of size 2 * <a href="#">HSE_BITS_TO_BYTES (pBitLen)</a> . |

### struct hseFormatKeyCatalogsSrv\_t

HSE "Format Key Catalogs" service.

Used to configure the NVM or RAM key catalogs. The catalogs format should be define according to the total number of groups ( [HSE\\_TOTAL\\_NUM\\_OF\\_KEY\\_GROUPS](#) ). and the maximum available memory for NVM or RAM keys handled by the HSE Firmware (see [HSE\\_MAX\\_NVM\\_STORE\\_SIZE](#) and [HSE\\_MAX\\_RAM\\_STORE\\_SIZE](#) ). If the catalog definition does not fit within the available memory, an error occurs and the key format fails. Each catalog should terminate with a zero filled entry.

The key catalogs (NVM and RAM) can only be formatted (or re-formatted) only if one of the following conditions is met:

- if the application has CUST\_DEL SuperUser rights (see [hseSysAuthorizationReqSrv\\_t](#)).
- if [HSE\\_STATUS\\_INSTALL\\_OK](#) is cleared (failure to read SYS-IMG from flash, or SYS-IMG missing in flash). In this case, after formatting the key catalogs, the application will be granted with CUST and OEM SU rights (ANY).

#### Note

- Each catalog entry represent a key group of the same key type.
- Each group is identified by its index within the catalog.
- Each group has an owner (see [hseKeyGroupOwner\\_t](#)). NVM keys can be owned by CUST or OEM, except the NVM SHE keys that shall have the owner HSE\_KEY\_OWNER\_ANY; RAM key owner is always [HSE\\_KEY\\_OWNER\\_ANY](#).
- Note that a key group can contain keys that have keybitLen <= maxKeyBitLen. For example, the group of key type [HSE\\_KEY\\_TYPE\\_AES](#) of 256bits can contain AES128, AES192 and AES256 keys. If there are not enough slots for an AES128 key in an AES128 group, the key can be store in an AES256 slot.
- At least one group should be defined for each catalog (NVM or RAM).
- [HSE\\_KEY\\_TYPE\\_SHARED\\_SECRET](#) key group can only be used for RAM key catalog.
- [HSE\\_KEY\\_TYPE\\_RSA\\_PAIR](#) key group can only be used for NVM key catalog.
- A key group can belong to one or more MUs.
- Both NVM and RAM catalogs shall be set in the same manner.

Example of NVM key catalog configuration.

```
{
    { HSE\_MU0\_MASK, HSE\_KEY\_OWNER\_CUST, HSE\_KEY\_TYPE\_AES,          20U,          HSE\_
      KEY128\_BITS },
    { HSE\_MU0\_MASK, HSE\_KEY\_OWNER\_CUST, HSE\_KEY\_TYPE\_ECC\_PAIR,      2U,          HSE\_
      KEY256\_BITS },
    { HSE\_MU1\_MASK, HSE\_KEY\_OWNER\_OEM,   HSE\_KEY\_TYPE\_AES,          20U,          HSE\_
      KEY256\_BITS },
}
```

```

    { HSE_ MU1_ MASK, HSE_ KEY_ OWNER_ OEM, HSE_ KEY_ TYPE_ HMAC, 10U, HSE_
KEY512_ BITS },
    { HSE_ MU1_ MASK, HSE_ KEY_ OWNER_ OEM, HSE_ KEY_ TYPE_ ECC_ PAIR, 2U, HSE_
KEY256_ BITS },
    { HSE_ MU1_ MASK, HSE_ KEY_ OWNER_ OEM, HSE_ KEY_ TYPE_ ECC_ PUB, 6U, HSE_
KEY256_ BITS },
    { HSE_ MU1_ MASK, HSE_ KEY_ OWNER_ OEM, HSE_ KEY_ TYPE_ ECC_ PUB_ EXT, 10U, HSE_
KEY256_ BITS },
    { 0U, 0U, 0U, 0U, 0U }
}

```

SHE Key catalog configuration (see below configuration):

- A maximum 5 NVM SHE groups are allowed.
- NVM SHE keys shall be mapped in the first 5 groups in NVM key Catalog.
- It is recommended to have the SHE key groups in the first groups of the NVM key catalog. Any other non-SHE key group can be added after SHE key groups in NVM/RAM Key Catalogs.
- RAM SHE key shall be mapped on key group 0 in RAM key Catalog.
- The owner for SHE key group shall be set to [HSE\\_KEY\\_OWNER\\_ANY](#).

NVM SHE Key Catalog Configuration:

- row0: MASTER\_ECU\_KEY, BOOT\_MAC\_KEY, KEY\_1 to KEY\_10
- row1: KEY\_11 to KEY\_20
- row2: KEY\_21 to KEY\_30
- row3: KEY\_31 to KEY\_40
- row4: KEY\_41 to KEY\_50

```

{
    { HSE_ MU0_ MASK, HSE_ KEY_ OWNER_ ANY, HSE_ KEY_ TYPE_ SHE, 12U, HSE_ KEY128_ BITS
    },
    { HSE_ MU0_ MASK, HSE_ KEY_ OWNER_ ANY, HSE_ KEY_ TYPE_ SHE, 10U, HSE_ KEY128_ BITS
    },
    { HSE_ MU0_ MASK, HSE_ KEY_ OWNER_ ANY, HSE_ KEY_ TYPE_ SHE, 10U, HSE_ KEY128_ BITS
    },
    { HSE_ MU0_ MASK, HSE_ KEY_ OWNER_ ANY, HSE_ KEY_ TYPE_ SHE, 10U, HSE_ KEY128_ BITS
    },
    { HSE_ MU0_ MASK, HSE_ KEY_ OWNER_ ANY, HSE_ KEY_ TYPE_ SHE, 10U, HSE_ KEY128_ BITS
    },
    { 0U, 0U, 0U, 0U, 0U }
}

```

RAM SHE Key Catalog Configuration

```

{
    { HSE_ MU0_ MASK, HSE_ KEY_ OWNER_ ANY, HSE_ KEY_ TYPE_ SHE, 1U, HSE_ KEY128_
    BITS },
    { 0U, 0U, 0U, 0U, 0U }
}

```

Data Fields

| Type     | Name              | Description                                                                                              |
|----------|-------------------|----------------------------------------------------------------------------------------------------------|
| uint32_t | pNvmKeyCatalogCfg | INPUT: Points to "NVM Key Catalog" table (table entries of type <a href="#">hseKeyGroupCfgEntry_t</a> ). |
| uint32_t | pRamKeyCatalogCfg | INPUT: Points to "RAM Key Catalog" table (table entries of type <a href="#">hseKeyGroupCfgEntry_t</a> ). |

## Key Management Services

### **struct hseEraseKeySrv\_t**

HSE "Extend Key Catalog" service.

Used to update the NVM or RAM key catalogs without reformatting and erasing the contents. The new key groups added via this services will be appended to the end of the groups already part of the targeted key catalog configuration. As a precondition, the key catalogs must be formatted before calling this service.

HSE Erase key.

This service can be used to erase RAM or NVM keys. The erase service depends on HSE access right (see [hseSysRights\\_t](#)):

1. SuperUser rights (CUST or OEM):
  - NVM CUST keys can be erased only if the CUST SuperUser rights were granted (see [hseSysAuthorizationReqSrv\\_t](#) service)
  - NVM OEM keys can be erased only if the OEM SuperUser rights were granted (see [hseSysAuthorizationReqSrv\\_t](#) service)
  - RAM keys can be erased
2. User rights:
  - NVM keys can NOT be erased.
  - RAM keys can be erased.

#### Note

- The MU mask of the key group(s) must match the MU interface on which the erase request was sent.
- For NVM key erase, the MU interface on which the host was authorized as SuperUser must match the MU interface on which erase service request has been sent.
- SHE keys cannot be erased individually (as single slot or as single NVM group). When [HSE\\_ERASE\\_ALL\\_NVM\\_SYM\\_KEYS\\_ON\\_MU\\_IF](#) or [HSE\\_ERASE\\_ALL\\_NVM\\_KEYS\\_ON\\_MU\\_IF](#) options are used, the SHE keys would be erased only if system authorization was performed beforehand using MASTER\_ECU key. Otherwise, the operation will be successful erasing other key types, but not SHE keys.

## Data Fields

| Type                                 | Name            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a>       | keyHandle       | <p>INPUT: The key handle. It is used if the erase option is <a href="#">HSE_ERASE_NOT_USED</a>, specifying the one key to be erased or if the erase option is <a href="#">HSE_ERASE_KEYGROUP_ON_MU_IF</a>, specifying the key catalog and group to be erased.</p> <p>Otherwise, it must be set to <a href="#">HSE_INVALID_KEY_HANDLE</a> when used with the other erase options ( <a href="#">HSE_ERASE_ALL_RAM_KEYS_ON_MU_IF</a>, <a href="#">HSE_ERASE_ALL_NVM_SYM_KEYS_ON_MU_IF</a>, <a href="#">HSE_ERASE_ALL_NVM_ASYM_KEYS_ON_MU_IF</a>, <a href="#">HSE_ERASE_ALL_NVM_KEYS_ON_MU_IF</a> ).</p> <p>Note</p> <p>A single write-protected NVM key cannot be deleted. Write-protected NVM keys can be deleted when multiple keys are erased (using <a href="#">HSE_ERASE_ALL_NVM_SYM_KEYS_ON_MU_IF</a>, <a href="#">HSE_ERASE_ALL_NVM_ASYM_KEYS_ON_MU_IF</a>, <a href="#">HSE_ERASE_ALL_NVM_KEYS_ON_MU_IF</a> or <a href="#">HSE_ERASE_KEYGROUP_ON_MU_IF</a> options).</p> |
| <a href="#">hseEraseKeyOptions_t</a> | eraseKeyOptions | INPUT: The Erase key options (see <a href="#">hseEraseKeyOptions_t</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| uint8_t                              | reserved[3]     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

**struct hseGetKeyInfoSrv\_t**

HSE Get Key Info service.

Return the key information (or properties) using the "key handle" as input parameter.

## Data Fields

| Type                           | Name      | Description                                                                                                                   |
|--------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a> | keyHandle | INPUT: The key handle.                                                                                                        |
| uint32_t                       | pKeyInfo  | OUTPUT: Address where to store <a href="#">hseKeyInfo_t</a> (Specifies usage flags, restriction access, key bit length etc ). |

**struct hseKeyVerifySrv\_t**

HSE Key Verify service.

This service is used to verify a CMAC, SHA256, SHA384, SHA512 (SHA384/SHA512 is not supported on HSE\_B) over a key stored inside HSE. The CMAC and SHA tag are provided by the application.

## Key Management Services

### Data Fields

| Type                            | Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a>  | keyHandle     | <p>INPUT: The key handle of the key that needs to be verified. The verification is performed on the following key formats/types:</p> <ul style="list-style-type: none"> <li>– symmetric key (HSE_KEY_TYPE_AES, HSE_KEY_TYPE_HMAC, HSE_KEY_TYPE_SHARED_SECRET, HSE_KEY_TYPE_SIPHASH): array of <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a> size</li> <li>– HSE_KEY_TYPE_ECC_PUB: <ul style="list-style-type: none"> <li>· Weierstrass curve keys: x-coordinate    y-coordinate (all in big endian); the length must be 2 * <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a></li> <li>· Twisted Edwards curve keys: point Y (with the sign bit of X), in big endian; the length must be <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a></li> <li>· Montgomery curve keys: the x-coordinate, in big endian; the length must be <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a></li> </ul> </li> <li>– HSE_KEY_TYPE_ECC_PAIR: <ul style="list-style-type: none"> <li>· Weierstrass curve keys: x-coordinate    y-coordinate    scalar (all in big endian); the length must be 3 * <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a></li> <li>· Twisted Edwards curve keys: point y (with the sign bit of X)    scalar, in big endian; the length must be 2 * <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a></li> <li>· Montgomery curve keys: the X coordinate  scalar, in big endian; the length must be 2 * <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a></li> </ul> </li> <li>– HSE_KEY_TYPE_RSA_PUB / HSE_KEY_TYPE_DH_PUB : modulus    public exponent (all in big endian)</li> <li>– HSE_KEY_TYPE_RSA_PAIR / HSE_KEY_TYPE_DH_PAIR: modulus    public exponent    private exponent (all in big endian)</li> <li>– HSE_KEY_TYPE_ECC_PUB_EXT/<br/>HSE_KEY_TYPE_RSA_PUB_EXT: verify the internal hash over the key container (e.g. certificate)</li> </ul> |
| <a href="#">hseKeyHandle_t</a>  | cmackeyHandle | <p>INPUT: The key handle used for CMAC operation. If the keyVerAlgo parameter uses a hash algorithm, this parameter is ignored.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <a href="#">hseKeyVerAlgo_t</a> | keyVerAlgo    | <p>INPUT: Key verification algorithm (see <a href="#">hseKeyVerAlgo_t</a>) Note:</p> <ul style="list-style-type: none"> <li>– If this parameter is set to <a href="#">HSE_KEY_VER_PUB_EXT_HASH</a>, the key slot corresponding to keyHandle must be a PUB_EXT key slot.</li> <li>– For HSE_B devices, the #HSE_KEY_VER_SHA_384 and #HSE_KEY_VER_SHA_512 options are not supported.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



## Data Fields

| Type     | Name         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint8_t  | tagLen       | INPUT: The provided tag length. It can be: <ul style="list-style-type: none"> <li>– a CMAC tag; the length must be between 8 - 16 bytes</li> <li>– a SHA2_256 hash; the length must be between 8 - 32 bytes</li> <li>– a SHA2_384 hash; the length must be between 8 - 48 bytes</li> <li>– a SHA2_512 hash; the length must be between 8 - 64 bytes</li> </ul> Note: If keyVerAlgo == HSE_KEY_VER_PUB_EXT_HASH, this parameter is ignored (the internal hash is used instead). |
| uint8_t  | reserved[2U] | Reserved bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| uint32_t | pTag         | INPUT: Address where tag is stored (CMAC tag or hash value). Note: If keyVerAlgo == HSE_KEY_VER_PUB_EXT_HASH, this parameter is ignored (the internal hash is used instead).                                                                                                                                                                                                                                                                                                   |

## Macro Definition Documentation

**HSE\_ERASE\_NOT\_USED**

```
#define HSE_ERASE_NOT_USED (0U)
```

Erase key options not used.

**HSE\_ERASE\_ALL\_RAM\_KEYS\_ON\_MU\_IF**

```
#define HSE_ERASE_ALL_RAM_KEYS_ON_MU_IF (1U)
```

Erase all RAM keys assigned to MU Interface on which the erase service is sent.

**HSE\_ERASE\_ALL\_NVM\_SYM\_KEYS\_ON\_MU\_IF**

```
#define HSE_ERASE_ALL_NVM_SYM_KEYS_ON_MU_IF (2U)
```

Erase all NVM symmetric keys assigned to MU Interface on which the erase service is sent (needs CUST/OEM SuperUser rights with [HSE\\_SYS\\_AUTH\\_KEY\\_MGMT](#) privileges).

**HSE\_ERASE\_ALL\_NVM\_ASYM\_KEYS\_ON\_MU\_IF**

```
#define HSE_ERASE_ALL_NVM_ASYM_KEYS_ON_MU_IF (3U)
```

## Key Management Services

Erase all NVM asymmetric keys assigned to MU Interface on which the erase service is sent (needs CUST/OEM SuperUser rights with [HSE\\_SYS\\_AUTH\\_KEY\\_MGMT](#) privileges).

### HSE\_ERASE\_ALL\_NVM\_KEYS\_ON\_MU\_IF

```
#define HSE_ERASE_ALL_NVM_KEYS_ON_MU_IF (4U)
```

Erase all NVM KEYS assigned to MU Interface on which the erase service is sent (needs CUST/OEM SuperUser rights with [HSE\\_SYS\\_AUTH\\_KEY\\_MGMT](#) privileges).

### HSE\_ERASE\_KEYGROUP\_ON\_MU\_IF

```
#define HSE_ERASE_KEYGROUP_ON_MU_IF (5U)
```

Erase all keys assigned to the key group referenced in the key handle. The MU Interface on which the erase service is sent to must be part of the group mask. CUST/OEM SuperUser rights with [HSE\\_SYS\\_AUTH\\_KEY\\_MGMT](#) privileges are needed to perform this operation. In case the key group has an owner (CUST/OEM) the SU rights must be provided for this owner.

### HSE\_KEY\_VER\_SHA256

```
#define HSE_KEY_VER_SHA256 ((hseKeyVerAlgo_t) HSE_HASH_ALGO_SHA2_256)  
SHA256.
```

### HSE\_KEY\_VER\_SHA384

```
#define HSE_KEY_VER_SHA384 ((hseKeyVerAlgo_t) HSE_HASH_ALGO_SHA2_384)  
SHA384.
```

### HSE\_KEY\_VER\_SHA512

```
#define HSE_KEY_VER_SHA512 ((hseKeyVerAlgo_t) HSE_HASH_ALGO_SHA2_512)  
SHA512.
```

**HSE\_KEY\_VER\_CMAC**

```
#define HSE_KEY_VER_CMAC ((hseKeyVerAlgo_t) HSE_MAC_ALGO_CMAC)
```

CMAC (AES)

**HSE\_KEY\_VER\_PUB\_EXT\_HASH**

```
#define HSE_KEY_VER_PUB_EXT_HASH ((hseKeyVerAlgo_t) 0xE5U)
```

Verify the internal hash of a PUB\_EXT key (e.g. external stored certificate)

**Typedef Documentation****hseEraseKeyOptions\_t**

```
typedef uint8_t hseEraseKeyOptions_t
```

Options to erase keys.

The erase key options are used only if the provided key handle is set to [HSE\\_INVALID\\_KEY\\_HANDLE](#).

**hseKeyVerAlgo\_t**

```
typedef uint8_t hseKeyVerAlgo_t
```

The algorithm used for key verification.

**5.3 HSE MSC Key Management Services****Data Structures**

- struct [hseKHTTEntry\\_t](#)
- struct [hseConfigKHTTSrv\\_t](#)
- struct [hsePushMscKeySrv\\_t](#)

**Macros**

| Type: <a href="#">hseKHTTRestrictFlags_t</a> |       |
|----------------------------------------------|-------|
| Name                                         | Value |
| <a href="#">HSE_KHTT_RESTRICT_PUSH_MANY</a>  | 1U    |

## Key Management Services

### Typedefs

- typedef uint8\_t [hseKHTTRestrictFlags\\_t](#)

### Data Structure Documentation

#### struct hseKHTTEntry\_t

The Key Handle Translation Table (KHTT) entry. Each KHTT entry contains the correspondence between HSE Key Handles and MSC key slots (e.g. ACE key slots).

##### Note

- The "Managed Security Component"(MSC) term is used bellow for any HW accelerator on host-side that uses keys managed by HSE FW.
- Each MSC key inherit the properties of its corresponding key in HSE.
- The only accepted values for keyInfo.keyFlags are HSE\_KF\_USAGE\_ENCRYPT, HSE\_KF\_USAGE\_DECRYPT, HSE\_KF\_USAGE\_SIGN or HSE\_KF\_USAGE\_VERIFY. Any other key usage flag set beside the accepted ones will lead to the key importing failure with HSE\_SRV\_RSP\_NOT\_ALLOWED status.
- If keyInfo.keyFlags is set to any of the HSE\_KF\_USAGE\_VERIFY and HSE\_KF\_USAGE\_SIGN flags, the key can only be used with CMAC operation. The rest of the key flags are ignored. For SHE RAM keys, they can only be used with CMAC operation.
- If the DID input received over ACE buses is not matching the DID flags configured for the entry, the key can not be used.
- MSC key store must be statically partitioned per host (e.g. each host has its own key slots)

##### Data Fields

| Type                           | Name         | Description                                                                                                                                                                               |
|--------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a> | hseKeyHandle | INPUT: The key handle used with HSE keystore.<br><br>Note<br><br>The key bit length of HSE key can only be 128 or 256 bits (e.g. the 256 bit key group cannot get a key of 128 bit size). |

## Data Fields

| Type                                   | Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint8_t                                | mscKeySlotIdx | <p>INPUT: The MSC key slot index. The MSC/ACE subsystem accesses a key selecting a key slot in the range 0 to 127.</p> <p>Note</p> <ul style="list-style-type: none"> <li>– The MSC keystore (e.g. ACE keystore) contains 128 X key slots of 16 bytes (or 80 X key slots on S32K388) and the associated key properties.</li> <li>– Without considering the key properties, it can be seen as an array of 128 elements (or 80 elements on S32K388), the size of each element being 16 bytes (e.g keystore[128][16]).</li> <li>– An AES128 key occupies only one key slot. An AES256 key occupies two key slots. In this case, the next key slot following an AES256 key must be placed at N+2 (e.g keystore[N+2]), where N is the AES256 key slot index.</li> <li>– The maximum number of MSC keys is 128 x 128-bit keys or 64 x 256-bit keys (on S32K388, 80 x 128-bit keys or 40 x 256-bit keys) or any combination in between.</li> </ul> |
| uint8_t                                | mscInstance   | <p>INPUT: MSC Instance (e.g. the instance of the HW accelerator on the host side).</p> <p>Note</p> <p>On S32ZE/S32K388, field is ignored; there is only one MSC instance (e.g. ACE HW accelerator).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| uint16_t                               | aceDidFlags   | <p>INPUT: 16 bits for Domain ID (DID) filtering. The didFlags is compared against "1&lt;&lt;bus DID value".</p> <p>Note</p> <p>On the buses the DID value is 4 bits only. The DID comparison works in this way: If bus DID = 4, it means bit 4 in didFlags must be set to be able to use the key.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <a href="#">hseKHTTRestrictFlags_t</a> | restrictFlags | INPUT: Restriction flags (see <a href="#">hseKHTTRestrictFlags_t</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| uint8_t                                | reserved[3]   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**struct hseConfigKHTTSrv\_t**

Configure the Key Handle Translation Table (KHTT) service. This service sends the Key Handle Translation Table (KHTT) which is statically defined by the application at configuration time.

## Key Management Services

### Note

- The "Managed Security Component"(MSC) term is used below for any HW accelerator on host-side that uses keys managed by HSE FW.
- Before configuring the MSC keys, the application shall enable the MSC clock (e.g ACE clock must be enabled by the application).
- The Key Handle Translation Table (KHTT) is a table stored in SYS-IMG with correspondences between HSE KeyHandles and MSC key slots.
- Using KHTT table, the HSE keys are mirrored by HSE subsystem in the MSC key store (e.g. ACE key store). This means that any key import/erase is automatically performed on both key stores (HSE and MSC keystores).
- The KHTT table maps each HSE keyHandle to MSC mscKeySlotIdx. The key handles can be NVM or RAM keys. Only the NVM keys are stored in SYS-IMG (the RAM keys are not persistent).
- The MSC keys can be loaded (at initialization) from SYS-IMG in MSC key store calling the [hsePushMscKeySrv\\_t](#) service with the hseKeyHandle parameter equal with HSE\_INVALID\_KEY\_HANDLE. The same service can be used to push only one HSE key slot to one MSC key slot, providing the corresponding hseKeyHandle.

### Data Fields

| Type     | Name             | Description                                                                                                                                              |
|----------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint16_t | numOfKHTTEntries | INPUT: Number of KHTT entries. It shall be maximum 200 entries(S32ZE) and 80 entries(S32K388).                                                           |
| uint8_t  | reserved[2]      |                                                                                                                                                          |
| uint32_t | pKHTT            | INPUT: Pointer to the Key Handle Translation Table(KHTT) that contains <a href="#">numOfKHTTEntries</a> entries of type <a href="#">hseKHTTEntry_t</a> . |

### struct hsePushMscKeySrv\_t

Push one or more MSC keys. This service requests HSE to push one or more HSE keys in the corresponding MSC slots. The HSE key handle must be found in the Key Handle Translation Table (KHTT) (see [hseConfigKHTTSrv\\_t](#)).

## Data Fields

| Type                           | Name         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a> | hseKeyHandle | <p>INPUT: The HSE key handle to be pushed to MSC key slot. If hseKeyHandle == HSE_INVALID_KEY_HANDLE, HSE pushes all key slots (not empty) from KHTT table that do not have the <a href="#">HSE_KHTT_RESTRICT_PUSH_MANY</a> flag set.</p> <p>Note</p> <ul style="list-style-type: none"> <li>– This option can be used to load (at initialization time) the NVM HSE keys whose handles are found in the KHTT and have the <a href="#">HSE_KHTT_RESTRICT_PUSH_MANY</a> flag set.</li> <li>– If hseKeyHandle != HSE_INVALID_KEY_HANDLE, HSE pushes a single key which must be in the KHTT table.</li> <li>– The pushed key(s) must not be empty</li> </ul> |

## Macro Definition Documentation

**HSE\_KHTT\_RESTRICT\_PUSH\_MANY**

```
#define HSE_KHTT_RESTRICT_PUSH_MANY ((hseKHTTRestrictFlags_t)1U)
```

The restriction used when more MSC keys are pushed at once:

- If it is set 0, HSE loads the MSC key when the [hsePushMscKeySrv\\_t](#) service is called with hseKeyHandle = HSE\_INVALID\_KEY\_HANDLE.
- If it is set 1, HSE does not load the MSC key when the [hsePushMscKeySrv\\_t](#) service is called with hseKeyHandle = HSE\_INVALID\_KEY\_HANDLE.

## Typedef Documentation

**hseKHTTRestrictFlags\_t**

```
typedef uint8_t hseKHTTRestrictFlags_t
```

The restriction flags for the Key Handle Translation Table (KHTT) entry.

## 5.4 HSE Key Import/Export Services

## Data Structures

- union [hseKeyFormat\\_t](#)
- struct [hseImportKeySrv\\_t](#)

## Key Management Services

- struct [hseExportKeySrv\\_t](#)
- struct [hseImportKeySrv\\_t.cipher](#)
- struct [hseImportKeySrv\\_t.keyContainer](#)
- struct [hseExportKeySrv\\_t.cipher](#)
- struct [hseExportKeySrv\\_t.keyContainer](#)

## Macros

| Type: <a href="#">hseEccKeyFormat_t</a>             |       |
|-----------------------------------------------------|-------|
| Name                                                | Value |
| <a href="#">HSE_KEY_FORMAT_ECC_PUB_RAW</a>          | 0U    |
| <a href="#">HSE_KEY_FORMAT_ECC_PUB_UNCOMPRESSED</a> | 1U    |
| <a href="#">HSE_KEY_FORMAT_ECC_PUB_COMPRESSED</a>   | 2U    |

## Typedefs

- typedef uint8\_t [hseEccKeyFormat\\_t](#)

## Data Structure Documentation

### union [hseKeyFormat\\_t](#)

HSE key format.

Includes additional information about the format of the key. Currently only used for ECC keys.

Data Fields

| Type                              | Name         | Description            |
|-----------------------------------|--------------|------------------------|
| <a href="#">hseEccKeyFormat_t</a> | eccKeyFormat | INPUT: ECC key format. |
| uint8_t                           | reserved[4]  |                        |

### struct [hseImportKeySrv\\_t](#)

HSE Import Key Service.

This service can be used to import a key in an empty slot or to update an existing key.

1. Common key restrictions (which apply for both SuperUser and User rights):
  - Key flags (of key properties) are always applied.
  - The NVM provisioning keys can be installed/updated without authentication only having SuperUser rights; they can also be updated having User rights using the pre-installed provision keys.



- The RAM provision keys can be imported only authenticated and can be used only to import RAM keys.
  - A key can be authenticated signing the key container (e.g. X.509 certificate or any container). The HOST shall provide a pointer to that key container, pointer(s) to key value(s) within the key container and pointer(s) to the tag/signature(s) (computed over the key container).
  - To import an encrypted/authenticated NVM key, the provided provision key(s) must have the same group owner as the imported NVM key.
  - To import an encrypted/authenticated NVM symmetric key using AEAD, the pointer to key info must be in the additional data
  - The key properties (keyInfo) along with the public key values are always imported in plain format.
2. Restrictions for SuperUser rights:
- NVM keys:
    - In empty slots, an encrypted key can be imported only authenticated, and a plain key can be imported with/without authentication (public keys must be imported in plain).
    - In non-empty slots, NVM keys can be imported(overwritten) in plain/encrypted, only authenticated.
  - RAM keys:
    - An encrypted key can be imported only authenticated. A plain key can be imported with/without authentication. Exception: RAM provision keys can be imported only authenticated.
3. Restrictions for User rights:
- NVM keys:
    - NVM secrets (symmetric keys and key pairs) can be imported only encrypted and authenticated. For key pair, private value must be encrypted and public value(s) unencrypted. NVM secrets imported from a signed key container MUST include the key properties (keyInfo) in the container (the provided key counter must be bigger than the previous one).
    - NVM public keys can be imported in plain, only authenticated. NVM public key imported from a signed key container can/cannot include the keyInfo in the container.
  - RAM keys:
    - An encrypted key can be imported only authenticated. A plain key can be imported with/without authentication.
    - key pairs can be imported only authenticated; private value encrypted and public value(s) unencrypted
    - public keys can be imported in plain, only authenticated.
- Note
- The key catalogs must have been formatted prior to provisioning the keys.
  - When AEAD is used to import a key, the container cannot be used.
  - The key types \*\_PUB\_EXT are stored in plain in the application NVM. For these key types, HSE stores only the key properties and the pointers to the public key values, as well as an authentication tag calculated over the key container: the authentication tag is verified by the HSE firmware whenever the related key is used by the host.
  - For HSE\_H/M devices, the SYS-IMAGE does not have to be written to application NVM after each key import operation; the SYS-IMAGE update process can be done at the end of the configuration process.

## Key Management Services

### Data Fields

| Type                           | Name            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a> | targetKeyHandle | INPUT: Specifies the slot where to add or updated a key. Note that the keyHandle identifies the key catalog, key group index and key slot index.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| uint32_t                       | pKeyInfo        | <p>INPUT: Specifies usage flags, restriction access, key length in bits, etc for the key (see <a href="#">hseKeyInfo_t</a>).</p> <p>Note</p> <ul style="list-style-type: none"><li>– Only keys that are not write protected can be updated with this service.</li><li>– NVM keys are secured against replay attacks by including a counter value stored within HSE. The anti-replay attack counter included in the key info header should be greater than the counter of the HSE key that will be updated (in case of key update). This mean that keyInfo MUST be included in the signed key container (when the Life Cycle is IN_FIELD).</li><li>– For RAM keys the key counter is ignored (keyInfo may not be in the key container).</li></ul> |

## Data Fields

| Type     | Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pKey[3]     | <p>INPUT: Pointer to key values. A RSA private key should always be imported together with the public key. An ECC private key can be imported standalone if the public key (pKey[0]) is NULL and the public key length (keyLen[0]) is zero. The public key will be computed internally from the private key.</p> <ul style="list-style-type: none"> <li>– pKey[0]: <ul style="list-style-type: none"> <li>· RSA public modulus n (big-endian).</li> <li>· ECC depends on the key format</li> <li>· Weierstrass curve keys: <ul style="list-style-type: none"> <li>· raw format: X    Y, in big endian; keyLen[0] must be 2 * <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a></li> <li>· uncompressed format: 0x04    X    Y, in big endian; keyLen[0] must be 1 + 2 * <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a></li> <li>· compressed format: 0x02 / x03    X, in big endian; keyLen[0] must be 1 + <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a></li> </ul> </li> <li>· Twisted Edwards curve keys: <ul style="list-style-type: none"> <li>· raw format: point Y with the sign bit of X, in big endian; keyLen[0] must be <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a></li> </ul> </li> <li>· Montgomery curve keys: <ul style="list-style-type: none"> <li>· raw format: the X coordinate, in big endian; keyLen[0] must be <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a></li> </ul> </li> <li>· Classic DH prime modulus p</li> </ul> </li> <li>– pKey[1]: <ul style="list-style-type: none"> <li>· RSA public exponent e (big-endian).</li> <li>· Classic DH public key</li> </ul> </li> <li>– pKey[2]: <ul style="list-style-type: none"> <li>· RSA private exponent d (big-endian).</li> <li>· ECC/ED25519 private scalar (big-endian).</li> <li>· The symmetric key (e.g AES, HMAC).</li> <li>· Classic DH private key</li> </ul> </li> </ul> |
| uint16_t | keyLen[3]   | <p>INPUT: The length in bytes for the above key values in the same order. Note that keyInfo.keyBitLen specifies the key length in bits.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| uint8_t  | reserved[2] |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Key Management Services

### Data Fields

| Type                                     | Name         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| struct <a href="#">hseImportKeySrv_t</a> | cipher       | <p>INPUT: Cipher parameters are used only if the cipherKeyHandle is not <a href="#">HSE_INVALID_KEY_HANDLE</a>.</p> <p>Note</p> <ul style="list-style-type: none"><li>– For AES-block cipher, if the keyBitLen is not multiple of AES block size (128bits), the key value have to be padded with zeros.</li><li>– For RSAES NO PADDING, the keyBitLen of the imported key must be less than or equal to <a href="#">HSE_BITS_TO_BYTES(cipherKey_keyBitLen)</a>, and the key is considered a big-endian integer.</li><li>– For RSAES-PKCS1-v1_5, the keyBitLen of the imported key shall not be greater than <a href="#">HSE_BITS_TO_BYTES(cipherKey_keyBitLen)</a> - 11 bytes.</li><li>– For RSAES-OAEP, the keyBitLen of the imported key shall not be greater than <a href="#">HSE_BITS_TO_BYTES(cipherKey_keyBitLen)</a> - 2 * hashLen - 2 bytes.</li></ul> |
| struct <a href="#">hseImportKeySrv_t</a> | keyContainer | <p>INPUT: The keyContainer parameters should be used if the key comes in a signed key container: pointers to key values within the key container should be provided. The signature/tag is assumed to be done over the key container.</p> <p>Note</p> <ul style="list-style-type: none"><li>– For NVM keys having User rights, the keyInfo MUST be included in the key container.</li><li>– If the HOST is authorized (SU rights), the *_PUB_EXT key type can be imported from an unauthenticated key container (providing the key container without the signature).</li></ul>                                                                                                                                                                                                                                                                                  |
| <a href="#">hseKeyFormat_t</a>           | keyFormat    | INPUT: Additional information about the format of the key. Key type specific.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### struct hseExportKeySrv\_t

#### HSE Export Key Service.

The key values and the key properties (optional) can be exported to the host via a key export service.

1. Common key restrictions (which apply for both SuperUser and User rights):
  - Key flags (of key properties) are always applied; this service can only be used if the key is exportable.
  - Provision/Authorization keys are NOT exportable ( [HSE\\_KF\\_ACCESS\\_EXPORTABLE](#) flag is

ignored).

- NVM keys can not be exported using RAM provision keys.
- NVM/RAM symmetric keys can be exported only encrypted with/without authentication.
- NVM/RAM public keys (from key pair or public key slots) can be exported in plain; keys may/may not be authenticated.
- The private part of a key pair can NOT be exported (the private part is never disclosed to the host).
- `_PUB_EXT` can NOT be exported.
- To export an encrypted/authenticated NVM key, the provided provision key must have the same group owner as the exported NVM key (not applicable for RAM keys).
- When AEAD is used to export a key, the container cannot be used.

#### Data Fields

| Type                           | Name            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a> | targetKeyHandle | INPUT: The key handle to be exported.<br>Note that the keyHandle identifies the key catalog, key group index and key slot index.                                                                                                                                                                                                                                                                                                                                            |
| uint32_t                       | pKeyInfo        | OUTPUT: Export the key information (see <a href="#">hseKeyInfo_t</a> ).<br><br>Note <ul style="list-style-type: none"> <li>– For symmetric keys exported in an authenticated key container, key information MUST be part of the key container;</li> <li>– For symmetric keys exported authenticated with AEAD, key information MUST be part of AAD (see <a href="#">hseAeadScheme_t</a>);</li> <li>– For public keys this parameter is optional. It can be NULL.</li> </ul> |

## Key Management Services

### Data Fields

| Type     | Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pKey[3]    | <p>OUTPUT: Addresses where to fill to key values.</p> <ul style="list-style-type: none"><li>– pKey[0]:<ul style="list-style-type: none"><li>· RSA public modulus n.</li><li>· ECC depends on the key format</li><li>· Weierstrass curve keys:<ul style="list-style-type: none"><li>· raw format: X    Y, in big endian; the HSE will output 2 * <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a> bytes</li><li>· uncompressed format: 0x04    X    Y, in big endian; the HSE will output 1 + 2 * <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a> bytes</li><li>· compressed format: 0x02 / x03    X, in big endian; the HSE will output 1 + <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a> bytes</li></ul></li><li>· Twisted Edwards curve keys:<ul style="list-style-type: none"><li>· raw format: point Y with the sign bit of X, in big endian; the HSE will output <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a> bytes</li></ul></li><li>· Montgomery curve keys:<ul style="list-style-type: none"><li>· raw format: the X coordinate, in big endian; the HSE will output <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a> bytes</li></ul></li><li>· Classic DH prime modulus p</li></ul></li><li>– pKey[1]:<ul style="list-style-type: none"><li>· RSA public exponent e.</li><li>· Classic DH public key</li></ul></li><li>– pKey[2]:<ul style="list-style-type: none"><li>· The symmetric key (e.g AES, HMAC).</li><li>· Classic DH private key</li></ul></li></ul> |
| uint32_t | pKeyLen[3] | <p>INPUT/OUTPUT: Addressed of uint16_t values of the length (in bytes) for the above buffers (INPUT). As output, it provides the lengths of the encrypted or unencrypted (only for public) keys.</p> <p>Note that the length in bits of the key is specified by <a href="#">hseKeyInfo_t</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Data Fields

| Type                                     | Name         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| struct <a href="#">hseExportKeySrv_t</a> | cipher       | <p>INPUT: Cipher parameters.</p> <p>Note</p> <ul style="list-style-type: none"> <li>– Only the private keys are encrypted and the encrypted value length is specified by the corresponding private key length (in bytes).</li> <li>– For AES-block cipher, if the keyBitLen of the exported is not multiple of AES block size (128bits), the key value will be padded with zeros.</li> <li>– For RSAES NO PADDING, the keyBitLen of the exported key must be less than or equal to <a href="#">HSE_BITS_TO_BYTES(cipherKey_keyBitLen)</a>, and the key is considered a big-endian integer.</li> <li>– For RSAES-PKCS1-v1_5, the keyBitLen of the exported key shall not be greater than <a href="#">HSE_BITS_TO_BYTES(cipherKey_keyBitLen)</a> - 11 bytes.</li> <li>– For RSAES-OAEP, the keyBitLen of the exported key shall not be greater than <a href="#">HSE_BITS_TO_BYTES(cipherKey_keyBitLen)</a> - 2 * hashLen - 2 bytes.</li> </ul> |
| struct <a href="#">hseExportKeySrv_t</a> | keyContainer | <p>INPUT: The keyContainer parameters should be used when the key have to be exported in a key container that will be authenticated: pointers to where key values will be exported should be provided within the key container. Optionally, the pKeyInfo may point inside the key container. The signature/tag is done over the key container.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <a href="#">hseKeyFormat_t</a>           | keyFormat    | <p>INPUT: Additional information about the format of the key. Key type specific.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**struct hseImportKeySrv\_t.cipher**

INPUT: Cipher parameters are used only if the cipherKeyHandle is not [HSE\\_INVALID\\_KEY\\_HANDLE](#).

## Note

- For AES-block cipher, if the keyBitLen is not multiple of AES block size (128bits), the key value have to be padded with zeros.
- For RSAES NO PADDING, the keyBitLen of the imported key must be less than or equal to [HSE\\_BITS\\_TO\\_BYTES\(cipherKey\\_keyBitLen\)](#), and the key is considered a big-endian integer.
- For RSAES-PKCS1-v1\_5, the keyBitLen of the imported key shall not be greater than [HSE\\_BITS\\_TO\\_BYTES\(cipherKey\\_keyBitLen\)](#) - 11 bytes.
- For RSAES-OAEP, the keyBitLen of the imported key shall not be greater than [HSE\\_BITS\\_TO\\_BYTES\(cipherKey\\_keyBitLen\)](#) - 2 \* hashLen - 2 bytes.

## Key Management Services

### Data Fields

| Type                              | Name            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a>    | cipherKeyHandle | INPUT: Decryption key handle. The cipherKeyHandle can only be a provisioning key ( <a href="#">HSE_KF_USAGE_KEY_PROVISION</a> and <a href="#">HSE_KF_USAGE_DECRYPT</a> flags are set).<br>Note that the key handle identifies the cipher scheme below. In case of symmetric cipher scheme and authenticated encryption scheme(AEAD) the differentiation is made using the first byte of cipherScheme. Must be set to <a href="#">HSE_INVALID_KEY_HANDLE</a> if not used. |
| <a href="#">hseCipherScheme_t</a> | cipherScheme    | Symmetric, asymmetric and AEAD cipher scheme.<br>Note<br>– Only the private keys are encrypted.                                                                                                                                                                                                                                                                                                                                                                          |

### struct hseImportKeySrv\_t.keyContainer

INPUT: The keyContainer parameters should be used if the key comes in a signed key container: pointers to key values within the key container should be provided. The signature/tag is assumed to be done over the key container.

#### Note

- For NVM keys having User rights, the keyInfo MUST be included in the key container.
- If the HOST is authorized (SU rights), the \*\_PUB\_EXT key type can be imported from an unauthenticated key container (providing the key container without the signature).

### Data Fields

| Type                           | Name            | Description                                                                                                                                                                                                                                                   |
|--------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint16_t                       | keyContainerLen | INPUT: The container length.<br><br>Note<br><br>The container includes only the signed block (without the signature).                                                                                                                                         |
| uint8_t                        | reserved[2]     |                                                                                                                                                                                                                                                               |
| uint32_t                       | pKeyContainer   | INPUT: Address of the key container; includes the key value(s) and other information used to authenticate the key. (e.g. TBSCertificate for a X.509 certificate).                                                                                             |
| <a href="#">hseKeyHandle_t</a> | authKeyHandle   | INPUT: Authentication key handle ( <a href="#">HSE_KF_USAGE_KEY_PROVISION</a> and <a href="#">HSE_KF_USAGE_VERIFY</a> flags are set). Must be set to <a href="#">HSE_INVALID_KEY_HANDLE</a> if not used. An encrypted key can be imported only authenticated. |



## Data Fields

| Type                            | Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseAuthScheme_t</a> | authScheme | INPUT: Authentication scheme.<br>Note that the key handle identifies the authentication scheme below.<br><br>Note<br><br>For the GMAC scheme, the minimum IV length is 12 bytes.                                                                                                                                                                                                                                                                                                                  |
| uint16_t                        | authLen[2] | INPUT: Byte length(s) of the authentication tag(s).<br><br>Note<br><br><ul style="list-style-type: none"> <li>– For MAC and RSA signature, only authLen[0] is used.</li> <li>– Both lengths are used for (R,S) (ECC or ED25519).</li> <li>– The MAC tag size must be minimum 16 bytes.</li> <li>– RSA signature size must be <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a>;</li> <li>– R or S size for ECDSA/EdDSA signature must be <a href="#">HSE_BITS_TO_BYTES(keyBitLength)</a></li> </ul> |
| uint32_t                        | pAuth[2]   | INPUT: Address(es) to authentication tag.<br><br>Note<br><br><ul style="list-style-type: none"> <li>– For MAC and RSA signature, only pAuth[0] is used.</li> <li>– Both pointers are used for (R,S) (ECC or ED25519).</li> </ul>                                                                                                                                                                                                                                                                  |

**struct hseExportKeySrv\_t.cipher**

INPUT: Cipher parameters.

## Note

- Only the private keys are encrypted and the encrypted value length is specified by the corresponding private key length (in bytes).
- For AES-block cipher, if the keyBitLen of the exported is not multiple of AES block size (128bits), the key value will be padded with zeros.
- For RSAES NO PADDING, the keyBitLen of the exported key must be less than or equal to [HSE\\_BITS\\_TO\\_BYTES\(cipherKey\\_keyBitLen\)](#), and the key is considered a big-endian integer.
- For RSAES-PKCS1-v1\_5, the keyBitLen of the exported key shall not be greater than [HSE\\_BITS\\_TO\\_BYTES\(cipherKey\\_keyBitLen\)](#) -11 bytes.
- For RSAES-OAEP, the keyBitLen of the exported key shall not be greater than [HSE\\_BITS\\_TO\\_BYTES\(cipherKey\\_keyBitLen\)](#) - 2 \* hashLen - 2 bytes.

## Key Management Services

### Data Fields

| Type                              | Name            | Description                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a>    | cipherKeyHandle | INPUT: Encryption key handle. The cipherKeyHandle can only be a provisioning key ( <a href="#">HSE_KF_USAGE_KEY_PROVISION</a> and <a href="#">HSE_KF_USAGE_ENCRYPT</a> flags are set).<br>Note that the key handle will identifies the cipher scheme below. Must be set to <a href="#">HSE_INVALID_KEY_HANDLE</a> if not used. |
| <a href="#">hseCipherScheme_t</a> | cipherScheme    | Symmetric, asymmetric and AEAD cipher scheme.<br><br>Note<br>– Only the private keys are encrypted.                                                                                                                                                                                                                            |

### struct hseExportKeySrv\_t.keyContainer

INPUT: The keyContainer parameters should be used when the key have to be exported in a key container that will be authenticated: pointers to where key values will be exported should be provided within the key container. Optionally, the pKeyInfo may point inside the key container. The signature/tag is done over the key container.

### Data Fields

| Type                            | Name            | Description                                                                                                                                                                                                                                                                    |
|---------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint16_t                        | keyContainerLen | INPUT: The container length.<br><br>Note<br>The key container length is the size of the byte block to be signed (without the signature).                                                                                                                                       |
| uint8_t                         | reserved[2]     |                                                                                                                                                                                                                                                                                |
| uint32_t                        | pKeyContainer   | INPUT: Address of the key container; includes the key value(s) and other information used to authenticate the key. (e.g. TBSCertificate for a X.509 certificate).                                                                                                              |
| <a href="#">hseKeyHandle_t</a>  | authKeyHandle   | INPUT: Authentication key handle ( <a href="#">HSE_KF_USAGE_KEY_PROVISION</a> and <a href="#">HSE_KF_USAGE_SIGN</a> flags are set).<br>Note that the key handle identifies the authentication scheme below. Must be set to <a href="#">HSE_INVALID_KEY_HANDLE</a> if not used. |
| <a href="#">hseAuthScheme_t</a> | authScheme      | INPUT: Authentication scheme.<br><br>Note<br>For the GMAC scheme, the minimum IV length is 12 bytes.                                                                                                                                                                           |

## Data Fields

| Type     | Name        | Description                                                                                                                                                                                                                                                         |
|----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pAuthLen[2] | OUTPUT: Address(es) for the length(s) (uin16_t values) of the authentication tag.<br><br>Note <ul style="list-style-type: none"> <li>– For MAC and RSA signature, only pAuthLen[0] is used.</li> <li>– Both lengths are used for (R,S) (ECC or ED25519).</li> </ul> |
| uint32_t | pAuth[2]    | OUTPUT: Address of authentication tag.<br><br>Note <ul style="list-style-type: none"> <li>– For MAC and RSA signature, only pAuth[0] is used.</li> <li>– Both pointers are used for (R,S) (ECC or ED25519).</li> </ul>                                              |

## Macro Definition Documentation

**HSE\_KEY\_FORMAT\_ECC\_PUB\_RAW**

```
#define HSE_KEY_FORMAT_ECC_PUB_RAW ((hseEccKeyFormat_t)0U)
```

Raw ECC public key: X || Y.

**HSE\_KEY\_FORMAT\_ECC\_PUB\_UNCOMPRESSED**

```
#define HSE_KEY_FORMAT_ECC_PUB_UNCOMPRESSED ((hseEccKeyFormat_t)1U)
```

Standard ECC uncompressed public key: 0x04 || X || Y.

**HSE\_KEY\_FORMAT\_ECC\_PUB\_COMPRESSED**

```
#define HSE_KEY_FORMAT_ECC_PUB_COMPRESSED ((hseEccKeyFormat_t)2U)
```

Standard ECC compressed public key: 0x02/0x03 || X.

## Typedef Documentation

## Key Management Services

### hseEccKeyFormat\_t

typedef uint8\_t hseEccKeyFormat\_t

HSE ECC key format.

Additional info for Ecc key format for import and export For Weierstrass curve public keys:

- the raw format is the X coordinate concatenated with the Y coordinate ( X || Y ), in big endian
  - the uncompressed format is a byte of 0x04, concatenated with the X coordinate and Y coordinates ( 0x04 || X || Y )
  - the compressed format is a byte of 0x02 or 0x03, depending on the (lsb) of Y, concatenated with the X coordinate
    - ( 0x02 || X ) if the lsb of Y is 0
    - ( 0x03 || X ) if the lsb of Y is 1
- For Twisted Edwards curve public keys:
- the raw format is the standard compressed format (point Y with the the sign bit of X), but in big endian
- For Montgomery curve public keys:
- the raw format is the X coordinate, in big endian

## 5.5 HSE Key Generate service

### Data Structures

- struct [hseKeyGenRsaScheme\\_t](#)
- struct [hseKeyGenEccScheme\\_t](#)
- struct [hseKeyGenTls12RsaPreMaster\\_t](#)
- struct [hseKeyGenerateSrv\\_t](#)
- struct [hseDHComputeSharedSecretSrv\\_t](#)
- union [hseKeyGenerateSrv\\_t.sch](#)

### Macros

| Type: (implicit C type)                             |       |
|-----------------------------------------------------|-------|
| Name                                                | Value |
| <a href="#">HSE_KEY_GEN_SYM_RANDOM_KEY</a>          | 1U    |
| <a href="#">HSE_KEY_GEN_RSA_KEY_PAIR</a>            | 2U    |
| <a href="#">HSE_KEY_GEN_ECC_KEY_PAIR</a>            | 3U    |
| <a href="#">HSE_TLS12_RSA_PRE_MASTER_SECRET_GEN</a> | 5U    |

### Typedefs

- typedef uint8\_t [hseKeyGenScheme\\_t](#)

### Data Structure Documentation

**struct hseKeyGenRsaScheme\_t**

RSA key generate scheme.

It generates a RSA key pair. Note that the public modulus can be exported to HOST via this service or using the export key service.

Data Fields

| Type     | Name         | Description                                                                                                                                                                         |
|----------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pubExpLength | INPUT: The length of public exponent "e". Should not be more than 16 bytes.                                                                                                         |
| uint32_t | pPubExp      | INPUT: The public exponent "e".                                                                                                                                                     |
| uint32_t | pModulus     | OUTPUT: The public modulus n. It can be NULL (the modulus is not provided using this service). The size of this memory area must be at least the byte length of the public modulus. |

**struct hseKeyGenEccScheme\_t**

ECC Key Generate scheme.

It generates a ECC key pair.

Note

- the curve ID is specified by the keyInfo.specific.eccCurveId parameter.
- Note that the public key can be exported to HOST via this service or using the export key service.

Data Fields

| Type     | Name    | Description                                                                                                                                                                                                                                                                                        |
|----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pPubKey | OUTPUT: Where to store the public key. If the public key is not needed at this point, pass a NULL pointer.<br>The x and y coordinate of the public key will be passed concatenated one after another, as big-endian strings. The size of the buffer must be double the byte length of the prime n. |

**struct hseKeyGenTls12RsaPreMaster\_t**

Generate the pre-master secret for TLS 1.2 RSA key exchange.

It computes the pre-master secret for TLS 1.2 RSA key exchange as specified by rfc5246(TLS 1.2):

- The [hseKeyGenerateSrv\\_t::targetKeyHandle](#) must be a [HSE\\_KEY\\_TYPE\\_SHARED\\_SECRET](#) key slot.

## Key Management Services

- The `hseKeyGenerateSrv_t::keyInfo` must have the following key flags set: `HSE_KF_USAGE_DERIVE`, `HSE_KF_ACCESS_EXPORTABLE`.
- The rfc5246 specification is used:
  - `keyInfo::keyBitLen` must be 384bits (48bytes)
  - The pre-master secret is computed as ProtocolVersion (2bytes) concatenated with 46 byte random number. The ProtocolVersion = {3,3} for TLS 1.2.
- To encrypt the generated pre-master secret, the `hseExportKeySrv_t` service with (the proper RSA scheme) must be used. The encrypted pre-master secret is sent to the peer node.
- To decrypt an encrypted pre-master secret, the `hseImportKeySrv_t` service (with the proper RSA scheme) must be used. The destination key slot can be a `HSE_KEY_TYPE_SHARED_SECRET` (with `HSE_KF_USAGE_DERIVE` key flag set) that further can be used to derive the TLS 1.2 key\_block.
- To generate the master secret the `hseKdfTLS12PrfScheme_t` service must be used.

### Note

- This service can also be used to perform the RSA\_PSK key exchange as specified by rfc4279. In the same manner as explained above, it can be used to generate the input needed for RSA encryption (see EncryptedPreMasterSecret). The EncryptedPreMasterSecret can be generated using the `hseExportKeySrv_t` service (on the client side), and imported using the `hseImportKeySrv_t` service (on the server side). In this case, to generate the master secret the `hseKdfTLS12PrfScheme_t` service must be executed using the `tlsPskUsage = HSE_TLS_KEY_EXCHANGE_RSA_PSK` option.

### Data Fields

| Type    | Name                | Description                                                                                                                                                                                                                       |
|---------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint8_t | protocolVersion[2U] | INPUT: The TLS or DTLS version. E.g. for TLS1.2 must be {3, 3}; for DTLS1.2 must be { 254, 253 }.<br><br>Note<br><br>HSE does not check the provided values;it just concatenates the protocol version with 46 byte random number. |
| uint8_t | reserved[2U]        | Reserved for future use.                                                                                                                                                                                                          |

## struct hseKeyGenerateSrv\_t

HSE Key generate service.

It can be used to generate a key pair (e.g. public and private RSA, ECC, classic DH) or a random symmetric key.

## Note

- Key flags (of key properties) are always applied.
- The keys can be generated as follow:
  1. Restrictions for SuperUser rights:
    - NVM keys can only be generated in empty slots (an erase shall be performed in advance)
    - RAM keys can always be generated (RAM keys can be overwritten)
  2. Restrictions for User rights:
    - NVM keys can NOT be generated.
    - RAM keys can always be generated (RAM keys can be overwritten)

## Data Fields

| Type                                      | Name            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a>            | targetKeyHandle | INPUT: The target key handle (where to store the new key).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <a href="#">hseKeyInfo_t</a>              | keyInfo         | INPUT: Specifies usage flags, restriction access, key bit length etc for the key.<br><br>Note <ul style="list-style-type: none"> <li>– For random symmetric key, the key length in bits should be specified by keyBitLen.</li> <li>– For RSA, keyBitLen specifies the bit length of the public modulus which shall be generated.</li> <li>– For ECC, the keyInfo should specify the ECC curve ID and the length of the base point order.</li> <li>– For RSA TLS 1.2 pre-master secret, see the <a href="#">hseKeyGenTls12RsaPreMaster_t</a> notes.</li> <li>– For classic DH, the keyBitLen specifies the bit length of the public modulus.</li> </ul> |
| <a href="#">hseKeyGenScheme_t</a>         | keyGenScheme    | INPUT: Specifies the key generation scheme (e.g random sym key, rsa key pair, ecc key pair, RSA TLS 1.2 pre-master secret, classic-DH key pair).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| uint8_t                                   | reserved[3]     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| union <a href="#">hseKeyGenerateSrv_t</a> | sch             | INPUT: The selected scheme parameters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**struct hseDHComputeSharedSecretSrv\_t**

DH Compute Shared Secret service.

Computes the Diffie-Hellman share secret for ECC or classic DH (e.g. the key exchange protocol). The share secret can only be computed in a shared secret slot, and can not be exported.

## Key Management Services

### Data Fields

| Type                           | Name             | Description                                                                                                                                                                                             |
|--------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a> | targetKeyHandle  | INPUT: The target key handle (where to store the shared secret). It must specify a <a href="#">HSE_KEY_TYPE_SHARED_SECRET</a> key slot.                                                                 |
| <a href="#">hseKeyHandle_t</a> | privKeyHandle    | INPUT: The private key.                                                                                                                                                                                 |
| <a href="#">hseKeyHandle_t</a> | peerPubKeyHandle | INPUT: The peer public key. Must be previously imported into the HSE.<br>Note that the peer public key can also be imported as a *_PUB_EXT key type (external public key stored on the application NVM) |

### union hseKeyGenerateSrv\_t.sch

INPUT: The selected scheme parameters.

### Data Fields

| Type                                         | Name         | Description                                                    |
|----------------------------------------------|--------------|----------------------------------------------------------------|
| <a href="#">hseNoScheme_t</a>                | symKey       | INPUT: No scheme (parameter) is used for random symmetric key. |
| <a href="#">hseKeyGenRsaScheme_t</a>         | rsaKey       | INPUT: The scheme used to generate a RSA key pair.             |
| <a href="#">hseKeyGenEccScheme_t</a>         | eccKey       | INPUT: The scheme used to generate a ECC key pair.             |
| <a href="#">hseKeyGenTls12RsaPreMaster_t</a> | rsaPreMaster | INPUT: The scheme used to generate the Rsa pre-master secret.  |

## Macro Definition Documentation

### HSE\_KEY\_GEN\_SYM\_RANDOM\_KEY

```
#define HSE_KEY_GEN_SYM_RANDOM_KEY 1U
```

Generate a random symmetric key (e.g AES, HMAC).

### HSE\_KEY\_GEN\_RSA\_KEY\_PAIR

```
#define HSE_KEY_GEN_RSA_KEY_PAIR 2U
```

Generate a RSA key pair.



**HSE\_KEY\_GEN\_ECC\_KEY\_PAIR**

```
#define HSE_KEY_GEN_ECC_KEY_PAIR 3U
```

Generate a ECC key pair.

**HSE\_TLS12\_RSA\_PRE\_MASTER\_SECRET\_GEN**

```
#define HSE_TLS12_RSA_PRE_MASTER_SECRET_GEN 5U
```

Generate the pre-master secret for TLS 1.2 RSA key exchange.

**Typedef Documentation****hseKeyGenScheme\_t**

```
typedef uint8_t hseKeyGenScheme_t
```

HSE Key Generate schemes.

**5.6 HSE Key Derivation Service****Data Structures**

- struct [hseKdfSalt\\_t](#)
- struct [hseKdfExtractStepScheme\\_t](#)
- struct [hseKdfCommonParams\\_t](#)
- struct [hseKdfNxpGenericScheme\\_t](#)
- struct [hseKdfSP800\\_56COneStepScheme\\_t](#)
- struct [hseKdfSP800\\_108Scheme\\_t](#)
- struct [hseKdfSP800\\_56CTwoStepScheme\\_t](#)
- struct [hsePBKDF2Scheme\\_t](#)
- struct [hseHKDF\\_ExpandScheme\\_t](#)
- struct [hseKdfTLS12PrfScheme\\_t](#)
- struct [hseKeyDeriveSrv\\_t](#)
- struct [hseKeyDeriveCopyKeySrv\\_t](#)
- union [hseKdfExtractStepScheme\\_t.prfAlgo](#)
- union [hseKdfCommonParams\\_t.prfAlgo](#)
- union [hseKeyDeriveSrv\\_t.sch](#)

**Macros**

| Type: <a href="#">hseKdfSP800_108Mode_t</a> |       |
|---------------------------------------------|-------|
| Name                                        | Value |
| <a href="#">HSE_KDF_SP800_108_COUNTER</a>   | 1U    |

## Key Management Services

| Type: <a href="#">hseKdfPrf_t</a>    |       |
|--------------------------------------|-------|
| Name                                 | Value |
| <a href="#">HSE_KDF_PRF_HASH</a>     | 1U    |
| <a href="#">HSE_KDF_PRF_HMAC</a>     | 2U    |
| <a href="#">HSE_KDF_PRF_CMAC</a>     | 3U    |
| <a href="#">HSE_KDF_PRF_XCBC_MAC</a> | 4U    |

| Type: <a href="#">hseKdfHashAlgo_t</a> |                                            |
|----------------------------------------|--------------------------------------------|
| Name                                   | Value                                      |
| <a href="#">HSE_KDF_SHA2_224</a>       | <a href="#">HSE_HASH_ALGO_SHA2_224</a>     |
| <a href="#">HSE_KDF_SHA2_256</a>       | <a href="#">HSE_HASH_ALGO_SHA2_256</a>     |
| <a href="#">HSE_KDF_SHA2_384</a>       | <a href="#">HSE_HASH_ALGO_SHA2_384</a>     |
| <a href="#">HSE_KDF_SHA2_512</a>       | <a href="#">HSE_HASH_ALGO_SHA2_512</a>     |
| <a href="#">HSE_KDF_SHA2_512_224</a>   | <a href="#">HSE_HASH_ALGO_SHA2_512_224</a> |
| <a href="#">HSE_KDF_SHA2_512_256</a>   | <a href="#">HSE_HASH_ALGO_SHA2_512_256</a> |

| Type: <a href="#">hseKdfAlgo_t</a>              |       |
|-------------------------------------------------|-------|
| Name                                            | Value |
| <a href="#">HSE_KDF_ALGO_NXP_GENERIC</a>        | 1U    |
| <a href="#">HSE_KDF_ALGO_EXTRACT_STEP</a>       | 2U    |
| <a href="#">HSE_KDF_ALGO_SP800_56C_ONE_STEP</a> | 3U    |
| <a href="#">HSE_KDF_ALGO_SP800_56C_TWO_STEP</a> | 4U    |
| <a href="#">HSE_KDF_ALGO_SP800_108</a>          | 5U    |
| <a href="#">HSE_KDF_ALGO_PBKDF2HMAC</a>         | 6U    |
| <a href="#">HSE_KDF_ALGO_HKDF_EXPAND</a>        | 7U    |
| <a href="#">HSE_KDF_ALGO_ANS_X963</a>           | 8U    |
| <a href="#">HSE_KDF_ALGO_ISO18033_KDF1</a>      | 9U    |
| <a href="#">HSE_KDF_ALGO_ISO18033_KDF2</a>      | 10U   |
| <a href="#">HSE_KDF_ALGO_TLS12PRF</a>           | 11U   |

| Type: <a href="#">hseTlsPskUsage_t</a>         |       |
|------------------------------------------------|-------|
| Name                                           | Value |
| <a href="#">HSE_TLS_PSK_NOT_USED</a>           | 0U    |
| <a href="#">HSE_TLS_KEY_EXCHANGE_PSK</a>       | 1U    |
| <a href="#">HSE_TLS_KEY_EXCHANGE_ECDHE_PSK</a> | 2U    |
| <a href="#">HSE_TLS_KEY_EXCHANGE_RSA_PSK</a>   | 3U    |
| <a href="#">HSE_TLS_KEY_EXCHANGE_DHE_PSK</a>   | 4U    |

|                                                   |
|---------------------------------------------------|
| Type: <a href="#">hseKdfSP800_108CounterLen_t</a> |
|---------------------------------------------------|

| Name                                                  | Value |
|-------------------------------------------------------|-------|
| <a href="#">HSE_KDF_SP800_108_COUNTER_LEN_DEFAULT</a> | 0U    |
| <a href="#">HSE_KDF_SP800_108_COUNTER_LEN_1</a>       | 1U    |
| <a href="#">HSE_KDF_SP800_108_COUNTER_LEN_2</a>       | 2U    |

| Type: <a href="#">hseIkev2Steps_t</a>   |       |
|-----------------------------------------|-------|
| Name                                    | Value |
| <a href="#">HSE_IKEV2_STEP_INIT_SA</a>  | 1U    |
| <a href="#">HSE_IKEV2_STEP_CHILD_SA</a> | 2U    |
| <a href="#">HSE_IKEV2_STEP_REKEY_SA</a> | 3U    |

## Typedefs

- typedef uint8\_t [hseKdfAlgo\\_t](#)
- typedef uint8\_t [hseKdfHashAlgo\\_t](#)
- typedef uint8\_t [hseKdfPrf\\_t](#)
- typedef [hseKdfHashAlgo\\_t](#) [hseHashPrfAlgo\\_t](#)
- typedef [hseKdfHashAlgo\\_t](#) [hseHmacPrfAlgo\\_t](#)
- typedef uint8\_t [hseNoPrfAlgo\\_t](#)
- typedef uint8\_t [hseKdfSP800\\_108Mode\\_t](#)
- typedef uint8\_t [hseKdfSP800\\_108CounterLen\\_t](#)
- typedef uint8\_t [hseIkev2Steps\\_t](#)
- typedef uint8\_t [hseTlsPskUsage\\_t](#)
- typedef [hseKdfCommonParams\\_t](#) [hseKdfANSX963Scheme\\_t](#)
- typedef [hseKdfCommonParams\\_t](#) [hseKdfISO18033\\_KDF1Scheme\\_t](#)
- typedef [hseKdfCommonParams\\_t](#) [hseKdfISO18033\\_KDF2Scheme\\_t](#)

## Data Structure Documentation

### struct [hseKdfSalt\\_t](#)

The KDF salt definition.

The salt is used as the MAC key during the execution of the randomness-extraction step (first step). The salt can be a secret (providing the key handle) or a non-secret (e.g. value computed from nonces exchanged as part of a key-establishment protocol).

#### Data Fields

| Type                           | Name                          | Description                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a> | <a href="#">saltKeyHandle</a> | INPUT: The salt key handle (when the salt is provided as a secret). If ( <a href="#">saltKeyHandle</a> == <a href="#">HSE_INVALID_KEY_HANDLE</a> ), the salt shall be specified by <a href="#">saltLength</a> and <a href="#">pSalt</a> parameters. If the <a href="#">saltKeyHandle</a> is valid, the salt length is the key size in bytes and should match the the input block size. |

## Key Management Services

### Data Fields

| Type     | Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | saltLength | INPUT: Length of the salt in bytes. Used only if <code>saltKeyHandle == HSE_INVALID_KEY_HANDLE</code> . The length of salt are determined by the PRF algorithm: <ul style="list-style-type: none"><li>– For HMAC-hash PRF, the <code>saltLength</code> should be equal with the input block size (e.g 64/128 bytes).<br/>If <code>saltLength</code> is shorter, it will be padded with zeros. The <code>saltLength</code> greater than input block size will be firstly hashed using HASH PRF and then use the resultant byte string.</li><li>– CMAC requires keys that are N bits long (for N = 128, 192, or 256). In this case, the salt should be 16, 24, or 32 bytes, depending upon the AES variant.<br/>Note that the <code>saltLength</code> can also be zero. In this case, the salt is an all-zero byte array whose length is equal to input block size (for hash or CMAC).</li></ul> |
| uint32_t | pSalt      | INPUT: The salt. Used only if <code>saltKeyHandle == HSE_INVALID_KEY_HANDLE</code> .<br>If <code>pSalt</code> is not passed ( <code>pSalt</code> is NULL), default_salt will be used (the default_salt is all-zero byte array of length determined by input block).<br><br>Note<br><br>If the HOST_ADDR is on 64 bits, the address must fall within the 32-bit address range.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### struct hseKdfExtractStepScheme\_t

#### KDF Extraction step.

The extraction step is a Pseudo-Random Function (PRF) that takes as inputs a shared secret (`secretKeyHandle`) and the salt which can be a secret (a key) or non-secret (a generated random number). From these inputs, the PRF generates a pseudo-random key (PRK). The PRK can be used for the Expansion phase. The size of the PRK is equal with the size of the PRF output.

The following PRFs can be performed:

- PRK = HMAC-hash(salt, secret);
- PRK = CMAC(salt, secret);

### Data Fields

| Type                        | Name            | Description                                            |
|-----------------------------|-----------------|--------------------------------------------------------|
| <code>hseKeyHandle_t</code> | secretKeyHandle | INPUT: The shared secret to be used for the operation. |

## Data Fields

| Type                                            | Name            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a>                  | targetKeyHandle | INPUT: The target key handle (where to store the new key). It should point to a <a href="#">HSE_KEY_TYPE_SHARED_SECRET</a> slot. The application can use the generated PRK for the Expand phase (using the same key handle) or it can extract the key(s) (in different slots) using the <a href="#">hseKeyDeriveCopyKeySrv_t</a> service. The size of the PRK is equal with the size of the PRF output (e.g. for hmac-sha256, the key bit length is 256 bits) |
| <a href="#">hseKdfPrf_t</a>                     | kdfPrf          | INPUT: Selected the PRF to be used.<br>Supported options: <a href="#">HSE_KDF_PRF_HMAC</a> , <a href="#">HSE_KDF_PRF_CMAC</a> .                                                                                                                                                                                                                                                                                                                               |
| union <a href="#">hseKdfExtractStepScheme_t</a> | prfAlgo         | INPUT: Selects the algorithm for the PRF.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| uint8_t                                         | reserved[2]     |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <a href="#">hseKdfSalt_t</a>                    | salt            | INPUT: The salt which is used as key. The saltLength should be equal with the input block size (e.g 16/64/128 bytes). See <a href="#">hseKdfSalt_t</a> comments.                                                                                                                                                                                                                                                                                              |

**struct hseKdfCommonParams\_t**

KDF Common parameters.

Common parameters for expansion step used for different KDFs (SP800\_56CTwoStep, HKDF-Expand, prf+ from IKEV2 etc). The expansion inputs are the output from the extractor (pseudo-random key from [hseKdfExtractStepScheme\\_t](#)) and the public context information ([pInfo](#)).

## Data Fields

| Type                                       | Name            | Description                                                                                                                                                                                                                                                                    |
|--------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a>             | srcKeyHandle    | INPUT: The source key to be used for the operation. For the expansion step, the source key handle should be a pseudorandom key (PRK) or a shared secret. (usually, the output from the extraction step; see <a href="#">hseKdfExtractStepScheme_t</a> ).                       |
| <a href="#">hseKeyHandle_t</a>             | targetKeyHandle | INPUT: The target key handle (where to store the new key).It should point to a <a href="#">HSE_KEY_TYPE_SHARED_SECRET</a> slot. The user can extract the key(s) (in different slots) from the derived key material using the <a href="#">hseKeyDeriveCopyKeySrv_t</a> service. |
| uint16_t                                   | keyMatLen       | INPUT: The key material length to be derived (it must be >= 16 bytes and <= slot size).                                                                                                                                                                                        |
| <a href="#">hseKdfPrf_t</a>                | kdfPrf          | INPUT: The PRFs used for KDF.<br>Supported options: <a href="#">HSE_KDF_PRF_HASH</a> , <a href="#">HSE_KDF_PRF_HMAC</a> , <a href="#">HSE_KDF_PRF_CMAC</a> .                                                                                                                   |
| union <a href="#">hseKdfCommonParams_t</a> | prfAlgo         | INPUT: Selects the algorithm for the PRF.                                                                                                                                                                                                                                      |

## Key Management Services

### Data Fields

| Type     | Name       | Description                                          |
|----------|------------|------------------------------------------------------|
| uint32_t | infoLength | INPUT: Length of the pInfo. It must be <= 256 bytes. |
| uint32_t | pInfo      | INPUT: The Info.                                     |

### struct hseKdfNxpGenericScheme\_t

KDF NXP generic scheme.

Used for deriving a cryptographic key from a source key and seed as described below:

```
{
    K[0]= NULL;
    key_mat[0]= NULL;
    iter = key_mat_len/prfOut_size;
    if(0 != (key_mat_len%prfOut_size))
    {
        iter = iter+1;
    }
    for(i = 1; i <= iter;i++)
    {
        step1: K[i] = Prf(srcKey, K[i-1] || seed)
        step2: key_mat[i]= key_mat[i-1] || K[i]
    }
    key_mat = truncate(key_mat_len, key_mat[iter]).
}
```

#### Note

- If the key\_mat\_len >= 32 bytes, the last 8 bytes from the key material can be exported to the HOST.
- For SHA PRF:
  - if srcKeyAfterSeed = FALSE, step1 is K[i] = SHA(srcKey || K[i-1] || seed)
  - if srcKeyAfterSeed = TRUE , step1 is K[i] = SHA(K[i-1] || seed || srcKey)

### Data Fields

| Type                                 | Name            | Description                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKdfCommonParams_t</a> | kdfCommon       | INPUT: KDF common parameters. Only HASH PRF is supported. <ul style="list-style-type: none"><li>– <a href="#">hseKdfCommonParams_t::kdfPrf</a> = HSE_KDF_PRF_HASH</li><li>– <a href="#">hseKdfCommonParams_t::pInfo</a> = Seed.</li><li>– <a href="#">hseKdfCommonParams_t::infoLength</a> = Seed length (must be &lt;= 256 bytes). Zero means the Seed is not used.</li></ul> |
| bool_t                               | srcKeyAfterSeed | INPUT: Concatenate the source key after the seed.                                                                                                                                                                                                                                                                                                                              |
| uint8_t                              | reserved        |                                                                                                                                                                                                                                                                                                                                                                                |
| uint16_t                             | outputLength    | INPUT: Output data length to be exported to the host. It should be <= 8 bytes and can be used only if <a href="#">hseKdfCommonParams_t::keyMatLen</a> >= 32 bytes.                                                                                                                                                                                                             |
| uint32_t                             | pOutput         | OUTPUT: Export outputLength bytes to host (only if the <a href="#">hseKdfCommonParams_t::keyMatLen</a> >= 32 bytes). It can be NULL.                                                                                                                                                                                                                                           |

**struct hseKdfSP800\_56COneStepScheme\_t**

SP800 56C One Step Key derivation.

Perform One step KDF specified in SP800-56C rev1.

Note

Length of the counter is always 32bits.

Data Fields

| Type                                 | Name      | Description                                                                                                                                                                                                                                                         |
|--------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKdfCommonParams_t</a> | kdfCommon | INPUT: KDF common parameters. Only HASH and HMAC are supported.<br>– kdfCommon::kdfPrf = <a href="#">HSE_KDF_PRF_HASH</a> or <a href="#">HSE_KDF_PRF_HMAC</a> .<br>– kdfCommon::pInfo = Fixed Info specified according to SP800_56C OneStep.                        |
| <a href="#">hseKdfSalt_t</a>         | salt      | INPUT: The salt. The salt is used only if HMAC PRF is selected (it's used as key). The saltLength should be equal with the input block size (e.g 64/128 bytes). If saltLength is shorter, it will be padded with zeros; if saltLength is longer, it will be hashed. |

**struct hseKdfSP800\_108Scheme\_t**

SP800 108 Key derivation.

The KDF(Counter mode) as defined by SP800-108.

Note

The key material length ([L]\_2) from SP800 108 is represented on 32 bits. The iteration counter ([i]\_2) can have 8, 16 or 32 bits.

## Key Management Services

### Data Fields

| Type                                        | Name              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKdfCommonParameters_t</a>    | kdfCommon         | INPUT: KDF common parameters. Only HMAC and CMAC are supported. <ul style="list-style-type: none"><li>– .kdfCommon.kdfPrf = <a href="#">HSE_KDF_PRF_HMAC</a> or <a href="#">HSE_KDF_PRF_CMAC</a>.</li><li>– .kdfCommon.pInfo = the context-specific data according to SP800_108: "Label  0x00  Context  [L]_2".</li></ul> Note<br><br>Source key should be a valid symmetric key of length that respects the constraints defined for kdf salt (see <a href="#">hseKdfSalt_t</a> ). |
| <a href="#">hseKdfSP800_108Mode_t</a>       | mode              | INPUT: Selects the SP800_108 mode: Counter (e.g. Feedback, Pipeline not supported)                                                                                                                                                                                                                                                                                                                                                                                                 |
| <a href="#">hseKdfSP800_108CounterLen_t</a> | counterByteLength | INPUT: Selects the length in bytes of the counter ([i]_2). The length of the counter can be 1, 2 or 4 bytes.<br><br>Note<br><br>Any other value will be treated as the default value (4 bytes)                                                                                                                                                                                                                                                                                     |
| uint8_t                                     | reserved[14]      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### struct hseKdfSP800\_56CTwoStepScheme\_t

SP800 56C Two-step Key derivation.

Perform Two step KDF specified in SP800-56C.

SP800\_56C Two Step includes SP800 108 parameters for Expansion Step, and additional the salt for Extraction Step.

Note

- OtherInput define by SP800 56C contains the salt, the key material length (L) and FixedInfo, which are provided as parameters by the service.
- Counter length [r] supported is 32 bits.

### Data Fields

| Type                                    | Name   | Description                                                                                                                                                                                                                                                                               |
|-----------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKdfSP800_108Scheme_t</a> | expand | INPUT: KDF common parameters. Only HMAC and CMAC are supported. <ul style="list-style-type: none"><li>– .expand.kdfCommon.kdfPrf = <a href="#">HSE_KDF_PRF_HMAC</a> or <a href="#">HSE_KDF_PRF_CMAC</a>.</li><li>– .expand.kdfCommon.pInfo = FixedInfo which follows SP800-56C.</li></ul> |
| <a href="#">hseKdfSalt_t</a>            | salt   | INPUT: The salt used for Extraction Step.                                                                                                                                                                                                                                                 |



**struct hsePBKDF2Scheme\_t**

Password Based Key Derivation Function 2.

Used for deriving a cryptographic key from a password

Data Fields

| Type                             | Name            | Description                                                                                                                                                                                                                                                                    |
|----------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a>   | srcKeyHandle    | INPUT: The source key to be used for the operation (shared secret).                                                                                                                                                                                                            |
| <a href="#">hseKeyHandle_t</a>   | targetKeyHandle | INPUT: The target key handle (where to store the new key).It should point to a <a href="#">HSE_KEY_TYPE_SHARED_SECRET</a> slot. The user can extract the key(s) (in different slots) from the derived key material using the <a href="#">hseKeyDeriveCopyKeySrv_t</a> service. |
| uint16_t                         | keyMatLen       | INPUT: The key material length to be derived (it must be <= slot size).                                                                                                                                                                                                        |
| <a href="#">hseHmacPrfAlgo_t</a> | hmacHash        | INPUT: The hash algorithm for HMAC PRF.<br><br>Note<br><br>HSE_M and HSE_B platforms do not support SHA2_384/512 hash algorithms for PBKDF2.                                                                                                                                   |
| uint8_t                          | reserved        |                                                                                                                                                                                                                                                                                |
| uint32_t                         | iterations      | INPUT: The number of iterations to be performed. Should take a value higher than 100.                                                                                                                                                                                          |
| uint32_t                         | saltLength      | INPUT: Length of the salt. It must be < 8192 bytes.                                                                                                                                                                                                                            |
| uint32_t                         | pSalt           | INPUT: A salt; 16 bytes or longer (randomly generated)                                                                                                                                                                                                                         |

**struct hseHKDF\_ExpandScheme\_t**

HKDF-Expand KDF Function.

It is suitable for deriving keys of a fixed size used for other cryptographic operations.

Note

For TLS1.3, the HKDF-Extract function (first step) can be performed using [hseKdfExtractStepScheme\\_t](#).

Data Fields

| Type                                     | Name      | Description                                                                                                                                                                        |
|------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKdfCommonParameters_t</a> | kdfCommon | INPUT: KDF common parameters. Only HMAC is supported.<br>– .kdfCommon.kdfPrf = <a href="#">HSE_KDF_PRF_HMAC</a><br>– .kdfCommon.pInfo = Application specific context. Can be NULL. |

## Key Management Services

### Data Fields

| Type     | Name      | Description                                                                                                                                                                                                                                                                                                                      |
|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pIvOutput | OUTPUT: The TLS1.3 IV output. HSE exports the HKDF expansion output only if the kdfCommon.pInfo starts with the following concatenation: kdfCommon.keyMatLen(2 bytes big-endian)   "tls13 iv" (string of 8 bytes).<br>The length of pIvOutput is the kdfCommon.keyMatLen.<br>In this case kdfCommon.targetKeyHandle is not used. |

### struct hseKdfTLS12PrfScheme\_t

TLS 1.2 PRF as specified by RFC 5246.

The PRF needed in TLS1.2 protocol to derive the master secret, the key block and the verify data.

### Data Fields

| Type     | Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint16_t | labelLength   | INPUT: The label length in bytes (without '\0' termination). Only the following labels are valid in case of TLS 1.2 PRF. <ul style="list-style-type: none"><li>– master secret label - "master secret"</li><li>– extended master secret - "extended master secret" (refer to rfc7627)</li><li>– key expansion label - "key expansion"</li><li>– client finished label - "client finished"</li><li>– server finished label - "server finished"</li></ul> Note <ul style="list-style-type: none"><li>– The above arrays do not contain the string termination character.</li><li>– The above label lengths are the only valid label lengths that must be provided by the Host Application (refer to RFC 5246).</li></ul> |
| uint8_t  | reserved1[2U] |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Data Fields

| Type                             | Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t                         | pLabel        | <p>INPUT: The label of the TLS1.2 PRF operations.</p> <ul style="list-style-type: none"> <li>– If pLabel = "master secret" or "extended master secret", HSE computes the master secret; the <a href="#">hseKdfTLS12PrfScheme_t::keyMatLength</a> must be 48 bytes.</li> <li>– If pLabel = "key expansion", HSE computes the key_block; the <a href="#">hseKdfTLS12PrfScheme_t::keyMatLength</a> must be &gt;= 32 bytes. HSE also outputs the client and server IVs (see <a href="#">pOutput</a>).</li> <li>– if pLabel = "client finished" or "server finished", HSE computes the verify_data (see <a href="#">pOutput</a>).</li> </ul> <p>Note</p> <ul style="list-style-type: none"> <li>– The pre-master shared secret ( <a href="#">HSE_KEY_TYPE_SHARED_SECRET</a> key slot) is deleted after master secret computation (see <a href="#">rfc5246</a>).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <a href="#">hseTlsPskUsage_t</a> | tlsPskUsage   | <p>INPUT: Selects TLS-PSK algorithm usage. Used only for master secret computation (label = "master secret"). Ignored for other labels.</p> <p>Note</p> <ul style="list-style-type: none"> <li>– <a href="#">HSE_TLS_PSK_NOT_USED</a> - pre-shared key not used</li> <li>– <a href="#">HSE_TLS_KEY_EXCHANGE_PSK</a> - pre-master secret is computed as: If the PSK is N octets long, concatenate a uint16 with the value N, N zero octets, a second uint16 with the value N, and the PSK itself (refer to <a href="#">rfc4279</a>)</li> <li>– <a href="#">HSE_TLS_KEY_EXCHANGE_ECDHE_PSK</a> - pre-master secret is computed as: Let Z be the octet string of ECDH shared secret. The pre-master is the concatenation of a uint16 containing the length of Z (in octets), Z itself, a uint16 containing the length of the PSK (in octets), and the PSK itself (refer to <a href="#">rfc5489</a>)</li> <li>– <a href="#">HSE_TLS_KEY_EXCHANGE_RSA_PSK</a> - pre-master secret is computed as: concatenate a uint16 with the value 48, the 2-byte version number and the 46-byte random value, a uint16 containing the length of the PSK (in octets), and the PSK itself (the pre-master secret is thus 52 octets longer than the PSK); refer to <a href="#">rfc4279</a>.</li> <li>– <a href="#">HSE_TLS_KEY_EXCHANGE_DHE_PSK</a> - let Z be the value produced by classic DH computation. The pre-master secret is computed: concatenate a uint16 containing the length of Z (in octets), Z itself, a uint16 containing the length of the PSK (in octets), and the PSK itself.</li> </ul> |
| uint8_t                          | reserved2[3U] |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Key Management Services

### Data Fields

| Type                             | Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a>   | pskKeyHandle  | INPUT: Pre-shared key handle. It can be any symmetric NVM key that has the <a href="#">HSE_KF_USAGE_DERIVE</a> flag set. Used only for master secret computation and <a href="#">tlsPskUsage</a> != <a href="#">HSE_TLS_PSK_NOT_USED</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <a href="#">hseKeyHandle_t</a>   | srcKeyHandle  | <p>INPUT: The source key handle (it must point to a <a href="#">HSE_KEY_TYPE_SHARED_SECRET</a> slot).</p> <ul style="list-style-type: none"> <li>For label = "master secret": <ul style="list-style-type: none"> <li>if <a href="#">tlsPskUsage</a> = <a href="#">HSE_TLS_PSK_NOT_USED</a>, it must be the pre-master secret (e.g DH shared secret).</li> <li>if <a href="#">tlsPskUsage</a> = <a href="#">HSE_TLS_KEY_EXCHANGE_PSK</a>, it is ignored (key handle is provided by <a href="#">pskKeyHandle</a>).</li> <li>if <a href="#">tlsPskUsage</a> = <a href="#">HSE_TLS_KEY_EXCHANGE_ECDHE_PSK</a>, it is the DH shared secret.</li> <li>if <a href="#">tlsPskUsage</a> = <a href="#">HSE_TLS_KEY_EXCHANGE_RSA_PSK</a>, the shared secret slot contains: ProtocolVersion (2bytes) concatenated with 46 byte random number.</li> </ul> </li> <li>For key_block or verify_data, it must be the master secret.</li> </ul> |
| <a href="#">hseHmacPrfAlgo_t</a> | hmacHash      | INPUT: The hash algorithm for HMAC PRF.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| uint8_t                          | reserved3[1U] |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| uint16_t                         | seedLength    | INPUT: The seed length. It must be <= 256 bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| uint32_t                         | pSeed         | <p>INPUT: The seed for TLS 1.2 PRF. In TLS, this is usually a combination of user and random data.</p> <p>This is the concatenation of Server and Client Hello random data.</p> <ul style="list-style-type: none"> <li>For master secret, it is concatenation of Server Random Data    Client Random Data.</li> <li>For extended master secret, it is the session_hash (refer to rfc7627).</li> <li>For Key Expansion, it is concatenation of Client Random Data    Server Random Data.</li> <li>For <a href="#">pLabel</a> = "client finished" or "server finished", pSeed must be Hash(handshake_messages). Refer to RFC 5246 for more details.</li> </ul>                                                                                                                                                                                                                                                                  |

## Data Fields

| Type                           | Name            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a> | targetKeyHandle | <p>INPUT: The target key handle (where to store the new key). It shall point to a <a href="#">HSE_KEY_TYPE_SHARED_SECRET</a> slot (this means <a href="#">HSE_KF_USAGE_DERIVE</a> flag is set by default).</p> <p>It can be:</p> <ul style="list-style-type: none"> <li>– the derived master secret</li> <li>– the derived key_block. The user can extract the key(s) using the <a href="#">hseKeyDeriveCopyKeySrv_t</a> service.</li> </ul> <p>The key_block is partitioned as follows:</p> <ul style="list-style-type: none"> <li>· client_write_MAC_key[]</li> <li>· server_write_MAC_key[]</li> <li>· client_write_key[]</li> <li>· server_write_key[]</li> <li>· client_write_IV[]; exported in pOutput below if <a href="#">pLabel</a> = "key expansion"</li> <li>· server_write_IV[]; exported in pOutput below if <a href="#">pLabel</a> = "key expansion"</li> <li>– not used for verify_data (<a href="#">pLabel</a> = "client finished" or <a href="#">pLabel</a> = "server finished")</li> </ul> |
| uint16_t                       | keyMatLength    | <p>INPUT: The key material length (in bytes).</p> <ul style="list-style-type: none"> <li>– If <a href="#">pLabel</a> = "master secret" or "extended master secret", the <a href="#">keyMatLength</a> must be 48 bytes.</li> <li>– If <a href="#">pLabel</a> = "key expansion" (key_block), the <a href="#">keyMatLength</a> must be <math>\geq 32</math> bytes. It must be the total length for Client and Server keys without the IVs (only the MAC and encryption keys).</li> <li>– Not used for verify_data (if the <a href="#">pLabel</a> = "client finished" or <a href="#">pLabel</a> = "server finished")</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                  |
| uint16_t                       | outputLength    | <p>INPUT: The length for output data (<a href="#">pOutput</a>) which can be:</p> <ul style="list-style-type: none"> <li>– For <a href="#">pLabel</a> = "key expansion", the total length for client and server Initialization Vectors from key_block. Can be 0. If it is provided, it must be <math>\leq 32</math> bytes (2*block size).</li> <li>– For <a href="#">pLabel</a> = "client finished" or "server finished", the verify_data length. Must be 12 bytes.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| uint32_t                       | pOutput         | <p>OUTPUT: The output data which can be:</p> <ul style="list-style-type: none"> <li>– For <a href="#">pLabel</a> = "key expansion", concatenated client and server IVs of totalIvLength (client_write_IV[]    server_write_IV[]). Can be NULL.</li> <li>– For <a href="#">pLabel</a> = "client finished" or "server finished", verify_data sent in the Finished message.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Key Management Services

### struct hseKeyDeriveSrv\_t

HSE Key Derive service.

The key derive service (KDF) derives one or more secret keys from a secret value.

Note

- The key material can be derived only in [HSE\\_KEY\\_TYPE\\_SHARED\\_SECRET](#) slots (specified as targetKeyHandle), which can not be exported outside HSE.

#### Data Fields

| Type                                    | Name        | Description                                   |
|-----------------------------------------|-------------|-----------------------------------------------|
| <a href="#">hseKdfAlgo_t</a>            | kdfAlgo     | INPUT: The key derivation algorithm.          |
| uint8_t                                 | reserved[3] |                                               |
| union <a href="#">hseKeyDeriveSrv_t</a> | sch         | INPUT: The selected key derivation algorithm. |

### struct hseKeyDeriveCopyKeySrv\_t

HSE Key Derive - Copy Key service.

This service can be used to extract keys (or a key) from the derived key material placed in a temporary shared secret slot ( [HSE\\_KEY\\_TYPE\\_SHARED\\_SECRET](#) ).

The key(s) can be copied in NVM/RAM slots as follow:

1. Restrictions for SuperUser rights:
  - keys can be copied in NVM key store from the derived key material only in empty slots (an erase shall be performed in advance if needed).
  - keys can be copied in RAM key store from the derived key material (RAM keys can be overwritten).
2. Restrictions for User rights:
  - keys can NOT be copied in NVM key store from the derived key material.
  - keys can be copied in RAM key store from the derived key material (RAM keys can be overwritten).

#### Data Fields

| Type                           | Name        | Description                                                                                                                                    |
|--------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a> | keyHandle   | INPUT: The key handle to be used to extract a key value. The key handle should point to a <a href="#">HSE_KEY_TYPE_SHARED_SECRET</a> key type. |
| uint16_t                       | startOffset | INPUT: Start offset from where to copy the key. The offset can be zero or a multiple of 4 bytes.                                               |

## Data Fields

| Type                           | Name            | Description                                                                                                                                                                                                                                   |
|--------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint8_t                        | reserved[2]     |                                                                                                                                                                                                                                               |
| <a href="#">hseKeyHandle_t</a> | targetKeyHandle | INPUT: The target key handle (where to store the new key).                                                                                                                                                                                    |
| <a href="#">hseKeyInfo_t</a>   | keyInfo         | INPUT: Specifies usage flags, restriction access, key bit length etc for the key. Note that the length of the copied key is considered to be <a href="#">hseKeyInfo_t::keyBitLen</a> . The minimum key length that can be copied is 16 bytes. |

**union hseKdfExtractStepScheme\_t.prfAlgo**

INPUT: Selects the algorithm for the PRF.

## Data Fields

| Type                             | Name     | Description                       |
|----------------------------------|----------|-----------------------------------|
| <a href="#">hseHmacPrfAlgo_t</a> | hmacHash | The hash algorithm used for HMAC. |
| <a href="#">hseNoPrfAlgo_t</a>   | cmac     | Dummy byte.                       |

**union hseKdfCommonParams\_t.prfAlgo**

INPUT: Selects the algorithm for the PRF.

## Data Fields

| Type                             | Name     | Description                       |
|----------------------------------|----------|-----------------------------------|
| <a href="#">hseHashPrfAlgo_t</a> | hash     | The KDF hash algorithm .          |
| <a href="#">hseHmacPrfAlgo_t</a> | hmacHash | The hash algorithm used for HMAC. |
| <a href="#">hseNoPrfAlgo_t</a>   | cmac     | Dummy byte.                       |

**union hseKeyDeriveSrv\_t.sch**

INPUT: The selected key derivation algorithm.

## Data Fields

| Type                                     | Name       | Description                    |
|------------------------------------------|------------|--------------------------------|
| <a href="#">hseKdfNxpGenericScheme_t</a> | nxpGeneric | INPUT: NXP generic KDF scheme. |

## Key Management Services

### Data Fields

| Type                                           | Name             | Description                                |
|------------------------------------------------|------------------|--------------------------------------------|
| <a href="#">hseKdfExtractStepScheme_t</a>      | extractStep      | Generic Extraction Step for Two-step KDFs. |
| <a href="#">hseKdfSP800_56COneStepScheme_t</a> | SP800_56COneStep | INPUT: One-Step SP800_56C KDF scheme.      |
| <a href="#">hseKdfSP800_56CTwoStepScheme_t</a> | SP800_56CTwoStep | INPUT: Two-Step SP800_56C KDF scheme.      |
| <a href="#">hseKdfSP800_108Scheme_t</a>        | SP800_108        | INPUT: SP800 108 KDF scheme.               |
| <a href="#">hsePBKDF2Scheme_t</a>              | PBKDF2           | INPUT: PBKDF2 scheme.                      |
| <a href="#">hseHKDF_ExpandScheme_t</a>         | HKDF_Expand      | INPUT: HKDF-Expand scheme.                 |
| <a href="#">hseKdfANSX963Scheme_t</a>          | ANS_X963         | INPUT: ANS_X963 KDF scheme.                |
| <a href="#">hseKdfISO18033_KDF1Scheme_t</a>    | ISO18033_KDF1    | INPUT: ISO18033 KDF1 scheme.               |
| <a href="#">hseKdfISO18033_KDF2Scheme_t</a>    | ISO18033_KDF2    | INPUT: ISO18033 KDF2 scheme.               |
| <a href="#">hseKdfTLS12PrfScheme_t</a>         | TLS12Prf         | INPUT: TLS 1.2 PRF.                        |

## Macro Definition Documentation

### HSE\_KDF\_ALGO\_NXP\_GENERIC

```
#define HSE_KDF_ALGO_NXP_GENERIC ((hseKdfAlgo_t)1U)
```

NXP Generic KDF.

### HSE\_KDF\_ALGO\_EXTRACT\_STEP

```
#define HSE_KDF_ALGO_EXTRACT_STEP ((hseKdfAlgo_t)2U)
```

Generic Extraction Step for Two-step KDFs.

### HSE\_KDF\_ALGO\_SP800\_56C\_ONE\_STEP

```
#define HSE_KDF_ALGO_SP800_56C_ONE_STEP ((hseKdfAlgo_t)3U)
```



One-step KDF as defined by SP800-56C rev1.

### **HSE\_KDF\_ALGO\_SP800\_56C\_TWO\_STEP**

```
#define HSE_KDF_ALGO_SP800_56C_TWO_STEP ((hseKdfAlgo_t) 4U)
```

Two-step KDF as defined by SP800-56C rev1.

### **HSE\_KDF\_ALGO\_SP800\_108**

```
#define HSE_KDF_ALGO_SP800_108 ((hseKdfAlgo_t) 5U)
```

KDF(Counter, Feedback, Pipeline) as defined by SP800-108.

### **HSE\_KDF\_ALGO\_PBKDF2HMAC**

```
#define HSE_KDF_ALGO_PBKDF2HMAC ((hseKdfAlgo_t) 6U)
```

PBKDF2HMAC as defined by PKCS#5 v2.1 and RFC-8018.

### **HSE\_KDF\_ALGO\_HKDF\_EXPAND**

```
#define HSE_KDF_ALGO_HKDF_EXPAND ((hseKdfAlgo_t) 7U)
```

HKDF Expand KDFs as defined by RFC-5869.

### **HSE\_KDF\_ALGO\_ANS\_X963**

```
#define HSE_KDF_ALGO_ANS_X963 ((hseKdfAlgo_t) 8U)
```

KDF as defined by ANS X9.63.

### **HSE\_KDF\_ALGO\_ISO18033\_KDF1**

```
#define HSE_KDF_ALGO_ISO18033_KDF1 ((hseKdfAlgo_t) 9U)
```

KDF1 as defined by ISO18033.

### HSE\_KDF\_ALGO\_ISO18033\_KDF2

```
#define HSE_KDF_ALGO_ISO18033_KDF2 ((hseKdfAlgo_t)10U)
```

KDF2 as defined by ISO18033.

### HSE\_KDF\_ALGO\_TLS12PRF

```
#define HSE_KDF_ALGO_TLS12PRF ((hseKdfAlgo_t)11U)
```

TLS 1.2 PRF as defined by RFC-5246.

### HSE\_KDF\_SHA2\_224

```
#define HSE_KDF_SHA2_224 ((hseKdfHashAlgo_t) HSE_HASH_ALGO_SHA2_224)
```

### HSE\_KDF\_SHA2\_256

```
#define HSE_KDF_SHA2_256 ((hseKdfHashAlgo_t) HSE_HASH_ALGO_SHA2_256)
```

### HSE\_KDF\_SHA2\_384

```
#define HSE_KDF_SHA2_384 ((hseKdfHashAlgo_t) HSE_HASH_ALGO_SHA2_384)
```

### HSE\_KDF\_SHA2\_512

```
#define HSE_KDF_SHA2_512 ((hseKdfHashAlgo_t) HSE_HASH_ALGO_SHA2_512)
```

### HSE\_KDF\_SHA2\_512\_224

```
#define HSE_KDF_SHA2_512_224 ((hseKdfHashAlgo_t) HSE_HASH_ALGO_SHA2_512_224)
```

**HSE\_KDF\_SHA2\_512\_256**

```
#define HSE_KDF_SHA2_512_256 ((hseKdfHashAlgo_t) HSE_HASH_ALGO_SHA2_512_256)
```

**HSE\_KDF\_PRF\_HASH**

```
#define HSE_KDF_PRF_HASH ((hseKdfPrf_t) 1U)
```

SHA2 families.

**HSE\_KDF\_PRF\_HMAC**

```
#define HSE_KDF_PRF_HMAC ((hseKdfPrf_t) 2U)
```

HMAC-SHA2 families.

**HSE\_KDF\_PRF\_CMAC**

```
#define HSE_KDF_PRF_CMAC ((hseKdfPrf_t) 3U)
```

CMAC.

**HSE\_KDF\_PRF\_XCBC\_MAC**

```
#define HSE_KDF_PRF_XCBC_MAC ((hseKdfPrf_t) 4U)
```

XCBC\_MAC (used only for IKEV2 KDF).

**HSE\_KDF\_SP800\_108\_COUNTER**

```
#define HSE_KDF_SP800_108_COUNTER ((hseKdfSP800_108Mode_t) 1U)
```

SP800 108 Counter step.

## Key Management Services

### HSE\_KDF\_SP800\_108\_COUNTER\_LEN\_DEFAULT

```
#define HSE_KDF_SP800_108_COUNTER_LEN_DEFAULT ((hseKdfSP800_108CounterLen_t) 0U)
```

SP800 108 default counter length (4 bytes)

### HSE\_KDF\_SP800\_108\_COUNTER\_LEN\_1

```
#define HSE_KDF_SP800_108_COUNTER_LEN_1 ((hseKdfSP800_108CounterLen_t) 1U)
```

SP800 108 1 byte counter length.

### HSE\_KDF\_SP800\_108\_COUNTER\_LEN\_2

```
#define HSE_KDF_SP800_108_COUNTER_LEN_2 ((hseKdfSP800_108CounterLen_t) 2U)
```

SP800 108 2 bytes counter length.

### HSE\_IKEV2\_STEP\_INIT\_SA

```
#define HSE_IKEV2_STEP_INIT_SA ((hseIkev2Steps_t) 1U)
```

IKE\_SA\_INIT step - Initial Keying Material for the IKE SA.

### HSE\_IKEV2\_STEP\_CHILD\_SA

```
#define HSE_IKEV2_STEP_CHILD_SA ((hseIkev2Steps_t) 2U)
```

CHILD\_SA step - Generating Keying Material for Child SAs.

### HSE\_IKEV2\_STEP\_REKEY\_SA

```
#define HSE_IKEV2_STEP_REKEY_SA ((hseIkev2Steps_t) 3U)
```

REKEY step - Rekeying IKE SAs Using a CREATE\_CHILD\_SA Exchange.

**HSE\_TLS\_PSK\_NOT\_USED**

```
#define HSE_TLS_PSK_NOT_USED ((hseTlsPskUsage_t) 0U)
```

TLS PSK is not used.

**HSE\_TLS\_KEY\_EXCHANGE\_PSK**

```
#define HSE_TLS_KEY_EXCHANGE_PSK ((hseTlsPskUsage_t) 1U)
```

Key Exchange PSK (refer to rfc4279)

**HSE\_TLS\_KEY\_EXCHANGE\_ECDHE\_PSK**

```
#define HSE_TLS_KEY_EXCHANGE_ECDHE_PSK ((hseTlsPskUsage_t) 2U)
```

Key Exchange ECDHE\_PSK (refer to rfc5489)

**HSE\_TLS\_KEY\_EXCHANGE\_RSA\_PSK**

```
#define HSE_TLS_KEY_EXCHANGE_RSA_PSK ((hseTlsPskUsage_t) 3U)
```

Key Exchange RSA\_PSK (refer to rfc4279)

**HSE\_TLS\_KEY\_EXCHANGE\_DHE\_PSK**

```
#define HSE_TLS_KEY_EXCHANGE_DHE_PSK ((hseTlsPskUsage_t) 4U)
```

Key Exchange DHE\_PSK (refer to rfc4279)

**Typedef Documentation****hseKdfAlgo\_t**

```
typedef uint8_t hseKdfAlgo_t
```

HSE Key derivation algorithms.

## Key Management Services

### **hseKdfHashAlgo\_t**

```
typedef uint8_t hseKdfHashAlgo_t
```

Hash algorithm available for KDF.

### **hseKdfPrf\_t**

```
typedef uint8_t hseKdfPrf_t
```

HSE KDF "Pseudo-Random Function" (PRF).

### **hseHashPrfAlgo\_t**

```
typedef hseKdfHashAlgo_t hseHashPrfAlgo_t
```

HSE PRF algorithm.

Algorithm for hash PRF (e.g SHA256)

### **hseHmacPrfAlgo\_t**

```
typedef hseKdfHashAlgo_t hseHmacPrfAlgo_t
```

Algorithm for hmac PRF (e.g SHA256)

### **hseNoPrfAlgo\_t**

```
typedef uint8_t hseNoPrfAlgo_t
```

No PRF algorithm.

### **hseKdfSP800\_108Mode\_t**

```
typedef uint8_t hseKdfSP800_108Mode_t
```

SP800-108 KDF modes (only Counter mode supported).

**hseKdfSP800\_108CounterLen\_t**

```
typedef uint8_t hseKdfSP800_108CounterLen_t
```

SP800-108 KDF counter length (only 1, 2 and 4 bytes supported).

**hseIkev2Steps\_t**

```
typedef uint8_t hseIkev2Steps_t
```

HSE IKEv2 exchange of messages steps.

**hseTlsPskUsage\_t**

```
typedef uint8_t hseTlsPskUsage_t
```

TLS PSK usage.

**hseKdfANSX963Scheme\_t**

```
typedef hseKdfCommonParams_t hseKdfANSX963Scheme_t
```

ANS X9.63 KDF as specified by SEC1-v2.

One-step KDF performed in the context of an ANS X9.63 key agreement scheme.

ANS X9.63 KDF supports:

- .kdfPrf = [HSE\\_KDF\\_PRF\\_HASH](#) (ANS X9.63 supports only hash PRF).
- .pInfo points to SharedInfo (optional, as defined by ANS X9.63).

**hseKdfISO18033\_KDF1Scheme\_t**

```
typedef hseKdfCommonParams_t hseKdfISO18033_KDF1Scheme_t
```

KDF1 as specified by ISO18033.

One-step KDF performed as specified by ISO18033.

ISO18033 KDF1 supports:

- .kdfPrf = [HSE\\_KDF\\_PRF\\_HASH](#) (ISO18033 supports only hash PRF).
- .pInfo = NULL.
- .infoLength = 0UL

### **hseKdfISO18033\_KDF2Scheme\_t**

```
typedef hseKdfCommonParams_t hseKdfISO18033_KDF2Scheme_t
```

KDF2 as specified by ISO18033.

One-step KDF performed as specified by ISO18033.

ISO18033 KDF2 supports:

- .kdfPrf = [HSE\\_KDF\\_PRF\\_HASH](#) (ISO18033 supports only hash PRF).
- .pInfo = NULL.
- .infoLength = 0UL





## 6 Boot and Memory Verification Services

### 6.1 HSE Core Reset And Secure Memory Region (SMR) Services

#### Data Structures

- struct [hseSmrDecrypt\\_t](#)
- struct [hseSmrEntry\\_t](#)
- struct [hseCrEntry\\_t](#)
- struct [hseSmrEntryInstallSrv\\_t](#)
- struct [hseSmrVerifySrv\\_t](#)
- struct [hseSmrEntryEraseSrv\\_t](#)
- struct [hseCrEntryInstallSrv\\_t](#)
- struct [hseCrOnDemandBootSrv\\_t](#)
- struct [hseCrEntryEraseSrv\\_t](#)
- struct [hseSmrEntryInstallSrv\\_t.cipher](#)

#### Macros

| Type: (implicit C type)                             |       |
|-----------------------------------------------------|-------|
| Name                                                | Value |
| <a href="#">HSE_SMR_DECRYPT_KEY_HANDLE_NOT_USED</a> | 0UL   |
| <a href="#">HSE_SMR_VERSION_NOT_USED</a>            | 0UL   |

| Type: <a href="#">hseCrStartOption_t</a> |         |
|------------------------------------------|---------|
| Name                                     | Value   |
| <a href="#">HSE_CR_AUTO_START</a>        | 0x35A5U |
| <a href="#">HSE_CR_ON_DEMAND</a>         | 0x5567U |

| Type: <a href="#">hseSmrVerificationOptions_t</a>   |                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                                                | Value                                                                                                                                                                                                                                                                                   |
| <a href="#">HSE_SMR_VERIFICATION_OPTION_NONE</a>    | 0UL /** @brief Default verification of the SMR at run-time. *                                                                                                                                                                                                                           |
| <a href="#">HSE_SMR_VERIFICATION_OPTION_NO_LOAD</a> | (3UL << 0U) /** @brief SMR is verified from the external flash (using pSmrSrc address) even if pSmrDest is specified or if already loaded. Can be used only if SMR is in a memory mapped external flash (e.g. QSPI/LPDDR4 and not SD/eMMC). Additionally the SMR cannot be encrypted. * |
| <a href="#">HSE_SMR_VERIFICATION_OPTION_RELOAD</a>  | (3UL << 2U) /** @brief SMR is loaded from the external flash and verified even if it is already loaded. Can be used only if SMR is in a memory mapped external flash (e.g. QSPI/LPDDR4 and not SD/eMMC). *                                                                              |

## Boot and Memory Verification Services

|                                                         |                                                                                                                                                           |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">HSE_SMR_VERIFICATION_OPTION_PASSIVE_MEM</a> | 3UL << 4U /** @brief Only for <a href="#">HSE_B</a> with A/B Swap Configuration. Verifies the SMR from the passive block, applying address translation. * |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|

| Type: <a href="#">hseSmrConfig_t</a>          |          |
|-----------------------------------------------|----------|
| Name                                          | Value    |
| <a href="#">HSE_SMR_CFG_FLAG_QSPI_FLASH</a>   | 0x0U     |
| <a href="#">HSE_SMR_CFG_FLAG_SD_FLASH</a>     | 0x2U     |
| <a href="#">HSE_SMR_CFG_FLAG_MMC_FLASH</a>    | 0x3U     |
| <a href="#">HSE_SMR_CFG_FLAG_INSTALL_AUTH</a> | 1U << 2U |
| <a href="#">HSE_SMR_CFG_FLAG_AUTH_AAD</a>     | 1U << 3U |

| Type: <a href="#">hseCrSanction_t</a>              |         |
|----------------------------------------------------|---------|
| Name                                               | Value   |
| <a href="#">HSE_CR_SANCTION_DIS_INDIV_KEYS</a>     | 0x7433U |
| <a href="#">HSE_CR_SANCTION_KEEP_CORE_IN_RESET</a> | 0x7455U |
| <a href="#">HSE_CR_SANCTION_RESET_SOC</a>          | 0x8B17U |
| <a href="#">HSE_CR_SANCTION_DIS_ALL_KEYS</a>       | 0x8B1EU |

## Typedefs

- typedef uint16\_t [hseCrSanction\\_t](#)
- typedef uint16\_t [hseCrStartOption\\_t](#)
- typedef uint8\_t [hseSmrConfig\\_t](#)
- typedef uint16\_t [hseSmrVerificationOptions\\_t](#)

## Data Structure Documentation

### struct hseSmrDecrypt\_t

Defines the parameters to decrypt an encrypted SMR.

The parameters below are used in the SMR entry only with an encrypted SMR.

#### Note

The following algorithms can be used:

- If pGmacTag == NULL, the SMR must be encrypted using AES-CTR
- If pGmacTag != NULL, the SMR must be encrypted using AEAD-GCM with AAD = NULL (pGmacTag shall point to the GMAC Tag).

## Data Fields

| Type                           | Name             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyHandle_t</a> | decryptKeyHandle | <p>The key handle referencing the decryption key.</p> <ul style="list-style-type: none"> <li>– If <a href="#">decryptKeyHandle</a> == <a href="#">HSE_SMR_DECRYPT_KEY_HANDLE_NOT_USED</a>, the SMR is not encrypted; all the fields below are ignored.</li> <li>– If <a href="#">decryptKeyHandle</a> != <a href="#">HSE_SMR_DECRYPT_KEY_HANDLE_NOT_USED</a>, the <a href="#">decryptKeyHandle</a> specifies the key used to decrypt the SMR.</li> </ul> <p>Note</p> <ul style="list-style-type: none"> <li>· The used algorithm is always AEAD-GCM, where GMAC and AAD are optional.</li> <li>· If the GMAC tag is provided (is not NULL), the same key is also used to verify the tag.</li> </ul> |
| uint32_t                       | pGmacTag         | <p>The Tag used for GCM. If it set NULL, AES-CTR (instead of GCM) is used for decryption.</p> <ul style="list-style-type: none"> <li>– If <a href="#">pGmacTag</a> == NULL, an internal hash is computed at installation over the encrypted SMR. This internal hash is used at verification phase.</li> <li>– If <a href="#">pGmacTag</a> != NULL, the external stored GMAC tag (in flash) is used to verify the encrypted SMR. The length considered in this case is 16 bytes.</li> </ul>                                                                                                                                                                                                          |
| uint8_t                        | aadLength        | <p>Optional - the length in bytes of the Authenticated Additional Data (AAD).</p> <ul style="list-style-type: none"> <li>– If not used, the length can be zero;</li> <li>– If used, the length can be either 64 or 128 bytes.</li> <li>– If used, <a href="#">pGmacTag</a> must also be provided.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                        |
| uint8_t                        | reserved[3U]     | Reserved - alignment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| uint32_t                       | pAAD             | <p>Optional - the AAD used for AEAD.</p> <ul style="list-style-type: none"> <li>– Ignored if aadLength is zero;</li> <li>– If provided, the AAD is NOT stored by HSE internally; pAAD address must point to an external flash location that HSE will use during verification.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                            |

## struct hseSmrEntry\_t

Define the parameters of a Secure Memory Region (SMR) entry in a SMR table.

## Boot and Memory Verification Services

The SMR entry is installed and verified in two phases:

1. "Installation Phase" (using [hseSmrEntryInstallSrv\\_t](#) service).
  - The parameters related to SMR authentication and encryption, namely [authScheme](#), [authKeyHandle](#) and if the SMR is encrypted, [hseSmrDecrypt\\_t::decryptKeyHandle](#) and [hseSmrDecrypt\\_t::pGmacTag](#) will be used by HSE at installation time from the [hseSmrEntry\\_t](#) structure referenced in the [hseSmrEntryInstallSrv\\_t::pSmrEntry](#).
  - This phase happens at run-time and as a consequence any data provided to HSE must be memory-mapped (QSPI/LPDDR4/RAM). In case an SMR lying in SD/eMMC is installed, a copy of the data that is not stored by the HSE internally must be done available in RAM (e.g. SMR source, signature, AAD, GMAC tag, etc.). At installation time HSE will use the matching pointer fields from the [hseSmrEntryInstallSrv\\_t](#) structure to access the data.
2. "Verification Phase" that can be configured to be performed in two modes:
  - Verify with the Original/Installation Authentication TAG over the plaintext ( [HSE\\_SMR\\_CFG\\_FLAG\\_INSTALL\\_AUTH](#) flag is set); the [pInstAuthTag](#) parameter must be provided and must point to original signature.
  - Verify using an internal computed hash ( [HSE\\_SMR\\_CFG\\_FLAG\\_INSTALL\\_AUTH](#) flag is cleared); [pInstAuthTag](#) is not used in this case.
  - In the same manner, if the SMR is encrypted, HSE can use the provided [hseSmrDecrypt\\_t::pGmacTag](#) (original) or an internally computed hash to verify the encrypted SMR before decryption.

### Data Fields

| Type                           | Name          | Description                                                                                                                                                                                                                                                              |
|--------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t                       | pSmrSrc       | Source address where the SMR needs to be loaded from. This address must be absolute address.                                                                                                                                                                             |
| uint32_t                       | smrSize       | The size in bytes of the SMR to be loaded/verified.                                                                                                                                                                                                                      |
| uint32_t                       | pSmrDest      | Destination address of SMR (where to copy the SMR after authentication).<br><br>Note <ul style="list-style-type: none"><li>– For HSE_B, if this parameter is specified (i.e. pSmrDest != NULL), pSmrDest and (pSmrDest + smrSize) must be aligned to 16 bytes.</li></ul> |
| <a href="#">hseSmrConfig_t</a> | configFlags   | Configuration flags of SMR entry (see <a href="#">hseSmrConfig_t</a> ).                                                                                                                                                                                                  |
| uint8_t                        | reserved0[3U] | Reserved for alignment.                                                                                                                                                                                                                                                  |

## Data Fields

| Type                            | Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t                        | checkPeriod   | <p>If <a href="#">checkPeriod</a> != 0, HSE verify the SMR entry periodically (in background). Specifies the verification period in x100 milliseconds when HSE is running at maximum frequency. Otherwise, the period is multiplied by the factor max_freq/actual_freq (e.g. 100ms at 400MHz, 200ms at 200MHz, etc).</p> <p>Note</p> <ul style="list-style-type: none"> <li>– The value 0xFFFFFFFFFUL invalid; the checkPeriod max value must be [MAX_UNSIGNED32_INT - 1].</li> <li>– If the checkPeriod is non zero, the <a href="#">pSmrDest</a> must be non zero and the <a href="#">configFlags</a> must be zero.</li> <li>– The SMR periodic verification will start on next boot after PRE and POST boot verification.</li> <li>– If the periodic SMR verification is used, the HSE firmware always uses the internal hash for verification.</li> </ul>                                                              |
| <a href="#">hseKeyHandle_t</a>  | authKeyHandle | <p>The key handle used to check the authenticity of the plaintext SMR.</p> <p>Note</p> <ul style="list-style-type: none"> <li>– If the <a href="#">HSE_SMR_CFG_FLAG_INSTALL_AUTH</a> flag is cleared, the authKeyHandle is used only in the Installation Phase.</li> <li>– The key flags must be configured as follow: <a href="#">HSE_KF_USAGE_VERIFY</a> must be set, <a href="#">HSE_KF_USAGE_SIGN</a> flag must NOT be set.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <a href="#">hseAuthScheme_t</a> | authScheme    | <p>The authentication scheme used to verify the SMR either during the Installation Phase or Verification phase.</p> <ul style="list-style-type: none"> <li>– If the <a href="#">HSE_SMR_CFG_FLAG_INSTALL_AUTH</a> flag is set (see <a href="#">hseSmrConfig_t</a>), the same authentication scheme (installation TAG) can be used to verify the authenticity of SMR during verification phase too;</li> <li>– Otherwise an internal authentication scheme is used.</li> </ul> <p>Note</p> <ul style="list-style-type: none"> <li>– The authKeyHandle must match the authentication scheme (e.g. a RSA key must be used for RSA signature).</li> <li>– Pure EDDSA scheme (eddsa.bHashEddsa != TRUE) is not supported for streaming installation.</li> <li>– Pure EDDSA scheme (eddsa.bHashEddsa != TRUE) is not supported with encrypted SMR.</li> <li>– EDDSA scheme Context (if used) can be maximum 16 bytes.</li> </ul> |

## Boot and Memory Verification Services

### Data Fields

| Type                            | Name            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t                        | pInstAuthTag[2] | <p>Optional - The location in external flash of the initial proof of authenticity over SMR.</p> <ul style="list-style-type: none"><li>– If the <a href="#">HSE_SMR_CFG_FLAG_INSTALL_AUTH</a> flag is set, it specifies the address(es) where the SMR original authentication TAG to be verified is located.</li><li>– If the <a href="#">HSE_SMR_CFG_FLAG_INSTALL_AUTH</a> flag is cleared, this field is not used (an internal authentication scheme is used).</li></ul> <p>Note</p> <ul style="list-style-type: none"><li>– The SMR authentication proof is always computed over the plain SMR.</li><li>– For MAC and RSA signature, only pInstAuthTag[0] is used.</li><li>– Both addresses are used for ECDSA and EDDSA signatures (specified by (r,s), with r at index 0, and s at index 1).</li></ul>                                                                                                                            |
| <a href="#">hseSmrDecrypt_t</a> | smrDecrypt      | Specifies the parameters for SMR decryption.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| uint32_t                        | versionOffset   | <p>Optional - The offset in SMR where the image version can be found. May be used to provide the SMR version which offers anti-rollback protection for the image against attacks during update.</p> <p>Note</p> <ul style="list-style-type: none"><li>– Ignored if set to <a href="#">HSE_SMR_VERSION_NOT_USED</a> (i.e. 0).</li><li>– If used, it must be a valid offset within the SMR in the range [4, <a href="#">hseSmrEntry_t::smrSize</a> - 4].</li><li>– Once used when installing an SMR, all subsequent updates of that SMR must have a version GREATER than the previous one.</li><li>– During SMR update the version offset can be modified only having SU rights. The version value must still be GREATER than the previous one.</li><li>– The version offset must be aligned to 4 bytes.</li><li>– Not used for SHE based secure boot (must be set to <a href="#">HSE_SMR_VERSION_NOT_USED</a> in this case).</li></ul> |

### struct hseCrEntry\_t

Define the parameters of a Core Reset entry in CR table.

The CR table contains the configurations for each Application Core that HSE will use to perform the advanced secure boot.

## Note

- SU right are needed to install/update a Core reset entry.
- Updating an existing CR entry is conditioned by having all preBoot and postBoot SMR(s) linked with the previous entry verified successfully (applicable only in OEM\_PROD/IN\_FIELD LCs).
- The core release strategy is defined by the [HSE\\_CORE\\_RESET\\_RELEASE\\_ATTR\\_ID](#) attribute ("ALL-AT-ONCE" or "ONE-BY-ONE")
- For devices with SD/eMMC support (e.g. HSE\_H), the SMR having the source address in SD/eMMC can be used only if the following conditions are met:
  - The release core strategy is either set to "ALL-AT-ONCE" or "ONE-BY-ONE", the SMR in SD/eMMC is linked only to the first entry in the CR table (see [hseAttrCoreResetRelease\\_t](#)).
  - The [startOption](#) is [HSE\\_CR\\_AUTO\\_START](#).
  - SMR type: either SMR is linked via [preBootSmrMap](#) or [altPreBootSmrMap](#) to the CR entry (i.e. will be loaded and verified in PRE-BOOT phase).
  - SMR type: or SMR is linked via [postBootSmrMap](#) when [preBootSmrMap](#) & [altPreBootSmrMap](#) are zero (i.e. will be used for parallel secure boot - loaded in PRE-BOOT phase and verified POST-BOOT).

## Data Fields

| Type                            | Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseAppCore_t</a>    | coreId        | Identifies the core Id to be started (see <a href="#">hseAppCore_t</a> for core mapping).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| uint8_t                         | reserved0[1U] |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <a href="#">hseCrSanction_t</a> | crSanction    | <p>The sanction applied if one of the SMR(s) linked to the CR entry failed the verification.</p> <p>Note</p> <ul style="list-style-type: none"> <li>– If at least one SMR from each PRE-BOOT bitfield (i.e. <a href="#">preBootSmrMap</a> and <a href="#">altPreBootSmrMap</a>) failed verification, the sanction will be applied prior to releasing the core from reset.</li> <li>– If on SMR specified by <a href="#">postBootSmrMap</a> failed, the sanction will be applied after the core is released from reset. In this case, the <a href="#">HSE_CR_SANCTION_KEEP_CORE_IN_RESET</a> option has no effect.</li> <li>– <a href="#">HSE_CR_SANCTION_DIS_INDIV_KEYS</a> option has no effect on the behavior of the core itself, but will take effect on the key usage at run-time (see SMR flags from <a href="#">hseKeyInfo_t</a>).</li> </ul> |
| uint32_t                        | preBootSmrMap | <p>The PRE-BOOT SMR(s) which need to be verified before releasing the core from <a href="#">pPassReset</a> address.</p> <p>It's a 32 bits value, each bit specifies the particular SMR entry index from 0-31. HSE loads and verifies each PRE-BOOT SMR entry specified by this bitfield.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## Boot and Memory Verification Services

### Data Fields

| Type     | Name             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pPassReset       | <p>The primary address of the first instruction after a regular reset. The core starts the execution from this address if all <a href="#">preBootSmrMap</a> SMR(s) have been successfully verified.</p> <p>Note</p> <ul style="list-style-type: none"> <li>– The <a href="#">pPassReset</a> must be within a SMR specified by <a href="#">preBootSmrMap</a>.</li> <li>– If <a href="#">preBootSmrMap</a> == 0, <a href="#">pPassReset</a> must be within a SMR specified by <a href="#">postBootSmrMap</a>. In this case, the HSE will attempt a "parallel secure boot" for this core (see <a href="#">postBootSmrMap</a> description below).</li> </ul>                                                                                                                                                                                                                                                                       |
| uint32_t | altPreBootSmrMap | <p>The ALT-PRE-BOOT SMR(s) which need to be verified before releasing the core from <a href="#">pAltReset</a> address. It's a 32 bits value, each bit specifying the particular SMR entry index from 0-31. HSE verifies each SMR entry specified by this bitfield. The <a href="#">altPreBootSmrMap</a> SMR(s) are verified ONLY if one of the SMR(s) specified by <a href="#">preBootSmrMap</a> failed.</p> <p>Note</p> <ul style="list-style-type: none"> <li>– Once <a href="#">altPreBootSmrMap</a> SMR(s) are loaded and the verification process is triggered, the <a href="#">preBootSmrMap</a> SMR(s) will be considered overwritten/not loaded (see <a href="#">hseSmrVerifySrv_t</a>).</li> <li>– If <a href="#">preBootSmrMap</a> == 0, <a href="#">altPreBootSmrMap</a> must be also 0 (cannot be used).</li> </ul>                                                                                                |
| uint32_t | pAltReset        | <p>The alternative address of the first instruction after a regular reset. The core starts the execution if all <a href="#">altPreBootSmrMap</a> SMR(s) have been successfully verified.</p> <p>Note</p> <ul style="list-style-type: none"> <li>– HSE will try to boot the core from the alternate address only if the <a href="#">preBootSmrMap</a> SMR(s) verification failed.</li> <li>– The <a href="#">pAltReset</a> must be within a SMR specified by <a href="#">altPreBootSmrMap</a>.</li> <li>– If <a href="#">altPreBootSmrMap</a> == 0, <a href="#">pAltReset</a> field is ignored (can not used).</li> <li>– If the conditions to boot from <a href="#">pAltReset</a> are not met (<a href="#">altPreBootSmrMap</a> == 0, <a href="#">pAltReset</a> == NULL or one of the <a href="#">altPreBootSmrMap</a> SMR(s) fails) HSE will apply the sanctions as specified in <a href="#">crSanction</a> field.</li> </ul> |



## Data Fields

| Type                               | Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t                           | postBootSmrMap | <p>The POST-BOOT SMR(s) which need to be loaded after verifying the <a href="#">preBootSmrMap</a> SMR(s) (if any). It's a 32 bits value, each bit specifying the particular SMR entry index from 0-31. HSE verifies each SMR entry specified by this bitfield.</p> <p>Note</p> <ul style="list-style-type: none"> <li>– If <a href="#">preBootSmrMap</a> == 0 (no PRE-BOOT SMR is specified), the SMR(s) specified by <a href="#">postBootSmrMap</a> will be loaded before the core is un-gated from <a href="#">pPassReset</a> address. In this case, only the verification is done after the core is released from reset (POST-BOOT). This is referenced as "parallel secure boot". It can be used only if the POST-BOOT SMRs are in memory mapped locations (e.g. QuadSPI Flash, and not in SD/eMMC).</li> </ul> |
| <a href="#">hseCrStartOption_t</a> | startOption    | Specifies if the Application Core is automatically released from reset or not.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| uint8_t                            | reserved1[6U]  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## struct hseSmrEntryInstallSrv\_t

HSE Secure Memory Region Installation service (update or add new entry).

This service installs a SMR entry which needs to be verify during boot or runtime phase. The installation can be done in one-pass or streaming mode. The streaming mode is useful when the SMR content to be install is not entirely available in the system memory when the installation starts (OTA use case). The table below summarizes the fields needed to be provided for each access mode. Unused fields are ignored by the HSE. SMR(s) can be installed only in sequence, one at a time. This service does not use a stream ID as HSE uses internal contexts when processing in streaming mode.

| Field \ Mode    | One-pass | Start | Update | Finish |
|-----------------|----------|-------|--------|--------|
| accessMode      | *        | *     | *      | *      |
| entryIndex      | *        | *     |        |        |
| pSmrEntry       | *        | *     |        |        |
| pSmrData        | *        | *     | *      | *      |
| smrDataLength   | *        | *     | *      | *      |
| pAuthTag        | *        |       |        | *      |
| authTagLength   | *        |       |        | *      |
| cipher.pIV      | *        | *     |        |        |
| cipher.pGmacTag | *        |       |        | *      |

## Boot and Memory Verification Services

### Note

- The provisioning of the original authentication tag shall be optional when `LC == CUST_DEL`. This allows to implement SHE use-case: autonomous bootstrap.
- In User mode, the SMR can be updated only changing the [hseSmrEntry\\_t::pSmrSrc](#), [hseSmrEntry\\_t::smrSize](#) and [hseSmrEntry\\_t::pInstAuthTag](#). Any other configuration fields (such as `keyHandle`, `configFlags`, `verifMethod`, etc.) of a SMR entry can only be updated if the host has SuperUser rights (for NVM Configuration).
- POST\_BOOT and periodic SMR(s) source addresses cannot be in SD/MMC or external flash memory.
- The keys linked with a SMR entry (through `smrFlags` in [hseKeyInfo\\_t](#)) will become unavailable after successful installation of the SMR entry. The SMR must be verified (automatically at boot-time, periodically or via verify request at run-time) before the key can be used again.
- If a periodic SMR is updated during runtime using this service, the periodic checks for this SMR entry are disabled till the next reset.
- The HSE firmware authenticate image without loading the image when `pSmrSrc` and `pSmrDest` address are the same.

(SHE boot):

The SMR #0 is the only SMR that can be associated to the SHE AES key `BOOT_MAC_KEY` as the SMR authentication key. In this case, the reference authentication tag is the CMAC value referred to as `BOOT_MAC`. The `BOOT_MAC` value can be initialized and updated via the SHE key update protocol.

In addition, when `LC` is set to `CUST_DEL`, `BOOT_MAC` can be automatically calculated as described below:

- On the first SMR #0 installation using `BOOT_MAC_KEY`, if `BOOT_MAC` is empty (i.e. not initialized) and if `BOOT_MAC_KEY` has been provisioned, the reference authentication tag is calculated by the HSE and saved in the `BOOT_MAC` slot. This specific installation process satisfies the SHE requirement referred to as "autonomous bootstrap configuration".
- When installing SMR #0 using the `BOOT_MAC_KEY` while the `BOOT_MAC` is already initialized, the `BOOT_MAC` value must be updated via the SHE key update protocol prior to issuing the SMR installation service.
- In all cases, the arrays [pAuthTag](#) and [authTagLength](#) are always discarded and should be set respectively to `NULL` and `0`.
- If SMR #0 installation using the `keyHandle` for `SHE(BOOT_MAC_KEY)`, [HSE\\_SMR\\_CFG\\_FLAG\\_INSTALL\\_AUTH](#) = 0 is not allowed.

Installing a NXP RFE SMRs entry:

For `SAFXXXX`, the protected NXP RFE images are installed configuring two image:

- one image (CODE) having the destination address only in RFE ITCM (2 SMRs, one as primary and one as back-up)
- and another image (configuration DATA) having the destination address RFE DTCM (2 SMRs, one as primary and one as back-up) All four SMR above are linked with the CR entry for RFE-M7 core (see the installation of NXP RFE CR entry). These images are encrypted and authenticated by NXP (using the NXP ROM keys) and have specific handling on installation (refer to HSE FW reference manual)

To install a single image one must (see the example code below):

- Program the image(s) to the external flash to a chosen location, e.g. ExternalFlashAddr.
- Provide the encryption and authentication key handles of the ROM keys (#HSE\_ROM\_KEY\_AES256\_KEY2 and #HSE\_ROM\_KEY\_RSA2048\_PUB\_KEY1).
- Provide the installation address of the image (can be the same as ExternalFlashAddr).
- Provide a SMR entryIndex for the installation Example of a single NXP RFE image installation (note that 4 SMRs must be installed):

```
smrEntry.pSmrSrc = ExternalFlashAddr;
smrEntry.authKeyHandle = HSE_ROM_KEY_RSA2048_PUB_KEY1;
smrEntry.smrDecrypt.decryptKeyHandle = HSE_ROM_KEY_AES256_KEY2;
hseDescriptor.srvId = HSE_SRV_ID_SMR_ENTRY_INSTALL;
hseDescriptor.smrEntryInstallReq.accessMode = HSE_ACCESS_MODE_ONE_PASS;
hseDescriptor.smrEntryInstallReq.entryIndex = Ind;
hseDescriptor.smrEntryInstallReq.pSmrEntry = HSE_PTR_TO_HOST_ADDR(&smrEntry);
hseDescriptor.smrEntryInstallReq.pSmrData = InstallationAddr;
response = SendDescToHse(&hseDescriptor);
```

Constraints and additional notes:

- Only [HSE\\_ACCESS\\_MODE\\_ONE\\_PASS](#) access mode can be used.
- All parameters not specified in the above example are ignored.

## Data Fields

| Type                            | Name         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseAccessMode_t</a> | accessMode   | INPUT: Specifies the access mode: ONE-PASS, START, UPDATE, FINISH.<br><br>Note <ul style="list-style-type: none"> <li>– Streaming is not supported for Pure EDDSA scheme (eddsa.bHashEddsa != TRUE). STREAMING USAGE: Used in all steps.</li> </ul>                                                                                                                                                                                                                                                                           |
| uint8_t                         | entryIndex   | INPUT: Identifies the index of SMR entry (in the SMR table) which has to be installed/updated. Refer to <a href="#">HSE_NUM_OF_SMR_ENTRIES</a> STREAMING USAGE: Used in START.                                                                                                                                                                                                                                                                                                                                                |
| uint8_t                         | reserved[2U] |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| uint32_t                        | pSmrEntry    | INPUT: Address of SMR entry structure containing the configuration properties to be installed (refer to <a href="#">hseSmrEntry_t</a> ).                                                                                                                                                                                                                                                                                                                                                                                      |
| uint32_t                        | pSmrData     | INPUT: The address where SMR data to be installed is located. STREAMING USAGE: Used in all steps, but ignored if smrDataLength is zero.<br><br>Note <ul style="list-style-type: none"> <li>– If SMR#0 is used for SHE-boot and the BOOT_MAC slot is empty then the BOOT_MAC is be calculated by HSE FW at the time of SMR installation.</li> <li>– For HSE_H/M devices, if the SMR is flashed in SD/eMMC, the application need to copy SMR data in System RAM (and pSmrData must point to that System RAM address)</li> </ul> |

## Boot and Memory Verification Services

### Data Fields

| Type     | Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | smrDataLength | <p>INPUT: The length of the SMR data. In case of streaming mode, the total size of SMR is computed by summing the length of SMR chunks provided during Update/Finish STREAMING USAGE: Used in all steps.</p> <ul style="list-style-type: none"><li>– START: Must be a multiple of 64/128 bytes, or zero. Cannot be zero for HMAC.</li><li>– UPDATE: Must be a multiple of 64/128 bytes. Cannot be zero. Refrain from issuing the service request, instead of passing zero.</li><li>– FINISH: Can be any value (For CMAC &amp; XCBC-MAC, zero length is invalid).</li></ul> <p>Note</p> <ul style="list-style-type: none"><li>– Depending on the algorithm used, the length must be:<ul style="list-style-type: none"><li>· Multiple of 64 bytes:<ul style="list-style-type: none"><li>· CMAC, GMAC, XCBC-MAC;</li><li>· HMAC, RSA, ECDSA with underlying hash: SHA1, SHA2_224, SHA2_256;</li></ul></li><li>· Multiple of 128 bytes:<ul style="list-style-type: none"><li>· HMAC, RSA, ECDSA with underlying hash: SHA2_384, SHA2_512, SHA2_512_224, SHA2_512_256;</li></ul></li></ul></li><li>– Miyaguchi-Preneel not supported as hash algorithm;</li><li>– HMAC: SHA3 not supported as hash algorithm.</li><li>– Pure EDDSA scheme (eddsa.bHashEdDSA != TRUE): not supported in streaming mode.</li></ul> |
| uint32_t | pAuthTag[2]   | <p>INPUT: The address where SMR Original authentication tag to be verify is located.</p> <p>Note</p> <ul style="list-style-type: none"><li>– The SMR authentication proof is always computed over the plain SMR.</li><li>– For MAC and RSA signature, only pAuthTag[0] is used.</li><li>– Both pointers are used for ECDSA and EDDSA signatures (specified as (r,s), with r at index 0, and s at index 1).</li><li>– ignored if SMR#0 is SHE-boot. STREAMING USAGE: Used in FINISH.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## Data Fields

| Type                                           | Name             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint16_t                                       | authTagLength[2] | <p>INPUT: The length of the SMR authentication proof (tag/signature).</p> <p>Note</p> <ul style="list-style-type: none"> <li>– For MAC and RSA signature, only authTagLength[0] is used.</li> <li>– Both pointers are used for ECDSA and EDDSA signatures (specified the length of (r,s), with r at index 0, and s at index 1).</li> <li>– Ignored if SMR#0 is used for SHE-boot.</li> <li>– The MAC tag size must be minimum 16 bytes.</li> <li>– RSA signature size must be <a href="#">HSE_BYTES_TO_BITS(keyBitLength)</a>;</li> <li>– R or S size for ECDSA/EDDSA signature must be <a href="#">HSE_BYTES_TO_BITS(keyBitLength)</a> STREAMING USAGE: Used in FINISH.</li> </ul>                                                                                                                  |
| struct <a href="#">hseSmrEntryInstallSrv_t</a> | cipher           | <p>INPUT: Optional - Cipher parameters used for installing encrypted SMR(s).</p> <p>Note</p> <ul style="list-style-type: none"> <li>– These parameters are use only if <a href="#">hseSmrDecrypt_t::decryptKeyHandle</a> != <a href="#">HSE_SMR_DECRYPT_KEY_HANDLE_NOT_USED</a> (see <a href="#">hseSmrDecrypt_t</a>).</li> <li>– The pointers that are specified in this structure shall be provided from a memory-mapped location (QSPI/LPDDR4/RAM).</li> <li>– In case an SMR lying in SD/eMMC external flash is installed, a copy of GMAC tag (if used) shall be done in RAM and provided via the fields below.<br/>The pointers provided via <a href="#">hseSmrEntryInstallSrv_t::pSmrEntry</a> shall point to the location in external flash that will be used by HSE at boot-time.</li> </ul> |

## struct hseSmrVerifySrv\_t

HSE Secure Memory Region verification service.

This service starts the on-demand verification of a secure memory region by specifying the index in the SMR table.

## Boot and Memory Verification Services

### Note

: For SAF86XX devices, if the Flashless Boot Mode is used (pSmrSrc == pSmrDest), for any "options" value, HSE will skip the loading (the SMR image is already loaded in SRAM)

### Data Fields

| Type                                        | Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint8_t                                     | entryIndex | INPUT: Specifies the entry in the SMR table to be verified (max <a href="#">HSE_NUM_OF_SMR_ENTRIES</a> ). This service loads and verifies on-demand an SMR entry (in SRAM).<br><br>Note<br><br>(HSE_H/M) <ul style="list-style-type: none"><li>– The SMR(s) used in CORE RESET table can be verified on-demand only if they were loaded before in SRAM or the BOOT_SEQ = 0. Otherwise, an error will be reported (NOT ALLOWED).</li><li>– The SMR(s) that are not part of the CORE RESET table configuration can be loaded and verified at run time. Note that on the second call of this service, the HSE will only performed the verification in SRAM. Using this service, the SMR(s) can not be loaded and verified from SD/MMC memory.</li></ul> |
| uint8_t                                     | reserved   | RFU. Set to 0 for compatibility with future updates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <a href="#">hseSmrVerificationOptions_t</a> | options    | INPUT: Options for customizing the on-demand SMR verification (see <a href="#">hseSmrVerificationOptions_t</a> ). Values not defined or not applicable are ignored.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## struct hseSmrEntryEraseSrv\_t

SMR entry erase service.

This service erases one SMR entry from the internal HSE memory.

### Note

- SuperUser (SU) access rights with privileges over HSE\_SYS\_AUTH\_NVM\_CONFIG data are required to perform this service.

### Data Fields

| Type    | Name        | Description                                                              |
|---------|-------------|--------------------------------------------------------------------------|
| uint8_t | smrEntryInd | INPUT: Identifies the index in the SMR table for the entry to be erased. |
| uint8_t | reserved[3] |                                                                          |

**struct hseCrEntryInstallSrv\_t**

Core Reset entry install (update or add new entry)

This service updates an existing or add a new entry in the Core Reset table.

## Note

- SMR entries that are linked with the installed CR entry (via preBoot/altPreBoot/postBoot SMR maps) must be installed in HSE prior to the CR installation.
- SuperUser rights (for NVM Configuration) are needed to perform this service.
- Updating an existing CR entry is conditioned by having all preBoot and postBoot SMR(s) linked with the previous entry verified successfully (applicable only in OEM\_PROD/IN\_FIELD LCs).

Installing a NXP RFE Core Reset entry:

For SAF85XX, SAF86XX, the HSE FW provides the functionality of installing the protected NXP RFE images (e.g. RFE - CORE1 on SAF85XX platform). This CR entry are linked with the NXP RFE images (4 SMRs, two for primary and two for back-up) and have a specific handling on installation (refer to HSE FW reference manual). To install such an entry one must (refer to the example code below):

- Install the corresponding NXP SMR images (see SMR installation NXP RFE images)
- Link the NXP RFE images (4 SMRs) to the CR entry to be installed.
- Provide the crEntryIndex index (CR\_IND\_RFE). Example of RFE CR installation for NXP RFE im:

```
crEntry.coreId = HSE_APP_CORE1;
crEntry.preBootSmrMap = ((1UL < SMR_IND_RFE_ITCM_PRIMARY) | (1UL < SMR_IND_RFE_DTCM_PRIMARY));
crEntry.altPreBootSmrMap = ((1UL < SMR_IND_RFE_ITCM_BACKUP) | (1UL < SMR_IND_RFE_DTCM_BACKUP));
desc.srvId = HSE_SRV_ID_CORE_RESET_ENTRY_INSTALL;
desc.crEntryInstallReq.crEntryIndex = CR_IND_RFE;
desc.crEntryInstallReq.pCrEntry = HSE_PTR_TO_HOST_ADDR(&crEntry);
response = SendHseDescriptor(&desc);
```

Constraints and additional notes:

- The referenced NXP SMR must be installed prior to CR entry installation.
- All parameters not specified in the above example are ignored.

## Data Fields

| Type     | Name         | Description                                                                                                                                |
|----------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| uint8_t  | crEntryIndex | INPUT: Identifies the index in the Core Reset table which has to be added/updated Refer to <a href="#">HSE_NUM_OF_CORE_RESET_ENTRIES</a> . |
| uint8_t  | reserved[3]  |                                                                                                                                            |
| uint32_t | pCrEntry     | INPUT: Address of Core Reset entry structure (refer to <a href="#">hseCrEntry_t</a> ).                                                     |

**struct hseCrOnDemandBootSrv\_t**

On-demand boot of a Core Reset entry.

This service triggers the loading, verification and reset release of a core that is not automatically started

## Boot and Memory Verification Services

(at boot time).

### Note

- This service can be called only once and only for the Core Reset entries that have the startOption option set to [HSE\\_CR\\_ON\\_DEMAND](#).
- Using this service, the SMR(s) can not be loaded and verified from SD/MMC memory.

### Data Fields

| Type    | Name         | Description                                                                                                                                                                      |
|---------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint8_t | crEntryIndex | INPUT: Identifies the index in the Core Reset table which has to be released from reset after loading and verification. Refer to <a href="#">HSE_NUM_OF_CORE_RESET_ENTRIES</a> . |
| uint8_t | reserved[3]  |                                                                                                                                                                                  |

### struct hseCrEntryEraseSrv\_t

Core Reset entry erase service.

This service erases one Core Reset entry from the internal HSE memory.

### Note

- SuperUser (SU) access rights with privileges over HSE\_SYS\_AUTH\_NVM\_CONFIG data are required to perform this service.

### Data Fields

| Type    | Name        | Description                                                                     |
|---------|-------------|---------------------------------------------------------------------------------|
| uint8_t | crEntryInd  | INPUT: Identifies the index in the Core Reset table for the entry to be erased. |
| uint8_t | reserved[3] |                                                                                 |

### struct hseSmrEntryInstallSrv\_t.cipher

INPUT: Optional - Cipher parameters used for installing encrypted SMR(s).

### Note

- These parameters are use only if [hseSmrDecrypt\\_t::decryptKeyHandle](#) != [HSE\\_SMR\\_DECRYPT\\_KEY\\_HANDLE\\_NOT\\_USED](#) (see [hseSmrDecrypt\\_t](#)).
- The pointers that are specified in this structure shall be provided from a memory-mapped location (QSPI/LPDDR4/RAM).



- In case an SMR lying in SD/eMMC external flash is installed, a copy of GMAC tag (if used) shall be done in RAM and provided via the fields below.  
The pointers provided via [hseSmrEntryInstallSrv\\_t::pSmrEntry](#) shall point to the location in external flash that will be used by HSE at boot-time.

### Data Fields

| Type     | Name     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pIV      | INPUT: Initialization Vector/Nonce. The length of the IV is 16 bytes. Will be stored by HSE internally. STREAMING USAGE: Used in START.                                                                                                                                                                                                                                                                                                                                          |
| uint32_t | pGmacTag | INPUT: Optional - tag used for AEAD. The length considered for the GMAC tag is 16 bytes (if used - see <a href="#">hseSmrDecrypt_t</a> ).<br><br>Note <ul style="list-style-type: none"> <li>– Used only if <a href="#">hseSmrDecrypt_t::pGmacTag</a> != NULL.</li> <li>– Must point to the same data as <a href="#">hseSmrDecrypt_t::pGmacTag</a>, however the memory location may differ (QSPI/LPDDR4/RAM vs QSPI/LPDDR4/SD/eMMC). STREAMING USAGE: Used in FINISH.</li> </ul> |
| uint32_t | pAAD     | INPUT: Optional - the AAD used for AEAD. The length considered for the AAD is specified via pSmrEntry->smrDecrypt.aadLength (see <a href="#">hseSmrDecrypt_t</a> ).<br><br>Note <ul style="list-style-type: none"> <li>– Used only if length is not zero.</li> <li>– Must point to the same data as pSmrEntry-&gt;smrDecrypt.pAAD, however the memory location may differ (QSPI/LPDDR4/RAM vs QSPI/LPDDR4/SD/eMMC). STREAMING USAGE: Used in START.</li> </ul>                   |

## Macro Definition Documentation

### HSE\_SMR\_DECRYPT\_KEY\_HANDLE\_NOT\_USED

```
#define HSE_SMR_DECRYPT_KEY_HANDLE_NOT_USED (0UL)
```

Decryption of SMR is not used.

### HSE\_SMR\_VERSION\_NOT\_USED

```
#define HSE_SMR_VERSION_NOT_USED (0UL)
```

## Boot and Memory Verification Services

SMR version is not used (value to ignore `hseSmrEntryInstallSrv_t::versionOffset` field).

### HSE\_CR\_SANCTION\_DIS\_INDIV\_KEYS

```
#define HSE_CR_SANCTION_DIS_INDIV_KEYS ((hseCrSanction_t) 0x7433U)
```

Disable individual keys; if at least one SMR entry specified by the key `smrFlags` (see [hseKeyInfo\\_t](#)) is not verified, the key can not be used.

### HSE\_CR\_SANCTION\_KEEP\_CORE\_IN\_RESET

```
#define HSE_CR_SANCTION_KEEP_CORE_IN_RESET ((hseCrSanction_t) 0x7455U)
```

The HSE keeps in reset the core (if the verification of at least one SMR entry fails)

### HSE\_CR\_SANCTION\_RESET\_SOC

```
#define HSE_CR_SANCTION_RESET_SOC ((hseCrSanction_t) 0x8B17U)
```

The HSE reset the SoC.

### HSE\_CR\_SANCTION\_DIS\_ALL\_KEYS

```
#define HSE_CR_SANCTION_DIS_ALL_KEYS ((hseCrSanction_t) 0x8B1EU)
```

Disable all keys.

### HSE\_CR\_AUTO\_START

```
#define HSE_CR_AUTO_START ((hseCrStartOption_t) 0x35A5U)
```

The Core is released from reset automatically at startup (if the corresponding SMR(s) are loaded and verified).

### HSE\_CR\_ON\_DEMAND

```
#define HSE_CR_ON_DEMAND ((hseCrStartOption_t)0x5567U)
```

The Core is not released from reset automatically; this can be triggered by another Application Core using [hseCrOnDemandBootSrv\\_t](#) service.

### HSE\_SMR\_CFG\_FLAG\_QSPI\_FLASH

```
#define HSE_SMR_CFG_FLAG_QSPI_FLASH ((hseSmrConfig_t)0x0U)
```

Identifies the Interface (where the SMR needs to be copied from)

### HSE\_SMR\_CFG\_FLAG\_SD\_FLASH

```
#define HSE_SMR_CFG_FLAG_SD_FLASH ((hseSmrConfig_t)0x2U)
```

Identifies the Interface (where the SMR needs to be copied from)

### HSE\_SMR\_CFG\_FLAG\_MMC\_FLASH

```
#define HSE_SMR_CFG_FLAG_MMC_FLASH ((hseSmrConfig_t)0x3U)
```

Identifies the Interface (where the SMR needs to be copied from)

### HSE\_SMR\_CFG\_FLAG\_INSTALL\_AUTH

```
#define HSE_SMR_CFG_FLAG_INSTALL_AUTH ((hseSmrConfig_t)(1U << 2U))
```

If it is set, the authentication scheme and tag provided during installation phase (installation TAG) are used also during the verification phase. If it is cleared, during installation HSE will compute and store an internal hash digest (SHA2-256) During verification phase, HSE will use this internal digest.

#### Note

- If the [HSE\\_SMR\\_CFG\\_FLAG\\_INSTALL\\_AUTH](#) flag is cleared and SHE-boot is used (SMR #0 with [BOOT\\_MAC\\_KEY](#)), HSE FW will return [HSE\\_SRV\\_RSP\\_NOT\\_ALLOWED](#) on SMR#0 installation request.

## Boot and Memory Verification Services

### HSE\_SMR\_CFG\_FLAG\_AUTH\_AAD

```
#define HSE_SMR_CFG_FLAG_AUTH_AAD ((hseSmrConfig_t)(1U << 3U))
```

If this bit is set, the authentication is computed over [AAD || Plain] image.

Note

- The SMR has to be configured with AEAD-GCM decryption (i.e. AAD and GMAC tag are provided as part of decryption parameters).

### HSE\_SMR\_VERIFICATION\_OPTION\_NONE

```
#define HSE_SMR_VERIFICATION_OPTION_NONE ((hseSmrVerificationOptions_t)0UL) /**  
@brief Default verification of the SMR at run-time. */
```

### HSE\_SMR\_VERIFICATION\_OPTION\_NO\_LOAD

```
#define HSE_SMR_VERIFICATION_OPTION_NO_LOAD ((hseSmrVerificationOptions_t)(3UL <<  
0U)) /** @brief SMR is verified from the external flash (using pSmrSrc address) even  
if pSmrDest is specified or if already loaded. Can be used only if SMR is in a  
memory mapped external flash (e.g. QSPI/LPDDR4 and not SD/eMMC). Additionally the  
SMR cannot be encrypted. */
```

### HSE\_SMR\_VERIFICATION\_OPTION\_RELOAD

```
#define HSE_SMR_VERIFICATION_OPTION_RELOAD ((hseSmrVerificationOptions_t)(3UL <<  
2U)) /** @brief SMR is loaded from the external flash and verified even if it is  
already loaded. Can be used only if SMR is in a memory mapped external flash (e.g.  
QSPI/LPDDR4 and not SD/eMMC). */
```

### HSE\_SMR\_VERIFICATION\_OPTION\_PASSIVE\_MEM

```
#define HSE_SMR_VERIFICATION_OPTION_PASSIVE_MEM ((hseSmrVerificationOptions_t)(3UL  
<< 4U)) /** @brief Only for HSE_B with A/B Swap Configuration. Verifies the SMR  
from the passive block, applying address translation. */
```

## Typedef Documentation

## **hseCrSanction\_t**

```
typedef uint16_t hseCrSanction_t
```

CORE sanctions to be applied if the verification of at least one SMR entry fails on both Primary and Backup SMR maps as defined in CR entry ([hseCrEntry\\_t::preBootSmrMap](#) and [hseCrEntry\\_t::altPreBootSmrMap](#))

## **hseCrStartOption\_t**

```
typedef uint16_t hseCrStartOption_t
```

The start option for a Core Reset Entry.

## **hseSmrConfig\_t**

```
typedef uint8_t hseSmrConfig_t
```

Specifies the boot interface (where the SMR needs to be copied from).

Note

- For HSE\_H/M, the SMR source memory can be:
  - QSPI Flash
  - SD card
  - MMC
  - LPDDR4 Flash (used only for S32ZE devices)
  - for different SMR(s), any combination of the above memory interfaces, except MMC and SD (e.g. QSPI/LPDDR4 Flash and SD, QSPI/LPDDR4 Flash and MMC).
- For HSE\_B, the source memory flags (QSPI/SD/MMC/LPDDR4) are not used.
- For SAF86XX devices, if the Flashless Boot Mode is used, The pSmrSrc and pSmrDest addresses must be equal. The SMR will be only authenticated and decrypted (only if encrypted) in place. BootROM copied the SMR images in SRAM.

## **hseSmrVerificationOptions\_t**

```
typedef uint16_t hseSmrVerificationOptions_t
```

Options for customizing SMR run-time verification.

## 7 SHE Specification

### 7.1 HSE SHE Specification Services

#### Data Structures

- struct [hseSheLoadKeySrv\\_t](#)
- struct [hseSheLoadPlainKeySrv\\_t](#)
- struct [hseSheExportRamKeySrv\\_t](#)
- struct [hseSheGetIdSrv\\_t](#)

#### Data Structure Documentation

##### struct hseSheLoadKeySrv\_t

SHE load key service.

Load a SHE key into the HSE according to the SHE memory update protocol.

##### Note

The SHE keys can be used for any supported AES operations (e.g. AES with all block modes, AEAD etc.) given the proper flags are set. One exception is BOOT\_MAC\_KEY, which can only be used with CMAC verify operation.

##### Data Fields

| Type                             | Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseKeyGroupIdx_t</a> | sheGroupIndex | Group Index for the SHE NVM catalog, ranging from 0 to 4. This parameter also decides the KDF input constants "CENC" & "CMAC" to be used in memory update protocol operation. <ul style="list-style-type: none"> <li>– For (1 &lt;= keyID &lt;= 3), this parameter is ignored and taken as zero to decide "CENC" and "CMAC".</li> <li>– For (keyID = 14) and (4 &lt;= authID &lt;= 13), this parameter is used to select auth-user-key (authID) group and to decide "CENC" &amp; "CMAC".</li> <li>– For (keyID = 14) and (authID = 0), this parameter is ignored &amp; taken as zero to decide "CENC" &amp; "CMAC".</li> <li>– For (4 &lt;= keyID &lt;= 13) and (4 &lt;= authID &lt;= 13), given that keyID = authID, this parameter is used to select both user-key (keyID) group &amp; auth-user-key (authID) group and to decide "CENC" &amp; "CMAC".</li> <li>– For (4 &lt;= keyID &lt;= 13) and (authID = 1), this parameter is used to select user-key (keyID) group and to decide "CENC" &amp; "CMAC".</li> </ul> |
| uint8_t                          | reserved[3]   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| uint32_t                         | pM1           | INPUT: Pointer to M1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| uint32_t                         | pM2           | INPUT: Pointer to M2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Data Fields

| Type     | Name | Description            |
|----------|------|------------------------|
| uint32_t | pM3  | INPUT: Pointer to M3.  |
| uint32_t | pM4  | OUTPUT: Pointer to M4. |
| uint32_t | pM5  | OUTPUT: Pointer to M5. |

**struct hseSheLoadPlainKeySrv\_t**

SHE load plain key service.

Load a SHE RAM key from plain text

## Data Fields

| Type     | Name | Description                            |
|----------|------|----------------------------------------|
| uint32_t | pKey | INPUT: Pointer to the unencrypted key. |

**struct hseSheExportRamKeySrv\_t**

SHE export RAM key service.

Export a SHE RAM key in the format used for re-loading with SHE Load key. This export can happen only if RAM key was loaded using SHE RAM plain key service.

## Data Fields

| Type     | Name | Description            |
|----------|------|------------------------|
| uint32_t | pM1  | OUTPUT: Pointer to M1. |
| uint32_t | pM2  | OUTPUT: Pointer to M2. |
| uint32_t | pM3  | OUTPUT: Pointer to M3. |
| uint32_t | pM4  | OUTPUT: Pointer to M4. |
| uint32_t | pM5  | OUTPUT: Pointer to M5. |

**struct hseSheGetIdSrv\_t**

SHE get ID service.

Returns the Identity (UID) and the value of the status register protected by a MAC over a challenge and the data. If MASTER\_ECU\_KEY is empty, the returned MAC has to be set to zero.

## SHE Specification

### Note

For HSE\_H/M, the UID is read by HSE from system fuses. The application must provide read-only access (xRDC restriction) to HSE to read the UID.

### Data Fields

| Type     | Name       | Description                                                                                                             |
|----------|------------|-------------------------------------------------------------------------------------------------------------------------|
| uint32_t | pChallenge | INPUT: Pointer to 128-bit Challenge.                                                                                    |
| uint32_t | pId        | OUTPUT: Pointer to 120-bit UID.                                                                                         |
| uint32_t | pSreg      | OUTPUT: Pointer to 8-bit Status Register (SREG). Refer to HSE Status for status related information (boot, debug, etc.) |
| uint32_t | pMac       | OUTPUT: Pointer to 128-bit CMAC(CHALLENGE   ID   SREG) using MASTER_ECU_KEY as key.                                     |





## 8 Monotonic Counters Services

### 8.1 HSE Monotonic Counters

#### Data Structures

- struct [hseIncrementCounterSrv\\_t](#)
- struct [hseReadCounterSrv\\_t](#)
- struct [hseConfigSecCounterSrv\\_t](#)

#### Note

For HSE\_B (devices with internal flash), the first service request after reset that depends on Monotonic Counters, i.e., [hseConfigSecCounterSrv\\_t](#), [hseReadCounterSrv\\_t](#), [hseIncrementCounterSrv\\_t](#) and #hseCmacWithCounterSrv\_t will take more time because of Monotonic Counters Initialization. Further requests will take usual time.

#### Data Structure Documentation

##### struct hseIncrementCounterSrv\_t

Increment a monotonic counter service with a specific value.

- For HSE\_H/M, the counters are volatile. Host application has to publish/load the monotonic counter table using hsePublishLoadCntTblSrv\_t service.
- For HSE\_B, the host application shall use the [hseConfigSecCounterSrv\\_t](#) service to initialize and configure the secure counters.
- If the counter is saturated, an error is reported.

#### Data Fields

| Type     | Name         | Description                   |
|----------|--------------|-------------------------------|
| uint32_t | counterIndex | INPUT: The counter Index.     |
| uint32_t | value        | INPUT: The value to be added. |

##### struct hseReadCounterSrv\_t

Read a monotonic counter service.

#### Data Fields

| Type     | Name         | Description                                                                 |
|----------|--------------|-----------------------------------------------------------------------------|
| uint32_t | counterIndex | INPUT: The counter Index.                                                   |
| uint32_t | pCounterVal  | OUTPUT: The address where the counter value is returned (a uint64_t value). |

## Monotonic Counters Services

### struct hseConfigSecCounterSrv\_t

Initialize and configure a secure counter.

HSE supports 16 X 64 bits secure counters, each counter having associated a CounterIndex from 0 to 15. By default, all the counters are disabled.

The secure counter (SC) consists of 2 separate bitfields: Rollover Protection (RP) + Volatile Counter (VC). HSE stores the secure counter in data flash each time the Rollover Protection (RP) is updated.

The purpose of this service is to enable the secure counter and configure the Rollover Protection bitfield size. The objective is to reduce the rate at which NVM programming operations occur.

If the secure counter is already configured and this service is called, HSE re-configures the counter with the new Rollover Protection (RP) and reset it to 0.

#### Note

- SuperUser rights are needed to configure/enable the monotonic counters.
- For HSE\_B (devices with internal flash)
  - **WARNING:** The HSE erases a flash sector after 511 Rollover Protection updates in data flash. The number of data flash erases is limited to 100.000. The application shall configure each secure counter depending on the SC update rate and the number of enabled counters.
  - The secure counter configuration is stored in data flash each time [hseConfigSecCounterSrv\\_t](#) is called.
  - If RPBitSize = 64bits, the HSE stores the SC in flash each time is updated.
- For HSE\_H/M (flashless devices)
  - The RPBitSize is configured for all the enabled secure counters. If the RP of a counter is updated, a warning event is trigger called #HSE\_WA\_PUBLISH\_COUNTER\_TBL through MUB\_GSR register. The application shall clear the warning bit (WIC) and use the #hsePublishLoadCntTblSrv\_t service to publish and store the counter table in the external flash. Note that the counter table must be loaded at initialization time by the application (anti-rollback protection is not supported).

Example: Let's consider the RPBitSize = 40bits and SC = 0x00000000000000001.

This means Rollover Protection (40bits) + Volatile Counter (24bits).

The secure counter (SC) will be stored in flash if the incremental value is  $\geq 0xFFFFFFFF$ . Otherwise, the counter will be updated but not stored.

SC = 0x00000000000000001 + 0xFFFFFFFF = 0x0000000001000000 (RP was changed)

#### Data Fields

| Type     | Name         | Description                                                                                                                                                                                              |
|----------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | counterIndex | INPUT: - For HSE_B, specifies the counter Index.<br>– For HSE_H/M, specifies the number of counters to be enabled (max 16). E.g. if it is set to 5, the counters with the index from 0 to 4 are enabled. |
| uint8_t  | RPBitSize    | INPUT: The Rollover Protection bit size (refer to service comments). It shall be $\geq 32$ bits and $\leq 64$ bits.                                                                                      |
| uint8_t  | reserved[3]  |                                                                                                                                                                                                          |

## 9 Random Number Generator Services

### 9.1 HSE Random Number Generator services

#### Data Structures

- struct [hseGetRandomNumSrv\\_t](#)

#### Macros

| Type: <a href="#">hseRngClass_t</a> |       |
|-------------------------------------|-------|
| Name                                | Value |
| <a href="#">HSE_RNG_CLASS_DRG3</a>  | 0U    |
| <a href="#">HSE_RNG_CLASS_DRG4</a>  | 1U    |
| <a href="#">HSE_RNG_CLASS_PTG3</a>  | 2U    |

#### Typedefs

- typedef uint8\_t [hseRngClass\\_t](#)

#### Data Structure Documentation

##### struct [hseGetRandomNumSrv\\_t](#)

Get random number service.

Note

- When the HSE\_STATUS\_RNG\_INIT\_OK status flag is cleared (failed), the call of any service that requires a random number (e.g. Get Random Number, ECDSA signature generate etc.) triggers a RNG re-initialization before requesting the random number. If the RNG re-initialization is executed successfully, the HSE\_STATUS\_RNG\_INIT\_OK status flag is set to 1.
- If the HSE\_SRV\_RSP\_RNG\_INIT\_IN\_PROGRESS status is received (RNG initialization is in progress), the application must try when the HSE\_STATUS\_RNG\_INIT\_OK status is set in the FSR register.

#### Data Fields

| Type                          | Name            | Description                                                                                            |
|-------------------------------|-----------------|--------------------------------------------------------------------------------------------------------|
| <a href="#">hseRngClass_t</a> | rngClass        | INPUT: The RNG class.                                                                                  |
| uint8_t                       | reserved[3]     |                                                                                                        |
| uint32_t                      | randomNumLength | INPUT: Length of the generated random number in bytes. The maximum value for one request is 512 bytes. |
| uint32_t                      | pRandomNum      | OUTPUT: The address where the random number will be stored.                                            |

### Macro Definition Documentation

#### HSE\_RNG\_CLASS\_DRG3

```
#define HSE_RNG_CLASS_DRG3 ((hseRngClass_t)0U)
```

DRG.3 class uses the RNG engine with prediction resistance disabled. This is the most efficient class in terms of performance.

#### HSE\_RNG\_CLASS\_DRG4

```
#define HSE_RNG_CLASS_DRG4 ((hseRngClass_t)1U)
```

DRG.4 (AIS-20/SP800-90A) class uses the RNG engine with prediction resistance enabled. Using the prediction resistance will impact the performance, as every call to Get Random invokes reseed internally.

#### HSE\_RNG\_CLASS\_PTG3

```
#define HSE_RNG_CLASS_PTG3 ((hseRngClass_t)2U)
```

PTG.3 (AIS 31/SP800-90B) class uses the RNG engine with prediction resistance enabled and will reseed for each 16 bytes of data. This is the most costly class in terms of performance.

### Typedef Documentation

#### hseRngClass\_t

```
typedef uint8_t hseRngClass_t
```

HSE RNG classes.

#### Note

Additional entropy (personalization string) is not needed to be provide by user. The entropy generated by the TRNG already ensures this with high probability.

## 10 Network Protocol Acceleration Services

## 11 Common Types and Definitions

### 11.1 HSE Common Types

#### Data Structures

- struct [hseSrvMetaData\\_t](#)
- struct [hseRsaOAEPScheme\\_t](#)
- struct [hseEcdsaScheme\\_t](#)
- struct [hseEddsaSignScheme\\_t](#)
- struct [hseRsaPssSignScheme\\_t](#)
- struct [hseRsaPkcs1v15Scheme\\_t](#)
- struct [hseSignScheme\\_t](#)
- struct [hseSymCipherScheme\\_t](#)
- struct [hseAeadScheme\\_t](#)
- struct [hseRsaCipherScheme\\_t](#)
- union [hseCipherScheme\\_t](#)
- struct [hseCmacScheme\\_t](#)
- struct [hseHmacScheme\\_t](#)
- struct [hseGmacScheme\\_t](#)
- struct [hseMacScheme\\_t](#)
- union [hseAuthScheme\\_t](#)
- struct [hseScatterList\\_t](#)
- union [hseSignScheme\\_t.sch](#)
- union [hseRsaCipherScheme\\_t.sch](#)
- union [hseMacScheme\\_t.sch](#)

#### Macros

| Type: (implicit C type)                          |                                                                              |
|--------------------------------------------------|------------------------------------------------------------------------------|
| Name                                             | Value                                                                        |
| <a href="#">HSE_MAX_DESCR_SIZE</a>               | 256U                                                                         |
| <a href="#">HSE_ALL_MU_MASK</a>                  | <a href="#">HSE_MU0_MASK</a>   <a href="#">HSE_MU1_MASK</a>                  |
| <a href="#">HSE_SGT_OPTION_INPUT_OUTPUT_MASK</a> | <a href="#">HSE_SGT_OPTION_INPUT</a>   <a href="#">HSE_SGT_OPTION_OUTPUT</a> |
| <a href="#">HSE_SGT_FINAL_CHUNK_BIT_MASK</a>     | 0x40000000UL                                                                 |

| Type: <a href="#">hseSGTOption_t</a>  |          |
|---------------------------------------|----------|
| Name                                  | Value    |
| <a href="#">HSE_SGT_OPTION_NONE</a>   | 0U       |
| <a href="#">HSE_SGT_OPTION_INPUT</a>  | 1U << 0U |
| <a href="#">HSE_SGT_OPTION_OUTPUT</a> | 1U << 1U |

| Type: <a href="#">hseMacAlgo_t</a> |       |
|------------------------------------|-------|
| Name                               | Value |
| <a href="#">HSE_MAC_ALGO_CMAC</a>  | 0x11U |

## Common Types and Definitions

|                                       |       |
|---------------------------------------|-------|
| <a href="#">HSE_MAC_ALGO_GMAC</a>     | 0x12U |
| <a href="#">HSE_MAC_ALGO_XCBC_MAC</a> | 0x13U |
| <a href="#">HSE_MAC_ALGO_HMAC</a>     | 0x20U |

| Type: <a href="#">hseCipherBlockMode_t</a> |       |
|--------------------------------------------|-------|
| Name                                       | Value |
| <a href="#">HSE_CIPHER_BLOCK_MODE_NULL</a> | 0U    |
| <a href="#">HSE_CIPHER_BLOCK_MODE_CTR</a>  | 1U    |
| <a href="#">HSE_CIPHER_BLOCK_MODE_CBC</a>  | 2U    |
| <a href="#">HSE_CIPHER_BLOCK_MODE_ECB</a>  | 3U    |
| <a href="#">HSE_CIPHER_BLOCK_MODE_CFB</a>  | 4U    |
| <a href="#">HSE_CIPHER_BLOCK_MODE_OFB</a>  | 5U    |

| Type: <a href="#">hseSignSchemeEnum_t</a> |       |
|-------------------------------------------|-------|
| Name                                      | Value |
| <a href="#">HSE_SIGN_ECDSA</a>            | 0x80U |
| <a href="#">HSE_SIGN_EDDSA</a>            | 0x81U |
| <a href="#">HSE_SIGN_RSASSA_PKCS1_V15</a> | 0x93U |
| <a href="#">HSE_SIGN_RSASSA_PSS</a>       | 0x94U |

| Type: <a href="#">hseAccessMode_t</a>    |       |
|------------------------------------------|-------|
| Name                                     | Value |
| <a href="#">HSE_ACCESS_MODE_ONE_PASS</a> | 0U    |
| <a href="#">HSE_ACCESS_MODE_START</a>    | 1U    |
| <a href="#">HSE_ACCESS_MODE_UPDATE</a>   | 2U    |
| <a href="#">HSE_ACCESS_MODE_FINISH</a>   | 3U    |

| Type: <a href="#">hseMuMask_t</a> |          |
|-----------------------------------|----------|
| Name                              | Value    |
| <a href="#">HSE_MU0_MASK</a>      | 1U << 0U |
| <a href="#">HSE_MU1_MASK</a>      | 1U << 1U |

| Type: <a href="#">hseAppCore_t</a> |       |
|------------------------------------|-------|
| Name                               | Value |
| <a href="#">HSE_APP_CORE0</a>      | 0U    |
| <a href="#">HSE_APP_CORE1</a>      | 1U    |
| <a href="#">HSE_APP_CORE2</a>      | 2U    |
| <a href="#">HSE_APP_CORE3</a>      | 3U    |
| <a href="#">HSE_APP_CORE4</a>      | 4U    |
| <a href="#">HSE_APP_CORE5</a>      | 5U    |

|                |     |
|----------------|-----|
| HSE_APP_CORE6  | 6U  |
| HSE_APP_CORE7  | 7U  |
| HSE_APP_CORE8  | 8U  |
| HSE_APP_CORE9  | 9U  |
| HSE_APP_CORE10 | 10U |
| HSE_APP_CORE11 | 11U |
| HSE_APP_CORE12 | 12U |
| HSE_APP_CORE13 | 13U |
| HSE_APP_CORE14 | 14U |
| HSE_APP_CORE15 | 15U |

| Type: hseAuthDir_t    |       |
|-----------------------|-------|
| Name                  | Value |
| HSE_AUTH_DIR_VERIFY   | 0U    |
| HSE_AUTH_DIR_GENERATE | 1U    |

| Type: hseCipherAlgo_t |       |
|-----------------------|-------|
| Name                  | Value |
| HSE_CIPHER_ALGO_NULL  | 0x00U |
| HSE_CIPHER_ALGO_AES   | 0x10U |

| Type: hseRsaAlgo_t           |       |
|------------------------------|-------|
| Name                         | Value |
| HSE_RSA_ALGO_NO_PADDING      | 0x90U |
| HSE_RSA_ALGO_RSAES_OAEP      | 0x91U |
| HSE_RSA_ALGO_RSAES_PKCS1_V15 | 0x92U |

| Type: hseCipherDir_t   |       |
|------------------------|-------|
| Name                   | Value |
| HSE_CIPHER_DIR_DECRYPT | 0U    |
| HSE_CIPHER_DIR_ENCRYPT | 1U    |

| Type: hseHashAlgo_t    |       |
|------------------------|-------|
| Name                   | Value |
| HSE_HASH_ALGO_NULL     | 0U    |
| HSE_HASH_RESERVED1     | 1U    |
| HSE_HASH_ALGO_SHA_1    | 2U    |
| HSE_HASH_ALGO_SHA2_224 | 3U    |
| HSE_HASH_ALGO_SHA2_256 | 4U    |
| HSE_HASH_ALGO_SHA2_384 | 5U    |

## Common Types and Definitions

|                                            |     |
|--------------------------------------------|-----|
| <a href="#">HSE_HASH_ALGO_SHA2_512</a>     | 6U  |
| <a href="#">HSE_HASH_ALGO_SHA2_512_224</a> | 7U  |
| <a href="#">HSE_HASH_ALGO_SHA2_512_256</a> | 8U  |
| <a href="#">HSE_HASH_ALGO_SHA3_224</a>     | 9U  |
| <a href="#">HSE_HASH_ALGO_SHA3_256</a>     | 10U |
| <a href="#">HSE_HASH_ALGO_SHA3_384</a>     | 11U |
| <a href="#">HSE_HASH_ALGO_SHA3_512</a>     | 12U |
| <a href="#">HSE_HASH_ALGO_MP</a>           | 13U |

| Type: <a href="#">hseAuthCipherMode_t</a> |       |
|-------------------------------------------|-------|
| Name                                      | Value |
| <a href="#">HSE_AUTH_CIPHER_MODE_CCM</a>  | 0x11U |
| <a href="#">HSE_AUTH_CIPHER_MODE_GCM</a>  | 0x12U |

## Typedefs

- typedef uint8\_t [hseMuMask\\_t](#)
- typedef uint8\_t [hseSGTOption\\_t](#)
- typedef uint8\_t [hseAccessMode\\_t](#)
- typedef uint8\_t [hseHashAlgo\\_t](#)
- typedef uint8\_t [hseCipherAlgo\\_t](#)
- typedef uint8\_t [hseCipherBlockMode\\_t](#)
- typedef uint8\_t [hseCipherDir\\_t](#)
- typedef uint8\_t [hseAuthCipherMode\\_t](#)
- typedef uint8\_t [hseAuthDir\\_t](#)
- typedef uint8\_t [hseMacAlgo\\_t](#)
- typedef uint8\_t [hseSignSchemeEnum\\_t](#)
- typedef uint8\_t [hseRsaAlgo\\_t](#)
- typedef uint8\_t [hseAppCore\\_t](#)
- typedef uint32\_t [hseSrvId\\_t](#)
- typedef uint8\_t [hseStreamId\\_t](#)
- typedef uint32\_t [hseKeyHandle\\_t](#)
- typedef uint8\_t [hseKeyGroupIdx\\_t](#)
- typedef uint8\_t [hseKeySlotIdx\\_t](#)
- typedef uint32\_t [hseNoScheme\\_t](#)

## Data Structure Documentation

### struct [hseSrvMetaData\\_t](#)

HSE service metadata.

Each service has a metadata (e.g. priority)



## Data Fields

| Type    | Name        | Description     |
|---------|-------------|-----------------|
| uint8_t | reserved[4] | For future use. |

**struct hseRsaOAEPScheme\_t**

RSAES OAEP Scheme.

Includes parameters needed for RSAES OAEP encryption/ decryption.

## Data Fields

| Type                          | Name        | Description                                                                                         |
|-------------------------------|-------------|-----------------------------------------------------------------------------------------------------|
| <a href="#">hseHashAlgo_t</a> | hashAlgo    | INPUT: The Hash algorithm for RSA OAEP padding.                                                     |
| uint8_t                       | reserved[3] |                                                                                                     |
| uint32_t                      | labelLength | INPUT: Optional OAEP label length (it can be 0). Must be less than 128.                             |
| uint32_t                      | pLabel      | INPUT: Optional OAEP label (it can be NULL if label length is 0). Must be less than 128 bytes long. |

**struct hseEcdsaScheme\_t**

ECDSA signature scheme.

Includes parameters needed for ECDSA signature generate/verify.

## Data Fields

| Type                          | Name        | Description                                                                                                                            |
|-------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseHashAlgo_t</a> | hashAlgo    | INPUT: The hash algorithm used to hash the input before applying the ECDSA operation. Must not be <a href="#">HSE_HASH_ALGO_NULL</a> . |
| uint8_t                       | reserved[3] |                                                                                                                                        |

**struct hseEdDSA\_SignScheme\_t**

EDDSA signature scheme.

Includes parameters needed for EDDSA signature generate/verify.

EdDSA signature mode vs. EdDDA sign scheme parameters:

| EdDSA signature mode | EdDsa sign scheme parameters                       |
|----------------------|----------------------------------------------------|
| PureEdDSA            | bHashEddsa == FALSE && contextLength/pContext == 0 |

## Common Types and Definitions

| EdDSA signature mode | EdDsa sign scheme parameters                       |
|----------------------|----------------------------------------------------|
| Context              | bHashEddsa == FALSE && contextLength/pContext != 0 |
| HashEdDSA            | bHashEddsa == TRUE                                 |

### Data Fields

| Type     | Name          | Description                                                                                                                                                                              |
|----------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bool_t   | bHashEddsa    | INPUT: Whether to pre-hash the input, and perform a HashEddsa signature.                                                                                                                 |
| uint8_t  | contextLength | INPUT: The length of the EDDSA context. Length of zero means no context.                                                                                                                 |
| uint8_t  | reserved[2]   |                                                                                                                                                                                          |
| uint32_t | pContext      | INPUT: The EDDSA context. Ignored if contextLength is zero. Must remain unchanged until the signing operation is finished (especially in streaming), or the signature will be incorrect. |

### struct hseRsaPssSignScheme\_t

RSASSA\_PSS signature scheme.

Includes parameters needed for RSASSA\_PSS signature generate/verify.

### Data Fields

| Type                          | Name        | Description                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hseHashAlgo_t</a> | hashAlgo    | INPUT: The hash algorithm used to hash the input before applying the RSA operation. Must not be <a href="#">HSE_HASH_ALGO_NULL</a> .                                                                                                                                                                                                                                           |
| uint8_t                       | reserved[3] |                                                                                                                                                                                                                                                                                                                                                                                |
| uint32_t                      | saltLength  | INPUT: The length of the salt in bytes. It must fulfill one of the following conditions: <ul style="list-style-type: none"><li>– 0 &lt;= saltLength &lt;= 62 if the key length is 128 bytes and SHA-512 is used as hash algorithm;</li><li>– 0 &lt;= saltLength &lt;= hashLength otherwise, where hashLength denotes the output length of the chosen hash algorithm.</li></ul> |

### struct hseRsaPkcs1v15Scheme\_t

RSASSA\_PKCS1\_V15 signature scheme.

Includes parameters needed for RSASSA\_PKCS1\_V15 signature generate/verify.

## Data Fields

| Type                          | Name        | Description                                               |
|-------------------------------|-------------|-----------------------------------------------------------|
| <a href="#">hseHashAlgo_t</a> | hashAlgo    | INPUT: The hash algorithm Must not be HSE_HASH_ALGO_NULL. |
| uint8_t                       | reserved[3] |                                                           |

**struct hseSignScheme\_t**

The HSE signature scheme.

Includes parameters needed for signature generate/verify.

## Data Fields

| Type                                  | Name        | Description                                                  |
|---------------------------------------|-------------|--------------------------------------------------------------|
| <a href="#">hseSignSchemeEnum_t</a>   | signSch     | INPUT: Signature scheme.                                     |
| uint8_t                               | reserved[3] |                                                              |
| union <a href="#">hseSignScheme_t</a> | sch         | INPUT: Additional information for selected Signature scheme. |

**struct hseSymCipherScheme\_t**

HSE symmetric cipher scheme.

Includes parameters needed for a symmetric cipher.

## Data Fields

| Type                                 | Name            | Description                                             |
|--------------------------------------|-----------------|---------------------------------------------------------|
| <a href="#">hseCipherAlgo_t</a>      | cipherAlgo      | INPUT: Select an symmetric cipher.                      |
| <a href="#">hseCipherBlockMode_t</a> | cipherBlockMode | INPUT: Specifies the cipher block mode.                 |
| uint8_t                              | reserved[2]     |                                                         |
| uint32_t                             | ivLength        | INPUT: Initialization Vector length(at least 16 bytes). |
| uint32_t                             | pIV             | INPUT: Initialization Vector/Nonce.                     |

**struct hseAeadScheme\_t**

## Common Types and Definitions

### Data Fields

| Type                                | Name           | Description                                                   |
|-------------------------------------|----------------|---------------------------------------------------------------|
| <a href="#">hseAuthCipherMode_t</a> | authCipherMode | INPUT: Specifies the authenticated cipher mode.               |
| uint8_t                             | reserved[1]    |                                                               |
| uint16_t                            | tagLength      | INPUT: Specifies the tag length.                              |
| uint32_t                            | pTag           | INPUT: Tag pointer.                                           |
| uint32_t                            | ivLength       | INPUT: Initialization Vector length(at least 12 bytes).       |
| uint32_t                            | pIV            | INPUT: Initialization Vector/Nonce.                           |
| uint32_t                            | aadLength      | INPUT: The length of Additional Data (in bytes). Can be zero. |
| uint32_t                            | pAAD           | INPUT: The AAD Header data. Ignored if aadLength is zero.     |

### struct hseRsaCipherScheme\_t

RSA cipher scheme.

Performs the RSA encryption/decryption).

### Data Fields

| Type                                       | Name        | Description                               |
|--------------------------------------------|-------------|-------------------------------------------|
| <a href="#">hseRsaAlgo_t</a>               | rsaAlgo     | INPUT: RSA algorithm.                     |
| uint8_t                                    | reserved[3] |                                           |
| union <a href="#">hseRsaCipherScheme_t</a> | sch         | INPUT: Scheme for selected RSA algorithm. |

### union hseCipherScheme\_t

HSE Cipher scheme.

Includes parameters needed for symmetric cipher/RSA encryption and decryption.

### Data Fields

| Type                                 | Name       | Description                                            |
|--------------------------------------|------------|--------------------------------------------------------|
| <a href="#">hseSymCipherScheme_t</a> | symCipher  | INPUT: Symmetric cipher scheme.                        |
| <a href="#">hseAeadScheme_t</a>      | aeAdCipher | INPUT: Authenticated encryption scheme (AEAD-GCM/CCM). |
| <a href="#">hseRsaCipherScheme_t</a> | rsaCipher  | INPUT: RSA cipher scheme.                              |

**struct hseCmacScheme\_t**

CMAC scheme.

Includes parameters needed for CMAC tag generation/verification.

Data Fields

| Type                            | Name        | Description                                |
|---------------------------------|-------------|--------------------------------------------|
| <a href="#">hseCipherAlgo_t</a> | cipherAlgo  | INPUT: Select a cipher algorithm for CMAC. |
| uint8_t                         | reserved[3] |                                            |

**struct hseHmacScheme\_t**

HMAC scheme.

Includes parameters needed for HMAC tag generation/verification.

Data Fields

| Type                          | Name        | Description                                                                                          |
|-------------------------------|-------------|------------------------------------------------------------------------------------------------------|
| <a href="#">hseHashAlgo_t</a> | hashAlgo    | INPUT: Specifies the hash algorithm for HMAC. SHA3 and Miyaguchi-Preneel are not supported for HMAC. |
| uint8_t                       | reserved[3] |                                                                                                      |

**struct hseGmacScheme\_t**

GMAC scheme (AES only).

Includes parameters needed for GMAC tag generation/verification.

Data Fields

| Type     | Name     | Description                                                                                |
|----------|----------|--------------------------------------------------------------------------------------------|
| uint32_t | ivLength | INPUT: Initialization Vector length. Zero is not allowed. Recommended 12 bytes or greater. |
| uint32_t | pIV      | INPUT: Initialization Vector/Nonce.                                                        |

**struct hseMacScheme\_t**

HSE MAC scheme.

Includes parameters needed for MAC computation.

## Common Types and Definitions

### Data Fields

| Type                                 | Name        | Description                                                       |
|--------------------------------------|-------------|-------------------------------------------------------------------|
| <a href="#">hseMacAlgo_t</a>         | macAlgo     | INPUT: Select an MAC algorithm.                                   |
| uint8_t                              | reserved[3] |                                                                   |
| union <a href="#">hseMacScheme_t</a> | sch         | INPUT: The scheme (or parameters) for the selected mac algorithm. |

### union hseAuthScheme\_t

HSE authentication scheme.

Includes parameters needed for authentication.

### Data Fields

| Type                            | Name      | Description              |
|---------------------------------|-----------|--------------------------|
| <a href="#">hseMacScheme_t</a>  | macScheme | INPUT: MAC scheme.       |
| <a href="#">hseSignScheme_t</a> | sigScheme | INPUT: Signature scheme. |

### struct hseScatterList\_t

HSE Scatter List .

The input and output data can be provided as a scatter list. A scatter list is used when the input/output is not a continuous buffer (the buffer is spread across multiple memory locations). The input and output pointers are specified as a list of entries as below.

### Data Fields

| Type     | Name   | Description                                                                                                                                                                                             |
|----------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uint32_t | length | The length of the chunk. Maximum size must be less than $2^{30}$ . The final chunk from scatter list must have bit30 set to 1 (e.g. length = chunk_len   <a href="#">HSE_SGT_FINAL_CHUNK_BIT_MASK</a> ) |
| uint32_t | pPtr   | Pointer to the chunk.                                                                                                                                                                                   |

### union hseSignScheme\_t.sch

INPUT: Additional information for selected Signature scheme.

## Data Fields

| Type                                   | Name        | Description                               |
|----------------------------------------|-------------|-------------------------------------------|
| <a href="#">hseEcdsaScheme_t</a>       | ecdsa       | INPUT: ECDSA signature scheme.            |
| <a href="#">hseEddsaSignScheme_t</a>   | eddsa       | INPUT: EDDSA signature scheme.            |
| <a href="#">hseRsaPssSignScheme_t</a>  | rsaPss      | INPUT: RSA PSS signature scheme.          |
| <a href="#">hseRsaPkcs1v15Scheme_t</a> | rsaPkcs1v15 | INPUT: RSASSA_PKCS1_V15 signature scheme. |

**union hseRsaCipherScheme\_t.sch**

INPUT: Scheme for selected RSA algorithm.

## Data Fields

| Type                               | Name        | Description                        |
|------------------------------------|-------------|------------------------------------|
| <a href="#">hseRsaOAEPScheme_t</a> | rsaOAEP     | INPUT: RSA-OAEP scheme.            |
| <a href="#">hseNoScheme_t</a>      | rsaPkcs1v15 | INPUT: No scheme for RSA-PKCS1V15. |

**union hseMacScheme\_t.sch**

INPUT: The scheme (or parameters) for the selected mac algorithm.

## Data Fields

| Type                            | Name    | Description                                        |
|---------------------------------|---------|----------------------------------------------------|
| <a href="#">hseCmacScheme_t</a> | cmac    | INPUT: CMAC scheme (AES).                          |
| <a href="#">hseHmacScheme_t</a> | hmac    | INPUT: HMAC scheme.                                |
| <a href="#">hseGmacScheme_t</a> | gmac    | INPUT: GMAC scheme. Supports only AES.             |
| <a href="#">hseNoScheme_t</a>   | xCbcmac | INPUT: No scheme parameters; supports only AES128. |

**Macro Definition Documentation****HSE\_MAX\_DESCR\_SIZE**

```
#define HSE_MAX_DESCR_SIZE (256U)
```

## Common Types and Definitions

Absolute maximum HSE service descriptor size. This is determined by the HSE-HOST shared memory size, the number of MUs and the number of channels per MU.

### HSE\_MU0\_MASK

```
#define HSE_MU0_MASK ((hseMuMask_t)1U << 0U)
```

MU Instance 0.

### HSE\_MU1\_MASK

```
#define HSE_MU1_MASK ((hseMuMask_t)1U << 1U)
```

MU Instance 1.

### HSE\_ALL\_MU\_MASK

```
#define HSE_ALL_MU_MASK ( HSE_MU0_MASK | HSE_MU1_MASK )
```

Mask for all MU Instances.

### HSE\_SGT\_OPTION\_NONE

```
#define HSE_SGT_OPTION_NONE ((hseSGTOption_t)0U)
```

Scatter list is not used.

### HSE\_SGT\_OPTION\_INPUT

```
#define HSE_SGT_OPTION_INPUT ((hseSGTOption_t)1U << 0U)
```

Input pointer is provided a scatter list.

### HSE\_SGT\_OPTION\_OUTPUT

```
#define HSE_SGT_OPTION_OUTPUT ((hseSGTOption_t)1U << 1U)
```



Output pointer is provided a scatter list.

### **HSE\_SGT\_OPTION\_INPUT\_OUTPUT\_MASK**

```
#define HSE_SGT_OPTION_INPUT_OUTPUT_MASK ( HSE_SGT_OPTION_INPUT | HSE_SGT_OPTION_OUTPUT )
```

Mask for input/output scatter-gather option.

### **HSE\_SGT\_FINAL\_CHUNK\_BIT\_MASK**

```
#define HSE_SGT_FINAL_CHUNK_BIT_MASK (0x40000000UL)
```

Scatter-gather Final chunk BIT. This bit is set in the "length" field of the chunk (see [hseScatterList\\_t](#)).

### **HSE\_ACCESS\_MODE\_ONE\_PASS**

```
#define HSE_ACCESS_MODE_ONE_PASS ((hseAccessMode_t) 0U)
```

ONE-PASS access mode.

### **HSE\_ACCESS\_MODE\_START**

```
#define HSE_ACCESS_MODE_START ((hseAccessMode_t) 1U)
```

START access mode

### **HSE\_ACCESS\_MODE\_UPDATE**

```
#define HSE_ACCESS_MODE_UPDATE ((hseAccessMode_t) 2U)
```

UPDATE access mode

## Common Types and Definitions

### HSE\_ACCESS\_MODE\_FINISH

```
#define HSE_ACCESS_MODE_FINISH ((hseAccessMode_t) 3U)
```

FINISH access mode

### HSE\_HASH\_ALGO\_NULL

```
#define HSE_HASH_ALGO_NULL ((hseHashAlgo_t) 0U)
```

None.

### HSE\_HASH\_RESERVED1

```
#define HSE_HASH_RESERVED1 ((hseHashAlgo_t) 1U)
```

Reserved (MD5 obsolete)

### HSE\_HASH\_ALGO\_SHA\_1

```
#define HSE_HASH_ALGO_SHA_1 ((hseHashAlgo_t) 2U)
```

SHA1 hash.

### HSE\_HASH\_ALGO\_SHA2\_224

```
#define HSE_HASH_ALGO_SHA2_224 ((hseHashAlgo_t) 3U)
```

SHA2\_224 hash.

### HSE\_HASH\_ALGO\_SHA2\_256

```
#define HSE_HASH_ALGO_SHA2_256 ((hseHashAlgo_t) 4U)
```

SHA2\_256 hash.

**HSE\_HASH\_ALGO\_SHA2\_384**

```
#define HSE_HASH_ALGO_SHA2_384 ((hseHashAlgo_t) 5U)  
SHA2_384 hash.
```

**HSE\_HASH\_ALGO\_SHA2\_512**

```
#define HSE_HASH_ALGO_SHA2_512 ((hseHashAlgo_t) 6U)  
SHA2_512 hash.
```

**HSE\_HASH\_ALGO\_SHA2\_512\_224**

```
#define HSE_HASH_ALGO_SHA2_512_224 ((hseHashAlgo_t) 7U)  
SHA2_512_224 hash.
```

**HSE\_HASH\_ALGO\_SHA2\_512\_256**

```
#define HSE_HASH_ALGO_SHA2_512_256 ((hseHashAlgo_t) 8U)  
SHA2_512_256 hash.
```

**HSE\_HASH\_ALGO\_SHA3\_224**

```
#define HSE_HASH_ALGO_SHA3_224 ((hseHashAlgo_t) 9U)  
SHA3_224 hash.
```

**HSE\_HASH\_ALGO\_SHA3\_256**

```
#define HSE_HASH_ALGO_SHA3_256 ((hseHashAlgo_t) 10U)  
SHA3_256 hash.
```

## Common Types and Definitions

### HSE\_HASH\_ALGO\_SHA3\_384

```
#define HSE_HASH_ALGO_SHA3_384 ((hseHashAlgo_t)11U)
```

SHA3\_384 hash.

### HSE\_HASH\_ALGO\_SHA3\_512

```
#define HSE_HASH_ALGO_SHA3_512 ((hseHashAlgo_t)12U)
```

SHA3\_512 hash.

### HSE\_HASH\_ALGO\_MP

```
#define HSE_HASH_ALGO_MP ((hseHashAlgo_t)13U)
```

Miyaguchi-Preneel compression using AES-ECB with 128-bit key size (SHE spec support).

### HSE\_CIPHER\_ALGO\_NULL

```
#define HSE_CIPHER_ALGO_NULL ((hseCipherAlgo_t)0x00U)
```

NULL cipher.

### HSE\_CIPHER\_ALGO\_AES

```
#define HSE_CIPHER_ALGO_AES ((hseCipherAlgo_t)0x10U)
```

AES cipher.

### HSE\_CIPHER\_BLOCK\_MODE\_NULL

```
#define HSE_CIPHER_BLOCK_MODE_NULL ((hseCipherBlockMode_t)0U)
```

NULL cipher.

**HSE\_CIPHER\_BLOCK\_MODE\_CTR**

```
#define HSE_CIPHER_BLOCK_MODE_CTR ((hseCipherBlockMode_t) 1U)
```

CTR mode (AES)

**HSE\_CIPHER\_BLOCK\_MODE\_CBC**

```
#define HSE_CIPHER_BLOCK_MODE_CBC ((hseCipherBlockMode_t) 2U)
```

CBC mode (AES)

**HSE\_CIPHER\_BLOCK\_MODE\_ECB**

```
#define HSE_CIPHER_BLOCK_MODE_ECB ((hseCipherBlockMode_t) 3U)
```

ECB mode (AES)

**HSE\_CIPHER\_BLOCK\_MODE\_CFB**

```
#define HSE_CIPHER_BLOCK_MODE_CFB ((hseCipherBlockMode_t) 4U)
```

CFB mode (AES)

**HSE\_CIPHER\_BLOCK\_MODE\_OFB**

```
#define HSE_CIPHER_BLOCK_MODE_OFB ((hseCipherBlockMode_t) 5U)
```

OFB mode (AES)

**HSE\_CIPHER\_DIR\_DECRYPT**

```
#define HSE_CIPHER_DIR_DECRYPT ((hseCipherDir_t) 0U)
```

Decrypt.

## Common Types and Definitions

### HSE\_CIPHER\_DIR\_ENCRYPT

```
#define HSE_CIPHER_DIR_ENCRYPT ((hseCipherDir_t)1U)
```

Encrypt.

### HSE\_AUTH\_CIPHER\_MODE\_CCM

```
#define HSE_AUTH_CIPHER_MODE_CCM ((hseAuthCipherMode_t)0x11U)
```

CCM mode.

### HSE\_AUTH\_CIPHER\_MODE\_GCM

```
#define HSE_AUTH_CIPHER_MODE_GCM ((hseAuthCipherMode_t)0x12U)
```

GCM mode.

### HSE\_AUTH\_DIR\_VERIFY

```
#define HSE_AUTH_DIR_VERIFY ((hseAuthDir_t)0U)
```

Verify authentication tag.

### HSE\_AUTH\_DIR\_GENERATE

```
#define HSE_AUTH_DIR_GENERATE ((hseAuthDir_t)1U)
```

Generate authentication tag.

### HSE\_MAC\_ALGO\_CMAC

```
#define HSE_MAC_ALGO_CMAC ((hseMacAlgo_t)0x11U)
```

CMAC (AES)

**HSE\_MAC\_ALGO\_GMAC**

```
#define HSE_MAC_ALGO_GMAC ((hseMacAlgo_t) 0x12U)
```

GMAC (AES)

**HSE\_MAC\_ALGO\_XCBC\_MAC**

```
#define HSE_MAC_ALGO_XCBC_MAC ((hseMacAlgo_t) 0x13U)
```

XCBC MAC (AES128)

**HSE\_MAC\_ALGO\_HMAC**

```
#define HSE_MAC_ALGO_HMAC ((hseMacAlgo_t) 0x20U)
```

HMAC.

**HSE\_SIGN\_ECDSA**

```
#define HSE_SIGN_ECDSA ((hseSignSchemeEnum_t) 0x80U)
```

ECDSA signature scheme.

**HSE\_SIGN\_EDDSA**

```
#define HSE_SIGN_EDDSA ((hseSignSchemeEnum_t) 0x81U)
```

EdDSA signature scheme.

**HSE\_SIGN\_RSASSA\_PKCS1\_V15**

```
#define HSE_SIGN_RSASSA_PKCS1_V15 ((hseSignSchemeEnum_t) 0x93U)
```

RSASSA\_PKCS1\_V15 signature scheme.

## Common Types and Definitions

### HSE\_SIGN\_RSASSA\_PSS

```
#define HSE_SIGN_RSASSA_PSS ((hseSignSchemeEnum_t) 0x94U)
```

RSASSA\_PSS signature scheme.

### HSE\_RSA\_ALGO\_NO\_PADDING

```
#define HSE_RSA_ALGO_NO_PADDING ((hseRsaAlgo_t) 0x90U)
```

The input will be treated as an unsigned integer and perform a modular exponentiation of the input

### HSE\_RSA\_ALGO\_RSAES\_OAEP

```
#define HSE_RSA_ALGO_RSAES_OAEP ((hseRsaAlgo_t) 0x91U)
```

RSAES OAEP cipher.

### HSE\_RSA\_ALGO\_RSAES\_PKCS1\_V15

```
#define HSE_RSA_ALGO_RSAES_PKCS1_V15 ((hseRsaAlgo_t) 0x92U)
```

ECDSA RSAES\_PKCS1\_V15 cipher.

### HSE\_APP\_CORE0

```
#define HSE_APP_CORE0 ((hseAppCore_t) 0U)
```

Core0.

### HSE\_APP\_CORE1

```
#define HSE_APP_CORE1 ((hseAppCore_t) 1U)
```

Core1.



**HSE\_APP\_CORE2**

```
#define HSE_APP_CORE2 ((hseAppCore_t) 2U)
```

Core2.

**HSE\_APP\_CORE3**

```
#define HSE_APP_CORE3 ((hseAppCore_t) 3U)
```

Core3.

**HSE\_APP\_CORE4**

```
#define HSE_APP_CORE4 ((hseAppCore_t) 4U)
```

Core4.

**HSE\_APP\_CORE5**

```
#define HSE_APP_CORE5 ((hseAppCore_t) 5U)
```

Core5.

**HSE\_APP\_CORE6**

```
#define HSE_APP_CORE6 ((hseAppCore_t) 6U)
```

Core6.

**HSE\_APP\_CORE7**

```
#define HSE_APP_CORE7 ((hseAppCore_t) 7U)
```

Core7.

## Common Types and Definitions

### HSE\_APP\_CORE8

```
#define HSE_APP_CORE8 ((hseAppCore_t) 8U)
```

Core8.

### HSE\_APP\_CORE9

```
#define HSE_APP_CORE9 ((hseAppCore_t) 9U)
```

Core9.

### HSE\_APP\_CORE10

```
#define HSE_APP_CORE10 ((hseAppCore_t) 10U)
```

Core10.

### HSE\_APP\_CORE11

```
#define HSE_APP_CORE11 ((hseAppCore_t) 11U)
```

Core11.

### HSE\_APP\_CORE12

```
#define HSE_APP_CORE12 ((hseAppCore_t) 12U)
```

Core12.

### HSE\_APP\_CORE13

```
#define HSE_APP_CORE13 ((hseAppCore_t) 13U)
```

Core13.

**HSE\_APP\_CORE14**

```
#define HSE_APP_CORE14 ((hseAppCore_t)14U)
```

Core14.

**HSE\_APP\_CORE15**

```
#define HSE_APP_CORE15 ((hseAppCore_t)15U)
```

Core15.

**Typedef Documentation****hseMuMask\_t**

```
typedef uint8_t hseMuMask_t
```

HSE Message Unite (MU) masks.

**hseSGTOption\_t**

```
typedef uint8_t hseSGTOption_t
```

HSE Scatter-Gather Option .

Specifies if the input or output data is provided a scatter list (see [hseScatterList\\_t](#)).

Note

The remaining bit are ignored when SGT option is used.

**hseAccessMode\_t**

```
typedef uint8_t hseAccessMode_t
```

HSE access modes.

## Common Types and Definitions

### **hseHashAlgo\_t**

```
typedef uint8_t hseHashAlgo_t
```

HASH algorithm types.

### **hseCipherAlgo\_t**

```
typedef uint8_t hseCipherAlgo_t
```

Symmetric Cipher Algorithms .

### **hseCipherBlockMode\_t**

```
typedef uint8_t hseCipherBlockMode_t
```

Symmetric Cipher Block Modes.

### **hseCipherDir\_t**

```
typedef uint8_t hseCipherDir_t
```

HSE cipher direction: encryption/decryption.

### **hseAuthCipherMode\_t**

```
typedef uint8_t hseAuthCipherMode_t
```

HSE Authenticated cipher/encryption mode (only AES supported).

### **hseAuthDir\_t**

```
typedef uint8_t hseAuthDir_t
```

HSE authentication direction: generate/verify.

**hseMacAlgo\_t**

```
typedef uint8_t hseMacAlgo_t
```

HSE MAC algorithm.

**hseSignSchemeEnum\_t**

```
typedef uint8_t hseSignSchemeEnum_t
```

Signature scheme enumeration.

**hseRsaAlgo\_t**

```
typedef uint8_t hseRsaAlgo_t
```

RSA algorithm types.

**hseAppCore\_t**

```
typedef uint8_t hseAppCore_t
```

The application core IDs (that can be started). Only the IDs for the table below must be provided for a specific platform; otherwise an error will be reported.

Core assignment table:

| CoreID | S32K311 | S32K312 | S32K342 | S32K344 | S32K396 | S32K364 | S32K366 | S32K358 | S32K388 |
|--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 0      | M7_0    | M7_0    | M7_0    | M7_0    | M7_0    | M7_0    | M7_0    | M7_0    | M7_0    |
| 1      |         |         | M7_1    | M7_1    | M7_1    |         |         | M7_1    | M7_1    |
| 2      |         |         |         |         | M7_2    | M7_2    | M7_2    | M7_2    | M7_2    |
| 3      |         |         |         |         |         |         |         |         | M7_3    |

**hseSrvId\_t**

```
typedef uint32_t hseSrvId_t
```

HSE Service IDs.

## Common Types and Definitions

### **hseStreamId\_t**

```
typedef uint8_t hseStreamId_t
```

Stream ID type.

The stream ID identifies the stream to be used in streaming operations.

### **hseKeyHandle\_t**

```
typedef uint32_t hseKeyHandle_t
```

Key Handle type.

The keyHandle identifies the key catalog(byte2), group index in catalog(byte1) and key slot index (byte0)

### **hseKeyGroupIdx\_t**

```
typedef uint8_t hseKeyGroupIdx_t
```

HSE key group index.

A group represents a set of keys of the same type. Each group is identified by its index within the catalog where it is declared

### **hseKeySlotIdx\_t**

```
typedef uint8_t hseKeySlotIdx_t
```

HSE key slot index.

A key slot represent a memory container for a single key. A group contains several key slots as defined during the key configuration

### **hseNoScheme\_t**

```
typedef uint32_t hseNoScheme_t
```

No scheme (or parameters) are defined.

## 11.2 HSE Defines

### Macros

| Type: (implicit C type) |       |
|-------------------------|-------|
| Name                    | Value |

|                                   |                                       |
|-----------------------------------|---------------------------------------|
| HSE_SRV_VER_0                     | 0x00000000UL                          |
| HSE_SRV_VER_1                     | 0x01000000UL                          |
| NUM_OF_ELEMS (x)                  | sizeof(x)/sizeof((x)[0])              |
| SIZE_OF_STRING (string)           | (sizeof(string) - 1U)                 |
| HSE_BITS_TO_BYTES (bitLen)        | ((((bitLen) + 7UL) >> 3UL))           |
| HSE_BITS_TO_BYTES_UINT16 (bitLen) | (uint16_t) HSE_BITS_TO_BYTES (bitLen) |
| HSE_BYTES_TO_BITS (byteLen)       | ((byteLen) << 3UL)                    |
| HOST_ADDR                         | uint32_t                              |
| NULL_HOST_ADDR                    | ( HOST_ADDR )0UL                      |
| HSE_PTR_TO_HOST_ADDR (ptr)        | ( HOST_ADDR )(uintptr_t)(ptr)         |
| HSE_HOST_ADDR_TO_PTR (type, val)  | ((type *)(uintptr_t)(val))            |

| Type: hseDigestLen_t  |       |
|-----------------------|-------|
| Name                  | Value |
| HSE_SHA1_DIGEST_LEN   | 20U   |
| HSE_SHA224_DIGEST_LEN | 28U   |
| HSE_SHA256_DIGEST_LEN | 32U   |
| HSE_SHA384_DIGEST_LEN | 48U   |
| HSE_SHA512_DIGEST_LEN | 64U   |
| HSE_MAX_DIGEST_LEN    | 64U   |

| Type: hseAlgoCapIdx_t        |                              |
|------------------------------|------------------------------|
| Name                         | Value                        |
| HSE_CAP_IDX_RANDOM           | 0U                           |
| HSE_CAP_IDX_SHE              | 1U                           |
| HSE_CAP_IDX_AES              | 2U                           |
| HSE_CAP_IDX_XTS_AES          | 3U                           |
| HSE_CAP_IDX_AEAD_GCM         | 4U                           |
| HSE_CAP_IDX_AEAD_CCM         | 5U                           |
| HSE_CAP_IDX_RESERVED1        | 6U /* Reserved MD5 obsolete* |
| HSE_CAP_IDX_SHA1             | 7U                           |
| HSE_CAP_IDX_SHA2             | 8U                           |
| HSE_CAP_IDX_SHA3             | 9U                           |
| HSE_CAP_IDX_MP               | 10U                          |
| HSE_CAP_IDX_CMAC             | 11U                          |
| HSE_CAP_IDX_HMAC             | 12U                          |
| HSE_CAP_IDX_GMAC             | 13U                          |
| HSE_CAP_IDX_XCBC_MAC         | 14U                          |
| HSE_CAP_IDX_RSAES_NO_PADDING | 15U                          |
| HSE_CAP_IDX_RSAES_OAEP       | 16U                          |
| HSE_CAP_IDX_RSAES_PKCS1_V15  | 17U                          |
| HSE_CAP_IDX_RSASSA_PSS       | 18U                          |
| HSE_CAP_IDX_RSASSA_PKCS1_V15 | 19U                          |

## Common Types and Definitions

|                                               |     |
|-----------------------------------------------|-----|
| <a href="#">HSE_CAP_IDX_ECDH</a>              | 20U |
| <a href="#">HSE_CAP_IDX_ECDSA</a>             | 21U |
| <a href="#">HSE_CAP_IDX_EDDSA</a>             | 22U |
| <a href="#">HSE_CAP_IDX_MONTDH</a>            | 23U |
| <a href="#">HSE_CAP_IDX_CLASSIC_DH</a>        | 24U |
| <a href="#">HSE_CAP_IDX_KDF_SP800_56C</a>     | 25U |
| <a href="#">HSE_CAP_IDX_KDF_SP800_108</a>     | 26U |
| <a href="#">HSE_CAP_IDX_KDF_ANS_X963</a>      | 27U |
| <a href="#">HSE_CAP_IDX_KDF_ISO18033_KDF1</a> | 28U |
| <a href="#">HSE_CAP_IDX_KDF_ISO18033_KDF2</a> | 29U |
| <a href="#">HSE_CAP_IDX_PBKDF2</a>            | 30U |
| <a href="#">HSE_CAP_IDX_KDF_TLS12_PRF</a>     | 31U |
| <a href="#">HSE_CAP_IDX_HKDF</a>              | 32U |
| <a href="#">HSE_CAP_IDX_KDF_IKEV2</a>         | 33U |

| Type: <a href="#">hseBlockLen_t</a> |       |
|-------------------------------------|-------|
| Name                                | Value |
| <a href="#">HSE_AES_BLOCK_LEN</a>   | 16U   |

## Typedefs

- typedef uint8\_t [hseDigestLen\\_t](#)
- typedef uint8\_t [hseBlockLen\\_t](#)
- typedef uint8\_t [hseAlgoCapIdx\\_t](#)

## Macro Definition Documentation

### HSE\_SRV\_VER\_0

```
#define HSE_SRV_VER_0 (0x00000000UL)
```

HSE Service versions.

### HSE\_SRV\_VER\_1

```
#define HSE_SRV_VER_1 (0x01000000UL)
```

### NUM\_OF\_ELEMS

```
#define NUM_OF_ELEMS( x ) (sizeof(x)/sizeof((x)[0]))
```



Compute the number of elements of an array.

### SIZE\_OF\_STRING

```
#define SIZE_OF_STRING( string ) (sizeof(string) - 1U)
```

Compute the size of a string initialized with quotation marks.

### HSE\_BITS\_TO\_BYTES

```
#define HSE_BITS_TO_BYTES( bitLen ) (((bitLen) + 7UL) >> 3UL)
```

Translate bits to bytes.

### HSE\_BITS\_TO\_BYTES\_UINT16

```
#define HSE_BITS_TO_BYTES_UINT16( bitLen ) ((uint16_t) HSE_BITS_TO_BYTES (bitLen))
```

Translate bits to bytes (uint16\_t)

### HSE\_BYTES\_TO\_BITS

```
#define HSE_BYTES_TO_BITS( byteLen ) ((byteLen) << 3UL)
```

Translate bytes to bits.

### HOST\_ADDR

```
#define HOST_ADDR uint32_t
```

Host address size.

### NULL\_HOST\_ADDR

```
#define NULL_HOST_ADDR (( HOST\_ADDR ) 0UL)
```

NULL host address.

## Common Types and Definitions

### HSE\_PTR\_TO\_HOST\_ADDR

```
#define HSE_PTR_TO_HOST_ADDR( ptr ) (( HOST_ADDR ) (uintptr_t) (ptr))
```

Pointer to Host address

### HSE\_HOST\_ADDR\_TO\_PTR

```
#define HSE_HOST_ADDR_TO_PTR( type, val ) ((type *) (uintptr_t) (val))
```

Host address to Pointer

### HSE\_SHA1\_DIGEST\_LEN

```
#define HSE_SHA1_DIGEST_LEN ((hseDigestLen_t) 20U)
```

SHA1 digest length in bytes.

### HSE\_SHA224\_DIGEST\_LEN

```
#define HSE_SHA224_DIGEST_LEN ((hseDigestLen_t) 28U)
```

SHA224 digest length in bytes.

### HSE\_SHA256\_DIGEST\_LEN

```
#define HSE_SHA256_DIGEST_LEN ((hseDigestLen_t) 32U)
```

SHA256 digest length in bytes.

### HSE\_SHA384\_DIGEST\_LEN

```
#define HSE_SHA384_DIGEST_LEN ((hseDigestLen_t) 48U)
```

SHA384 digest length in bytes.

### **HSE\_SHA512\_DIGEST\_LEN**

```
#define HSE_SHA512_DIGEST_LEN ((hseDigestLen_t) 64U)
```

SHA512 digest length in bytes.

### **HSE\_MAX\_DIGEST\_LEN**

```
#define HSE_MAX_DIGEST_LEN ((hseDigestLen_t) 64U)
```

Max digest buffer in bytes.

### **HSE\_AES\_BLOCK\_LEN**

```
#define HSE_AES_BLOCK_LEN ((hseBlockLen_t) 16U)
```

AES block length in bytes

### **HSE\_CAP\_IDX\_RANDOM**

```
#define HSE_CAP_IDX_RANDOM ((hseAlgoCapIdx_t) 0U)
```

### **HSE\_CAP\_IDX\_SHE**

```
#define HSE_CAP_IDX_SHE ((hseAlgoCapIdx_t) 1U)
```

### **HSE\_CAP\_IDX\_AES**

```
#define HSE_CAP_IDX_AES ((hseAlgoCapIdx_t) 2U)
```

## Common Types and Definitions

### HSE\_CAP\_IDX\_XTS\_AES

```
#define HSE_CAP_IDX_XTS_AES ((hseAlgoCapIdx_t) 3U)
```

### HSE\_CAP\_IDX\_AEAD\_GCM

```
#define HSE_CAP_IDX_AEAD_GCM ((hseAlgoCapIdx_t) 4U)
```

### HSE\_CAP\_IDX\_AEAD\_CCM

```
#define HSE_CAP_IDX_AEAD_CCM ((hseAlgoCapIdx_t) 5U)
```

### HSE\_CAP\_IDX\_RESERVED1

```
#define HSE_CAP_IDX_RESERVED1 ((hseAlgoCapIdx_t) 6U) /* Reserved (MD5 obsolete)*/
```

### HSE\_CAP\_IDX\_SHA1

```
#define HSE_CAP_IDX_SHA1 ((hseAlgoCapIdx_t) 7U)
```

### HSE\_CAP\_IDX\_SHA2

```
#define HSE_CAP_IDX_SHA2 ((hseAlgoCapIdx_t) 8U)
```

### HSE\_CAP\_IDX\_SHA3

```
#define HSE_CAP_IDX_SHA3 ((hseAlgoCapIdx_t) 9U)
```

### HSE\_CAP\_IDX\_MP

```
#define HSE_CAP_IDX_MP ((hseAlgoCapIdx_t) 10U)
```

**HSE\_CAP\_IDX\_CMAC**

```
#define HSE_CAP_IDX_CMAC ((hseAlgoCapIdx_t)11U)
```

**HSE\_CAP\_IDX\_HMAC**

```
#define HSE_CAP_IDX_HMAC ((hseAlgoCapIdx_t)12U)
```

**HSE\_CAP\_IDX\_GMAC**

```
#define HSE_CAP_IDX_GMAC ((hseAlgoCapIdx_t)13U)
```

**HSE\_CAP\_IDX\_XCBC\_MAC**

```
#define HSE_CAP_IDX_XCBC_MAC ((hseAlgoCapIdx_t)14U)
```

**HSE\_CAP\_IDX\_RSAES\_NO\_PADDING**

```
#define HSE_CAP_IDX_RSAES_NO_PADDING ((hseAlgoCapIdx_t)15U)
```

**HSE\_CAP\_IDX\_RSAES\_OAEP**

```
#define HSE_CAP_IDX_RSAES_OAEP ((hseAlgoCapIdx_t)16U)
```

**HSE\_CAP\_IDX\_RSAES\_PKCS1\_V15**

```
#define HSE_CAP_IDX_RSAES_PKCS1_V15 ((hseAlgoCapIdx_t)17U)
```

## Common Types and Definitions

### HSE\_CAP\_IDX\_RSASSA\_PSS

```
#define HSE_CAP_IDX_RSASSA_PSS ((hseAlgoCapIdx_t)18U)
```

### HSE\_CAP\_IDX\_RSASSA\_PKCS1\_V15

```
#define HSE_CAP_IDX_RSASSA_PKCS1_V15 ((hseAlgoCapIdx_t)19U)
```

### HSE\_CAP\_IDX\_ECDH

```
#define HSE_CAP_IDX_ECDH ((hseAlgoCapIdx_t)20U)
```

### HSE\_CAP\_IDX\_ECDSA

```
#define HSE_CAP_IDX_ECDSA ((hseAlgoCapIdx_t)21U)
```

### HSE\_CAP\_IDX\_EDDSA

```
#define HSE_CAP_IDX_EDDSA ((hseAlgoCapIdx_t)22U)
```

### HSE\_CAP\_IDX\_MONTDH

```
#define HSE_CAP_IDX_MONTDH ((hseAlgoCapIdx_t)23U)
```

### HSE\_CAP\_IDX\_CLASSIC\_DH

```
#define HSE_CAP_IDX_CLASSIC_DH ((hseAlgoCapIdx_t)24U)
```

### HSE\_CAP\_IDX\_KDF\_SP800\_56C

```
#define HSE_CAP_IDX_KDF_SP800_56C ((hseAlgoCapIdx_t)25U)
```

**HSE\_CAP\_IDX\_KDF\_SP800\_108**

```
#define HSE_CAP_IDX_KDF_SP800_108 ((hseAlgoCapIdx_t)26U)
```

**HSE\_CAP\_IDX\_KDF\_ANS\_X963**

```
#define HSE_CAP_IDX_KDF_ANS_X963 ((hseAlgoCapIdx_t)27U)
```

**HSE\_CAP\_IDX\_KDF\_ISO18033\_KDF1**

```
#define HSE_CAP_IDX_KDF_ISO18033_KDF1 ((hseAlgoCapIdx_t)28U)
```

**HSE\_CAP\_IDX\_KDF\_ISO18033\_KDF2**

```
#define HSE_CAP_IDX_KDF_ISO18033_KDF2 ((hseAlgoCapIdx_t)29U)
```

**HSE\_CAP\_IDX\_PBKDF2**

```
#define HSE_CAP_IDX_PBKDF2 ((hseAlgoCapIdx_t)30U)
```

**HSE\_CAP\_IDX\_KDF\_TLS12\_PRF**

```
#define HSE_CAP_IDX_KDF_TLS12_PRF ((hseAlgoCapIdx_t)31U)
```

**HSE\_CAP\_IDX\_HKDF**

```
#define HSE_CAP_IDX_HKDF ((hseAlgoCapIdx_t)32U)
```

## Common Types and Definitions

### HSE\_CAP\_IDX\_KDF\_IKEV2

```
#define HSE_CAP_IDX_KDF_IKEV2 ((hseAlgoCapIdx_t) 33U)
```

## Typedef Documentation

### hseDigestLen\_t

```
typedef uint8_t hseDigestLen_t
```

### hseBlockLen\_t

```
typedef uint8_t hseBlockLen_t
```

### hseAlgoCapIdx\_t

```
typedef uint8_t hseAlgoCapIdx_t
```

The capabilities indices for each enabled algorithm.



## 12 Features Implementation

### 12.1 HSE Platform

#### Macros

| Type: (implicit C type)     |       |
|-----------------------------|-------|
| Name                        | Value |
| <a href="#">HSE_B</a>       | -     |
| <a href="#">HSE_SPT_CMU</a> | -     |

#### Macro Definition Documentation

##### HSE\_B

```
#define HSE_B
```

##### HSE\_SPT\_CMU

```
#define HSE_SPT_CMU
```

Support CMU tamper for HSE on CMU0 (clock integrity verification)

### 12.2 HSE Basic Features

#### Macros

| Type: (implicit C type)                       |       |
|-----------------------------------------------|-------|
| Name                                          | Value |
| <a href="#">HSE_SPT_INTERNAL_FLASH_DEV</a>    | -     |
| <a href="#">HSE_SPT_RANDOM</a>                | -     |
| <a href="#">HSE_SPT_SHE</a>                   | -     |
| <a href="#">HSE_SPT_AES</a>                   | -     |
| <a href="#">HSE_SPT_CIPHER_BLOCK_MODE_CFB</a> | -     |
| <a href="#">HSE_SPT_CIPHER_BLOCK_MODE_CTR</a> | -     |
| <a href="#">HSE_SPT_CIPHER_BLOCK_MODE_ECB</a> | -     |
| <a href="#">HSE_SPT_CIPHER_BLOCK_MODE_OFB</a> | -     |
| <a href="#">HSE_SPT_AEAD_GCM</a>              | -     |
| <a href="#">HSE_SPT_AEAD_CCM</a>              | -     |
| <a href="#">HSE_SPT_HASH</a>                  | -     |
| <a href="#">HSE_SPT_SHA1</a>                  | -     |
| <a href="#">HSE_SPT_SHA2_224</a>              | -     |

## Features Implementation

|                               |   |
|-------------------------------|---|
| HSE_SPT_SHA2_256              | - |
| HSE_SPT_SHA2_384              | - |
| HSE_SPT_SHA2_512              | - |
| HSE_SPT_SHA2_512_224          | - |
| HSE_SPT_SHA2_512_256          | - |
| HSE_SPT_SHA3                  | - |
| HSE_SPT_MIYAGUCHI_PRENEEL     | - |
| HSE_SPT_MAC                   | - |
| HSE_SPT_FAST_CMAC             | - |
| HSE_SPT_CMAC                  | - |
| HSE_SPT_HMAC                  | - |
| HSE_SPT_GMAC                  | - |
| HSE_SPT_RSA                   | - |
| HSE_SPT_RSAES_NO_PADDING      | - |
| HSE_SPT_RSAES_OAEP            | - |
| HSE_SPT_RSAES_PKCS1_V15       | - |
| HSE_SPT_RSASSA_PSS            | - |
| HSE_SPT_RSASSA_PKCS1_V15      | - |
| HSE_SPT_ECC                   | - |
| HSE_SPT_ECDH                  | - |
| HSE_SPT_ECDSA                 | - |
| HSE_SPT_EDDSA                 | - |
| HSE_SPT_MONTDH                | - |
| HSE_SPT_ECC_USER_CURVES       | - |
| HSE_SPT_EC_SEC_SECP256R1      | - |
| HSE_SPT_EC_SEC_SECP384R1      | - |
| HSE_SPT_EC_SEC_SECP521R1      | - |
| HSE_SPT_EC_BRAINPOOL_         | - |
| BRAINPOOLP256R1               |   |
| HSE_SPT_EC_BRAINPOOL_         | - |
| BRAINPOOLP320R1               |   |
| HSE_SPT_EC_BRAINPOOL_         | - |
| BRAINPOOLP384R1               |   |
| HSE_SPT_EC_BRAINPOOL_         | - |
| BRAINPOOLP512R1               |   |
| HSE_SPT_EC_25519_ED25519      | - |
| HSE_SPT_EC_25519_CURVE25519   | - |
| HSE_SPT_KEY_GEN               | - |
| HSE_SPT_SYM_RND_KEY_GEN       | - |
| HSE_SPT_ECC_KEY_PAIR_GEN      | - |
| HSE_SPT_RSA_KEY_PAIR_GEN      | - |
| HSE_SPT_TLS12_RSA_PRE_MASTER_ | - |
| SECRET_GEN                    |   |
| HSE_SPT_KEY_DERIVE            | - |
| HSE_SPT_KDF_NXP_GENERIC       | - |

|                                           |       |
|-------------------------------------------|-------|
| HSE_SPT_KDF_SP800_56C_ONESTEP             | -     |
| HSE_SPT_KDF_SP800_56C_TWOSTEP             | -     |
| HSE_SPT_KDF_SP800_108                     | -     |
| HSE_SPT_KDF_ANS_X963                      | -     |
| HSE_SPT_KDF_ISO18033_KDF1                 | -     |
| HSE_SPT_KDF_ISO18033_KDF2                 | -     |
| HSE_SPT_PBKDF2                            | -     |
| HSE_SPT_KDF_TLS12_PRF                     | -     |
| HSE_SPT_HKDF                              | -     |
| HSE_SPT_NXP_ROM_KEYS                      | -     |
| HSE_SPT_NXP_ROM_RSA_PUB_KEYS              | -     |
| HSE_SPT_NXP_ROM_ECC_PUB_KEYS              | -     |
| HSE_SPT_FORMAT_KEY_CATALOGS               | -     |
| HSE_SPT_GET_KEY_INFO                      | -     |
| HSE_SPT_KEY_VERIFY                        | -     |
| HSE_SPT_IMPORT_KEY                        | -     |
| HSE_SPT_EXPORT_KEY                        | -     |
| HSE_SPT_KEY_MGMT_POLICIES                 | -     |
| HSE_SPT_PUBLISH_NVM_KEYSTORE_RAM_TO_FLASH | -     |
| HSE_SPT_MONOTONIC_COUNTERS                | -     |
| HSE_NUM_OF_MONOTONIC_COUNTERS             | 16U   |
| HSE_NVM_SPT_MONOTONIC_CNT_NO_OF_SECTORS   | 1U    |
| HSE_SPT_NXP_KEY_STORE_NO_OF_SECTORS       | 1U    |
| HSE_SECTOR_SIZE                           | 8192U |
| HSE_SPT_BOOTDATASIGN                      | -     |
| HSE_SPT_BSB                               | -     |
| HSE_SPT_SMR_CR                            | -     |
| HSE_NUM_OF_SMR_ENTRIES                    | 8U    |
| HSE_NUM_OF_CORE_RESET_ENTRIES             | 4U    |
| HSE_SPT_SMR_DECRYPT                       | -     |
| HSE_SPT_STREAM_CTX_IMPORT_EXPORT          | -     |
| HSE_SPT_MU_CONFIG                         | -     |
| HSE_SPT_CUST_SEC_POLICY                   | -     |
| HSE_SPT_OEM_SEC_POLICY                    | -     |
| HSE_SPT_PHYSICAL_TAMPER_CONFIG            | -     |
| HSE_NUM_OF_PHYSICAL_TAMPER_INSTANCES      | 1U    |
| HSE_SPT_MEM_REGION_PROTECT                | -     |
| HSE_MAX_NUM_OF_MEM_REGIONS                | 6U    |
| HSE_SPT_OTA_FIRMWARE_UPDATE               | -     |
| HSE_SPT_OTA_SBAF_UPDATE                   | -     |
| HSE_SPT_FW_INTEGRITY_CHECK                | -     |

## Features Implementation

|                                      |                                                                                   |
|--------------------------------------|-----------------------------------------------------------------------------------|
| HSE_SPT_ERASE_FW                     | -                                                                                 |
| HSE_SPT_SGT_OPTION                   | -                                                                                 |
| HSE_MAX_NUM_OF_SGT_ENTRIES           | 8U                                                                                |
| HSE_NUM_OF_MU_INSTANCES              | 2U                                                                                |
| HSE_NUM_OF_CHANNELS_PER_MU           | 4U                                                                                |
| HSE_STREAM_COUNT                     | 2U                                                                                |
| HSE_NUM_OF_USER_ECC_CURVES           | 1U                                                                                |
| HSE_TOTAL_NUM_OF_KEY_GROUPS          | 32U                                                                               |
| HSE_MAX_NVM_STORE_SIZE               | $7768U + ((HSE\_SPT\_NXP\_KEY\_STORE\_NO\_OF\_SECTORS - 1U) * HSE\_SECTOR\_SIZE)$ |
| HSE_MAX_RAM_STORE_SIZE               | 6144U                                                                             |
| HSE_AES_KEY_BITS_LENS                | {128U, 192U, 256U}                                                                |
| HSE_MAX_SHARED_SECRET_BITS_LEN       | 4096U                                                                             |
| HSE_MIN_HMAC_KEY_BITS_LEN            | 128U                                                                              |
| HSE_MAX_HMAC_KEY_BITS_LEN            | 1024U                                                                             |
| HSE_MIN_ECC_KEY_BITS_LEN             | 192U                                                                              |
| HSE_MAX_ECC_KEY_BITS_LEN             | 521U                                                                              |
| HSE_MIN_RSA_KEY_BITS_LEN             | 1024U                                                                             |
| HSE_MAX_RSA_KEY_BITS_LEN             | 4096U                                                                             |
| HSE_MAX_RSA_PUB_EXP_SIZE             | 8U                                                                                |
| HSE_DEFAULT_MIN_FAST_CMAC_TAG_BITLEN | 32U                                                                               |
| HSE_SPT_SIGN                         | -                                                                                 |
| HSE_SPT_AEAD                         | -                                                                                 |
| HSE_SPT_COMPUTE_DH                   | -                                                                                 |
| HSE_SPT_SHA2                         | -                                                                                 |

## Macro Definition Documentation

### HSE\_SPT\_INTERNAL\_FLASH\_DEV

```
#define HSE_SPT_INTERNAL_FLASH_DEV
```

Device has internal flash.

### HSE\_SPT\_RANDOM

```
#define HSE_SPT_RANDOM
```

Support for Random Number Generation.

**HSE\_SPT\_SHE**

```
#define HSE_SPT_SHE
```

Support for SHE specification.

Note

AES and CMAC features must be enabled.

**HSE\_SPT\_AES**

```
#define HSE_SPT_AES
```

Support for AES\_(128, 192, 256)\_(ECB, CBC, CFB, OFB, CTR). AES-CBC is supported on all platforms by default.

**HSE\_SPT\_CIPHER\_BLOCK\_MODE\_CFB**

```
#define HSE_SPT_CIPHER_BLOCK_MODE_CFB
```

AES-CFB cipher mode supported.

**HSE\_SPT\_CIPHER\_BLOCK\_MODE\_CTR**

```
#define HSE_SPT_CIPHER_BLOCK_MODE_CTR
```

AES-CTR cipher mode supported.

**HSE\_SPT\_CIPHER\_BLOCK\_MODE\_ECB**

```
#define HSE_SPT_CIPHER_BLOCK_MODE_ECB
```

AES-ECB cipher mode supported.

**HSE\_SPT\_CIPHER\_BLOCK\_MODE\_OFB**

```
#define HSE_SPT_CIPHER_BLOCK_MODE_OFB
```

## Features Implementation

AES-OFB cipher mode supported.

### **HSE\_SPT\_AEAD\_GCM**

```
#define HSE_SPT_AEAD_GCM
```

Support for AEAD AES GCM as defined in FIPS PUB 197, NIST SP 800-38D, RFC-5288 and RFC-4106.

### **HSE\_SPT\_AEAD\_CCM**

```
#define HSE_SPT_AEAD_CCM
```

Support for AEAD AES CCM as defined in FIPS PUB 197, NIST SP 800-38C, RFC-6655 and RFC-4309.

### **HSE\_SPT\_HASH**

```
#define HSE_SPT_HASH
```

Hash support.

### **HSE\_SPT\_SHA1**

```
#define HSE_SPT_SHA1
```

Support for SHA-1 as defined in FIPS PUB 180-4.

### **HSE\_SPT\_SHA2\_224**

```
#define HSE_SPT_SHA2_224
```

Support for SHA2\_224 in FIPS PUB 180-4.

### **HSE\_SPT\_SHA2\_256**

```
#define HSE_SPT_SHA2_256
```

Support for SHA2\_256 in FIPS PUB 180-4.

### **HSE\_SPT\_SHA2\_384**

```
#define HSE_SPT_SHA2_384
```

Support for SHA2\_384 in FIPS PUB 180-4. Scatter gather feature is not supported.

### **HSE\_SPT\_SHA2\_512**

```
#define HSE_SPT_SHA2_512
```

Support for SHA2\_512 in FIPS PUB 180-4. Scatter gather feature is not supported.

### **HSE\_SPT\_SHA2\_512\_224**

```
#define HSE_SPT_SHA2_512_224
```

Support for SHA2\_512\_224 in FIPS PUB 180-4. Scatter gather feature is not supported.

### **HSE\_SPT\_SHA2\_512\_256**

```
#define HSE_SPT_SHA2_512_256
```

Support for SHA2\_512\_256 in FIPS PUB 180-4. Scatter gather feature is not supported.

### **HSE\_SPT\_SHA3**

```
#define HSE_SPT_SHA3
```

Support for SHA3\_(224, 256, 384, 512) as defined in FIPS PUB 202.

### **HSE\_SPT\_MIYAGUCHI\_PRENEEL**

```
#define HSE_SPT_MIYAGUCHI_PRENEEL
```

Miyaguchi-Preneel compression function (SHE spec support)

## Features Implementation

### HSE\_SPT\_MAC

```
#define HSE_SPT_MAC
```

MAC support.

### HSE\_SPT\_FAST\_CMACE

```
#define HSE_SPT_FAST_CMACE
```

Support for AES fast CMACE (optimized)

### HSE\_SPT\_CMACE

```
#define HSE_SPT_CMACE
```

Support for AES CMACE as defined in NIST SP 800-38B.

### HSE\_SPT\_HMAC

```
#define HSE_SPT_HMAC
```

Support for HMAC\_SHA1 and HMAC\_SHA2\_(224, 256, 384, 512) as defined in FIPS PUB 198-1 and SP 800-107.

### HSE\_SPT\_GMAC

```
#define HSE_SPT_GMAC
```

Support for AES GMACE as defined in NIST SP 800-38D.

### HSE\_SPT\_RSA

```
#define HSE_SPT_RSA
```

RSA support.



**HSE\_SPT\_RSAES\_NO\_PADDING**

```
#define HSE_SPT_RSAES_NO_PADDING
```

RSA modular exponentiation operations( RSAEP and RSADP).

**HSE\_SPT\_RSAES\_OAEP**

```
#define HSE_SPT_RSAES_OAEP
```

Support for RSAES\_OAEP as defined by RFC-8017.

**HSE\_SPT\_RSAES\_PKCS1\_V15**

```
#define HSE_SPT_RSAES_PKCS1_V15
```

Support for RSAES\_PKCS1\_V15 as defined by PKCS#1 v2.2.

**HSE\_SPT\_RSASSA\_PSS**

```
#define HSE_SPT_RSASSA_PSS
```

Support for RSASSA\_PSS as defined by FIPS 186-4.

**HSE\_SPT\_RSASSA\_PKCS1\_V15**

```
#define HSE_SPT_RSASSA_PKCS1_V15
```

Support RSASSA\_PKCS1\_V15 as defined by PKCS#1 v2.2.

**HSE\_SPT\_ECC**

```
#define HSE_SPT_ECC
```

Support for ECC.

## Features Implementation

### HSE\_SPT\_ECDH

```
#define HSE_SPT_ECDH
```

ECDH support.

### HSE\_SPT\_ECDSA

```
#define HSE_SPT_ECDSA
```

ECDSA support.

### HSE\_SPT\_EDDSA

```
#define HSE_SPT_EDDSA
```

Twisted Edwards EDDSA (e.g. ED25519) support.

### HSE\_SPT\_MONTDH

```
#define HSE_SPT_MONTDH
```

Montgomery DH (e.g X25519 curve) support.

### HSE\_SPT\_ECC\_USER\_CURVES

```
#define HSE_SPT_ECC_USER_CURVES
```

Support to set ECC curve (not supported by default)

### HSE\_SPT\_EC\_SEC\_SECP256R1

```
#define HSE_SPT_EC_SEC_SECP256R1
```

Support Ecc p256v1.

**HSE\_SPT\_EC\_SEC\_SECP384R1**

```
#define HSE_SPT_EC_SEC_SECP384R1
```

Support Ecc SECP p384r1.

**HSE\_SPT\_EC\_SEC\_SECP521R1**

```
#define HSE_SPT_EC_SEC_SECP521R1
```

Support Ecc SECP p521r1.

**HSE\_SPT\_EC\_BRAINPOOL\_BRAINPOOLP256R1**

```
#define HSE_SPT_EC_BRAINPOOL_BRAINPOOLP256R1
```

Support Ecc BrainPool p256r1.

**HSE\_SPT\_EC\_BRAINPOOL\_BRAINPOOLP320R1**

```
#define HSE_SPT_EC_BRAINPOOL_BRAINPOOLP320R1
```

Support Ecc BrainPool p320r1.

**HSE\_SPT\_EC\_BRAINPOOL\_BRAINPOOLP384R1**

```
#define HSE_SPT_EC_BRAINPOOL_BRAINPOOLP384R1
```

Support Ecc BrainPool p384r1.

**HSE\_SPT\_EC\_BRAINPOOL\_BRAINPOOLP512R1**

```
#define HSE_SPT_EC_BRAINPOOL_BRAINPOOLP512R1
```

Support Ecc BrainPool p521r1.

## Features Implementation

### **HSE\_SPT\_EC\_25519\_ED25519**

```
#define HSE_SPT_EC_25519_ED25519
```

Twisted Edwards ED25519 curve support (used with EdDSA )

### **HSE\_SPT\_EC\_25519\_CURVE25519**

```
#define HSE_SPT_EC_25519_CURVE25519
```

Montgomery X25519 curve support (used with MONTDH)

### **HSE\_SPT\_KEY\_GEN**

```
#define HSE_SPT_KEY_GEN
```

Key Generate support.

### **HSE\_SPT\_SYM\_RND\_KEY\_GEN**

```
#define HSE_SPT_SYM_RND_KEY_GEN
```

Support for symmetric random key generation.

### **HSE\_SPT\_ECC\_KEY\_PAIR\_GEN**

```
#define HSE_SPT_ECC_KEY_PAIR_GEN
```

Support for ECC key-pair generation.

### **HSE\_SPT\_RSA\_KEY\_PAIR\_GEN**

```
#define HSE_SPT_RSA_KEY_PAIR_GEN
```

Support for RSA key-pair generation.

**HSE\_SPT\_TLS12\_RSA\_PRE\_MASTER\_SECRET\_GEN**

```
#define HSE_SPT_TLS12_RSA_PRE_MASTER_SECRET_GEN
```

Support for RSA key exchange.

**HSE\_SPT\_KEY\_DERIVE**

```
#define HSE_SPT_KEY_DERIVE
```

KDF support.

**HSE\_SPT\_KDF\_NXP\_GENERIC**

```
#define HSE_SPT_KDF_NXP_GENERIC
```

NXP Generic KDF.

**HSE\_SPT\_KDF\_SP800\_56C\_ONESTEP**

```
#define HSE_SPT_KDF_SP800_56C_ONESTEP
```

Support for KDF One-step as defined by SP800-56C rev1. HMAC and XCBC\_MAC options as PRF algorithm are not supported.

**HSE\_SPT\_KDF\_SP800\_56C\_TWOSTEP**

```
#define HSE_SPT_KDF_SP800_56C_TWOSTEP
```

Support for KDF Two-step as defined by SP800-56C rev1. HMAC and XCBC\_MAC options as PRF algorithm are not supported.

**HSE\_SPT\_KDF\_SP800\_108**

```
#define HSE_SPT_KDF_SP800_108
```

Support for KDF(Counter, Feedback, Pipeline) as defined by SP800-108. Only counter mode is supported. HMAC and XCBC\_MAC options as PRF algorithm are not supported.

## Features Implementation

### **HSE\_SPT\_KDF\_ANS\_X963**

```
#define HSE_SPT_KDF_ANS_X963
```

Support for KDF as defined by ANS X9.63. HMAC and XCBC\_MAC options as PRF algorithm are not supported.

### **HSE\_SPT\_KDF\_ISO18033\_KDF1**

```
#define HSE_SPT_KDF_ISO18033_KDF1
```

Support for KDF1 as defined by ISO18033. HMAC and XCBC\_MAC options as PRF algorithm are not supported.

### **HSE\_SPT\_KDF\_ISO18033\_KDF2**

```
#define HSE_SPT_KDF_ISO18033_KDF2
```

Support for KDF2 as defined by ISO18033. HMAC and XCBC\_MAC options as PRF algorithm are not supported.

### **HSE\_SPT\_PBKDF2**

```
#define HSE_SPT_PBKDF2
```

Support for PBKDF2 as defined as defined by PKCS#5 v2.1 and RFC-8018. Support of HMAC is provided with Hash algos SHA-1, SHA-224, and SHA-256.

### **HSE\_SPT\_KDF\_TLS12\_PRF**

```
#define HSE_SPT_KDF_TLS12_PRF
```

KDF Support for TLS 1.2 as defined by RFC-5246. Support of HMAC is provided with Hash algos SHA-1, SHA-224, and SHA-256.

### **HSE\_SPT\_HKDF**

```
#define HSE_SPT_HKDF
```

Support for HMAC-based Extract-and-Expand KDF as defined by RFC-5869. Support of HMAC is provided with Hash algos SHA-1, SHA-224, and SHA-256.

### **HSE\_SPT\_NXP\_ROM\_KEYS**

```
#define HSE_SPT_NXP_ROM_KEYS
```

Support NXP ROM keys.

### **HSE\_SPT\_NXP\_ROM\_RSA\_PUB\_KEYS**

```
#define HSE_SPT_NXP_ROM_RSA_PUB_KEYS
```

Support NXP RSA ROM public keys.

### **HSE\_SPT\_NXP\_ROM\_ECC\_PUB\_KEYS**

```
#define HSE_SPT_NXP_ROM_ECC_PUB_KEYS
```

Support NXP ECC ROM public keys.

### **HSE\_SPT\_FORMAT\_KEY\_CATALOGS**

```
#define HSE_SPT_FORMAT_KEY_CATALOGS
```

Support Format Key Catalogs service.

### **HSE\_SPT\_GET\_KEY\_INFO**

```
#define HSE_SPT_GET_KEY_INFO
```

Support Get Key Info Service.

## Features Implementation

### **HSE\_SPT\_KEY\_VERIFY**

```
#define HSE_SPT_KEY_VERIFY
```

Support Key Verify Service.

### **HSE\_SPT\_IMPORT\_KEY**

```
#define HSE_SPT_IMPORT_KEY
```

Support Import Key Service.

### **HSE\_SPT\_EXPORT\_KEY**

```
#define HSE_SPT_EXPORT_KEY
```

Support Export Key Service.

### **HSE\_SPT\_KEY\_MGMT\_POLICIES**

```
#define HSE_SPT_KEY_MGMT_POLICIES
```

Support Key Management configurable policies.

### **HSE\_SPT\_PUBLISH\_NVM\_KEYSTORE\_RAM\_TO\_FLASH**

```
#define HSE_SPT_PUBLISH_NVM_KEYSTORE_RAM_TO_FLASH
```

Support Publishing the keystore from RAM to data flash.

### **HSE\_SPT\_MONOTONIC\_COUNTERS**

```
#define HSE_SPT_MONOTONIC_COUNTERS
```

Monotonic Counter support.



**HSE\_NUM\_OF\_MONOTONIC\_COUNTERS**

```
#define HSE_NUM_OF_MONOTONIC_COUNTERS (16U)
```

The supported number of monotonic counters.

**HSE\_NVM\_SPT\_MONOTONIC\_CNT\_NO\_OF\_SECTORS**

```
#define HSE_NVM_SPT_MONOTONIC_CNT_NO_OF_SECTORS (1U)
```

The supported number of passive sectors used by monotonic counters.

**HSE\_SPT\_NXP\_KEY\_STORE\_NO\_OF\_SECTORS**

```
#define HSE_SPT_NXP_KEY_STORE_NO_OF_SECTORS (1U)
```

The supported number of active sectors used by NXP key store.

**HSE\_SECTOR\_SIZE**

```
#define HSE_SECTOR_SIZE (8192U)
```

The size of Flash Sector in S32K3xx Devices.

**HSE\_SPT\_BOOTDATASIGN**

```
#define HSE_SPT_BOOTDATASIGN
```

Boot Data Sign Support

**HSE\_SPT\_BSB**

```
#define HSE_SPT_BSB
```

Basic Secure Booting(ASB) Support

## Features Implementation

### HSE\_SPT\_SMR\_CR

```
#define HSE_SPT_SMR_CR
```

Advance Secure Booting(ASB) Secure memory regions verification (SMR) & Core Reset(CR) Table Support.

### HSE\_NUM\_OF\_SMR\_ENTRIES

```
#define HSE_NUM_OF_SMR_ENTRIES (8U)
```

The supported number of SMR entries.

### HSE\_NUM\_OF\_CORE\_RESET\_ENTRIES

```
#define HSE_NUM_OF_CORE_RESET_ENTRIES (4U)
```

The supported number of CORE RESET entries.

### HSE\_SPT\_SMR\_DECRYPT

```
#define HSE_SPT_SMR_DECRYPT
```

Support encrypted SMRs.

### HSE\_SPT\_STREAM\_CTX\_IMPORT\_EXPORT

```
#define HSE_SPT_STREAM_CTX_IMPORT_EXPORT
```

Support Import/Export of streaming context for symmetric operations.

### HSE\_SPT\_MU\_CONFIG

```
#define HSE_SPT_MU_CONFIG
```

Support MU configuration.

**HSE\_SPT\_CUST\_SEC\_POLICY**

```
#define HSE_SPT_CUST_SEC_POLICY
```

Support of Customer Security Policy.

**HSE\_SPT\_OEM\_SEC\_POLICY**

```
#define HSE_SPT_OEM_SEC_POLICY
```

Support of Oem Security Policy.

**HSE\_SPT\_PHYSICAL\_TAMPER\_CONFIG**

```
#define HSE_SPT_PHYSICAL_TAMPER_CONFIG
```

Support of active tamper.

**HSE\_NUM\_OF\_PHYSICAL\_TAMPER\_INSTANCES**

```
#define HSE_NUM_OF_PHYSICAL_TAMPER_INSTANCES (1U)
```

Number of Physical Tamper Instances.

**HSE\_SPT\_MEM\_REGION\_PROTECT**

```
#define HSE_SPT_MEM_REGION_PROTECT
```

Support memory region protection.

**HSE\_MAX\_NUM\_OF\_MEM\_REGIONS**

```
#define HSE_MAX_NUM_OF_MEM_REGIONS (6U)
```

Maximum number of memory regions configurable through [HSE\\_SPT\\_MEM\\_REGION\\_PROTECT](#) service.

## Features Implementation

### **HSE\_SPT\_OTA\_FIRMWARE\_UPDATE**

```
#define HSE_SPT_OTA_FIRMWARE_UPDATE
```

Support OTA Firmware Update.

### **HSE\_SPT\_OTA\_SBAF\_UPDATE**

```
#define HSE_SPT_OTA_SBAF_UPDATE
```

Support SBAF update.

### **HSE\_SPT\_FW\_INTEGRITY\_CHECK**

```
#define HSE_SPT_FW_INTEGRITY_CHECK
```

Support HSE flash memory integrity check.

### **HSE\_SPT\_ERASE\_FW**

```
#define HSE_SPT_ERASE_FW
```

Erase Active HSE FW, Backup (if present) and SYS-IMG from internal secure flash memory in CUST\_DEL lifecycle.

### **HSE\_SPT\_SGT\_OPTION**

```
#define HSE_SPT_SGT_OPTION
```

Enable support for Scatter Gatter Table.

### **HSE\_MAX\_NUM\_OF\_SGT\_ENTRIES**

```
#define HSE_MAX_NUM_OF_SGT_ENTRIES (8U)
```

Maximum number for SGT entries.

**HSE\_NUM\_OF\_MU\_INSTANCES**

```
#define HSE_NUM_OF_MU_INSTANCES (2U)
```

The maxim number of MU interfaces.

**HSE\_NUM\_OF\_CHANNELS\_PER\_MU**

```
#define HSE_NUM_OF_CHANNELS_PER_MU (4U)
```

The maxim number of channels per MU interface

**HSE\_STREAM\_COUNT**

```
#define HSE_STREAM_COUNT (2U)
```

HSE stream count per interface.

**HSE\_NUM\_OF\_USER\_ECC\_CURVES**

```
#define HSE_NUM_OF_USER_ECC_CURVES (1U)
```

The number of ECC curves the user can load into the HSE.

**HSE\_TOTAL\_NUM\_OF\_KEY\_GROUPS**

```
#define HSE_TOTAL_NUM_OF_KEY_GROUPS (32U)
```

The total number of catalog configuration entries for both NVM and RAM catalogs.

**HSE\_MAX\_NVM\_STORE\_SIZE**

```
#define HSE_MAX_NVM_STORE_SIZE (7768U + (( HSE_SPT_NXP_KEY_STORE_NO_OF_SECTORS -  
1U) * HSE_SECTOR_SIZE))
```

NVM key store size (in bytes)

## Features Implementation

### HSE\_MAX\_RAM\_STORE\_SIZE

```
#define HSE_MAX_RAM_STORE_SIZE (6144U)
```

RAM key store size (in bytes)

### HSE\_AES\_KEY\_BITS\_LEN

```
#define HSE_AES_KEY_BITS_LEN {128U, 192U, 256U}
```

AES key bit length (set to zero to disable a AES key size)

### HSE\_MAX\_SHARED\_SECRET\_BITS\_LEN

```
#define HSE_MAX_SHARED_SECRET_BITS_LEN (4096U)
```

Max shared secret bit length.

### HSE\_MIN\_HMAC\_KEY\_BITS\_LEN

```
#define HSE_MIN_HMAC_KEY_BITS_LEN (128U)
```

Min HMAC key bit length

### HSE\_MAX\_HMAC\_KEY\_BITS\_LEN

```
#define HSE_MAX_HMAC_KEY_BITS_LEN (1024U)
```

Max HMAC key bit length

### HSE\_MIN\_ECC\_KEY\_BITS\_LEN

```
#define HSE_MIN_ECC_KEY_BITS_LEN (192U)
```

Min ECC key bit length

### **HSE\_MAX\_ECC\_KEY\_BITS\_LEN**

```
#define HSE_MAX_ECC_KEY_BITS_LEN (521U)
```

Max ECC key bit length.

### **HSE\_MIN\_RSA\_KEY\_BITS\_LEN**

```
#define HSE_MIN_RSA_KEY_BITS_LEN (1024U)
```

Min RSA key bit length.

### **HSE\_MAX\_RSA\_KEY\_BITS\_LEN**

```
#define HSE_MAX_RSA_KEY_BITS_LEN (4096U)
```

Max RSA key bit length

### **HSE\_MAX\_RSA\_PUB\_EXP\_SIZE**

```
#define HSE_MAX_RSA_PUB_EXP_SIZE (8U)
```

Max RSA public exponent size (in bytes)

### **HSE\_DEFAULT\_MIN\_FAST\_CMACE\_TAG\_BITLEN**

```
#define HSE_DEFAULT_MIN_FAST_CMACE_TAG_BITLEN (32U)
```

FAST CMACE default min bit length.

### **HSE\_SPT\_SIGN**

```
#define HSE_SPT_SIGN
```

## Features Implementation

### HSE\_SPT\_AEAD

```
#define HSE_SPT_AEAD
```

### HSE\_SPT\_COMPUTE\_DH

```
#define HSE_SPT_COMPUTE_DH
```

### HSE\_SPT\_SHA2

```
#define HSE_SPT_SHA2
```







### **How to Reach Us:**

#### **Home Page:**

[nxp.com](http://nxp.com)

#### **Web Support:**

[nxp.com/support](http://nxp.com/support)

Information in this document is provided solely to enable system and software implementers to use NXP products. There are no express or implied copyright licenses granted hereunder to design or fabricate any integrated circuits based on the information in this document. NXP reserves the right to make changes without further notice to any products herein.

NXP makes no warranty, representation, or guarantee regarding the suitability of its products for any particular purpose, nor does NXP assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation consequential or incidental damages. "Typical" parameters that may be provided in NXP data sheets and/or specifications can and do vary in different applications, and actual performance may vary over time. All operating parameters, including "typicals", must be validated for each customer application by customer's technical experts. NXP does not convey any license under its patent rights nor the rights of others. NXP sells products pursuant to standard terms and conditions of sale, which can be found at the following address:

[nxp.com/SalesTermsandConditions](http://nxp.com/SalesTermsandConditions).

NXP, the NXP logo, NXP SECURE CONNECTIONS FOR A SMARTER WORLD, COOLFLUX, EMBRACE, GREENCHIP, HITAG, I2C BUS, ICODE, JCOP, LIFE VIBES, MIFARE, MIFARE CLASSIC, MIFARE DESFire, MIFARE PLUS, MIFARE FLEX, MANTIS, MIFARE ULTRALIGHT, MIFARE4MOBILE, MIGLO, NTAG, ROADLINK, SMARTLX, SMARTMX, STARPLUG, TOPFET, TRENCHMOS, UCODE, Freescale, the Freescale logo, Altivec, C-5, CodeTEST, CodeWarrior, ColdFire, ColdFire+, C-Ware, the Energy Efficient Solutions logo, Kinetis, Layerscape, MagniV, mobileGT, PEG, PowerQUICC, Processor Expert, QorIQ, QorIQ Qonverge, Ready Play, SafeAssure, the SafeAssure logo, StarCore, Symphony, VortiQa, Vybrid, Airfast, BeeKit, BeeStack, CoreNet, Flexis, MXC, Platform in a Package, QUICC Engine, SMARTMOS, Tower, TurboLink, and UMEMS are trademarks of NXP B.V. All other product or service names are the property of their respective owners.

ARM, AMBA, ARM Powered, Artisan, Cortex, Jazelle, Keil, SecurCore, Thumb, TrustZone, and  $\mu$  Vision are registered trademarks of ARM Limited (or its subsidiaries) in the EU and/or elsewhere. ARM7, ARM9, ARM11, big.LITTLE, CoreLink, CoreSight, DesignStart, Mali, mbed, NEON, POP, Sensinode, Socrates, ULINK and Versatile are trademarks of ARM Limited (or its subsidiaries) in the EU and/or elsewhere. All rights reserved. Oracle and Java are registered trademarks of Oracle and/or its affiliates. The Power Architecture and Power.org word marks and the Power and Power.org logos and related marks are trademarks and service marks licensed by Power.org.